

30 Hour General Industry Outreach Study Guide



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Module 1: Introduction to OSHA

Module Description

The Occupational Safety and Health Administration (OSHA) was established to protect the health of the American workers. In 1971, the Occupational Safety and Health Act was created to give structure to the worker protection activities. OSHA holds the employer responsible for providing a workplace that is free from recognized hazards. This module will give an understanding of OSHA, employer responsibilities, and employee rights in the workplace.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Explain why OSHA is important to workers.
- Explain worker rights under OSHA.
- Discuss employer responsibilities under OSHA.
- Discuss the use of OSHA standards.
- Explain how OSHA inspections are conducted.
- Utilize helpful worker safety and health resources.

Lesson 1: Introduction to OSHA

Lesson Focus

This lesson focuses on the following topics:

- OSHA's Mission
- State Plans
- OSHA Standards
- Employer Responsibilities
- Workers' Rights & Responsibilities
- Enforcing Standards
- Reporting Safety Hazards
- Whistleblower Protections
- Worker Resources

OSHA's Mission

History of OSHA

OSHA stands for the Occupational Safety and Health Administration, an agency of the U.S. Department of Labor. OSHA's responsibility is worker safety and health protection. The U.S. Congress created OSHA under the Occupational Safety and Health Act of 1970 (the OSH Act). Congress passed the law and established OSHA "to assure so far as possible every working man and woman in the nation safe and healthful working conditions and to preserve our human resources."

The current mission of OSHA is "to save lives, prevent injuries, and protect the health of America's workers". It is the employer's responsibility to keep the workplace free from any known or recognized hazard that is likely to cause injury or illness to their worker.



Job Safety and Health IT'S THE LAW!

All workers have the right to:

- A safe workplace.
- Raise a safety or health concern with your employer or OSHA, or report a work-related injury or illness, without being retaliated against.
- Receive information and training on job hazards, including all hazardous substances in your workplace.
- Request a confidential OSHA inspection of your workplace if you believe there are unsafe or unhealthy conditions. You have the right to have a representative contact OSHA on your behalf.
- Participate (or have your representative participate) in an OSHA inspection and speak in private to the inspector.
- File a complaint with OSHA within 30 days (by phone, online or by mail) if you have been retaliated against for using your rights.
- See any OSHA citations issued to your employer.
- Request copies of your medical records, tests that measure hazards in the workplace, and the workplace injury and illness log.

This poster is available free from OSHA.

Employers must:

- Provide employees a workplace free from recognized hazards. It is illegal to retaliate against an employee for using any of their rights under the law, including raising a health and safety concern with you or with OSHA, or reporting a work-related injury or illness.
- Comply with all applicable OSHA standards.
- Notify OSHA within 8 hours of a workplace fatality or within 24 hours of any work-related inpatient hospitalization, amputation, or loss of an eye.
- Provide required training to all workers in a language and vocabulary they can understand.
- Prominently display this poster in the workplace.
- Post OSHA citations at or near the place of the alleged violations.

On-Site Consultation services are available to small and medium-sized employers, without citation or penalty, through OSHA-supported consultation programs in every state.



Contact OSHA. We can help.

1-800-321-OSHA (6742) • TTY 1-877-889-5627 • www.osha.gov

<https://www.osha.gov/Publications/poster.html>

The OSH Act is also known as Public Law 91-596. It covers all private sector employers and their workers in the 50 states and all territories and jurisdictions under federal authority. Employers and workers in many fields, including but not limited to manufacturing, construction, long shoring, agriculture, law, medicine, charity and disaster relief are covered by OSHA. Religious groups are covered if they employ workers for secular purposes, such as maintenance or gardening.

Who is Covered by the OSH Act?

- OSHA covers all employees and their employers in the 50 states and certain territories and jurisdictions under federal government authority. Those jurisdictions include the District of Columbia, Puerto Rico, the Virgin Islands, American Samoa, Guam, Northern Mariana Islands, Wake Island, Johnston Island, and the Outer Continental Shelf Lands as defined in the Outer Continental Shelf Lands Act.
- Coverage is provided either directly by federal OSHA or through an OSHA-approved state program.
- Section 19 of the OSH Act makes federal agency heads responsible for providing safe and healthful working conditions for their employees. OSHA conducts federal workplace inspections in response to employee reports of hazards.
- The OSH Act also requires federal agencies to comply with standards consistent with those for private sector employers. Under a 1998 amendment to the Act, it covers the U.S. Postal Service the same as any private sector employer.

Which Groups Do Not Come Under OSHA's Coverage?

Those groups that are not covered by OSHA include:

- The self-employed
- Immediate members of farming families not employing outside workers
- Mine workers, certain truckers and transportation workers, and atomic energy workers who are covered by other federal agencies
- Public employees in state and local governments, although some states have their own plans that cover these workers

More Information:

OSHA provisions cover the private sector only. However, some federal agencies have created their own health and safety programs, which are at least as stringent as U.S. OSHA. These state programs cover state and local government employees.

OSHA does not cover the self-employed or immediate members of farm families that do not employ outside workers; worker conditions that are regulated under worker safety or health requirements of other federal agencies; or employees of state and local governments, although some states have their own occupational safety and health plans that cover these workers.

To achieve this, federal and state governments work together with more than 100 million working men and women and eight million employers. Some of the things OSHA does to carry out its mission are:

- Developing job safety and health standards and enforcing them through worksite inspections
- Maintaining a reporting and recordkeeping system to keep track of job-related injuries and illnesses
- Providing training programs to increase knowledge about occupational safety and health

OSHA also assists the States in their efforts to assure safe and healthful working conditions, through OSHA-approved job safety and health programs operated by individual states. State plans are OSHA-approved job safety and health programs created by individual states instead of federal OSHA.

States with approved plans cover most private sector employees as well as state and local government workers in the state. State plan programs respond to accidents and employee complaints and conduct unannounced inspections, just like federal OSHA. And, some states have OSHA-approved plans that cover only state and local government workers.

State Plans

Safety and Health Programs

State plans are OSHA-approved job safety and health programs operated by individual states instead of federal OSHA. The OSH Act encourages states to develop and operate their own job safety and health plans and precludes state enforcement of OSHA standards, unless the state has an approved plan. OSHA approves and monitors all state plans. The state plans must be at least as effective as federal OSHA requirements.

State plans covering the private sector also must cover state and local government employees. OSHA rules also permit states and territories to develop plans that cover only public sector (state and local government) employees. In these cases, private sector employment remains under federal OSHA jurisdiction. Twenty-two states and territories operate complete plans and six cover only the public sector.

For more information, see also: <http://www.osha.gov/dcsp/osp/index.html>

OSHA's Impact

Since OSHA's creation in 1970, the nation has made substantial progress in occupational safety and health. OSHA and its many partners in the public and private sectors have, for example:

- Cut the work-related fatality rate to historic lows for 2002 to 2004.

- From 2003 to 2004, reduced the number of workplace injuries and illnesses by 4 percent and lost workday case rates dropped by 5.8 percent in that same period.
- In 2005, OSHA conducted close to 39,000 inspections and issued just over 85,000 citations for violations.
- In 2004, the Consultation Program made over 31,000 visits to employers.

OSHA Standards

OSHA standards are rules that describe the methods that employers must use to protect their employees from hazards. There are OSHA standards for Construction work, Agriculture, Maritime operations, and General Industry, which are the standards that apply to most worksites. These standards limit the amount of hazardous chemicals workers can be exposed to, require the use of certain safe practices and equipment, and require employers to monitor hazards and keep records of workplace injuries and illnesses.

Examples of OSHA standards include, but are not limited to 29 CFR 1910, 1926, and 1928 (OSHA standards are [online](#)). General Duty Clause, Section 5(a) (1) explain that this is used when there is not a specific OSHA standard that applies to the situation.

Establishing a safe and healthful workplace requires every employer to make safety and health a priority. In general, OSHA requires employers to:

- Maintain conditions and adopt practices reasonably necessary to protect workers on the job. The first and best strategy is to control the hazard at its source. Engineering controls do this, unlike other controls that generally focus on the worker who is exposed to the hazard. The basic concept behind engineering controls is that, to the extent feasible, the work environment and the job itself should be designed to eliminate hazards or reduce exposure to hazards.
- Be familiar with the standards that apply to their workplaces, and comply with these standards.

- Ensure that workers are provided with, and use, personal protective equipment, when needed. When exposure to hazards cannot be engineered completely out of normal operations or maintenance work, and when safe work practices and other forms of administrative controls cannot provide sufficient additional protection, an additional method of control may be the use of protective clothing or equipment. This is collectively called personal protective equipment, or PPE. PPE may also be appropriate for controlling hazards while engineering and work practice controls are being installed.
- Comply with the OSH Act's "General Duty Clause" where no specific standards apply. The general duty clause, or Section 5(a)(1) of the Act requires each employer to "furnish a place of employment which is free from recognized hazards that are causing or are likely to cause death or serious physical harm to employees."

An example of the OSHA standard is 29 CFR 1910. 95 Occupational Noise Exposure. Noise in the workplace can cause adverse effects to the worker's hearing when the noise reaches over 85db for an 8 hour Time Weighted Average (TWA). The damage from continuous loud noise exposure is permanent. OSHA requires that hearing conservations programs are used for workplaces that exceed the 85 dB TWA and for employers to provide Personal Protective Equipment at no cost to the workers. Earplugs/Earmuffs must be used to reduce noise hazards in the workplace.

Though PPE is provided for workers at no cost to them, the employer should look for engineering controls such as sound proofing or using sound barriers to reduce the noise. Additionally, the standard requires that the worker will have annual Audiometric Testing and training regarding occupational noise. Other standards address issues such as:

- Chemicals
- Protective equipment
- Fall protection
- Guarding of open sided platform
- Scaffolding
- Right to know

Employer Responsibilities

OSHA standards mandate that employers must:

- Provide a workplace free from recognized hazards and comply with OSHA standards
- Provide training required by OSHA standards
- Keep records of injuries and illnesses
- Set up a reporting system;
 - Provide copies of logs (i.e., OSHA 300), upon request;
 - Post the annual summary;
 - Report within 8 hours any work-related fatalities and within 24 hours, all work-related: inpatient hospitalizations, amputations, and losses of an eye.
- Provide medical exams when required by OSHA standards and provide workers access to their exposure and medical records
- Not discriminate against workers who exercise their rights under the Act
- Post OSHA citations and abatement verification notices
- Provide and pay for most Personal Protective Equipment (PPE)



HANDOUT #2
Employers Must Provide and Pay for PPE



Personal Protective Equipment (PPE)

The Occupational Safety and Health Administration (OSHA) requires that employers protect you from workplace hazards that can cause injury or illness. Controlling a hazard at its source is the best way to protect workers. However, when engineering, work practice and administrative controls are not feasible or do not provide sufficient protection, employers must provide personal protective equipment (PPE) to you and ensure its use.

PPE is equipment worn to minimize exposure to a variety of hazards. Examples include items such as gloves, foot and eye protection, protective hearing protection (earplugs, muffs), hard hats and respirators.

Employer Obligations	Workers should:
✓ Performing a "hazard assessment" of the workplace to identify and control physical and health hazards.	✓ Properly wear PPE
✓ Identifying and providing appropriate PPE for employees.	✓ Attend training sessions on PPE
✓ Training employees in the use and care of the PPE.	✓ Care for, clean and maintain PPE, and
✓ Maintaining PPE, including replacing worn or damaged PPE.	✓ Inform a supervisor of the need to repair or replace PPE.
✓ Periodically reviewing, updating and evaluating the effectiveness of the PPE program.	

Employers Must Pay for Personal Protective Equipment (PPE)

On May 15, 2008, a new OSHA rule about employer payment for PPE went into effect. With few exceptions, OSHA now requires employers to pay for personal protective equipment used to comply with OSHA standards. The final rule does not create new requirements regarding what PPE employers must provide.

The standard makes clear that employers cannot require workers to provide their own PPE and the worker's use of PPE they already own must be completely voluntary. Even when a worker provides his or her own PPE, the employer must ensure that the equipment is adequate to protect the worker from hazards at the workplace.



Examples of PPE that Employers Must Pay for Include:

- Metatarsal foot protection
- Rubber boots with steel toes
- Non-prescription eye protection
- Prescription eyewear inserts/lenses for full face respirators
- Goggles and face shields
- Fire fighting PPE (helmet, gloves, boots, proximity suits, full gear)
- Hard hats
- Hearing protection
- Welding PPE



Payment Exceptions under the OSHA Rule

Employers are not required to pay for some PPE in certain circumstances:

- Non-specialty safety-toe protective footwear (including steel-toe shoes or boots) and non-specialty prescription safety eyewear provided that the employer permits such items to be worn off the job site. (OSHA based this decision on the fact that this type of equipment is very personal, is often used outside the workplace, and that it is taken by workers from jobsite to jobsite and employer to employer.)
- Everyday clothing, such as long-sleeve shirts, long pants, street shoes, and normal work boots.
- Ordinary clothing, skin creams, or other items, used solely for protection from weather, such as winter coats, jackets, gloves, parkas, rubber boots, hats, raincoats, ordinary sunglasses, and sunscreen
- Items such as hair nets and gloves worn by food workers for consumer safety.
- Lifting belts because their value in protecting the back is questionable.
- When the employee has lost or intentionally damaged the PPE and it must be replaced.

OSHA Standards that Apply

OSHA General Industry PPE Standards

- 1910.132: General requirements and payment
- 1910.133: Eye and face protection
- 1910.134: Respiratory protection
- 1910.135: Head protection
- 1910.136: Foot protection
- 1910.137: Electrical protective devices
- 1910.138: Hand protection

OSHA Construction PPE Standards

- 1926.28: Personal protective equipment
- 1926.95: Criteria for personal protective equipment
- 1926.96: Occupational foot protection
- 1926.100: Head protection
- 1926.101: Hearing protection
- 1926.102: Eye and face protection
- 1926.103: Respiratory protection

There are also PPE requirements in shipyards and marine terminals and many standards on specific hazards, such as 1910.1030: Bloodborne pathogens and 1910.146: Permit-required confined spaces.

OSHA standards are online at www.osha.gov.

Workers' Rights & Responsibilities

Workers' Rights

Most importantly, the creation of OSHA provided workers the right to a safe and healthful workplace. Section 5(a)(1) of the OSH Act states: "Each employer shall furnish to each of his employees' employment and a place of employment which are free from recognized hazards that are causing or are likely to cause death or serious physical harm to his employees."

A safe and healthful workplace means that hazards are removed and workers are trained. If a hazard cannot be removed completely, protection (for example, respirators or earplugs) must be provided.

OSHA Provides Workers the Right to:	
WORKERS RIGHTS	A safe and healthful workplace
	Know about hazardous conditions
	Information about injuries and illnesses in your workplace
	Complain or request hazard correction from employer
	Training as provided in the OSHA standards
	Hazard exposure and medical records
	File a complaint with OSHA
	Participate in an OSHA inspection
	Be free from retaliation for exercising safety and health rights

Workers' Rights: Right to Know About Hazardous Chemicals

Another important right is the Right to Know about hazardous substances in your workplace. Employers must have a written, complete hazard communication program that includes information on:

- Container labeling
- Safety Data Sheets (SDSs)

Worker training must include the physical and health hazards of the chemicals and how workers can protect themselves, including specific procedures the employer has implemented to protect workers, such as work practices, emergency procedures, and personal protective equipment

The program must also include a list of the hazardous chemicals in each work area and the means the employer uses to inform workers of the hazards of non-routine tasks. In addition, the program must explain how the employer will inform other employers of hazards to which their workers may be exposed (for example, contract workers).

Safety Data Sheets

Safety data sheets provide invaluable information about hazardous substances. SDS contain the following sections:

- Section 1—Identification of the Substance or Mixture and of the Supplier
- Section 2—Hazard(s) Identification
- Section 3—Composition and Information on Ingredients
- Section 4—First Aid Measures
- Section 5—Fire-Fighting Measures
- Section 6—Accidental Release Measures
- Section 7—Handling and Storage
- Section 8—Exposure Controls and Personal Protection
- Section 9—Physical and Chemical Properties
- Section 10—Stability and Reactivity
- Section 11—Toxicological Information
- Section 12—Ecological Information (Non-Mandatory)
- Section 13—Disposal Considerations (Non-Mandatory)
- Section 14—Transport Information (Non-Mandatory)
- Section 15—Regulatory Information
- Section 16—Other Information

Workers' Rights: Right to Information about Injuries and Illnesses in your Workplace

OSHA's Recordkeeping rule requires most employers with more than 10 workers to keep a log of injuries and illnesses. The log, which is also called the OSHA 300, must contain all work-related injuries and illnesses resulting in lost workdays, restricted work or transfer to another job, as well as any incident requiring more than first aid treatment.

You have the right to review the current log, as well as the logs stored for the past 5 years. The employer must provide this by the end of the next workday. The names and other information on the log may not be removed, unless the case is a "privacy concern case."

You also have the right to view the annually posted summary of the injuries and illnesses (OSHA 300A).

More Information: "Privacy concern cases" are those involving an intimate body part, mental illness, HIV, etc. For more detail, see 1904.29(b)(7).

The right to review the log includes former employees, their personal representatives, and authorized employee representatives.

Workers' Rights: Anti-Discrimination Provisions

The OSH Act prohibits employment retaliation against an employee who complains to an employer regarding a workplace safety issue or condition, files a complaint related to workplace safety or health conditions, initiates a proceeding, contests an abatement date, requests information from OSHA, or testifies under the Act. In certain circumstances, an employee may refuse to work under seriously threatening health or safety conditions.

Workers' Rights: Right to Training

You have a right to get training from your employer on a variety of health and safety hazards and standards that your employer must follow. We've already discussed the training required under OSHA's Hazard Communication (Right to Know) standard. Other required training may include lockout-tagout, bloodborne pathogens, noise, confined spaces, fall hazards in construction, personal protective equipment, and a variety of other subjects.

Some examples of a safe and healthful workplace when training is in place include:

- Trenches are inspected and have appropriate protective systems in place when necessary.
- Proper confined space entry procedures, testing, equipment is present and use appropriately.
- Noise levels are controlled. When levels remain at unsafe levels, workers are given hearing tests and are provided training and hearing protection.
- Protection from chemical hazards is provided, including an evaluation of chemicals used, a written program, Safety Data Sheets, worker protection (for example, respirators or gloves), and information and training.

Workers' Rights: Right to Hazardous Exposure Records and Medical Record

Under OSHA's standard 1910.1020, you have the right to examine and copy exposure and medical records, including records of workplace monitoring or measuring a toxic substance. This is important if you have been exposed to toxic substances or harmful physical agents in the workplace, as this regulation may help you detect, prevent, and treat occupational disease.

Examples of toxic substances and harmful physical agents are:

- Some metals and dusts, such as, lead, cadmium, and silica
- Biological agents, such as bacteria, viruses, and fungi
- Physical stress, such as noise, heat, cold, vibration, repetitive motion, and ionizing and non-ionizing radiation

OSHA standards require employers to measure exposure to potentially harmful substances, and workers or their representatives have the right to observe the testing and examine the results. If the exposure levels are above the limit set by the standard, the employer must tell workers what will be done to reduce their exposure.

Workers' Rights: Right to File a Complaint with OSHA

You may file a complaint with OSHA if you believe a violation of a safety or health standard or an imminent danger situation exists in your workplace. You may request that your name not be revealed to your employer. You can file a complaint on OSHA's web site, in writing or by telephone, to the nearest OSHA area office. You may also call the office and speak with an OSHA compliance officer about a hazard, violation, or the process for filing a complaint. You can contact OSHA by calling 1-800-321-OSHA.

If you file a complaint, you have the right to find out OSHA's action on the complaint and request a review if an inspection is not made.

Workers' Rights: Right to Participate in an OSHA Inspection

If an OSHA inspection is conducted in your workplace, you have the right to have your representative accompany the inspector on the inspection.

You also have the right to talk to the inspector privately. You may point out hazards, describe injuries, illnesses or near misses that resulted from those hazards, and describe any concerns you have about a safety or health issue.

You also have the right to find out about inspection results and abatement measures, and get involved in any meetings or hearings related to the inspection. You may also object to the date set for the violation to be corrected and be notified if the employer files a contest.

Contest: If an employer disagrees with the results of the OSHA inspection, he or she may submit a written objection to OSHA, called a Notice of Contest.

Enforcing Standards

OSHA enforces standards through inspections. The current budget is small —all together nationwide (with our State partners) there are only about 2,400 inspectors for over 7 million workplaces. At this rate, it would take about 100 years for us to inspect every workplace once.

Therefore, OSHA targets the most dangerous workplaces; industries with fatalities and serious injuries (e.g. grain handling in Colorado) and construction (i.e., falls).

The OSHA inspection process consists of an opening conference, a walkthrough and a closing conference with the employer. Results can take up to 6 months, after which OSHA may issue citations. These may include fines and will include dates by which hazard must be abated.

When an OSHA inspection is conducted in the workplace, workers have the right to have a worker representative accompany the inspector on the inspection. Workers can talk to the inspector privately. They may point out hazards, describe injuries, illnesses or near misses that resulted from those hazards and describe any concern you have about a safety or health issue. Workers also can find out about inspection results and abatement measures, and get involved in any meetings or hearings related to the inspection. Workers may also object to the date set for the violation to be corrected and be notified if the employer files a contest.

Reporting Safety Hazards

Workers have a right to seek safety and health on the job without fear of punishment. That right is spelled out in Section 11(c) of the OSH Act. The law says the employer shall not punish or discriminate against employees for exercising such rights as complaining to the employer, union, OSHA, or any other government agency about safety and health hazards. Workers are also protected for participation in OSHA inspections, conferences, hearings, and other OSHA-related activities.

Workers also have the right to refuse to do a job if they believe in good faith that they are exposed to an imminent danger. "Good faith" means that even if an imminent danger is not found to exist, the worker had reasonable grounds to believe that it did exist. Since the conditions necessary to justify a work refusal are very stringent, refusing work should be an action taken as a last resort.

If time permits, the condition should be reported to OSHA or the appropriate government agency. If you believe you have been punished for exercising your safety and health rights, you must contact OSHA within 30 days in order to preserve the protections provided under the OSHA Act. The *How OSHA Responds to a Complaint* table provides information on how OSHA responds to the way a complaint is reported.

Ways to Report Workplace Hazards	
Contact 	Contact your team leader, supervisor, manager, safety committee, etc.
Phone 	Call the Regional or local office or 1-800-321-OSHA. [Provide your local office number.]
Fax or Mail 	Obtain the complaint form online at www.osha.gov or from the local OSHA office.
In Person 	Visit the local OSHA Office. [A current list of OSHA offices is at www.osha.gov]
Online 	File complaints online at the OSHA website at www.osha.gov

How OSHA Responds to a Complaint	
<i>If the Complaint is...</i>	<i>Then OSHA will...</i>
Filed over the phone, is not signed, or is not a serious hazard	Contact the employer by phone, fax, or email
About a serious hazard	Conduct an inspection at the worksite
Written, signed and submitted to the OSHA area or State Plan office	Most likely, conduct an onsite inspection
Submitted online	Handle most complaints using the OSHA phone/fax system, which means they may even be resolved by phone

Whistleblower Protections

The OSH Act protects workers who complain to their employer, OSHA or other government agencies about unsafe or unhealthful working conditions in the workplace or environmental problems. Workers cannot be transferred, denied a raise, have their hours reduced, be fired, or punished in any other way because they exercised any right given to them under the OSHA Act. Help is available from OSHA for whistleblowers.

OSHA recommend that the following message be posted for OSHA outreach students:

If you have been punished or discriminated against for using your rights, you must file a complaint with OSHA within 30 days of the alleged reprisal for most complaints. No form is required, but you must send a letter or call the OSHA Area Office nearest you to report the discrimination within 30 days of the alleged discrimination.

Handout [3] https://www.osha.gov/OshDoc/data_General_Facts/whistleblower_rights.pdf

Worker Resources

There are many resources available to workers who want to find out more information about a safety or health issues both inside and outside of their workplace.

Because of the rights provided to workers, they can utilize some sources inside the workplace, such as:

- Employer or supervisor, co-workers and union representatives;
- Safety Data Sheet (SDS);
- Labels and warning signs; and/or
- Employee orientation manuals or other training materials.

Handout [4] <https://www.osha.gov/Publications/3334we-can-help-sm.pdf>

Samples of outside of the workplace worker safety and health resources are:

- Many at OSHA online at: www.osha.gov such as, OSHA Compliance Assistance Specialists in the area offices.
- NIOSH is OSHA's sister agency. Workers can request NIOSH conduct a Health Hazard Evaluations (HHEs) of workplaces in cases where workers are getting sick from an unknown cause or are exposed to an agent or working condition that is not regulated by OSHA.
- Health care providers can be a resource on the health effects of toxic substances, proper medical and first aid treatment, and other health-related issues.

Lesson Summary

In 1971, the Occupational Safety and Health Act was created to give structure to the worker protection activities. OSHA holds the employer responsible for providing a workplace that is free from recognized hazards.

The current mission of OSHA is "to save lives, prevent injuries, and protect the health of America's workers". It is the employer's responsibility to keep the workplace free from any known or recognized hazard that is likely to cause injury or illness to their worker.

Workers have the right to know what hazardous materials they are required to work around. Workers must have a way to report hazardous conditions, injuries and illnesses.

Employers must provide training for the workers to inform them on hazard recognition and hazard control. Additional to training, the employers must have medical screening and monitoring when employees are exposed to certain hazards.

Module 2: Managing Safety and Health

Module Description

This module begins with an overview of accident costs, direct and indirect. Then we explore the elements and processes of safety programs and worksite analysis. The job hazard analysis is commonly used for training workers as to how to perform a job in the safest manner possible. Hazard recognition and the hierarchy of hazard control are also discussed.

Next, the module explains how to understand accident causation and the various accident theories. If you learn the cause and effect relationship of accident causation and the human element, then you can reduce injuries and illnesses. The final lesson concludes with an overview of incident investigation techniques.

Module Learning Objectives

At the conclusion of this module, you will be able to:

- Examine the direct cost of an accident
- Evaluate the indirect cost of an accident
- Describe the true cost of an accident related to human error factors
- Summarize the components of a safety program
- Explain how hazards are identified, prevented, and controlled
- Discuss safety training and program improvement
- Analyze worksites and tasks and identify hazards
- Identify the hierarchy of hazard control
- Introduce accident causation theories
- Explain how unsafe behaviors contribute to accidents
- Examine the role of a safety management system in preventing accidents and incidents
- Identify incident investigation techniques to find a root cause of an event

Lesson 1: Accident Costs and Prevention

Lesson Focus

This lesson focuses on the following topics:

- Accident Costs
- Safety Programs
- Worksite Analysis
- Hazard Recognition
- Hierarchy of Hazard Control

Accident Costs

Many accidents are expensive when considering lost time events. However, there are many more cost factors related to accidents that can be direct or indirect. To evaluate the total cost of the accident, you must combine both of these costs.

In order for the cost to be paid, the organization must use the profits of the company. All profits are derived after the operational costs of the company have been calculated. Accidents affect the organization's profitability because the costs of the accidents must be paid from increased revenue. A company's profit margin is calculated by **Profit Margin=Total Profits/Total Sales**. The revenue required for funds to offset an injury are: **Revenue Required=Total Cost of Incident/Profit Margin**.

Additionally, the Business Roundtable publication, Improving Construction Safety Performance, published a study conducted by Stanford University Department of Civil Engineering that provides an indirect cost estimator for accidents as it relates to direct costs of an accident.

Direct Cost of an Accident

The direct costs of an accident are directly associated with the event and are easily quantifiable. Most direct costs are paid by the insurance company of the employer. Examples of some of these costs are:

- Physical therapy
- Medical expenses
- Repair fees for damaged equipment
- Increase in workers' compensation premium
- Continuation of pay
- Compensatory damages

Indirect Cost of an Accident

The indirect costs of an accident are not paid through the insurance and therefore are unrecoverable. While the direct costs are easy to quantify, the indirect costs are often unseen or impossible to quantify. The relationship between direct cost and indirect cost is the indirect cost is greater than the direct cost for the company. Examples of indirect cost are:

- Wages paid to injured workers for absences not covered by Worker's Compensation;
- Lost wages and work stoppage associated with the worker injury;
- Over time due to the accident;
- Administrative costs and time spent by safety personnel, clerical workers, and other employees after the injury;
- Training for replacement workers;
- Lost productivity due to the work unit separation from the injury;
- New employee learning curve;
- Accommodation of the injured employee within the organization;
- Clean up, preparation and replacement cost of damaged material, machinery, and property.

The National Council Compensation Insurance, Inc. (NCCI) collected statistics and data from insurance claims between policy periods 2011 through 2013. This data was incorporated by OSHA in the Safety Pays cost estimator for accidents. The NCCI manages the nation's largest database of workers compensation insurance information.

Indirect costs have a measurable relationship to the direct cost of accidents, which was discovered by the business Roundtable publication, improving construction safety performance, and the Stanford University Department of civil engineering. The magnitude of indirect costs is inversely related to the severity of an accident. Using these numbers, OSHA created the [Safety Pays Cost Calculator](#).

Cost Estimate Calculator

The following chart will give the relationship between direct costs and indirect costs in a ratio that is used to calculate the total accident cost.

Direct Costs	Indirect Cost Ratio
\$0-\$2,999	4.5
\$3000-\$4,999	1.6
\$5,000-\$9,999	1.2
\$10,000 or more	1.1

To use the cost estimate calculator, you need to know either the injury type or the workers compensation direct cost of an accident and the company's profit margin. However, if the profit margin is not known to the officer using the cost estimator then OSHA will give a default 3% profit margin for calculation of the direct and indirect costs in relation to the additional sales needed to pay total cost of the accident.

As an example of this calculator, if you select an amputation from the entry type menu and give the company a profit margin of 10%, the calculator will give you the following information:

- Amputation: (1) Instance

- Direct Cost: \$77,995
- Indirect Cost: \$85,794
- Total Cost: \$163,789
- Additional Sales needed to recuperate cost (Indirect Cost): \$857,945
- Additional Sales needed to recuperate cost (Total Cost); \$1,637,890

Therefore, this example illustrates that just one instance of an amputation through most likely an at-risk behavior has historically cost \$77,995 of direct costs and \$85,794 of indirect costs yielding a total injury cost of \$163,789.

Organizations will have to use funds from the profits to cover both direct and indirect cost of the accident. However, up to \$77,995 can be recoupable through workers' compensation; then sales must increase \$2,859,816 to cover the indirect costs of the injury that they cannot get reimbursed at a 3% profit margin.

Safety Programs

The components for an injury and illness program include:

- Management Leadership
- Worker Participation
- Hazard Identification & Assessment
- Hazard Prevention & Control
- Education & Training
- Program Evaluation & Improvement
- Communication & Coordination for Host Employers, Contractors, and Staffing Agencies

Management Leadership

Leaders are all considered managers in some form or fashion. A committed management unit provides clearly defined objectives and goals for organizational safety behavior. They finance the safety activities through purchases and resource allocations.

Every level of management value safety practices and accomplishments as much as regulatory compliance and water quality.

Steps to implement leadership commitment to safety are:

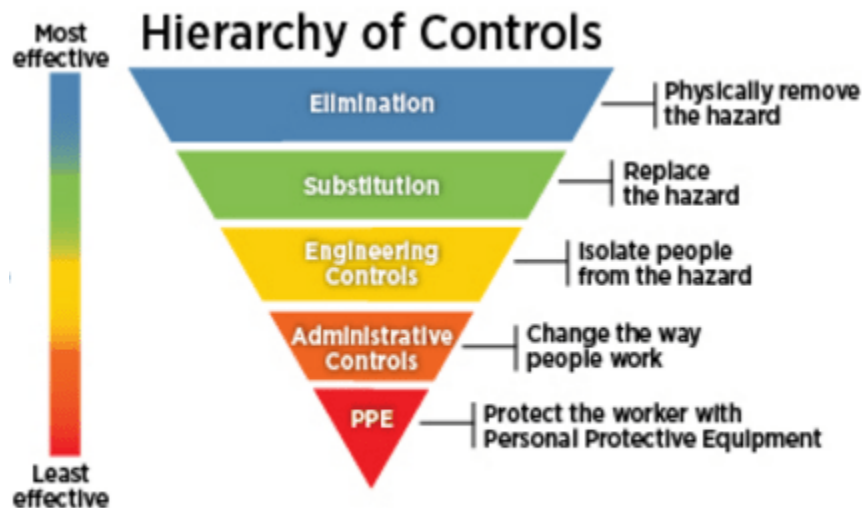
- Writing or personally signing a clearly defined safety policy that acknowledges safety and health as important as productivity, water quality, regulatory compliance, and customer service.
- Communicating the policy and values to all levels of the organization
- Visually set examples of safety behavior and demonstrate actions consistent with a safety culture.
- Allocate resources for safety and health
- Hold all levels of the organization accountable for safety performance

Hazard Identification & Assessment

A hazard is any condition or action that can cause an organizational loss. An organizational loss can come in the form of an injury, illness, damaged equipment, or even worker turnover. When a loss occurs, the organization must determine the root cause of the loss and not just the symptoms leading to the loss event. The assessment process must be structured, detailed, and deliver actionable measures to address the root cause. Hazard identification and assessment can be accomplished by:

- Worksite analysis of past, present, and predictive data from reports, instrumentation and maintenance logs, even worker injury and illness records
- Worksite inspections for safety hazards
- Investigate each accident until the root cause is completely disclosed
- Identify hazards that may arise outside of normal operating conditions including emergencies, start-up, or shut-down operations.
- Characterize the true composition of a hazard, give a priority value to them, and identify appropriate hazard controls

Hazard Prevention & Control



Source: NIOSH

The prevention and control of hazards protect the worker from injury and illness, but also give employees a clear sign that the company cares about their wellbeing.

Elimination of hazards is the best way to avoid an organizational loss. However, that may not be possible in all situations. Therefore, hazard control is appropriate for some hazards that are still present when workers are performing their daily tasks. Although some worksites are practicing substitution of highly hazardous chemicals such as gas chlorine to liquid chlorine, it is mostly because they are trying to avoid the Risk Management Program, regulated by the EPA, and not primarily worker safety.

The hierarchy of hazard controls after elimination and substitution are:

1. Engineering (Physical barrier or device such as a machine guard)
2. Administrative (Work rule such as work rotation)
3. PPE (Protection worn by workers as a barrier to hazards such as a hardhat)

Tips for implementing hazard prevention and controls are as follows:

- Identifying what controls are available for each type of hazard

- Selecting the proper controls by doing a detailed hazard assessment
- Develop, maintain, and update a hazard control plan
- Select controls that are applicable for all aspects of the organization and conditions
- Implement the selected hazard controls with a priority on elimination and substitution of hazards
- Follow up on all hazard controls for each task to make sure they are protective enough

Education & Training

Education and training can be thought of as a tool that binds each step together to keep the efforts cohesive. Some companies have relied on safety training from organizations or even video tapes with outdated material. The role of education and training must be a factor in developing both management and workers to meet the overall safety culture. General workers should have safety awareness training with regular operations or maintenance training. However, if they work in a specialized area that exposes them to unique hazards, then training must be applicable to that hazard. Effective training can be done peer-to-peer, formal classrooms, online, or at the worksite. Some **actions items**

OSHA suggests are:

- Provide program awareness training
- Train employers, managers, supervisors on their individual safety roles
- Train workers on their specific role in the safety program
- Train workers on hazard identification and controls

Program Evaluation & Improvement

Every program in an organization must be vetted and improved in order to stay viable and productive; safety programs are no different. Program evaluation must be done at given intervals by a competent group. If there are deficiencies found in a program, then the corrections must be made in a systematic way by high risk issues being fixed first then lower risk areas lastly.

Probability

Risk can be calculated as $\text{Probability} \times \text{Severity} = \text{Risk}$. The probability of a loss event occurring can be broken down into 5 categories:

1. Improbable
2. Unlikely
3. Probable
4. Likely
5. Frequent

Severity

Severity speaks of the consequence of a loss event when it does occur:

1. Minor
2. Marginal
3. Serious
4. Catastrophic

If your risk assessment tells you that a task is a $P4 \times S3 = R12$, then it should get your attention over a $R3$ item.

Program evaluation and improvement must include the following areas:

- Monitoring performance and progress
- Verifying the program is implemented and is operating
- Correct program shortcomings and identify opportunities to improve (**OSHA, 2016**)

Communication & Coordination for Host Employers, Contractors, and Staffing Agencies

The host company must take responsibility for all workers including contract and staffing agency workers. Many public sector businesses are not under the jurisdiction of federal OSHA or even a state OSHA, but the contract companies are under an occupational safety agency that will regulate and cite them for violations. However, local government officials have a moral obligation to make sure that workers of all types who do business with them are protected from hazards. To keep the workers safe, the company should:

- Communicate with all outside contractors the importance of worker safety
- Coordinate with supervisors, owners, and workers throughout the project to make sure the worksite is safe
- Hold all workers and agencies accountable for operating a safe worksite
- Verify that the bids and contracts specify that safe work practices are a must for working with the company

A safety culture will protect the workers from injury and illness because the company places a value on the lives of the workers. This is a deposit into the “good will” bank of the worker and will be rewarded with loyalty. A deep commitment to a safety culture will lead to worker retention and organizational benefits far beyond regulatory compliance.

Worksite Analysis

Hazard prevention and control comes from all levels of the organization including the chief executive officer all the way down to individual working at the tasks themselves. A safety management system is built on the foundation that not only is everyone responsible for their own safety, but a system can be incorporated in all task that will protect workers from contacting the hazard.

One of the most common ways for workers to visualize hazards in a job prior to doing the job is through a Job Hazard Analysis (JHA).

What is a JHA?

A job hazard analysis is a tool that is used throughout organizations to help the workers visualize a job that needs to be done and break it into steps. It is different than the standard operating procedure because each step will have an identified hazard and an identified control.

The job hazard analysis is commonly used for training workers as to how to perform a job in the safest manner possible. A standard operating procedure will only show the person how to do the job by breaking down its components. The components of any job are usually communicated through instructions by people who have done it in the past. These can be frontline supervisors or the manufacturers of the equipment.

Using the manufacturer's recommendation, you will see how the equipment is supposed to work, but it may not detail the safety requirements of the job. In a standard operating procedure (SOP), the task is detailed from the beginning to the end of the activity.

Many JHA can be started by using the SOP to get the steps outlined in chronological order as to the way it should be done. Then the safety committee with a good selection of the working crew will dissect the steps to see if and where a hazard may present itself.

Once the hazard or hazards for a step are identified, then the workers will look to the hierarchy of hazard control to see if they can either eliminate the hazard, substitute it, engineer out a control, use existing or new work rules as an administrative control, or finally wear personal protective equipment during that step.

The workers can go back and forth reviewing the JHA to determine completeness of tasks and total identification of hazards. The JHA is also formalized by writing on a JHA worksheet and kept for the review of the employees prior to doing the task.

Some JHA's can be done in advance for jobs that are routine, while others are done onsite prior to the beginning of the task. However, even JHA's that are done prior should be revisited by the work team before any work begins. If there is any doubt as to the steps, hazards, or hazard control, then the work should not progress until all questions have been answered.

Job Hazard Analysis Form

JOB TITLE:

DATE OF ANALYSIS:

JOB LOCATION:

STEP	HAZARD	NEW PROCEDURE OR PROTECTION

Hazard Recognition

To understand the behavior-based safety program, it is necessary to understand what a hazard is first. A hazard is anything that can injure or hurt an individual. While you know a workplace can have various hazards that does not mean that there will be an injury. Hazards must have exposure in order for there to be an injury or an incident.

Hazard identification is primary to understanding how an accident can happen. Later in this program, we will review accident causation from a few different models.

The hazard itself may be the issue for workers when it comes to safety or it may be the exposure to that hazard that is the issue for the employer. For instance, if there is a cave-in hazard for trenching and excavation the cave-in is possible even without the workers being present. The hazard becomes an accident or incident when workers are present, meaning exposure.

Hazard recognition should be first and foremost taught to the front-line supervisors. If the front-line supervisors, who control their work conditions, are aware of hazards then they will protect themselves and the people they supervise. Hazard recognition secondarily must be taught at the level of each worker as they become employed at the facility.

There are obvious hazards such as working at heights, but there are also hazards that may not be as obvious. Workers may be exposed to an odorless toxic fume, so they are not aware that they are being poisoned. This may be the case in many permit-required confined spaces because they often have hazardous atmospheres.

Hazard Types

Hazard types are ways to classify what specifically the hazard is to the worker. Examples of hazard types are:

- ergonomics
- caught in
- contact with
- chemical exposure
- flammable liquids
- laceration
- Falls to below
- Falls to the same level
- mechanical hazards
- engulfment

There are many more hazard types that are job specific, so the worker must be trained to recognize these hazards from the first day of work. Once the hazards are identified, the next stage is to learn how to control the hazard and prevent it from injuring a worker.

Hierarchy of Hazard Control

There is a hierarchy of hazard control beginning with:

1. **Elimination** is the primary way to protect the worker from any hazard. If workers do not have any exposure to that hazard because it is not present, then they will not be injured or harmed.
2. **Substitution** is a secondary way to protect the worker from injury or illness. If there is work with the gas Chlorine and the workplace decides to go to liquid bleach, that substitution would protect the workers from exposure to a hazardous gas.
3. **Risk Transfer:** Another type of hazard control would be to transfer exposure using a contractor. When the contractor is used to perform jobs that your workers would have been doing, then the exposure will go to the contractor and not to your employees. Many companies decide to use this method when they hire specialist such as mold remediation companies, asbestos removal companies, or private fire rescue companies.
4. **Engineering controls:** An engineering control is one that creates a physical barrier between the worker and the hazard point. This physical barrier is not something that can easily be removed or bypassed without worker intervention. There are several types of engineering controls; they should be used primarily for keeping the worker protected.
 - a. An example of engineering control is a machine guard on a table saw. If the workers want to use the table saw without the machine guard, they would have to physically take off that guard and then operate the machinery. The behavior would be an at-risk behavior creating exposure to a low hazard that can cause a laceration or amputation.
5. **Administrative controls:** An administrative control is a work rule set by the employer to protect workers from injuries or illnesses. A work rule, policies,

procedures, and employer's safety culture are company norms that will translate into absolutes. An example of an administrative control is to have workers read the procedure on how to safely do a task. This procedure would be detailed, identify the hazards, and help the worker select the appropriate control to that hazard.

6. **Personal Protective Equipment (PPE):** PPE is the very last line of defense for worker safety and health because they are still exposed to that hazard. If an engineering control is being manufactured and the worker must still be exposed to the hazard, then the PPE should be selected for protection.
 - a. In general industry, personal protective equipment must be accompanied with a PPE hazard assessment. The PPE hazard assessment is the workplace choosing the right protective equipment to match the hazard that the workers are exposed to. For instance, disposable gloves can be worn to protect against biological hazards. If a worker is cleaning up a blood-borne pathogen spill from human fluids, then he or she would use gloves. The PPE hazard assessment would tell him or her what types of gloves to use. It could be latex gloves, nitrile gloves, or even thicker gloves depending on if there is also a needlestick hazard.

Lesson Summary

There are many cost factors related to accidents that can be direct or indirect. The direct costs of an accident are directly associated with the event and are easily quantifiable. Most direct costs are paid by the insurance company of the employer. The indirect costs of an accident are not paid through the insurance and therefore are unrecoverable.

The components for an injury and illness program include:

- Management Leadership
- Worker Participation
- Hazard Identification & Assessment
- Hazard Prevention & Control
- Education & Training
- Program Evaluation & Improvement

- Communication & Coordination for Host Employers, Contractors, and Staffing Agencies

One of the most common ways for workers to visualize hazards in a job prior to doing the job is through a Job Hazard Analysis (JHA). The job hazard analysis is commonly used for training workers as to how to perform a job in the safest manner possible.

A hazard is anything that can injure or hurt an individual. Hazard identification is primary to understanding how an accident can happen. Hazard types are ways to classify what specifically the hazard is to the worker.

Once the hazards are identified, the next stage is to learn how to control the hazard and prevent it from injuring a worker.

There is a hierarchy of hazard control:

- Elimination
- Substitution
- Risk transfer
- Engineering controls
- Administrative controls
- Personal Protective Equipment (PPE)

Lesson 2: Accident Causation and Investigation

Lesson Focus

This lesson focuses on the following topics:

- Understanding Accident Causation
- Accident Theories
- Incident Investigation Techniques

Understanding Accident Causation

To understand accident causation, investigation relies on the idea that most accidents are caused by human error. Therefore, if you learn the cause and effect relationship of accident causation and the human element, then you can reduce injuries and illnesses. Although, there are additional factors that cause accidents, the human aspect is a leading cause of incidents.

The difference between an accident and an incident are slight but worth noting. Accidents denote a loss-producing, unintended event, and an incident denotes an unintended event that possibly doesn't result in a loss (Friend & Kohn, London). In this session, we will refer to incidents and accidents as interchangeable because, the outcome, in this context, is not as relevant as the causes.

In order for an accident to happen, there must be two factors present: the hazard and exposure to the hazard. The hazard is anything that will hurt or harm you through contact. Exposure is how susceptible you are to an injury or illness due to being near to the hazardous substance or material. Accident causation models deal with controlling/eliminating exposure or eliminating, substituting, or controlling the hazard itself.

Types of Hazards

There are several different types of hazards, including but not limited to:

- Chemical
- Caught in between
- Contact by
- Electrical
- Ergonomics
- Excavation
- Slips, trips, and falls
- Mechanical
- Overexertion
- Struck by
- Struck against

Hazardous Control Techniques

If safety professionals do not have the option to eliminate or substitute hazard, then they must control the hazard to which their employees are exposed. There are three types of hazardous control techniques:

1. Engineering controls are the most preferred way to control hazards from affecting a worker. The engineering control is a physical device or barrier that prevents the worker from coming into contact with a point of operation or uncontrolled energy. An example of engineering controls are machine guards, locking devices, and the fume exhaust systems.
2. Administrative controls (or work rules) are controls that create a policy and procedure to protect worker from the hazard. Examples include the safety management system, lockout tag out policies, and job hazard analysis.
3. Personal protective equipment is the last line of defense because the worker is still exposed to the hazard. Some examples of personal protective equipment are hardhats, gloves, and primary eye protectors.

These hazard controls are set up in the hierarchy scenario, therefore hazardous material must be controlled through engineering controls first, then administrative controls, and finally personal protective equipment. However, in most scenarios one or more of these controls are used to protect the worker from the hazardous condition or material.

Hazard Analysis

Hazard analysis is a way to understand the nature and complexity of a hazardous condition. A proper analysis is done through assessing the risk involved for the workers when confronted with this hazard. Usually hazard analysis includes the probability of the hazard creating an accident and the likelihood that the incident will result in an injury or illness to a certain degree. Therefore, a calculation of probability times severity equals risk.

Probability categories are often delineated into five cases improbable, remote, occasional, probable, and frequent. In most risk analyses, each probability category can be assigned a number beginning with the lowest category improbable being assigned one and the highest category frequent being assigned to five.

The severity factors also are delineated by the characteristics negligible, marginal, critical, and catastrophic. Similar to the probability factor, numbers are assigned to the severity factors beginning with negligible as a one and catastrophic events as a four.

With this understanding of risk, an assessment can be performed using the number system of probability times severity and give a class of hazard. The higher the hazard the more probable that there will be an accident. Combined with the human error factor, the probability and severity may both increase any hazard that is a high risk.

An example of conducting a risk analysis is as follows:

Risk = Probability x Severity

Frequency of Occurrence-Probability	Catastrophic (Fatality) (4)	Critical (OSHA Recordable) (3)	Marginal (First-Aid) (2)	Negligible (Near Miss) (1)
Frequent (5)				
Probable (4)				
Occasional (3)				
Remote (2)				
Improbable (1)				

Example:

A worker working in a laboratory is using acid without any hand protection.

The probability of a chemical burn is high enough to warrant the top level of likelihood of an event occurring, which is a 5. The consequence of a chemical burn will yield a severity of 3. In calculating this risk, the use of acid in a laboratory without hand protection is a risk factor 15. Using the maximum scale of 20, the highest probability 5 and the highest severity as 4, then this action by the worker is a high-risk activity.

In this example, the use of chemicals without proper personal protective equipment has an element of behavioral deficiency. This at-risk behavior can be identified through a behavioral-based safety inspection and coached. Although, the worker did not have gloves on when handling the chemical, it is unlikely that an accident will be the cause of the single factor.

Accident Theories

Single Factor Theory

The Single Factor Theory for accident investigation is the most basic of the accident causation models. In the single factor theory, it is thought that there is one event solely responsible for an accident or incident.

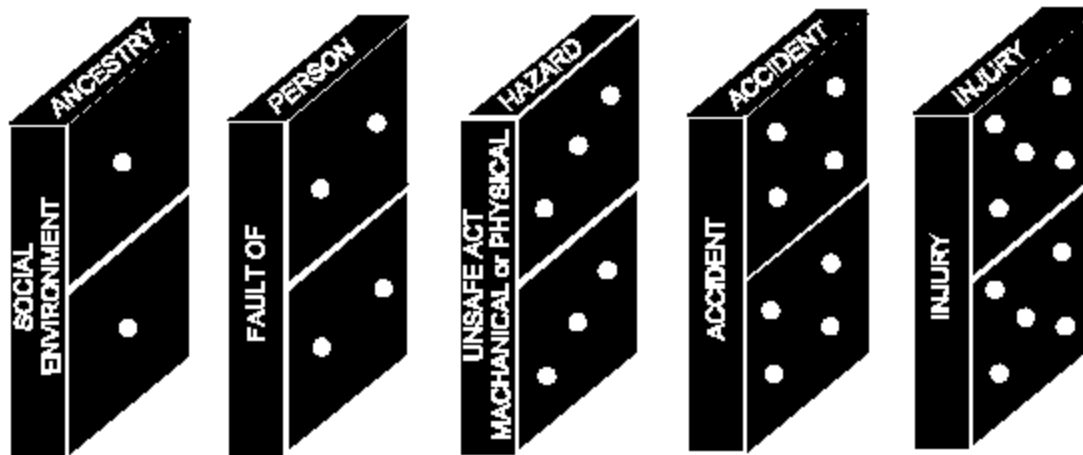
It is akin to the “pilot error syndrome” that states that the cause of the accident was solely due to an error caused by the pilot. In modern accident causation models, the single factor theory has little to no value.

H.W. Heinrich's Domino Theory

In 1931, HW Heinrich presented a theory called the axiom of industrial safety. The first axiom dealt with accident causation, which states “the occurrence of the injury invariably results from the complicated sequences of factor, the last one of which being the accident itself” (Heinrich et al, 1980). The model that was used to illustrate this point was a lineup of dominoes, which became known as the domino theory. In all the domino theories, there are 3 phases that are influencers of an accident. These phases are:

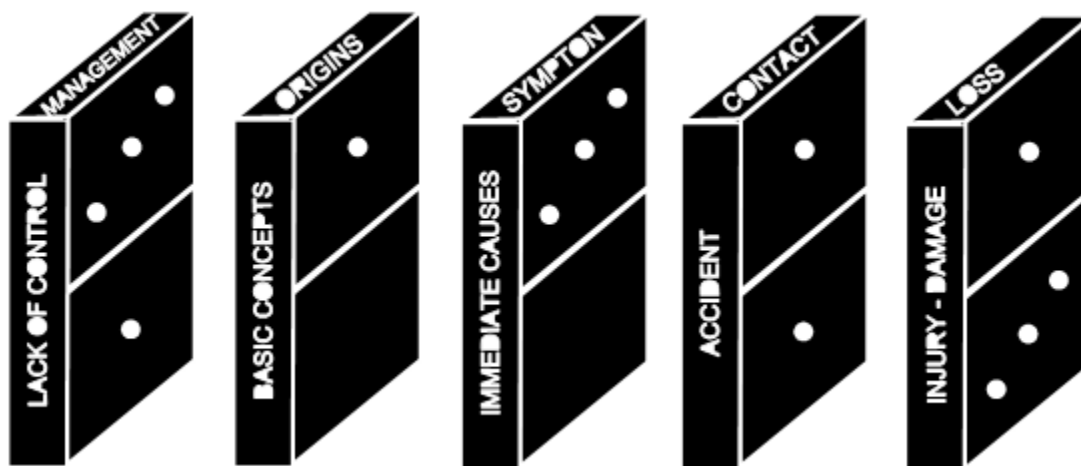
1. Pre-Contact phase prior to the event happening
2. Contact phase, which refers to the event as it's happening.
3. Post contact phase, which refers to after the release of unplanned energy, the product downtime, or the injury or illness

The domino theory is summarized as an injury is caused by an accident due to an unsaved act and/or mechanical or physical hazards due to the fault of the person, caused by their ancestry and social environment. (Heinrich et al, 1980)



(Heinrich, D, & Roos, 1980)

In this model, Heinrich states that by removing one of the dominoes, preferably the middle unsaved act domino, then the accident would have never occurred. The later model of the Domino, Bird and Loftus added managerial influence and managerial error dominoes. In the later model, loss was organizational loss as opposed to just an accident. An example of organizational loss could be property damage, loss of public trust, as well as an illness or injury.



(Heinrich, D, & Roos, 1980)

A point of contention of the Domino theories was that they don't glean your nature of the incidents. As the accident theories developed, more dimensional approaches became common, indicating that the cause of an incident may be multi-factual with events happening in a nonlinear format.

Multiple Factor Theory

The multiple factor theory looks at not just the 3 phases in the domino theory of pre-contact — contact — post contact, it analyzes other factors that influence an accident. Multiple factor theories analyze other things such as management, the machines, the media, and the man. The role of management would be to create the organizational structures, the policy, and procedures. The machinery includes the design, shape, size, and the type of energy sources of the equipment. The role of media can be described as environmental factors. The factor of the man includes gender, age, mentality, fatigue factors, height, weight, etc.

In analyzing all these factors, it becomes apparent that the weaning your model does not include the influencers I can appear nonlinearly. Unlike previous theories, management's role is more concrete in the multiple factor theories. Strong management that respects the role of occupational safety will support and lead the safety effort of the organization.

Human Factor Theory

The human factor theory states that the accident is caused by human error. This theory analyzes the factors that lead to the human error. Such factors are overload, inappropriate activities, and inappropriate responses.

Overload doesn't always mean that the worker is overburdened, but the job itself may have excess stressors such as noise, heat, or unclear instructions. Workers can often feel overloaded when given too many tasks at a time or work instructions do not clearly define the goals. Front-line supervisors are charged with making sure that the workers understand the steps of each task.

There are two types of failures that cause human errors: active failures and latent failures (Reason J. , 1990). An active failure is usually caused by the worker or person engaged in an activity. The actions give an immediate consequence, which is the direct cause of the accident. Latent failures lay the foundation of an active failure due to such issues as:

- Ineffective training

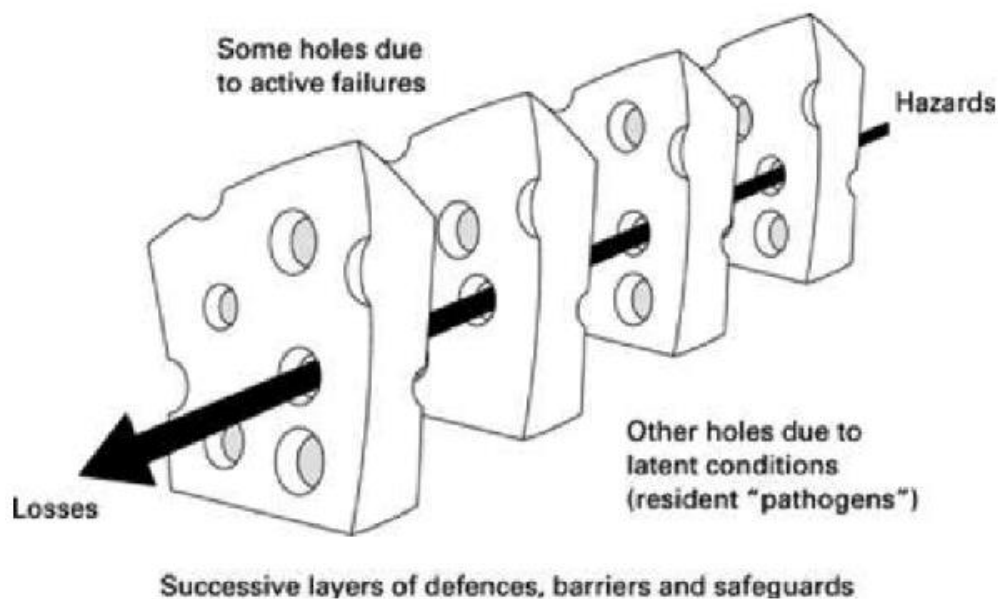
- Poor engineering of the equipment or location
- Poor or inadequate supervision
- Ineffective communication
- Unclear roles for the worker

The observation of the human factor theory through a behavioral-based safety analysis can help improve the safety system. The BBS is designed to highlight the behavior of the worker that is causing the accident and to address them systematically. The key to systematic intervention is having a well-planned and thought out BBS program.

James Reason's Swiss Cheese Theory

In recent years, a professor from Manchester University in the United Kingdom, James Reason, developed an accident causation model based on the analogy of holes in Swiss cheese. In his analysis of an accident, he uses a visualization of a complete system with no holes. In a complete system, management is involved, employees are motivated, there is a safety management system in place, and a safety culture exists in the organization.

Fig. (1): The Swiss Cheese Model by James Reason published in 2000. Adopted from Perneger BMC Health Services Research 2005 5:71.



A breakdown of any one of these components, the blanks, will create a whole similar to the hole in a slice of Swiss cheese. As mortals developed, the system becomes less than ideal and dysfunctional. Eventually each one of the systems will have their holes line up and allow a release of unplanned energy to create an accident condition. In this model, it does have a linear component, but it also includes multiple factors that influence the accident.

An example of how each one of the blanks can develop holes can be found in the following scenario:

Upper management has a change in leadership with the new ownership group. The ownership group values production more than safety. Therefore, the workers are not given adequate time for safety briefings in the morning. Production becomes the main factor for the organization. Quality and safety may suffer due to the production quotas. Eventually workers will become fatigued, equipment fails, or safety training is omitted entirely.

During a busy shift when the worker is trying to meet production quota, he decides not to report a broken guard on a piece of equipment and comes in contact with the unguarded point of operation. The worker sustains a laceration that could have easily been an amputation.

Incident Investigation Techniques

Root Cause Analysis

A root cause analysis is a detailed procedure used by safety professionals to look beyond the initial knee-jerk reaction to assign blame for an incident. First, the investigator will notice direct causes to any accident, but he or she cannot see indirect causes or latent causes that became activated to create the accident.

A root cause analysis must be conducted to determine all factors and variables responsible for an incident. Most of these types of analysis will require a team of trained

professionals alert to accident causation models, as well as good investigators and critical thinkers. There are many types of root cause analysis, but they all seek to do the same thing: address the source cause of any accident.

The 5 Whys

The 5 Whys is an accident investigation technique used by safety professionals in developing the root cause analysis. This technique will help the data entry investigator pass simple questions to date beyond the superficial yes or no answer. They are quantifiable questions that build one upon the next. As each question gets answered, the investigator will uncover more reasons for an accident. Although this technique is called the 5 Whys, it often takes more questionings to get to the root cause.

An example of the 5Y technique for incident investigation is as follows:

A worker receives a laceration from touching unguarded blade on a handfed ripsaw. Using the 5 Whys technique for questioning, the investigator can begin with these questions:

1. Why did the worker receive a laceration? -- Because, he touched unguarded blade.
2. Why did the worker touch the unguarded blade? --Because the guard was not placed on the saw.
3. Why was the guard not on the saw? --Because the worker believed he can work faster without the guard.
4. Why did the worker believe it is better to work faster and not safer? --Because the worker gets paid on a quota basis.
5. Why is management placing a greater importance on production quotas over safety? --Because management receives a bonus for achieving production quotas.

In this brief example, the use of production quotas indirectly led to the worker valuing speed over safety. There may be many other factors related to why the worker received a laceration in the scenario. Such factors can include but are not limited to:

- Poor organizational safety culture
- Untrained worker
- Management and supervision has unfair expectation of production quotas
- Worker fears retribution for slowing down production to address safety concerns
- Upper management has condoned and/or rewarded production goals over safety concerns.

When conducting a 5 Whys accident analysis, the investigator will dig deeper to uncover an accident causation. This new revelation may lead the organization to address gaps in their safety management system. The value of operating safely to reduce or eliminate injury and illness must be made apparent to management in order for them to value safety equal with production.

Lesson Summary

Managing safety and health is a key part of keeping workers free from injury and illness. The company will save money by investing in the safety needs of the employees. An organization that values low risk will have a safety culture that will promote safe behaviors in all level of the organization. Additionally, through accident investigations, the company will get a good understanding of what hazards created the incident and how to stop it from hurting future workers.

To understand accident causation, investigation relies on the idea that most accidents are caused by human error. Therefore, if you learn the cause and effect relationship of accident causation and the human element, then you can reduce injuries and illnesses. In order for an accident to happen, there must be two factors present: the hazard and exposure to the hazard.

Accident theories include:

- Single Factor Theory
- H.W. Heinrich's Domino Theory
- Multiple Factor Theory
- Human Factor Theory
- James Reason's Swiss Cheese Theory

A root cause analysis is a detailed procedure used by safety professionals to look beyond the initial knee-jerk reaction to assign blame for an incident. It is conducted to determine all factors and variables responsible for an incident.

The 5 Whys is an accident investigation technique used by safety professionals in developing the root cause analysis by asking questions. As each question gets answered, the investigator will uncover more reasons for an accident.

Module 3: Walking & Working Surfaces, including Fall Protection

Module Description

Every walking-working surface in the workplace can be potentially hazardous to workers. Hazardous surfaces include loose carpets, ramps, stairways, ladders, and scaffolds, as well as slippery floors resulting from liquid or other spills. Resulting slips, trips and falls can cause cuts and bruises, sprains and strains, broken bones, and various injuries to internal organs. They can also result in fatalities.

This module alerts participants to the hazards of walking-working surfaces and provides information about the workplace actions required to eliminate or control these hazards.

Module Learning Objectives

At the conclusion of this module, you will be able to:

- Summarize key elements of OSHA's Walking-Working Surfaces Regulation (29 CFR 1910.21-30, Subpart D).
- Discuss key terminology and concepts.
- Examine the consequences of poor housekeeping and hazards of wall and floor openings.
- Discuss the specifications for safe design and construction of fixed general industry stairs.
- Explain the minimum requirements for proper care and use of portable ladders.
- Discuss general scaffolding requirements.
- Elaborate on fall protection and related hazards.

Lesson 1: Wall and Floor Openings and Holes

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- General Requirements
- Guarding Floor and Wall Openings and Holes

Introduction

Slips, trips, and falls account for just over one-third of all recordable general industry injuries. They cause 15% of all accidental deaths in the workplace and are second only to motor vehicles as a cause of fatalities. The OSHA standard for walking-working surfaces applies to all permanent places of employment, except where only domestic, mining or agricultural work is performed.

General Requirements

The General Requirements section is found in 29 CFR 1910.22. It contains four elements:

- Housekeeping
- Aisles and Passages
- Covers and Guardrails
- Floor Loading Protection

In this section, each topic is presented with guidance information so that you will be able to anticipate and prevent accidents.

Housekeeping

Housekeeping rules include:

- Keep all places of employment, passageways, storerooms, and service areas orderly and in a sanitary condition.
- Keep the floor of every work area maintained in a clean and, so far as possible, dry condition.
- Where wet working conditions exist, provide gratings, mats, raised platforms, or other engineering controls to minimize the potential for slips.
- Keep every floor, working surface, and passageway free from protruding nails, splinters, holes, or loose boards.

Aisles and Passageways

Rules for aisles and passageways include:

- Keep aisles and passageways clear of any obstructions that create a hazard.
- Mark permanent aisles and passageways.
- Aisles must be sufficiently wide where mechanical handling equipment is used.

Improper aisle widths coupled with poor housekeeping and vehicle traffic can:

- Cause injury to employees.
- Cause damage to the equipment and material.
- Limit egress in emergencies.

Covers and Guardrails

Clean and orderly work areas are only a start. Covers and guardrails are required in certain situations. Here are some hazard areas where covers and guardrails are required:

- Open pits
- Tanks
- Vats
- Ditches

Floor Loading Protection

For floor loading protection, load ratings must be marked on plates and be conspicuously posted, and the load-rating limit must not be exceeded.

Guarding Floor and Wall Openings and Holes

Holes and openings present extreme hazards that must be addressed immediately. The hazards are obvious: Workers may fall through the openings or over a leading edge to a level below. Tools or materials may fall through the holes and strike workers or damage equipment on a lower level.

The three major elements of the guarding floor and wall openings and holes section are:

1. Protection for Floor Openings
2. Protection of Open-sided Floors, Platforms, and Runways
3. Stairway Railings and Guards

More Information: More detailed information concerning the guarding of floor and wall openings and holes can be found in [29 CFR 1910.23](#).

A floor opening is an opening measuring 12 inches or more in its least dimension in a floor, platform, or other work surface through which persons may fall.

- Standard railings shall be provided on all exposed sides of a stairway opening, except at the stairway entrance.
- For infrequently used stairways, where traffic across the opening prevents the use of a fixed standard railing, the guard shall consist of a hinged floor opening cover of standard strength and construction along with removable standard railings on all exposed sides, except at the stairway entrance.
- Floor openings may be covered rather than guarded with rails. When the floor opening cover is removed, a temporary guardrail shall be in place or an attendant shall be stationed at the opening to warn personnel.

Every floor hole, (a hole less than 12" but more than 1" at its least dimension) shall be guarded by either a:

- Standard railing with toe board, or
- Floor hole cover of standard strength and construction.

While the cover is not in place, the floor hole shall be constantly attended by someone or shall be protected by a removable standard railing.

Case Study

Laborer Killed by Falling Through Roof

A 40-year-old laborer/helper died when he fell through an opening in a warehouse roof. He fell approximately 27 feet to the floor below.

The employer was demolishing the roof of the warehouse portion of a commercial building. Work was done at night because the coal tar on the roof would release hazardous gases if disturbed in the heat of the day.

The site had adequate halogen lighting. None of the workers on the job were using fall protection.

After the roofing material was removed, 4x8 foot sheets of plywood were exposed. Any damaged sheets needed replacing. The helper's job was to follow the workers who were replacing the plywood, and to pick up the damaged sheets of plywood they had removed. He would then dispose of them in a chute.

On this evening, one worker had removed a sheet of damaged plywood, but had run out of nails to attach the replacement plywood. He walked away to get more nails. The opening where the damaged plywood had been was left unguarded. The crew was not informed that it was temporarily unguarded. The opening was covered by silver-colored insulation inside the roof.

The helper came along, picked up the sheet of damaged plywood, and headed for the chute. He stepped into the opening, ripped through the insulation, and fell.

Preventive Measures

Employers should:

- Have Site-specific Safety Plans addressing potential hazards that could lead to injury or death.
- Ensure that holes and openings are not left unprotected, unguarded, or uncovered.
- Equip all workers on the roof with fall protection (such as harnesses and lanyards). A retractable lanyard would allow the worker greater mobility and the worker would still have fall protection.
- Require that all hazards on the site be communicated on an ongoing basis to all workers in the area.

Protection of Open-sided Floors, Platforms, and Runways

Guard every open-sided floor or platform, four feet or more above adjacent floor or ground level, on all open sides by a standard railing.

Every ladder way, floor opening, or platform shall be guarded by a standard railing with standard toe board on all exposed sides (except at entrance to opening), with the passage through the railing either provided with a swinging gate or so offset that a person cannot walk directly into the opening.

Provide standard railing with a standard toe board on the open sides wherever:

- Persons may pass.
- There is moving equipment.
- There are tools or materials that could fall to the lower level.

Stairway Railings and Guards

Stairway openings must be guarded by a standard railing on all exposed sides (except at the entrance).

Every flight of stairs with four or more risers shall have standard stair railings or standard handrails. Stair width shall be a minimum of 22 inches.

Lesson Summary

Because slips, trips, and falls cause 15% of all accidental deaths in the workplace, one should:

- Keep all walking surfaces orderly and in sanitary condition.
- Keep the floor of every work area in a maintained, dry condition.
- Provide gratings, mats, raised platforms, etc., to avoid potential slippage.
- Keep flooring free from protruding objects that may catch shoes or clothing.

Standard railings and toe boards, or coverings of recommended strength and construction, should guard floor openings. While uncovered, the floor hole should be constantly attended to or protected by a temporary standard railing.

Every open-sided floor or platform that is four feet or more above adjacent floor or ground level must be guarded, but remember standard railings are not required where there is an entrance to a ramp, stairway, or fixed ladder. Instead, one must provide a standard railing with a standard toe board on open sides wherever:

- People may pass.
- Moving equipment is involved.
- Tools or materials could fall to a lower level.

Additionally, every flight of stairs with four or more risers should have standard stair railings or handrails.

Lesson 2: Stairs, Ladders, PFAS and Scaffolds

Lesson Focus

This lesson focuses on the following topics:

- Stairs
- Ladders
- Personal Fall Arrest Systems
- Scaffolds

Stairs

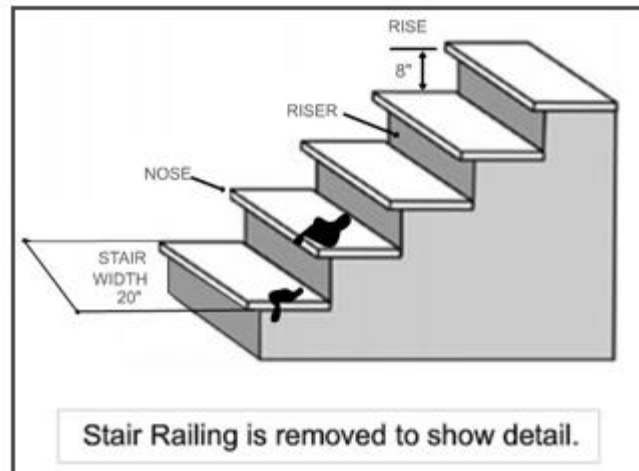
Fixed Industrial Stairs

Fixed industrial stairs are used for access to and from places of work where operations necessitate regular travel between levels; however, the standard does not apply to fire exits.

Requirements

The requirements for stairs include:

- Treads of Fixed Industrial Stairs must be slip resistant with uniform rise height and tread width.
- They must be able to carry five times expected load, with a minimum of 1000 pounds.
- They must have a minimum width of 22 inches.



Question: Besides the cited violation, what's wrong with this picture?

Answer: There is an oil spill on the stairs.

Inspection

When inspecting the condition of stairways in a workplace, you should inspect the following.

- Handrails and stair rails
- Treads
- Other factors (risers, housekeeping, width, absence of oil or liquids, etc.)



Question: Besides the cited violation, what's wrong with this picture?

Answer: There are no handrails, and tools are scattered on the stairs.

Ladders

A ladder is a tool consisting of two side rails joined at regular intervals by crosspieces on which a person may step to ascend or descend. Ladders are either "fixed" and remain in place or are portable and provide access to locations that are not readily accessible.

The chief hazard when using a ladder is falling. A poorly designed, maintained, or improperly used ladder may collapse under the load placed upon it and cause the employee to fall. Ladders must be inspected regularly, kept in good condition at all times, and must be free from sharp edges, splinters, oil, grease, or other defects that would affect their use.

Portable Ladders

The various types of portable ladders include the following:

- Step-ladder
- Single ladder
- Extension ladder

Portable Ladder Types

Step-Ladder

A self-supporting portable ladder, non-adjustable in length, having flat steps and hinged back, may not exceed 20' in length. Stepladders must be equipped with a spreader or locking device to securely hold the front and back sections in an open position.

Note: The top of a regular stepladder shall not be used as a step.

Single Ladder

A non-self-supporting portable ladder, or single ladder, is non-adjustable in length, consisting of one section. Its size is determined by overall length of the side rail but may not exceed 30' in length.

Extension Ladder

A non-self-supporting portable ladder adjustable in length may not exceed 60'.

Proper Use

Proper use of ladders is essential in preventing accidents. Even a ladder in good condition can be a serious safety hazard when used by workers in a dangerous way.

Portable ladders used to gain access to a roof or other elevated level must extend at least three feet above the highest, or last, point of support.

Remove defective ladders from service and tag or mark "Dangerous, Do Not Use" or properly dispose of the ladder. Never use ladders in a horizontal position as scaffolds or work platforms. Never use metal ladders near electrical equipment.

When climbing or descending ladders, workers should always maintain a 3-point (two hands and a foot, or two feet and a hand) contact. Also, workers must be facing the ladder.

Place the ladder on a secure footing. Ensure the feet of the ladder have secure contact with a level, stable surface.

The pitch of a single or lean ladder should always be one foot out from the base for every four feet of elevation, a one-to-four ratio.

Case Study

Billboard Worker Dies of Electrocution When Metal Ladder Contacts 7,200-Volt Power Line

A 28-year-old male billboard worker was electrocuted when a 24-foot hook ladder he was using contacted an overhead power line. At the time of the accident, the billboard worker was in the process of scraping a billboard.

Discussions: The victim had eight years of experience and had worked on this billboard before. At the time of the accident he was scraping an outdated poster from the 12 x 24 foot billboard. The bottom of the billboard was four feet off the ground, and the top was about 16 feet high. The victim removed the paper he could reach with a 14-foot ladder from the left side of the billboard, and then moved the 14-foot ladder to the right side of the billboard. He retrieved a 24-foot hook ladder from his truck and contacted the power line while moving the ladder into position. A 32-foot extension ladder was also on the truck and may have been a better choice, because its length could have been adjusted. The power line was located about eight feet above the top of the billboard. Two passing motorists saw the incident, summoned help, and administered CPR, but the victim was DOA at a local hospital.

Recommendations: Employers must identify all safety hazards at a work site. In the case of the billboard company, this would mean hazards such as power lines, uneven terrain, traffic, and deteriorated boards. Workers should then be informed of the hazards.

Employers must provide proper equipment to perform all job-related tasks. In the case of billboards, scaffolding or long-handled brushes would have minimized the power line hazard.

These methods were used in the past but have been replaced by the use of aluminum hook ladders. The employer should consider the use of fiberglass ladders when working around electrical hazards.

Employers should develop and enforce written safe procedures and practices that are site and task-specific.

Fixed Ladder

In many locations, workers use fixed ladders. A fixed ladder is a ladder permanently attached to a structure or building.

Always remember that fixed ladders with a length of more than 20 feet to a maximum unbroken length of 30 feet must be equipped with cages or a ladder safety device.

Another feature of fixed ladders is the landing platform, which provides a resting place during long climbs or in the event of an emergency.

When fixed ladders are used to ascend to heights exceeding 20 feet (except on chimneys), landing platforms shall be provided for each 30 feet of height or fraction thereof.

Ladder safety devices may be used on tower ladders, water tank ladders, and chimney ladders over 20 feet in unbroken length in lieu of cage protection. No landing platform is required in these cases.

The preferred pitch of a fixed ladder is between 75 degrees and 90 degrees with the horizontal.

Fixed ladders shall be considered to be substandard if they are installed within the pitch range of 60 and 75 degrees with the horizontal. Substandard fixed ladders are permitted only where it is found necessary to meet conditions of installation. This substandard pitch shall be considered as a critical range to be avoided, if possible.

Personal Fall Arrest Systems

In general industry, falls are the second leading cause of accidental deaths. If employees follow safe fall protection practices, they can avoid accidental deaths caused by falls.

Preventing employees from the dangers of falling provides for the highest level of safety. In those instances, where an employee is exposed to fall hazards, personal fall arrest systems provide a high level of protection from falls, as they are designed for each individual employee.

A personal fall arrest system includes:

- A body harness that is worn around the torso.
- An overhead anchor to which the lanyard is connected. This anchor must be strong enough to support 5,000 pounds of weight per employee or an anchorage designed to meet specific OSHA standards.
- A lanyard, which connects the harness to the anchor.

Scaffolds

Scaffolds are designed to provide employees with a safe elevated working surface. Only competent persons may select and direct the employees who erect, dismantle, move or alter scaffolding. Scaffolds should be able to support at least four times their expected loads. A competent person must inspect all scaffolds before employees can use them.

Those scaffolds that have been approved should be marked with green tags. Yellow tags are used if there are any limitations associated with the scaffold. Scaffolds unsuitable for use should be marked with red tags.

Employees must always check a scaffold before climbing onto it. If they find anything wrong with it, they must immediately inform their supervisor.

Safety Requirements for Scaffolds

The footing or anchorage for scaffolds must be sound, rigid, and capable of carrying the maximum intended load without settling or displacement. Do not use unstable objects,

such as barrels, boxes, loose brick, or even concrete blocks to support scaffolds or planks. Scaffolds and their components should be capable of supporting at least four times the maximum intended load.

Maintain scaffolds in a safe condition, and do not alter or move them horizontally while they are in use or occupied.

Repair damaged or weakened scaffolds immediately, and do not use them until repairs have been completed.

A safe means must be provided to gain access to the working platform level through the use of a ladder, stairs, ramp, or other device.

Overhead protection must be provided for personnel on a scaffold exposed to overhead hazards.

Fall protection must be provided on all open sides and ends of platforms more than 10 feet above the ground or floor. When guardrails are used, proper top rails, mid-rails, and toe boards must be provided. Wire mesh or other appropriate means of protection must be installed between the toe board and the guardrail along the entire opening, where persons are required to work or pass under the scaffolds.

Employees should not work on scaffolds during storms or high winds or when scaffolds are covered with ice or snow.

Case Study

One Killed, Three Injured in Scaffold Accident

Workers had been spraying fireproof insulation onto the structural steel frame of a building. They used a rolling tower scaffold to gain access to the structural steel overhead.

Putlogs (types of trusses) had been added to the sides of the rolling tower scaffold, and an extension platform had been built there. This platform was used to reach the outer side of the structural steel.

On this day, a supervisor said a guardrail was needed on the scaffold. A hod carrier joined three co-workers on the extension platform to help install the guardrail. Their combined weight caused the scaffold to tip. They were all thrown to the concrete deck 44 feet below.

The scaffold had not been engineered for the extension platform. No counterweights, anchorage, or bracing were used. Neither the hod carrier nor his co-workers were wearing personal fall protection, and the scaffold and platform had been constructed using parts from different manufacturers. As a result, one person was killed and three others were injured.

Preventive Measures

Employers must:

- Ensure that scaffolds are assembled according to the manufacturer's recommendations. If locally built, they must be properly designed and engineered.
- Ensure that no extensions or auxiliary parts are added to scaffolds unless designed and approved by a competent person.
- Ensure that workers follow safe work practices when constructing scaffolds.
- Ensure that scaffold load limits given by the manufacturer or engineer are not exceeded.

Lesson Summary

Safety should be a primary consideration in any work situation. For instance, treads of fixed industrial stairs should be slip resistant and have uniform height and tread width. Furthermore, some points to remember about ladders and ladder use, include the following:

- Portable ladders used to gain access to a roof or other elevated level must extend, at a minimum, three feet above the highest or last point of support.
- The pitch of a single or lean ladder should always be one foot from the base of the structure for every four feet of elevation, or a one-to-four ratio.

- Fixed ladders more than 20 feet and up to 30 feet must be equipped with cages or a ladder safety device.
- The preferred pitch of a fixed ladder is between 75 and 90 degrees.

Never use unstable objects to support a scaffold. Scaffolds and their fixtures should be capable of supporting at least four times their maximum intended load. In addition, overhead protection must be provided for personnel using scaffolds potentially exposed to overhead hazards.

Under scaffolds in areas where persons are required to work or pass, wire mesh or appropriate protection must be installed between the toe boards and a guardrail along the entire opening. Finally, one should ensure that scaffolds are assembled according to the manufacturer's recommendations, but if necessarily custom built, they must be properly designed and engineered.

Module 4: Emergency Action Plan

Module Description

Every year workers are injured or die because of nonexistent or inappropriate exit routes. Too often, inadequate emergency action plans are responsible for more casualties and fatalities than the actual emergency.

This module gives you a basic understanding of means of exit, emergency action plans, fire protection plans, and emergencies. Familiarity with these plans in any workplace can save lives.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Explore general requirements of exit routes.
- Describe emergency action plans.
- Emphasize the essential aspects of a Fire Protection Plan.
- Describe various evacuation elements and shelter-in-place
- Explain training and other essential factors associated with means of exit and fire protection.

Lesson 1: Exit and Its Standards

Lesson Focus

This lesson focuses on the following topics:

- Means of Exit
- The Coverage of Standards
- Reasons to Exit
- Fire, Rescue, and Medical Services
- Reporting Emergencies
- The Definitions

Means of Exit

A means of exit is a continuous and unhindered route from any point in a building or structure to any area of safety. It consists of three separate and distinct parts: the exit access, the exit, and the exit discharge.

A means of exit refers to both horizontal and vertical ways of travel; this includes corridors, hallways, ramps, stairs, yards, doorways, lobbies, escalators, balconies, courts, passageways, and enclosures.

The Coverage of Standards

Subpart E covers issues related to means of exit, emergency action plans, and fire prevention plans.

The means of exit standard does not apply to mobile workplaces such as vehicles or vessels.

Reasons to Exit

There are many reasons for emergency action plans in virtually every workplace. One of the most common reasons for needing to evacuate a workplace is a fire, but the following hazards should also be considered serious reasons for emergency escape:

- Earthquakes
- Smoke
- Bomb threats
- Chemical spills
- Explosions
- Flash floods
- Toxic vapors

All of the above hazards could occur individually or may be the result of other hazards.

Severity

The severity of an emergency depends upon the configuration of a building, the nature of the emergency and its intensity, the quality of the Emergency Action Plan (EAP), and the employees who will make the plan work.

More Information: A lack of information regarding the facility's EAP, a poorly planned EAP, and confusion or panic often make the situation worse; whereas, a well-planned EAP may save lives.

Fire, Rescue, and Medical Services

Although most of us quickly move away from the hazardous environments created during emergency situations, a group of dedicated and well-trained professional emergency responders and medical service personnel are tasked with containing and mitigating these incidents, rescuing individuals at-risk, and providing medical assistance to the injured.

Reporting Emergencies

Dialing "911" is a common method for reporting emergencies if external emergency personnel are used at your workplace. Internal numbers may be used for reporting emergencies. If they are, they should be posted on, or near, each phone. Internal numbers sometimes are connected to intercom systems so that coded announcements may be made. In some cases, employees are requested to activate manual pull stations or other alarm systems. No matter what system is used, it is imperative that emergency situations be immediately reported. Fires and other emergency situations can reach dangerous levels in seconds and any delay in getting emergency responders to the scene can result in additional loss of life and property.

Once an emergency that requires a response from employees is reported, a system - typically an alarm system - must be in place to notify employees. Alarms must be distinctive and recognized by all employees as a signal to evacuate the work area or perform other actions identified in your emergency action plan. The alarm must be capable of being perceived above ambient noise and light levels. Sequences of horn blows or different types of alarms (bells, horns, etc.) can be used to signal different responses or actions from employees. Ideally, alarms will be able to be heard, seen, or otherwise perceived by everyone in the workplace including those who may be blind or deaf. Otherwise, floor wardens or others should be tasked with ensuring all employees are notified. You might want to consider providing an auxiliary power supply to your alarm system in the event of an electrical failure. Consider making available an emergency communications system, such as a public address system, for broadcasting emergency information to employees.

The Definitions

Exit Route

An exit route is the route employees are directed to follow in the event of any emergency where evacuation of the facility is the best course of action.

Occupant Load

The occupant load refers to the total number of persons that may occupy a workplace or a portion of a workplace at a time. The occupant load is typically determined by local or state fire codes, so be sure you are aware of all local, county, and state requirements. They may be more stringent, thereby superseding, U.S. OSHA.

Refuge Area

The refuge area is a fire and smoke protected space within a building along an exit route. The refuge area must have a fire resistance-rating of at least one hour.

Lesson Summary

A means of exit consists of three separate and distinct parts: the exit access, the exit, and the exit discharge.

One of the most common reasons for needing to evacuate a workplace is fire, but there are other hazards that should be considered serious reasons for emergency escape, such as chemical spills and toxic vapors. Remember that a lack of information regarding the facility's EAP, a poorly planned EAP, and confusion or panic often make the situation worse, whereas a well-planned EAP may save lives.

Lesson 2: The Plans

Lesson Focus

This lesson focuses on the following topics:

- Emergency Action Plan
- Develop & Implement an EAP
- Evacuation Elements
- Shelter In-Place
- Fire Protection Plan
- Housekeeping
- Exit Routes

Emergency Action Plan

An emergency action plan (EAP) is a written document required by particular OSHA standards. [29 CFR 1910.38(a)] The purpose of an EAP is to facilitate and organize employer and employee actions during workplace emergencies. Well-developed emergency plans and proper employee training (such that employees understand their roles and responsibilities within the plan) will result in fewer and less severe employee injuries and less structural damage to the facility during emergencies. A poorly prepared plan, likely will lead to a disorganized evacuation or emergency response, resulting in confusion, injury, and property damage.

Minimum Requirements

Putting together a comprehensive emergency action plan that deals with those issues specific to your worksite is not difficult. It involves taking what was learned from your workplace evaluation and describing how employees will respond to different types of emergencies, taking into account your specific worksite layout, structural features, and emergency systems.

Develop & Implement an EAP

Drafting an emergency action plan (EAP) is not enough to ensure the safety of your employees. When an evacuation is necessary, you will need responsible, trained individuals who can supervise and coordinate activities to ensure a safe and successful evacuation.

A very simple plan will suffice in offices, small retail shops, and small manufacturing settings where there are few or no hazardous materials or processes, and employees evacuate when alarms sound or when notified by public address systems. More complex plans are required in workplaces containing hazardous materials or workplaces where employees fight fires, perform rescue and medical tasks, or delay evacuation after alarms sound to shut down critical equipment.

It is essential that the emergency action plan developed be site specific with respect to emergency conditions evaluated, evacuation policies and procedures, emergency reporting mechanisms, and alarm systems. To assist you in your planning, a checklist is provided that identifies issues that must be considered when drafting a comprehensive emergency action plan. An explanation of each issue and/or examples of how each issue might be addressed in typical workplaces are provided.

The best emergency action plans include employees in the planning process, specify what employees should do during an emergency, and ensure that employees receive proper training for emergencies. When you include your employees in your planning, encourage them to offer suggestions about potential hazards, worst-case scenarios, and proper emergency responses. After you develop the plan, review it with your employees to make sure everyone knows what to do before, during, and after an emergency. Keep a copy of your emergency action plan in a convenient location where employees can get to it or provide a copy to all employees.

An Emergency Action Plan (EAP) must be established in writing for any facility with 10 or more employees. However, a facility with fewer than 10 employees must still implement an EAP and communicate it verbally to each employee.

The EAP must include:

- Evacuation procedures and identification of all exit routes.
- First-aid and rescue duty assignments.
- Methods of reporting emergencies by any employee.
- Procedures for employees who remain behind.
- Procedures for an accounting of all employees after evacuation, or upon a shelter-in-place emergency.
- Procedures for critical operation shutdown.
- Clear identification of the 'person-in-charge'—of the EAP, and the chain-of-command.

It is critical that any facility's EAP addresses all potential emergencies whether they should come from regular operations or from outside the facility.

Alarm System

The alarm system must be in good working condition and able to be seen or heard by all employees in the facility. The alarm system's power supply must be self-contained. If audible alarms are used to initiate the EAP, the sound must be a "unique sound," one different from any other sound an employee might hear during normal operation. The sound can only mean the EAP has been initiated.

Evacuation

The establishment of an EAP is a responsibility of the employer. The EAP informs all employees of the type of evacuation procedure to follow in the event of an emergency, and which exit routes will be used.

In those cases, where an emergency is of an internal nature and serious enough to affect the entire facility, complete and immediate evacuation of all employees may be required. In non-severe emergencies, partial evacuation of non-essential employees could be enough. Employees must be aware of their duties and roles in case of an emergency.

This level of awareness can be attained as a result of drills and learning from any failures in the EAP process.

More Information: Depending on the size and intensity of a fire, those employees in the immediate area of a fire may be evacuated to a refuge area, if not entirely outside of the facility.

The evacuation assembly areas should be capable of accommodating all employees. These assembly areas may be near-by structures, open fields, streets, or parking lots located away from the emergency site.

Employees must be instructed to move away from the exit discharge in order to prevent a backup of other employees still inside an exit route.

Evacuation Elements

A disorganized evacuation can result in confusion, injury, and property damage. When developing your emergency action plan, it is important to determine the following:

- Conditions under which an evacuation would be necessary.
- Conditions under which it may be better to shelter-in-place.
- A clear chain of command and designation of the person in your business authorized to order an evacuation or shutdown.
- Specific evacuation procedures, including routes and exits.
- Specific evacuation procedures for high-rise buildings.
- Procedures for assisting visitors and employees to evacuate, particularly those with disabilities or who do not speak English.
- Designation of what, if any, employees will remain after the evacuation alarm to shut down critical operations or perform other duties before evacuating.
- A means of accounting for employees after an evacuation.
- Special equipment for employees.

- Appropriate respirators.

Fight or Flee?

A fire is the most common type of emergency for which small businesses must plan. A critical decision when planning is whether or not employees should fight a small fire with a portable fire extinguisher or simply evacuate. Small fires can often be put out quickly by a well-trained employee with a portable fire extinguisher. However, to do this safely, the employee must understand the use and limitation of a portable fire extinguisher and the hazards associated with fighting fires. Evacuation plans that designate or require some or all of the employees to fight fires with portable fire extinguishers increase the level of complexity of the plan and the level of training that must be provided to employees.

Shelter-in-Place

Chemical, biological, or radiological contaminants may be released into the environment in such quantity and/or proximity to a place of business that it is safer to remain indoors rather than to evacuate employees.

Such releases may be either accidental or intentional. Examples of situations that might result in a decision by an employer to institute "shelter-in-place" include an explosion in an ammonia refrigeration facility across the street, or a derailed and leaking tank car of chlorine on the rail line behind your place of business.

"Shelter-in-place" means selecting an interior room or rooms within your facility, or ones with no or few windows, and taking refuge there. In many cases, local authorities will issue advice to shelter-in-place via TV or radio.

Depending on your circumstances and the type of emergency, the first important decision is whether you stay put or get away. You should understand and plan for both possibilities. Use common sense and available information, including what you are learning here, to determine if there is immediate danger. In any emergency, local authorities may or may

not immediately be able to provide information on what is happening and what you should do. Use available information to assess the situation. If you see large amounts of debris in the air, or if local authorities say the air is badly contaminated, you may want to "shelter-in-place."

However, you should watch TV, listen to the radio, or check the Internet often for information or official instructions as it becomes available. If you are specifically told to evacuate or seek medical treatment, do so immediately.

If you intend to include a shelter-in-place option in your emergency plan, be sure to keep the following in mind:

- Implement a means of alerting your employees to shelter-in-place that is easily distinguishable from that used to signal an evacuation.
- Train employees in the shelter-in-place procedures and their roles in implementing them.

Specific procedures for shelter-in-place at a worksite may include the following:

- Close the business.
- If there are customers, clients, or visitors in the building, provide for their safety by asking them to stay - not leave. When authorities provide directions to shelter-in-place, they want everyone to take those steps immediately. Do not drive or walk outdoors.
- Unless there is an imminent threat, ask employees, customers, clients, and visitors to call their emergency contact to let them know where they are and that they are safe.
- Turn on call-forwarding or alternative telephone answering systems or services. If the business has voice mail or an automated attendant, change the recording to indicate that the business is closed, and that staff and visitors are remaining in the building until authorities advise it is safe to leave.
- Quickly lock exterior doors and close windows, air vents, and fireplace dampers. Have employees familiar with your building's mechanical systems turn off all fans,

heating and air conditioning systems, and clothes dryers. Some systems automatically provide for exchange of inside air with outside air. These systems, in particular, need to be turned off, sealed, or disabled.

- If you are told there is danger of explosion, close the window shades, blinds, or curtains.
- Gather essential disaster supplies, such as nonperishable food, bottled water, battery-powered radios, first-aid supplies, flashlights, batteries, duct tape, plastic sheeting, and plastic garbage bags.

Shelter-in-Place Selection and Supplies

Select interior room(s) above the ground floor, with the fewest windows or vents. The room(s) should have adequate space for everyone to be able to sit. Avoid overcrowding by selecting several rooms if necessary. Large storage closets, utility rooms, pantries, and copy and conference rooms without exterior windows will work well. Avoid selecting a room with mechanical equipment like ventilation blowers or pipes because this equipment may not be able to be sealed from the outdoors.

It is ideal to have a hard-wired telephone in the room(s) you select. Call emergency contacts and have the phone available if you need to report a life-threatening condition. Cellular telephone equipment may be overwhelmed or damaged during an emergency.

Take your emergency supplies and go into the room you have designated. Seal all windows, doors, and vents with plastic sheeting and duct tape or anything else you have on hand.

Consider precutting plastic sheeting (heavier than food wrap) to seal windows, doors, and air vents. Each piece should be several inches larger than the space you want to cover so that it lies flat against the wall. Label each piece with the location of where it fits.

Write down the names of everyone in the room and call your business' designated emergency contact to report who is in the room with you, and their affiliation with your business (employee, visitor, client, customer).

Listen to the radio, watch television, or use the Internet for further instructions until you are told all is safe or to evacuate. Local officials may call for evacuation in specific areas at greatest risk in your community.

Authority

It is common practice to select a responsible individual to lead and coordinate your emergency plan and evacuation. It is critical that employees know who the coordinator is and understand that this person has the authority to make decisions during emergencies. The coordinator should be responsible for assessing the situation to determine whether an emergency exists requiring activation of the emergency procedures, overseeing emergency procedures, notifying and coordinating with outside emergency services, and directing shutdown of utilities or plant operations if necessary.

In other instances, local emergency officials, such as the local fire department, may order you to evacuate your premises. If you have access to radio or television, listen to newscasts to keep informed and follow whatever official orders you receive.

Training

Training plays an important role in the success of every emergency plan. It may be necessary to train a sufficient number of employees in assisting other employees to evacuate safely from the facility.

The plan must be reviewed with each employee on the following occasions:

- When the plan is developed.
- When the employee's responsibility is changed under the action plan.
- When the plan is changed in any significant way.

When evacuation wardens are used, OSHA has indicated that a ratio of approximately one evacuation warden for 20 employees may be appropriate. However, circumstances in different facilities might require that ratio be increased or decreased.

When used, evacuation wardens must be thoroughly trained and well aware of the facility layout and exit routes. Additionally, the wardens and fellow employees must be aware of employees who may need special assistance in an emergency situation.

It is the employer's option as to whether wardens will be called upon to volunteer to act as "sweepers," meaning they will check all rooms and enclosed spaces for employees who may be trapped or are unaware the EAP has been initiated.

Fire Protection Plan

Subpart E is the requirement for employers to develop a Fire Prevention Plan. The Fire Prevention Plan must be in writing, kept in the facility, and available for review by any employee.

The Elements of Fire Protection Plan

The following elements must be included in a Fire Protection Plan:

- A list of all major fire hazards.
- Procedures of the proper handling and storage of all hazardous materials, potential ignition sources, and their control.
- A list of the types of equipment/systems used to control the fire with their handling and storage procedures.
- Procedures to control the accumulation of flammable and combustible waste material.
- Procedures for the regular maintenance of safeguards installed on heat-producing equipment to prevent the accidental ignition of combustible materials.
- Job titles and/or names of employees responsible for maintaining equipment to prevent or control sources of ignition or fires.
- The names and/or job titles of those employees responsible for the control of fuel source hazards.

Note: The written Fire Prevention Plan must be kept in the workplace and made available to every employee.

More Information: It is the employer's responsibility to ensure that every employee is familiar with his or her role (if any) in the Fire Prevention Plan, and that this role is reviewed.

Maintenance

Maintenance of all equipment and systems installed on heat producing units must be performed in order to prevent an accident. The procedure of their maintenance should be included in the Fire Prevention Plan.

Housekeeping

Proper housekeeping is very important in order to perform a safe evacuation. The exit routes must be clear from any obstruction that might make an evacuation more hazardous.

Exit routes must be kept free of explosive and highly flammable furnishings and other decorations. They must be arranged so the employees will not have to travel towards a high hazard area unless the path of travel is effectively shielded from the high hazards.

Exit Routes

The Requirements

Exit routes must meet the following requirements:

- Every exit route must be a part of the workplace.
- Exit route must be at least 28 inches wide at all points.
- An exit route must be designed using fire resistant materials.

- There must be an adequate number of exit routes to accommodate the number of employees needing access to them.
- Exit routes must be remote from each other so that if one route is inaccessible, employees can use an alternate route.
- Depending on the size of the facility and number of employees, local fire codes may require more than two exits. (Frequently, local code is more stringent, thereby superseding OSHA.

Exit Discharge

Each exit discharge must open directly outside or to a refuge area, walkway, street, or any area with access to the outside. An exit discharge must be large enough to safely accommodate all the occupants using the exit route.

All Exit Doors must be Unlocked

Exit doors must be unlocked at all times while the facility is occupied and should be able to be opened from inside without the use of keys or tools. These doors must be free from any instrument or device that could hinder their use at the time of an emergency. The doors must open outward with the direction of exit travel.

More Information: A side-hinged door should be used in order to connect any room to an exit route.

Case Study

Poisonous Fumes Killed Scientists

A chemical laboratory employed 20 scientists for research, 12 of which were present at the time of the incident. The laboratory had only one door for access and exit.

One morning scientists were performing a routine experiment when suddenly, due to the negligence of a one, some lethal chemical reactions occurred, and poisonous fumes covered the whole workplace within a few seconds.

The scientists tried to evacuate the workplace, but had a hard time doing so due to the heavy smoke and pungent smell. As they reached the exit, they found that the door was locked, and its key was not available. The poisonous fumes took only a few minutes to kill them; experts declared that they died due to the toxic fumes.

Reasons

- The negligence of a scientist caused the lethal chemical reactions.
- One door was not sufficient for 20 employees.
- No safety measures or standards of exit had been followed.
- The exit door was locked, which trapped the employees in the room with the toxic fumes.

The Outdoor Exit Route

An outdoor exit route is permitted. It must be protected with guardrails if a fall hazard exists and should be covered where snow and/or ice present an additional hazard. If no cover is provided, employers assume the responsibility of clearing the exit routes from snow and/or ice as it falls. The outdoor exit route must not lead to a dead-end and should have a safe path with a smooth and solid surface.

Exit Markings

There are two kinds of exit markings:

- Markings that clearly identify an exit or the way to an exit
- Markings that clearly identify the areas or doors that are not an exit

With exit markings, the word "EXIT" must be clearly visible with plain legible letters. Whenever the exit route is not clearly identified, use signs, markings, and/or symbols similar to exit signs in such a way as to provide proper directional information.

If passages, stairways, and doors could be mistaken for an exit, then that route should be clearly marked "Not an Exit" in legible letters or with a symbol that can convey the message to workers. Furthermore, in most instances, the exit signs must be illuminated with auxiliary power so that they continue to work in case of an electrical failure.

Case Study

Fire in a Footwear Factory

A footwear-manufacturing factory employing 30 workers used only one door for both an entrance and an exit. The factory also contained seven windows; five of them were in the front of the factory and two were in the back. The back windows were blocked and secured with wire mesh.

Due to sparks in the electrical wiring, the shoe factory caught fire one morning. The only door for access and exit was locked from the inside, and, as a result, a number of workers died or suffered injuries in that devastating accident.

Reasons

1. Only one door was not sufficient for 30 workers.
2. The only door used for an entrance and an exit was locked, preventing the workers from being able to leave.
3. No fire safety measures or standards of exit had been followed.
4. The workers had not received any safety and health or education.

Lesson Summary

Employees must be aware of their duties and roles in case of an emergency. This level of awareness can be attained as a result of drills and learning from any failures in the EAP process.

An Emergency Action Plan (EAP) must be established in writing for any facility with 10 or more employees. The EAP must include:

- Evacuation procedures and identification of all exit routes.
- First-aid and rescue duty assignments.
- Methods of reporting emergencies by any employee.
- Procedures for employees who remain behind.
- Procedures for an accounting of all employees after evacuation, or upon a shelter-in-place emergency.
- Procedures for critical operation shutdown.
- Clear identification of the 'person-in-charge'—of the EAP, and the chain-of-command.

A fire is the most common type of emergency for which small businesses must plan. A critical decision when planning is whether or not employees should fight a small fire with a portable fire extinguisher or simply evacuate.

A Fire Protection Plan should include:

- A list of all major fire hazards.
- Procedures of the proper handling and storage of all hazardous materials, potential ignition sources, and their control.
- A list of the types of equipment/systems used to control the fire with their handling and storage procedures.
- Procedures to control the accumulation of flammable and combustible waste material.

- Procedures for the regular maintenance of safeguards installed on heat-producing equipment to prevent the accidental ignition of combustible materials.
- Job titles and/or names of employees responsible for maintaining equipment to prevent or control sources of ignition or fires.
- The names and/or job titles of those employees responsible for the control of fuel source hazards.

Additionally, the following points are essential to consider when dealing with exit routes:

- Exit routes must be easily and readily accessible at all times for each employee.
- Access to the exit route and exit must be easily identified.
- Always use side-hinged doors when connecting any occupied room to an exit route.
- Exit doors must be able to open outward with the direction of exit travel.
- Exit routes must not open to high hazard areas, unless the route is protected from those hazards.

Lesson 3: Fire Detection and Extinguishing Systems— General

Lesson Focus

This lesson focuses on the following topics

- Fire Detection Systems
- Fixed Extinguishing Systems
- Total Flooding Applications

Fire Detection Systems

When combined with other elements of an emergency response and evacuation plan, automatic fire and smoke detection systems can significantly reduce property damage, personal injuries, and loss of life from fire in the workplace. Their main function is to identify a developing fire quickly and alert building occupants and emergency response personnel before extensive damage occurs.

How Do Fire Detectors Work?

Fire detectors work by sensing one or more products of fire. The three most common detectors are:

1. Smoke detectors
2. Heat detectors
3. Flame detectors

Smoke Detectors

Smoke detectors detect the smoke particles created from combustion. The two main types are ionization detectors and photoelectric detectors.

Ionization Detectors

The ionization detector contains a small radioactive source that is used to charge the air inside a small chamber. The charged air allows a small current to cross through the chamber and complete an electrical circuit. When smoke enters the chamber, it shields the radiation, which stops the current and triggers an alarm.

Photoelectric Detectors

In a photoelectric smoke detector, a light source and light sensor are arranged so that the rays from the light source do not hit the light sensor. When smoke particles enter the light path, some of the light is scattered and redirected onto the sensor, causing the detector to activate an alarm.

Heat Detectors

Heat detectors are normally used in dirty environments or where dense smoke is occasionally produced. Heat detectors may be less sensitive but may be more appropriate than a smoke detector in these environments.

Flame Detectors

Flame detectors are best for protecting:

- Areas with high ceilings and open-spaces, such as warehouses and auditoriums.
- Outdoor or semi-enclosed areas, where winds or draughts can prevent smoke from reaching a heat or smoke detector.
- Areas where rapidly developing flaming fires can occur, such as petrochemical production, fuel storage areas, paint shops, and solvent areas.
- Environments that are unsuitable for other types of detectors.

Installation and Restoration

General guidelines for placing fire detectors:

- Always follow the manufacturer's guidelines when selecting and installing any fire or smoke detection equipment.
- Install at least one detector in each room, storage area, and hallway. You may need more than one detector per room for those areas that exceed the manufacturer's spacing requirements. For example, if your detector is rated for 30 feet, install detectors so they are evenly spaced with no more than 30 feet between detectors.
- Place the detector as close to the center of the ceiling as possible when only one detector is required in a room or space.
- Put at least one detector in each closet, elevator or dumbwaiter shaft, stairwell, and other enclosed spaces.
- Place a detector at the top of each flight of stairs.
- Place the detectors in the path of the air flow toward the return air duct when air supply or return ducts are present in a room or space.
- Place all smoke detectors the proper distance from ceiling fans.

Maintenance and Testing

Over time, dust, dirt, and other foreign material can build up inside a detector's sensing elements, resulting in reduced sensitivity, which can limit the amount of warning time given during a fire. Dirty or dusty detectors can also result in unwanted alarms that can desensitize occupants to the alarm system or produce more serious behavior (such as disconnecting the system altogether).

To avoid malfunctions and unwanted alarms and to make sure your fire detection system will perform as expected in the event of a fire, you are required to:

- Follow all manufacturer's guidelines for the testing, maintenance, and cleaning of the equipment.

- Operate and maintain your system in a working condition, making sure it is always turned on, except during repairs or maintenance.
- Test and adjust fire detectors and fire detection systems often to ensure that they operate correctly and maintain reliability. Detectors found to be unreliable and/or with reduced sensitivity must be replaced or cleaned and recalibrated.
- Assure that pneumatic and hydraulic operated detection systems installed after January 1, 1981, are equipped with supervised systems.
- Have a qualified person service, maintain, and test all fire detection systems, including cleaning and necessary sensitivity adjustments.
- Have fire detectors cleaned of dust, dirt, or other particulates at periodic intervals to assure their proper operation.

All fire detection equipment must be returned to normal operation as soon as possible after being tested, used, or accidentally activated.

Note: You are also required to have spare detection devices and components readily available in the workplace or from a local supplier to ensure prompt restoration of the system.

Specific Applications or Uses

When a fire detection system is installed for the purpose of activating a fire extinguishing or suppression system, it must:

- Be designed to respond in time to control or extinguish the potential fire.
- Be designed and installed to provide a warning for emergency action and safe escape of employees.

Do not delay alarms or devices actuated by fire detectors for more than 30 seconds, unless the delay is necessary for the safety of employees.

For example, if a fire suppression system uses a compressed gas that will flood an occupied area, it will be necessary to give employees time to escape. If such a delay is necessary, it must be addressed in an emergency action plan.

Fixed Extinguishing Systems

Fixed fire extinguishing/suppression systems are commonly used to protect areas containing valuable or critical equipment such as data processing rooms, telecommunication switches, and process control rooms and areas of high hazard such as heat-treating and cooking. Their main function is to quickly extinguish a developing fire and alert occupants before extensive damage occurs by filling the protected area with a gas or chemical extinguishing agent.

System Components Review

In the system component diagram, there is a cutaway 3-D image of a room with an entrance door and an exit door on the opposite wall, with two discharge nozzles, one on the wall between the doors and one above the exit door (marked with an arrow and a number 1). Piping (marked with an arrow and a number 2) extends across the top of the room to the opposite wall with a set of four agent containers on the outer side of the wall attached to a control panel (marked with an arrow and a number 3).

Outside the room at the entrance door is a control panel (marked with an arrow and a number 4), and on the door, there is a hazard warning sign (marked with an arrow and a number 5). Above the exit door inside the room, there is a fire detector mounted adjacent to the exit sign (marked with an arrow and a number 6). To the right of the exit door is a manual discharge station (marked with an arrow and a number 7). Outside the room on the outer wall are the four agent storage containers connected to the piping (marked with an arrow and a number 8).

System Design and Application

Typical elements and components include:



System Components

- 1 Discharge nozzles
- 2 Piping
- 3 Control panel
- 4 Discharge or warning alarm(s)
- 5 Hazard warning or caution signs
- 6 Automatic fire detection device(s)
- 7 Manual discharge station(s)
- 8 Storage container(s) & extinguishing agent

System Design and Application

Typical elements and components include:

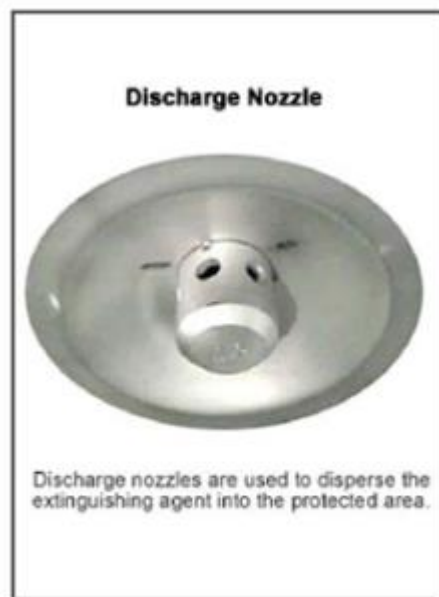
1. **Discharge Nozzles:** Discharge nozzles are used to disperse the extinguishing agent into the protected area.
2. **Piping:** The piping system is used to transport the extinguishing agent (carbon dioxide, halon, argon, etc.) from its storage container to the discharge nozzle.
3. **Control Panel:** The control panel integrates all devices and displays their operational status and condition.
4. **Warning Alarm:** Electronic devices that provide an audible or visual alarm when detected.
5. **Warning and Caution Signs:** Hazard warning signs must be posted at the entrance to, and inside, areas protected by fixed extinguishing system.
6. **Fire Detectors:** A device that detects fire and causes an alarm signal to be generated.
7. **Pull Station:** A device that provides a way to manually discharge the fire extinguishing system.

8. **Agent Storage Containers:** The storage system discharges agent into the piping and through the discharge nozzles when activated by a manual or automatic device.

Discharge Nozzles

A discharge nozzle is a device designed to release the extinguishing agent at a specific rate and pattern to quickly extinguish a fire.

The nozzle pictured next is designed for total flooding applications. Its bell shape and multiple small discharge openings will create a widely dispersed and diffused discharge pattern to flood an enclosed area quickly with an extinguishing agent. This type of nozzle is common in paint spray booths, flammable liquid storage rooms, or other similar areas.



More Information: Nozzles designed for local application may usually consist of a cylindrical body with a straight unobstructed center opening.

This type of nozzle produces a small discharge pattern that directs the agent toward a local application such as paint dip tanks, quench tanks, or areas where a flammable liquid spill may occur.

Piping

The piping network is designed to properly distribute the extinguishing agent to the protected areas.

Control Panel

The control panel monitors and integrates all components and controls the audible and visual alarms and discharge functions. When an automatic or manual device is activated, it sends a signal to the control panel where, depending on the type of system and hazards, can be programmed to:

- Activate a pre-discharge alarm
- Initiate agent release
- Shutdown ventilation systems
- Shutdown machinery or equipment
- Activate visual and audible fire alarms
- Notify emergency response personnel

Warning Alarm (Discharge Alarm or Signaling Devices)

Each area protected by a fixed extinguishing system must have a distinctive alarm or signal to alert occupants that the system is discharging. Unless the discharge is immediately recognizable, these alarms must be able to be heard or seen above ambient noise or light levels.

If the alarm or signaling device is used on a total flooding system, it must also:

- Alarm before the system discharges to give employees sufficient time to exit the space safely.
- Be connected to an approved fire detection device that automatically activates the pre-discharge alarm before the system discharge.
- Be addressed in an emergency action plan in accordance with each area that is protected.

More Information: Areas protected by a total flooding system where employees cannot enter during or after the system's operation are exempt from the requirements.

Warning and Caution Signs

Warning signs must be posted to warn employees in advance about the hazards associated with the extinguishing agent. Hazard warning or caution signs must be posted at the entrance to and inside of areas protected by fixed extinguishing systems that use agents in concentrations known to be hazardous to employee safety and health.

Recommended guidelines for posting:

- Avoid confusion. Keep hazard warning signs consistent by using standardized patterns, phrases, colors, shapes, and pictures.
- Wording on signs should be concise, easy to read, and contain sufficient information that is easily understood. Warning signs must be in English and should also be posted in the native language spoken by the employees.

Note: In addition to warning signs, the employer must also provide effective safeguards to warn employees when the atmosphere of a protected area remains hazardous to their safety or health as a result of a system discharge.

Fire Detectors (Automatic)

Automatic detection devices sense the smoke, heat, or flames from a fire and initiate an alarm.

Automatic detection devices used on a total flooding system must automatically activate the pre-discharge alarm before the system discharges to give employees time to exit from the protected area safely.

Pull Station (Manual Discharge Station)

A manual discharge station is a device usually mounted on a wall near the emergency exit(s) that will automatically sound an alarm and release the extinguishing agent when activated by personnel. If your workplace is equipped with a fixed suppressant system, there must be at least one manual station for each protected area.

Pull station guidelines include:

- Properly mark and/or label pull stations and other activation devices to indicate their function and the potential hazard to personnel.
- Pull stations must be clearly visible (not blocked or covered) and located near exits or escape routes.
- Post operating instructions at each station.
- Train employees to know when, where, and how to use pull stations.
- Determine, if possible, if someone is in the area and get the person out before using a manual discharge station.

Operation, Maintenance, and Testing

Automatic fire suppression systems must be operated properly and regularly maintained and tested to guarantee worker safety and system effectiveness. To ensure your fire extinguishing system will perform as expected in the event of a fire, you are required to:

- Inspect each system in accordance with the manufacturer's guidelines, and operate and maintain them in working condition, making sure they are always turned on, except during repairs or maintenance.
- Notify employees and establish measures to guarantee their safety if a fixed extinguishing system becomes inoperable. Fix any defects or impairments by trained personnel.
- Check the weight and pressure of refillable containers in accordance with the manufacturer's guidelines.
- Assure that factory charged non-refillable containers that have no means of pressure indication are weighed in accordance with the manufacturer's guidelines.
- Assure that inspection and maintenance dates are recorded and kept until the container is checked again or the life of the container expires, whichever comes first.
- Train employees designated to inspect, maintain, operate, or repair fixed extinguishing systems. Review their training at least annually to keep them up-to-date.
- Train all employees with respect to the type of systems installed in the workplace, the hazards involved, proper activation in case of emergency, and the correct response to audible and visual pre-discharge alarms. Provide training for non-English speaking employees in languages understood by the affected employees and other individuals that may be exposed to the hazard.
- Provide and assure the use of Personal Protective Equipment (PPE) for rescuing employees trapped in areas that have become hazardous due to an agent discharge or ensure that emergency rescue personnel are able to respond in an appropriate amount of time.

Total Flooding Applications

Safety and Health Hazards

The Safety Data Sheet for each extinguishing agent should be available in the workplace.

Carbon Dioxide

Carbon dioxide, under normal conditions, is a colorless, odorless, electrically nonconductive gas that is approximately 1.5 times heavier than air.

More Information: Specific hazards of carbon dioxide include:

- **Asphyxiation:** Exposure to concentrations of CO₂ may create an oxygen-deficient atmosphere.
- **Cold Temperature:** Direct contact with the vaporizing liquid during discharge can cause frostbite burns to the skin.

Halon Systems

Halon is a bromo freon that works by replacing oxygen, thus creating an oxygen-deficient atmosphere. This can be very dangerous to employees. The reaction products of halon on hot surfaces may result in the release of toxic substances and carcinogens. These areas should be clearly labeled as follows.

Specific halon hazards include:

- **Asphyxiation:** Exposure to concentrations of halon can create an oxygen-deficient atmosphere.
- **Cold Temperature:** Direct contact with the vaporizing liquid during discharge can cause frostbite burns to the skin.

- **Central Nervous System (CNS):** Inhaling concentrations of halon gas can cause dizziness, tingling in extremities and, in severe cases, unconsciousness.
- **Cardiovascular Effect:** In some people, exposure to halon can cause an increased sensitivity of the heart to adrenaline resulting in irregular heartbeats and, in severe cases, heart attack.

When halon is exposed to temperatures above 900 degrees Fahrenheit, it could break down and create high concentrations of toxic gases.

Lesson Summary

The employer shall provide a distinctive alarm or signaling system capable of being perceived above ambient noise or light levels to indicate when an extinguishing system is discharging; however, discharge alarms may not be required on systems where discharge is immediately recognizable.

Fire detectors work by sensing one or more products of fire. The three most common detectors are:

1. Smoke detectors
2. Heat detectors
3. Flame detectors

Fixed fire extinguishing/suppression systems are commonly used to protect areas containing valuable or critical equipment. Their main function is to quickly extinguish a developing fire and alert occupants before extensive damage occurs by filling the protected area with a gas or chemical extinguishing agent. Typical elements and components include:

- Discharge Nozzles
- Piping

- Control Panel
- Warning Alarm
- Warning and Caution Signs
- Fire Detectors
- Pull Station
- Agent Storage Containers

Module 5: Electrical Safety

Module Description

Every year, tens of thousands of people are injured or killed by electrical shocks/contacts in the United States. Employees are exposed to dangers such as electric shock, electrocution, burns, fires and explosions. It is essential to understand how electricity is potentially lethal for us and how we can save lives. This module gives a basic understanding of how to prevent or eliminate work-related injuries.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Identify electrical hazards that may be present at a workplace.
- Discuss types of injuries that may result from contact with electricity.
- Describe methods of protection from electrical hazards.
- Summarize training and other essential factors associated with electrical safety.

Lesson 1: Electricity, Hazards & Safety, & Tools

Lesson Focus

This lesson focuses on the following topics:

- Electricity
- Accidents and Injuries
- Hazards
- Safety Measures
- Grounding Tools and Equipment
- Personal Protective Equipment (PPE)

Electricity

Electricity is the flow of electrons, a form of energy, which flows from one place to another. The flow of electrons (current) travels through a conductor as opposed to an insulator, through which electricity will not flow. Electricity is considered one of the most serious workplace hazards; it exposes employees to electric shock, burn, electrocution, fires, and explosion. Whenever you are working with electrical circuits or power tools there is a risk of injury, especially of electrical shock.

Severity of the Shock

Shock severity depends upon the amount of the current, the presence of moisture, the path of the current through the body, and the health of the victim prior to the shock. The intensity of the injury increases depending on how long a body is in contact with a circuit.

Severe Electric Shock

In severe shock, breathing may become difficult—a sign of possible respiratory arrest—and the heart may stop pumping.

Medical attention should be provided immediately, as there is a likelihood of internal hemorrhaging and destruction of tissue, nerves, and muscle. The chance of death in a case of severe shock is high.

Note: Low voltage does not mean low hazard.

Current and Our Body

Our body is comprised of approximately 70 percent water; unfortunately for us, water can be a good conductor of electricity. Therefore, whenever you touch a live current, it completes its path rapidly due to the conductivity of the body, which could lead to serious injuries or death.

More Information: Pure water is a poor conductor, but small amounts of impurities, like salt and acid (both are in perspiration), make it a ready conductor.

Accidents and Injuries

When a current enters the body, it can produce several types of injuries, such as falls, burns, fires, etc.

Accidents

Major causes of electrical accidents include:

- Use of defective, damaged, and improper tools.
- Use of tools or equipment in close proximity to energized parts.
- Overloading outlets.
- Repairing tools or equipment while the power is on.
- Working near power lines.

Metal Parts

Electric shock can also occur from contact with metal surfaces or parts, as they may become energized if there is any break in the machine's wiring insulation.

Burns

Burns are the most common type of shock-related injury. They can occur when your body touches live wire or electric current.

Burns caused by electrical incidents can be classified into three types:

1. **Electrical Burns:** Electrical burns are the result of current flow in the body, which can damage body tissue or the heart.
2. **Arc and Flash Burns:** Arc and flash burns are caused by the high temperature generated by an electrical arc or explosion near the body.
3. **Thermal Contact Burns:** Thermal contact burns are the result of skin contact with overheated electric equipment or by ignited clothing.

Falls

Electric shocks can also be responsible for indirect injuries such as falls. Workers who experience shock may do so from a height, from which falling could also lead to serious injury or death.

Case Study

Fall Due to Electrical Contact

A victim was holding a roofing mop at the top of a ladder; the mop consisted of a new, heavy nylon mop head attached to a six feet long aluminum rod. As he climbed to the top of the ladder to access a 30-foot-high roof, the handle of the mop contacted a 3600-volt power line.

A witness described that the victim first shook for few seconds then fell backwards off the ladder to the ground below. The worker was taken to the hospital where he died the next day due to fall-related injuries.

Reasons

The victim moved the mop, which had an aluminum handle, only a few feet beneath power lines thereby ignoring the risks associated with power lines.

As the aluminum handle touched the high-voltage lines, the worker was immediately shocked and fell backwards off the ladder.

The victim died by indirect injuries of electric shock—injuries resulting from his fall were the actual causes of death.

More Information: Had the victim not received adequate safety and health training, as evidenced by the lack of protective safety equipment, his working in close proximity to power lines, and his use of a conductive tool? He didn't survive to respond to questioning.

Hazards

Electrical hazards are associated with inadequate or damaged wiring, exposed electrical parts, wires with bad insulation, ungrounded electrical systems and tools, overloaded circuits, damaged power tools and equipment, and power lines.

Protection measures include proper grounding, use of fuses and circuit breakers, use of GFCIs, and guarding live parts. Safety-related work practices include:

- Keeping working places and walkways clear of electrical cords.
- Using barriers and guards to prevent injuries and accidents from exposed energized equipment or parts.

- Pre-planning work, using PPE, and posting warnings.

Note: Electricity can be extremely dangerous in wet working conditions.

Case Study

No Grounding and No GFCI

A company owner directed his worker to check some microwave ovens. The worker knew about repairing microwaves but had not worked on similar types of equipment for a year. The victim was engaged in repairing a microwave oven, which was connected to an ungrounded extension cord. The victim opened the case of the microwave, exposing its inner components. At some point, he energized the appliance with his left hand to check something and held a pair of needle-nose pliers in his right hand. He was using pliers to wire a high-voltage transformer to the magnetron (a component of microwave ovens). The rubber insulation on the handle of the pliers was abraded. The victim inadvertently touched a live wire with the pair of pliers and was immediately electrocuted.

Reasons

- The appliance was being repaired while it was energized.
- The appliance was connected with an ungrounded extension cord.
- The rubber insulation on the handle of the pliers was abraded, allowing current to enter the worker's body.
- The victim had not received adequate safety training.

Electrical Hazards

Electrical shocks, fires, or falls result from these hazards:

- Exposed electrical parts
- Overhead power lines

- Inadequate wiring
- Defective insulation
- Improper grounding
- Overloaded circuits
- Wet working conditions
- Damaged tools and equipment
- Improper PPE

Prevention of Electrical Hazards

Consider the following points for safe use of tools:

- Inspect tools before use.
- Use the correct tool for the job and use it correctly.
- Store tools in a safe manner.
- Use double insulated tools.
- Use tools and equipment according to the instructions provided by the manufacturer.
- Visually inspect all electrical equipment before use. Remove from service any equipment with frayed cords, missing ground prongs, cracked tool casings, etc.
- Apply a warning tag to any defective tool and do not use it until the problem has been corrected.

Overloaded Circuits

Overloaded circuits can be very dangerous. This happens when:

- Too many devices/appliances are plugged into a circuit.
- Damaged tools are overheating.
- Wire insulation melts.

Safety Measures

Workplace Safety

The following are some essentials of workplace safety:

- Planning for every job is vital.
- Always use the right tools for the job.
- Follow procedures, drawings, diagrams, and other documentation to complete the job.
- Equipment isolation from energy sources is essential (Lockout/Tagout).
- You should be trained to complete the job in the proper way.
- Use appropriate Personal Protective Equipment (PPE) for the work being done.

Power Tools

Many power tools must have a three-wire cord plugged into a grounded receptacle or they must be double insulated. Some are powered by a low voltage transformer.

Power tools safety tips are as follows:

- Use gloves and appropriate footwear.
- Store tools in a dry place when not in use.
- Do not use in wet/damp conditions unless designed specifically for this environment.
- Keep working areas well lit.
- Ensure stable footing to avoid tripping hazards.
- Do not carry a tool by the cord.
- Do not yank the cord to disconnect it.
- Keep cords away from heat, oil, and sharp objects.
- Disconnect when not in use and when changing accessories such as blades and bits.
- Remove damaged tools from service.

Equipment Not Used in Manner Prescribed

Am I In Danger?

If electrical equipment is used in ways for which it is not designed, you can no longer depend on safety features built in by the manufacturer. This may damage your equipment and cause employee injuries.

Common examples of misused equipment include:

- Using multi-receptacle boxes designed to be *mounted* by fitting them with a power cord and placing them on the floor.
- Attaching ungrounded, two-prong adapter plugs to three-prong cords and tools.

Temporary Lights

Do not suspend temporary lights by cords unless they are designed to do so. Protect your body from contact with lights.

Lockout and Tagging of Circuits

Lockout and tagout is essential to prevent workplace electrical hazards. It includes applying locks to power sources after deenergizing and tagging deactivated controls. Tagging should identify the worker and the equipment or circuits being worked on.

Grounding Tools and Equipment

Grounding

Grounding helps safely route unwanted current through a path of least resistance to the earth or ground. It is designed so that when an electrical short occurs, electricity flows to the ground rather than back into the equipment and to the operator, thus protecting the worker from electrical shock, injury, and possible death.

Ground Fault Circuit Interrupters (GFCI)

A Ground Fault Circuit Interrupter (GFCI) is a fast-acting circuit breaker that senses small imbalances in a circuit caused by loss of ground or fluctuations in the electric current. It helps prevent major shocks by promptly shutting off the flow of electricity.

Note: If a ground fault is detected, GFCI shuts off electricity almost instantaneously.

The Control

To properly ground tools and equipment you must consider the following elements:

- Ground power supply systems, electrical circuits, and electrical equipment.
- Frequently inspect electrical systems to insure the path to ground is continuous.
- Inspect electrical equipment before use.
- Don't remove ground prongs from tools or extension cords.
- Ground exposed metal parts of equipment.

The Components—Extension Cords

A typical industrial extension cord grounding system has four components.

1. A third wire in the cord (the ground wire)
2. A three-prong plug with a grounding prong
3. Circuit wires
4. Insulation

Note: Extension cords may only be used in a properly grounded outlet.

The Kinds of Grounds

Two kinds of grounding are available as secondary protective measures to protect against electrical shock.

Service or System Ground

In this instance, one wire—called the neutral conductor or grounded conductor—is grounded. In an ordinary low-voltage circuit, the white (or gray) wire is grounded at the generator or transformer and again at the service entrance of the building. This type of ground is primarily designed to protect machines, tools, and insulation against damage.

Equipment Ground

For enhanced worker protection, an additional ground called the equipment ground is furnished by providing another path from the tool or machine through which the current can flow to the ground. This additional ground safeguards the electric equipment operator if a malfunction causes the metal frame of the tool to become energized.

Personal Protective Equipment (PPE)

Personal Protective Equipment (PPE) refers to items typically worn by a worker to provide protection from recognized hazards. Depending on the job task to be performed, PPE generally includes safety glasses, face shields, hard hats, safety shoes, insulating (rubber) gloves with leather protectors, insulating sleeves, and flame-resistant (FR) clothing. Additional PPE, such as fall protection equipment, respirators, chemical-resistant or cut-resistant gloves, and chaps, may be required, depending on the results of the hazard assessment required under 1910.132

Note: Standing on wet surfaces or in water, wearing wet clothes, working in areas of high humidity, and perspiring, may also increase the chances of being electrocuted.

Safety Shoes

Only use nonconductive safety shoes where a chance of electrocution exists. They help protect your feet from completing an electrical circuit to the ground.

These shoes should be used with other insulating equipment and in conjunction with other precautions to reduce or eliminate the potential for providing a path for hazardous electrical energy.

When it is necessary to handle or come close to live wires, it is essential to use proper insulating PPE to protect workers from contact with hazardous electrical energy.

Hard Hats

Hard hats are required when performing electrical work. An ANSI specification "Class E" Electrical Utility type hard hat provides protection protects against falling objects and high-voltage shocks and burns.

Gloves

Insulating gloves and sleeves are critical PPE for electrical work on or near exposed energized parts.

Additionally, 1910.137 provides specific design, care, and use requirements for rubber electrical protective equipment. Insulating gloves and sleeves must be rated for the voltage to which a worker will be exposed (phase to ground or phase to phase) and marked to indicate their rating. 1910.137 recognizes Class 0 (up to 1KV) through Class 4 (up to 36KV) rubber equipment.

Testing and Inspection

Gloves and sleeves must be electrically tested before being issued for use. They must also be visually inspected, and gloves need to be air tested for any possible defects (for example, cuts, holes, tears, embedded objects, changes in texture) before each day's use and whenever there is a reason to believe they may have been damaged. Best practice is to inspect PPE and air test the gloves and sleeves before each use.

Insulating equipment may not be used if any of the following defects are present: holes, tears, punctures or cuts, ozone cutting or ozone checking, embedded foreign objects, texture changes, including swelling, softening, hardening, or becoming sticky or inelastic, and any other defect that damages the insulating properties. [See 1910.137(b)(2)(iii) and *ASTM F1236-96*, Standard Guide for Visual Inspection of Electrical Protective Rubber Products].

Insulating equipment failing to pass inspection must be removed from service and may not be used by workers.

In addition, the gloves and sleeves must be electrically tested at regular intervals of not more than 6 months for gloves and 12 months for sleeves. (See *ASTM F496*, Standard Specification for In-Service Care of Insulating Gloves and Sleeves for some appropriate test methods.) When gloves and sleeves are used regularly, best practice is to test as frequently as monthly.

Protector Gloves and Storage

To ensure worker safety and the integrity of the gloves and sleeves, insulating gloves need to be worn along with protector gloves (such as leather), and both insulating gloves and sleeves need to be stored properly when not in use. Proper storage means that gloves must not be folded and need to be kept out of excessive heat, sunlight, humidity, ozone, and any chemical or substance that could damage the rubber.

Case Study

A Man Is Electrocuted while Removing a Building Fuse with a Bare Hand

An accident occurred in the electrical room of a building maintenance company. During maintenance work, the power supply was left on in the building. The victim opened the door of a high voltage control panel, held the frame of the panel case with his bare left

hand, and with the right hand tried to remove a fuse for a test. The victim suddenly fell to the floor and died about an hour later.

Reasons

- The victim tried to remove the fuse while the high-voltage current was passing from an isolator through the fuse and the instrument transformer.
- Gloves were not used (Personal Protective Equipment).
- Proper tools were not used.
- The victim had not received any safety training.

Lesson Summary

Electricity is considered one of the most serious workplace hazards; it exposes employees to electric shock, burn, electrocution, fires, and explosion. In severe shock, breathing may become difficult—a sign of possible respiratory arrest—and the heart may stop pumping.

Major causes of electrical accidents include:

- Use of defective, damaged, and improper tools.
- Use of tools or equipment in close proximity to energized parts.
- Overloading outlets.
- Repairing tools or equipment while the power is on.
- Working near power lines.

Burns caused by electrical incidents can be classified into three types:

1. **Electrical Burns:** Current flow in the body, which can damage body tissue or the heart.
2. **Arc and Flash Burns:** High temperature generated by an electrical arc or explosion near the body.

3. **Thermal Contact Burns:** Skin contact with overheated electric equipment or by ignited clothing.

Electric shocks can also be responsible for indirect injuries such as falls.

Safety-related work practices include:

- Keeping working places and walkways clear of electrical cords.
- Using barriers and guards to prevent injuries and accidents from exposed energized equipment or parts.
- Pre-planning work, using PPE, and posting warnings.

Overloaded circuits can be very dangerous. This happens when:

- Too many devices/appliances are plugged into a circuit.
- Damaged tools are overheating.
- Wire insulation melts.

Power tools safety tips are as follows:

- Use gloves and appropriate footwear.
- Store tools in a dry place when not in use.
- Do not use in wet/damp conditions.
- Keep working areas well lit.
- Ensure stable footing to avoid tripping hazards.
- Do not carry a tool by the cord.
- Do not yank the cord to disconnect it.
- Keep cords away from heat, oil, and sharp objects.
- Disconnect when not in use
- Remove damaged tools from service.

Grounding helps safely route unwanted current through a path of least resistance to the earth or ground. It is designed so that when an electrical short occurs, electricity flows to the ground rather than back into the equipment and to the operator, thus protecting the worker from electrical shock, injury, and possible death.

Personal Protective Equipment (PPE) refers to items typically worn by a worker to provide protection from recognized hazards. Depending on the job task to be performed, PPE generally includes safety glasses, face shields, hard hats, safety shoes, insulating (rubber) gloves with leather protectors, insulating sleeves, and flame-resistant (FR) clothing.

Lesson 2: Lines and Wires, First Aid, and Training

Lesson Focus

This lesson focuses on the following topics:

- Power Lines
- The Use of Wire
- First Aid
- Training

Power Lines

Buried and overhead power lines are very dangerous because they contain extremely high voltage. Workers coming in contact with energized power lines account for a significant number of electrocutions. Equipment and tools that can also come in contact with power lines increase the risk of an accident.

Some examples of equipment that can contact power lines follow:

- Ladders
- Cranes
- Scaffolds
- Aluminum paint rollers

Case Study

Power Lines Accident—An Illustration

An accident occurred when a dump truck driver raised the dump bed beneath power lines to dump the bed's contents. When the bed contacted overhead power lines, the truck's operation stopped, prompting the driver to see what happened.

As his foot touched the ground, his hand was on the door and he was immediately electrocuted. Unfortunately, the current had completed its path.

How to Minimize Power Line Accidents

To minimize the risk of accidents, remember the following guidelines:

- Post warning signs.
- Always assume that power lines are energized.
- Use wood or fiberglass ladders; avoid metal ladders.
- Stay at least 10 feet away from power lines.

Case Study

A Mobile Crane Boom Contacted High-Voltage Power Lines

The victim was engaged in transporting containers of corn from a farm to a warehouse by truck on which a mobile crane was installed. At around 6:00 P.M., as the victim extended the mobile crane boom to hoist the containers off the truck, the boom contacted a 6600-volt high-voltage power line. The victim was supporting the steel container with his bare hands to steady its position, which caused his immediate death by electrocution.

Reasons

- The mobile crane boom was extended beneath a high-voltage power line.
- The worker had not received any safety training.
- The work was carried out in dangerous circumstances where the hazards were misjudged or ignored.

Caught between Power Lines and Equipment

In case of contact between power lines and equipment, remember the following guidelines:

Stay in Equipment

Never touch equipment and the ground at the same time. Touching anything in contact with the ground could be lethal.

Keep Others Away

Immediately warn everyone not to touch the equipment or its load.

Call Local Utility

Ask someone to call 911 or a local electricity utility for help. Give complete details of the incident to the utility so that they can check wires that could fall later.

Contact with Power Lines

Am I In Danger?

Overhead and buried power lines at your site are especially hazardous because they carry extremely high voltage. Fatal electrocution is the main risk, but burns and falls from elevations are also hazards. Using tools and equipment that can contact power lines increases the risk.

How Do I Avoid Hazards?

- Look for overhead power lines and buried power line indicators. Post warning signs.
- Contact utilities for buried power line locations.

- Stay at least 10 feet away from overhead power lines.
- Unless you know otherwise, assume that overhead lines are energized.
- De-energize and ground lines when working near them. Other protective measures include guarding or insulating the lines.
- Use non-conductive wood or fiberglass ladders when working near power line

Lack of Ground-fault Protection

Am I In Danger?

Normal use of electrical equipment at some worksites cause wear and tear that results in insulation breaks, short-circuits, and exposed wires. If there is no ground-fault protection, these can cause a ground-fault that sends current through the worker's body, resulting in electrical burns, explosions, fire, or death.

How Do I Avoid Hazards?

- Use ground-fault circuit interrupters (GFCI)s on all 120-volt, single-phase, 15- and 20-ampere receptacles, *or* have an assured equipment grounding conductor program (AEGCP).
- Follow manufacturers' recommended testing procedure to ensure the GFCI is working correctly.
- Use double-insulated tools and equipment, distinctively marked.
- Use tools and equipment according to the instructions included in their listing, labeling or certification.
- Visually inspect all electrical equipment before use. Remove from service any equipment with frayed cords, missing ground prongs, cracked tool casings, etc. Apply a warning tag to any defective tool and do not use it until the problem has been corrected.

Path to Ground Missing or Discontinuous

Am I In Danger?

If the power supply to the electrical equipment at your site is not grounded or the path has been broken, fault current may travel through a worker's body, causing electrical burns or death. Even when the power system is properly grounded, electrical equipment can instantly change from safe to hazardous because of extreme conditions and rough treatment.

How Do I Avoid Hazards?

- Ground all power supply systems, electrical circuits, and electrical equipment.
- Frequently inspect electrical systems to ensure that the path to ground is continuous.
- Visually inspect all electrical equipment before use. Take any defective equipment out of service.
- Do not remove ground prongs from cord- and plug-connected equipment or extension cords.
- Use double-insulated tools and equipment, distinctively marked.
- Ground all exposed metal parts of equipment.
- Ground metal parts of the following non-electrical equipment, as specified by the OSHA standard [29 CFR 1926.404(f)(7)(v)]:
 - Frames and tracks of electrically operated cranes.
 - Frames of non-electrically driven elevator cars to which electric conductors are attached.
 - Hand-operated metal shifting ropes or cables of electric elevators.
 - Metal partitions, grill work, and similar metal enclosures around equipment of over 1kV between conductors.

The Use of Wire

The use of the correct wire is essential. The selection of wire depends on what type of operation you are going to perform, the building material, the load of electricity, and other environmental factors. Always choose a wire size that can handle the anticipated current load; and remember, the larger the gauge number, the smaller the wire.

Note: Hazards may exist in the wire when plastic or rubber insulation and covering is missing.

Cords and Wires

Essential points for the safe use of cords and wires follow:

- Use only insulated wires except in approved areas.
- Check before use.
- Use only cords of the proper type.
- Use only cords rated for the anticipated usage.
- Use only cords, connection devices, and fittings connected to circuits equipped with fuses or breakers.
- Remove cords from receptacles by pulling on the plugs, not the cords.
- Cords not rated for the anticipated load, or which have been modified, must be taken out of service immediately.

Proper Wiring and Connectors

Always consider the following points when using electrical wires:

- Use and test GFCIs when necessary.
- Check switches and insulation.
- Use three prong plugs where required.

- Use extension cords only when necessary and ensure that they are in acceptable condition and rated for job to be done.
- Use correct connectors.

Avoid Wet Conditions

A damaged tool may not be grounded properly, possibly allowing the housing of the tool to become energized. This may cause you to receive a shock. Improperly grounded switch plates and ceiling lights are especially hazardous in wet conditions. If you touch a live electrical component with an un-insulated hand tool, you are more likely to receive a shock when standing on a wet working surface.

Use of Flex Cords (Extension cords)

Flexible cords must not be:

- Run through holes in walls, ceilings, or floors.
- Run through doorways, windows, or similar openings (unless physically protected).
- Attached to building surfaces.
- Used as a substitute for permanent wiring.

Wire Size as a Hazard

An electrical hazard exists when wire is too small a gauge for the current it will carry. Normally, the circuit breaker in a circuit is matched to the wire size. However, in older wiring, branch lines to permanent appliances and receptacles could be wired with a smaller gauge than the supply line.

First Aid

If a worker's clothing should catch fire and an extinguisher is not available, wrap or smother the victim with blankets or towels after the dangers of electricity have been addressed. Water may be used if there is no danger of exposure to electrical hazards.

In the event a worker's clothing catches fire, use a properly rated portable fire extinguisher carefully, avoiding the victim's head.

Training

Training plays a very important role in the prevention of employees' injuries and deaths. Training must include instructions on how to de-energize electric equipment before inspecting and repairing, lockout/tag out procedures, using personal protective equipment, and how to use cords, cables, and electric tools properly. Employees must be trained with safety-related work practices according to their respective job assignments.

Lesson Summary

Buried and overhead power lines may contain extremely high voltage. Workers coming in contact with these energized power lines may suffer shock or electrocution.

Equipment and tools that come in contact with power lines can also increase the risk of accident.

Some examples of equipment that can contact power lines follow:

- Ladders
- Cranes
- Scaffolds
- Aluminum paint rollers

In case of contact between power lines and equipment, remember the following guidelines:

- Never touch equipment and the ground at the same time.
- Immediately warn everyone not to touch the equipment or its load.
- Ask someone to call 911 or a local electricity utility for help.

Even when the power system is properly grounded, electrical equipment can instantly change from safe to hazardous because of extreme conditions and rough treatment.

The selection of wire depends on what type of operation you are going to perform, the building material, the load of electricity, and other environmental factors. Always choose a wire size that can handle the anticipated current load; and remember, the larger the gauge number, the smaller the wire.

Essential points for the safe use of cords and wires include:

- Use only insulated wires except in approved areas.
- Check before use.
- Use only cords of the proper type and anticipated usage.
- Use only cords, connection devices, and fittings connected to circuits equipped with fuses or breakers.
- Remove cords from receptacles by pulling on the plugs, not the cords.
- Cords not rated for the anticipated load, or which have been modified, must be taken out of service immediately.

Lesson 3: Introduction to NFPA 70E

Lesson Focus

This lesson focuses on the following topics:

- OSHA Requests an Electrical Safety Standard
- The Birth of NFPA 70E
- OSHA Connection
- Safety-Related Work Practices

OSHA Requests an Electrical Safety Standard

A committee was formed by 1976 to help OSHA prepare electrical safety standards that would serve OSHA's needs. OSHA needed electrical safety standards that addressed the employer and employees in the workplace.

OSHA found it confusing when they tried to use the NEC as an overall electrical safety standard because the Code primarily addressed those who design, inspect, and install electrical installations.

The Birth of NFPA 70E

It became apparent that a new book of electrical safety standards was needed that fulfilled OSHA's responsibilities while remaining consistent with the National Electrical Code. This was the birth of NFPA 70E.

The 2018 NFPA 70E is the most current edition.

NFPA 70E Scope

This standard address electrical safety requirements for employee workplaces (including jobsites) that are necessary for the practical safeguarding of employees during activities such as the installation, operation, maintenance, and demolition of electric equipment, etc.

Intent of NFPA 70E

The intent of NFPA 70E is to provide a practical safe working area for employees exposed to hazards related to the use of electricity in the workplace.

OSHA Connection

NFPA 70E was originally developed at OSHA's request to address electrical hazards in the workplace. OSHA bases many of its electrical safety requirements on the comprehensive information on NFPA 70E.

Even though OSHA does not mandate compliance with NFPA 70E itself, it considers NFPA 70E to be an effective how-to manual for OSHA regulation compliance. An employee and others may be cited against the requirements of NFPA 70E by OSHA.

NFPA 70E Compliments OSHA Regulations

In lieu of detailed specifications, OSHA recognizes, and in some cases refers to, industry consensus standards such as NFPA 70E as a tool for assisting with regulatory compliance.

Safety-Related Work Practices

This part is the core of the NFPA 70E and is one of the primary sections that needs to be covered when training in NFPA 70E.

Incident Energy

Incident energy is the amount of thermal energy impressed on a surface, located a certain distance from the source of energy, generated during an electrical arc event.

Incident energy is usually measured in calories per square centimeter, or cal/cm².

Arc Flash Boundary

When an arc flash hazard exists, an arc flash boundary is an approach limit at a distance from a prospective arc source within which a person could receive a second degree burn if an electrical arc flash were to occur.

Limited Approach Boundary

A limited approach boundary is an approach limit at a distance from an exposed energized part within which a shock hazard exists.

Restricted Approach Boundary

This is an approach limit at a distance from an exposed energized electrical conductor or circuit part within which there is an increased likelihood of electric shock, due to electrical arc-over combined with inadvertent movement, for personnel working in close proximity to the energized electrical conductor or circuit part.

Qualified Person

A qualified person is someone who has demonstrated skills and knowledge related to the construction and operation of the electrical equipment and installations and who has received safety training to recognize and avoid the hazards involved.

Shock Hazard

A shock hazard is a dangerous condition associated with the possible release of energy caused by contact or approach to energized electrical conductors or parts.

Lesson Summary

It became apparent that a new book of electrical safety standards was needed that fulfilled OSHA's responsibilities while remaining consistent with the National Electrical Code. This was the birth of NFPA 70E.

This standard addresses electrical safety requirements for employee workplaces (including jobsites) that are necessary for the practical safeguarding of employees during activities such as the installation, operation, maintenance, and demolition of electric equipment, etc. The intent of NFPA 70E is to provide a practical safe working area for employees exposed to hazards related to the use of electricity in the workplace.

Incident energy is the amount of thermal energy impressed on a surface, located a certain distance from the source of energy, generated during an electrical arc event.

There are 3 boundaries used to warn and/or keep workers away from electrical hazards:

1. Arc Flash Boundary
2. Limited Approach Boundary
3. Restricted Approach Boundary

Lesson 4: Electrical Safety Program

Lesson Focus

This lesson focuses on the following topics:

- Electrical Safety Program
- Job Briefing
- General Electrical Safety-Related Work Practices
- Electrical Hazard Analysis
- Electrical Shock Hazard
- Arc Flash Hazard

Electrical Safety Program

According to NFPA 70E Section 110.1, it is the duty of the employer to implement and document an overall electrical safety program that is appropriate to the risk associated with the related hazards their employees may encounter. This program shall be included as a part of their comprehensive safety program and ensure that each employee has the requisite knowledge and skills to identify hazards and to perform their assigned work in a safe manner.

Components of the Electrical Safety Program

The electrical safety program must contain hazard or risk evaluation procedures that should be used before any work is started on or near live parts. The electrical safety program shall also identify the safety-related work practices and procedures and tools and equipment to be used when working on or near these live parts.

Job Briefing

In addition, the electrical safety program should require that job briefings be held for those workers involved prior to working on or near live parts.

The job briefings must cover the hazards associated with the job to be performed, work procedures involved, special precautions, energy source controls and PPE required.

General Electrical Safety-Related Work Practices

NFPA 70E covers work practices used to safeguard employees from electrical injuries while these employees are working on or near exposed parts that could become energized.

Safe Work Condition

Parts that could become energized must be put into an electrically safe work condition before any employee is permitted to work on or near them.

One Element of an Electrically Safe Work Condition

One primary element incorporated in an electrically safe work condition is that a lockout/tagout device must be applied to any circuit part that could become energized prior to working on or near these parts.

There are a few cases in which work on energized parts may be justified. These cases will be covered later. However, only qualified persons are permitted to work on any electrical components, specifically those parts that have not been put into an electrically safe work condition.

Electrical Hazard Analysis

In general, if a part or circuit cannot be put into an electrically safe work condition, an electrical hazard analysis must be performed to determine the electrical hazards that do exist and what must be done to protect the employee from these hazards.

A Lockout/Tagout Program requires the following:

- Application of locks to all power sources, including all potential sources of electrical energy, after each source has been de-energized.
- Application of a tag on each de-energized control identifying who has locked the control out and instructing others to not unlock or re-energize the control.
- Proper training for all workers involved in or potentially impacted by the de-energization of the equipment. The following groups of employees should be included in this training :
 - Authorized employees—those who lock out and/or tag out machines or equipment in order to perform maintenance or servicing
 - Affected employees—those whose job requires them to use or operate equipment or machines being maintained or serviced
 - All other employees who work or operate in areas where lockout/tagout procedures are used

Electrical Shock Hazard

OSHA estimates that approximately 5 workers are electrocuted each week in the United States each year from electric shocks.

The primary causes of electrocution are:

- Contact with overhead power lines
- Contact with live circuits
- Improperly maintained cords, tools, and equipment
- Lightning strikes

"Electrocution" by definition means being killed by electricity.

An employee can receive an electrical shock without being electrocuted, but an electrocution is fatal.

Death occurs when voltage pushes electrons through the human body, particularly the heart.

Electric current passing through the body, even at levels as low as 3 milliamps, can cause injury. Secondary injuries caused by this level of shock can result from falls from ladders and elevations, contact with dangerous surfaces, and other actions that may result in serious injury or even death.

An electric shock as little as 50 milliamps (mA) can disrupt the heart's rhythm and be lethal. It is the amount of current/amperes passing through the body that can cause death in this manner, not the voltage.

The severity of a shock is dependent on numerous factors, including:

- Path of the current through the body. Even a very small shock passing through the heart can actually cause it to stop
- Amount of current/amps flowing through the body
- Duration of exposure to the shock through the body

In spite of what many people believe, LOW VOLTAGE DOES NOT MEAN LOW HAZARD!

Levels of Electric Shock

mA	Affect
0.5–3	Tingling sensation
3–20	Muscle contractions and pain
10–40	"Let go" threshold may be exceeded. Worker may be unable to release a live circuit.
20–150	Painful shock with severe muscle contraction, breathing may become difficult

30–75	Possible respiratory paralysis
100–200	Possible ventricular fibrillation affecting the heart
200–4000	Likely heart damage or stoppage
500+	Tissue, nerve, and organ damage and burning
10,000+	Cardiac arrest, severe burns, death probable

The overcurrent at which a typical fuse or circuit breaker opens is 15,000 milliamps (15 amps). These devices are designed to protect the electrical system, not people! By the time these devices open, death or very serious injury is likely to have occurred.

Shock Risk Assessment

A shock risk assessment must determine:

- The voltage to which personnel will be exposed
- Approach boundary requirements
- The personal protective equipment necessary in order to minimize the possibility of electrical shock to personnel

Arc Flash Hazard

An arc flash occurs when a flashover of electric current leaves the intended path and travels through the air from one conductor to another, or to ground. The results of an arc flash are often very violent. Within an arc flash incident, a large amount of concentrated energy explodes outward from electrical equipment, creating pressure waves that can damage a person's hearing, a high-intensity flash that can damage eyesight, and a superheated ball of gas that can severely burn a worker's body and melt metal.

Origination of Arc Flash Energy

An arc flash, and its resulting release of energy, can only occur if an arc between two differences of potential occurs.

A difference of potential (voltage reading) exists between any two phase conductors, or any phase conductor and a grounded part (grounded systems only).

This can be caused by many different things, including: accidental contact with electrical system components, accumulation of dust, corrosion, dropped tools, improper installation of equipment, and improper work procedures.

Characteristics of an Arc Flash

When an arc occurs, current that is available from the source of electrical energy passes from one conductor to the other conductor at the point of the arc fault.

Because the travel of current is not contained within a conductor, but travels through free air, the effects of the energy are not contained. This energy is referred to as "incident energy."

Lesson Summary

According to NFPA 70E Section 110.1, it is the duty of the employer to implement and document an overall electrical safety program that is appropriate to the risk associated with the related hazards their employees may encounter. The electrical safety program must contain hazard or risk evaluation procedures that should be used before any work is started on or near live parts. In addition, the electrical safety program should require that job briefings be held for those workers involved prior to working on or near live parts.

Parts that could become energized must be put into an electrically safe work condition before any employee is permitted to work on or near them.

In general, if a part or circuit cannot be put into an electrically safe work condition, an electrical hazard analysis must be performed to determine the electrical hazards that do exist and what must be done to protect the employee from these hazards.

The primary causes of electrocution (killed by electricity) are:

- Contact with overhead power lines
- Contact with live circuits
- Improperly maintained cords, tools, and equipment
- Lightning strikes

It is the amount of current/amperes passing through the body that can cause death in this manner, not the voltage. Low voltage does not mean low hazard.

An arc flash occurs when a flashover of electric current leaves the intended path and travels through the air from one conductor to another, or to ground. The results of an arc flash are often very violent.

Lesson 5: Energized Electrical Work Permit

Lesson Focus

This lesson focuses on the following topics:

- Energized Electrical Work Permit
- Electrical Interlocks
- Test Instruments and Equipment
- Handling Portable Electrical Equipment
- Chapter 1—Article 120
- Verifying an Electrically Safe Work Condition
- Testing for the Absence of Voltage
- Grounding Phase Conductors or Circuit Parts

Energized Electrical Work Permit

The NFPA 70E Standard states that if live parts cannot be placed in an electrically safe work condition, then the work to be performed is considered as "energized work" and can only be performed when all required controls are in place, including a written permit.

Electrical Interlocks

Only a qualified person may bypass an electrical safety interlock and then only temporarily while he or she is working performing work on the equipment which requires it to be energized

Test Instruments and Equipment

Many personal injuries result from testing meters that may disintegrate into dangerous shrapnel when exposed to higher voltages or currents for which they are not rated.

All test equipment should be designed, used, and maintained for the specific purposes identified by the manufacturer.

Inspecting Test Equipment

Not only should the meter itself be visually inspected before each use, any leads or test probes that will be used along with the meter should also be inspected. Damaged meters or accessories must not be used and must be repaired or replaced.

Replacement of defective components is required. Taping over damaged leads is not considered a safe repair.

Integrity must be maintained in meter test leads and probe insulation.

Handling Portable Electrical Equipment

Portable electrical equipment must be safely and properly handled. Flexible electrical cords connected to the equipment must not be used to raise or lower the equipment.

Flexible cords may not be fastened in place with staples or hung in a way that could damage the outer insulation jacket.

Flexible Cord Grounding Conductor

Flexible cords that are used with equipment that requires grounding must be equipped with an equipment grounding conductor.

Altering Attachment Plugs and Receptacles

Any attachment plug or receptacle must not be altered if the alteration causes the grounding conductor to be interrupted. Likewise, the prongs of any attachment plug must not be altered to allow the grounding pole of the plug to be inserted into slots intended for current-carrying conductors.

Adapters that serve to interrupt the continuity of the equipment grounding conductor must not be used.

Inspection

Visually inspect cords and cord-and-plug-connected equipment in the following manner:

- Inspect cords, plugs, and attached equipment before each use on any shift.
- Remove defective cords, plugs, and equipment until repairs and tests have been performed on damaged parts.
- Make sure that the plug and receptacle have the same prong attachment and mate properly.

Cords, plugs, and equipment used in wet damp areas must be rated for these areas.

In locations where it is likely that employees will contact or be drenched with water or conductive liquids, ground-fault-circuit-interrupter (GFCI) protection for personnel must be used. The use of GFCI protection also provides additional protection in many instances.

Connecting Attachment Plugs

Employees' hands must not be wet when plugging and unplugging attachment plugs on flexible cords. If a receptacle remains energized while connecting a plug to it and a conductive path could develop between the energized part and the employee's hand due to water, etc., then insulating protective equipment (such as linemen gloves) must be used.

Chapter 1—Article 120

To review, we learned that an electrically safe work condition is defined as a state in which the conductor or circuit part that is to be worked on or near has been:

- Disconnected from energized parts.

- Locked/tagged in accordance with established standards.
- Tested to ensure the absence of voltage.
- Grounded if necessary.

Verifying an Electrically Safe Work Condition

- To verify an electrically safe work condition:
- Determine all possible sources of electrical supply.
- Open all disconnecting devices for each source of electrical supply.
- Visually verify that the disconnecting device is fully open in the disconnected position.
- Apply a lockout/tagout device.
- Test for the presence or absence of voltage.
- Ground the phase conductors in cases of induced voltages or stored energy.

Identifying Sources of Electrical Supply

Often, various parts of circuits and equipment are supplied by different electrical sources or control circuits. In order to identify all possible sources of electrical supply, you must examine all necessary electrical drawings, schematics, control drawings, or any identification tag that may indicate sources of energy to the equipment or circuit to be put in an electrically safe work condition.

Opening Disconnecting Devices

After all sources of electricity are determined, the next step in establishing an electrically safe work condition is opening the disconnecting devices for each source.

Running or operating equipment should be stopped before opening disconnecting devices to avoid arcing within the device, especially on large loads that demonstrate relatively high current flow levels.

Visually Verify Disconnect Blade Position

Disconnect knife blades sometimes become defective and inoperative due to excessive heat and usage. Visually verify, whenever possible, that all phase conductor blades have been fully disconnected, and are visibly separated, from the line side of the switch.

Draw-Out Circuit Breaker

If the disconnecting means is a draw-out type circuit breaker, NFPA 70E says that you should visibly verify that the circuit breaker is properly withdrawn to its fully disconnected position.

Applying Lockout/Tagout Devices

Once all disconnecting devices have been opened or withdrawn and verified, it is now time to apply an approved lockout/tagout device according to a documented and established policy.

Testing for the Absence of Voltage

Just because an electrical supply has been presumably opened and locked and/or tagged out, a circuit or equipment must still be considered as energized (and all required PPE used) until a test has been made using an adequately rated voltage meter to verify the absence of voltage.

Proper procedures for testing a circuit for the absence of voltage include:

- Verify the proper operation of the tester by testing for the presence of voltage on a known voltage value.
- Test each phase conductor (phase-to-phase and phase-to-ground) or circuit part in question to verify they are de-energized.

- Re-verify the proper operation of the tester on the same known voltage value as in Step 1 to verify that the meter was not made inoperative (fuse blown) during Step 2.

Grounding Phase Conductors or Circuit Parts

Even though the source of electrical energy may be disconnected, some circuit conductors may have voltage induced onto them by nearby energized conductors or voltage spikes. Other types of circuits may store electrical energy for quite a long period of time, even after the electrical source has been disconnected.

In these cases, grounding straps should be connected from the conductor or conductors in question to an effective grounding point in order to dissipate this energy.

Lesson Summary

The NFPA 70E Standard states that if live parts cannot be placed in an electrically safe work condition, then the work to be performed is considered as "energized work" and can only be performed when all required controls are in place, including a written permit.

All test equipment should be designed, used, and maintained for the specific purposes identified by the manufacturer.

Portable electrical equipment must be safely and properly handled. Flexible cords that are used with equipment that requires grounding must be equipped with an equipment grounding conductor.

- Any attachment plug or receptacle must not be altered if the alteration causes the grounding conductor to be interrupted. Cords, plugs, and equipment used in wet damp areas must be rated for these areas.
-

An electrically safe work condition is defined as a state in which the conductor or circuit part that is to be worked on or near has been:

- Disconnected from energized parts.
- Locked/tagged in accordance with established standards.
- Tested to ensure the absence of voltage.
- Grounded if necessary.

In order to identify all possible sources of electrical supply, you must examine all necessary electrical drawings, schematics, control drawings, or any identification tag that may indicate sources of energy.

Just because an electrical supply has been presumably opened and locked and/or tagged out, a circuit or equipment must still be considered energized (and all required PPE used) until a test has been made using an adequately rated voltage meter to verify the absence of voltage.

Module 6: Personal Protective Equipment

Module Description

This module will provide employers and employees alike with knowledge concerning the proper selection, care, and use of Personal Protective Equipment (PPE). They also will be informed of the requirements for compliance with OSHA requirements.

Module Learning Objectives

At the conclusion of this module, you will be able to:

- Describe the role of PPE in controlling exposure to hazards in the workplace.
- Identify parts of the body most vulnerable to injury and name hazards associated with each
- Describe the appropriate PPE to be worn for worker safety.

Lesson 1: Introduction to Personal Protective Equipment

Lesson Focus

This lesson focuses on the following topics:

- What is Personal Protective Equipment?
- Personal Protective Equipment Standards
- Employer Responsibilities
- Employee Responsibilities

Overview

What Is Personal Protective Equipment?

Personal Protective Equipment (PPE) means any type of device or clothing (i.e., head protection, eye/face protection, protective footwear, respiratory protection, etc.) worn for protection against biological, chemical, thermal, physical, or other hazards on the job.

All employees who are exposed to hazards like these, and others, may be required to wear appropriate PPE to reduce or eliminate their exposure to hazards.

Personal Protective Equipment Standard

What Is the Personal Protective Equipment Standard?

The Occupational Safety and Health Administration requires that, when necessary, employers establish and administer an effective Personal Protective Equipment (PPE) program for employees in order to reduce the effects of workplace hazard exposure and, as a result, the workplace accident rate in the U.S.

Engineering Controls

If . . .

The machine or work environment can be physically changed to prevent employee exposure to the potential hazard,

Then . . .

The hazard should be eliminated with an engineering control.

Payment for PPE

When PPE is required to protect employees, it must be provided by the employer at no cost to employees, except for specific items, such as:

- Safety-toe footwear not limited to the worksite
- Prescription safety eyewear
- Everyday clothing and weather-related gear and
- Logging boots

Employer Responsibilities

Hazard Assessment

The employer shall assess the workplace to determine if hazards are present, or likely to be present, which may necessitate the use of personal protective equipment. While an employer's first responsibility upon identification of a workplace hazard is to eliminate it through the use of engineering controls, PPE is often considered the last line of defense against the remaining hazards.

Physical Hazards

Physical hazards include:

- Motion
- High and low temperatures
- Light radiation:
 - Welding
 - Brazing
 - Heat treating
 - High intensity lighting
- Falling objects
- Sharp objects
- Sources of rolling or pinching objects
- Electric hazards
- Hazardous floor conditions

Health Hazards

Health hazards include:

- Types of chemicals an employee could be exposed to
- Harmful dusts, fumes, and mists
- Nuclear radiation
- Noise

Selection of PPE

The selection of personal protective equipment depends on the hazard the worker needs to be protected against, the level of hazard present, and the availability of suitable equipment. Individual components of clothing and equipment should be assembled that both protects the worker from the specific hazard and minimizes the hazard and potential

drawbacks of the PPE itself. PPE shall, wherever possible, be provided for the exclusive use of a single employee.

Periodic reevaluation of the selection is necessary in order to deal with changes in workplace conditions or wearer activities. The type of PPE selected is very important; different brands of PPE should be tried by workers to get the best possible protection. For example, using the wrong types of gloves to work with solvents can lead to ineffective protection.

Training

The employer shall provide training to each employee using PPE. Each employee shall be trained in at least the following:

- When and why PPE is necessary,
- What PPE is required for certain jobs.
- How to properly put on, take off, adjust, and wear PPE.
- The limitations of the PPE.
- The proper care, maintenance, useful life and disposal of PPE.
- How damaged, worn out, or defective PPE can be replaced.

Maintenance

All PPE should be inspected for tears, leaks, punctures, breaks, contamination, or signs of wear before use. Damaged or defective equipment should not be used.

PPE should be stored carefully and kept clean to prevent damage. Contaminated PPE that cannot be decontaminated should be disposed of properly.

Recordkeeping

The employer must maintain records of the workplace hazard assessment and employee training.

Employee Responsibilities

Employees are responsible to:

- Attend all required training sessions regarding PPE.
- Wear PPE as required and trained.
- Clean, maintain, and care for PPE as required.
- Report potential hazards they identify to their supervisors.
- Inform their supervisors or safety managers of the need to repair or replace PPE.
- Follow ALL warnings and precautions.
- Listen and follow the directions from supervisors or safety managers.

Case Study

This accident occurred in a cellophane-tape manufacturing factory during work to wash a drum that contained an adhesive.

The victim who was on the night shift was working with a group leader to monitor the machine that applies a coat of adhesive and to conduct sampling inspections. While they were taking turns working in thirty-minute shifts, the victim did not return to work after a break. When the group leader searched for the victim, he found him unconscious having fallen head first into an empty drum that previously contained adhesive.

Although the victim was immediately taken to the hospital for emergency treatment, he never regained consciousness.

Incidentally, a group leader on the day shift had washed the drum with toluene, before the task was transferred to the night shift workers at the time of the shift-change meeting.

What do you think some of the causes were?

- Ventilation was not used at a site where toluene was used.
- The victim bent forward to wash the inside of the drum that still contained toluene vapor.
- The victim did not use a respirator during the washing work.

Lesson Summary

Personal Protective Equipment (PPE) protects against biological, chemical, or physical hazards on the job. The employer will assess the workplace to determine if hazards are present, or likely to be present, which may necessitate the use of PPE. This protective equipment is often the final solution for hazards that cannot be eliminated through the use of engineering controls. Also, wherever possible, PPE will be provided for the exclusive use of a single employee.

Lesson 2: Eye, Face, and Respiratory Protection

Lesson Focus

This lesson focuses on the following topics:

- Eye and Face Protection
- Respiratory Protection

Eye and Face Protection

Every day an estimated 1,000 eye injuries occur in American workplaces.

The financial cost of these injuries is enormous—more than \$300 million per year is lost in production time, medical expenses, and workers compensation. No dollar figure can adequately reflect the personal toll these accidents take on injured workers.

What contributes to eye injuries at work?

Take a moment to think about possible eye hazards at your workplace. The Labor Department's Bureau of Labor Statistics (BLS) survey of about 1,000 minor eye injuries reveals how and why many on-the-job accidents occur.

1. **Not Wearing Eye Protection.** The BLS reports that nearly three out of every five workers injured were not wearing eye protection at the time of their accidents.
2. **Wearing the Wrong Kind of Eye Protection for the Job.** About 40% of the injured workers were wearing some form of eye protection when the accident occurred, but often, it was not the correct eye protection for the job being done.

What causes eye injuries?

1. **Flying Particles:** The BLS found that almost 70% of the accidents studied resulted from flying or falling objects or sparks striking the eye. Injured workers estimated that nearly three-fifths of the objects were smaller than pin heads. Most of the

particles were said to be traveling faster than hand-thrown objects when accidents occurred.

2. **Contact with Chemicals:** Chemicals caused one-fifth of the injuries.
3. **Other accidents:** Miscellaneous accidents were caused by objects swinging from a fixed or attached position—like tree limbs, ropes, chains, or tools pulled into an eye while a worker was using them.

Where do accidents occur most often?

Potential eye hazards can be found in nearly every industry, but BLS reported that more than 40% of injuries occurred among craft workers, like mechanics, repairers, carpenters, and plumbers.

Over a third of the 40% injured operated machinery, such as assemblers, sanders, and grinding machine operators. Laborers suffered about one-fifth of the eye injuries. Almost half the injured workers were employed in manufacturing; slightly more than 20% were in construction.

How can eye injuries be prevented?

1. **Always wear effective eye protection:** To be effective, eyewear must be appropriate for the hazard and also must be properly fitted. All eye-protective equipment provided by an employer must meet ANSI Z-87.1 standards. (American National Standards Institute)
2. **Better training and education:** The BLS reported that most workers were hurt while doing their regular jobs. Workers injured while not wearing protective eyewear most often said they believed it was not required by the situation. Even though the vast majority of employers furnished eye protection at no cost to employees, about 40% of the workers received no information on when and what kind of eyewear should be used.
3. **Maintenance:** Eye protection devices must be properly maintained. Scratched and dirty devices reduce vision, cause glare, and may contribute to accidents.

Eyewash Stations

Eyewash stations should be located within 10 seconds of a hazard area. If employees accidentally get something into their eyes, they must go directly to the eyewash station and flush their eyes with water for at least 15 minutes. The employee should hold the eyelids open and "look" directly into the water streams. They should NOT rub their eyes. Rubbing the eyes may scratch or embed particles. Employees should seek medical attention immediately.

Eye and Face Protectors

Face and eye protection includes:

1. **Goggles:** Goggles of soft, pliable body design provide adequate eye protection from many hazards. These goggles are available with clear or tinted lenses, perforated, port vented, or non-vented frames. Single lens goggles provide similar protection to spectacles and may be worn in combination with spectacles or corrective lenses to ensure protection along with proper vision.
2. **Face shields:** Face shields may be used in operations when the entire face needs protection and should be worn to protect eyes and face against flying particles and metal sparks. Specifically designed shields also provide protection from chemical and biological splash. Wearing a face shield does not necessarily protect from impact hazards.
3. **Welding shields:** Tinted shields will be provided to protect workers' eyes and face from infrared or radiant light burns, flying sparks, metal spatter and slag chips encountered during welding, brazing, soldering, resistance welding, bare or shielded electric arc welding and oxyacetylene welding and cutting operations. Tinted lenses are available in varying shades or degrees of tinting, and it's the employer's responsibility to provide the appropriate lenses for the hazard to be encountered.

Safe Work Practices

As an employee works:

- He or she should read and follow all warnings and precautions that may be found on equipment and hazardous materials.
- He or she should avoid throwing tools or participating in horseplay.
- He or she should keep sharp or pointed objects away from his or her eyes.
- He or she should follow the supervisor's or safety manager's suggestions and recommendations for working safely.

Respiratory Protection

Health hazards in the workplace are a major concern for both employers and employees.

Hazardous materials can enter our body in four ways:

- Ingestion
- Skin Absorption
- Inhalation
- Injection

Because many substances that are health hazards can become airborne, knowing how to protect ourselves is very important.

There are four basic methods of controlling inhalation hazards:

1. **Substitution** involves replacing the hazardous material or process with a non-hazardous or less hazardous one.
2. **Engineering controls** include enclosing the process so that contaminants do not get into the workspace, improving the ventilation, and changing the equipment or processes.

3. **Administrative controls** include restricting access to contaminated areas, limiting the total time workers are exposed, and establishing housekeeping procedures to control exposure.
4. **Personal protective equipment** includes the use of respirators. In some cases, however, respirators are the only means to protect workers.

What is a Respirator and When is it Needed?

A respirator is a protective device that covers the worker's nose and mouth or the entire face and head to keep airborne contaminants out of the worker's respiratory system and to provide a safe air supply.

There are two major categories of respirators:

1. **Air Purifying Respirators:** These types of respirators include:
 - Air Purifying Disposable Particulate Masks
 - Air Purifying Half Mask Respirators
 - Air Purifying Full Face Mask Respirators
 - Gas Masks
 - Powered Air Purifying Respirators
2. **Supplied Air Respirators:** These types of respirators include:
 - Airline Respirators
 - Emergency Escape Breathing Apparatus
 - Self-Contained Breathing Apparatus (SCBA)

Selecting the Correct Respirator

The first step in selecting the correct respirator is to determine the level of hazard that is posed by the environment in which one will be working. To do this, one must be able to answer five basic questions:

1. What type of contaminant is present?

2. What is the form of the contaminant?
3. How toxic is the contaminant?
4. What is the concentration of the contaminant?
5. What will be the length and duration of the exposure?

Employees should always work with their supervisor or safety professional to determine the correct answers to these questions. Without the technical knowledge to make correct decisions, it's best to consult with an industrial hygienist or safety professional who is trained to provide professional guidance on proper respirator selection and use.

It should be noted that before an employer provides any employee with a respirator to use in a workplace, the employer must create a formal written respiratory protection program and have every employee who will wear a respirator medically evaluated by a licensed healthcare professional.

Inspection before Use

Every time an employee uses a respirator, he or she must first inspect it. To properly inspect a respirator before using it, one should look for:

- Chips or cracks in the faceplate
- Cracks or holes in the breathing tube or airlines
- Worn or frayed straps
- Worn or damaged fittings
- Bent or corroded buckles
- Dirty or improperly seated valves

If you find anything wrong with your respirator, do not use it. Have it repaired or replaced immediately.

Donning (putting on) a Respirator

Each respirator should be donned making certain to follow the manufacturer's instructions carefully. You must be able to demonstrate proper donning of the respirator to your supervisor or safety professional.

Case Study

This accident occurred during work to lift a drain pump through a manhole in a sewerage construction worksite.

The work was to replace the pipes for rainwater. As another drain pump became necessary, it was decided to use the drain pump in another manhole for which work was almost completed. Accordingly, two workers opened the cover of the manhole and entered, using a ladder.

After a short while, a colleague who was going to the material shed spotted the two workers lying at the bottom when he looked into the manhole and raised an alarm with other workers. One of the workers who ran to the spot entered the manhole and called out to them but received no answer. Immediately after this, this worker also collapsed.

Although the three were taken to a hospital by a rescue team, the two workers died, and the would-be rescuer was hospitalized with brain damage caused by hypoxia (a lack of oxygen).

The two fatalities were attributed to anoxia (meaning, without oxygen).

What do you think were some of the causes?

- Although survey results had been provided by the client, neither the general contractor nor the subcontractor assessed the worksite as presenting the hazard of anoxia.
- The employer and supervisor failed to measure the oxygen content of the hole and to carry out ventilation measures before allowing any employee to enter.
- The project should have been declared a Permit Required, Confined Space, and all elements of OSHA's standard for PRCS should have been implemented.
- Neither special education nor rescue training concerning work at sites with the danger of anoxia was provided to workers.

Lesson Summary

To be effective, eyewear must be appropriate for the encountered hazard and also be fitted properly. All eye-protective equipment provided by an employer must meet ANSI standards.

Not wearing eye protection and wearing the wrong kind of eye protection for the job are common factors in eye injuries on the job. Causes of eye injuries include flying particles, contact with chemicals, and objects swinging from a fixed or attached position.

Eye injuries be prevented with:

- Effective eye protection
- Better training and education
- Maintenance

Face and eye protection includes:

- Goggles
- Face shields

- Welding shields

Because many substances that are health hazards can become airborne, knowing how to protect ourselves is very important. A respirator is a protective device that covers the worker's nose and mouth or the entire face and head to keep airborne contaminants out of the worker's respiratory system and to provide a safe air supply. There are two major categories of respirators air purifying respirators and supplied air respirators.

Every time an employee uses a respirator, he or she must first inspect it.

When necessary, workers must be able to demonstrate proper donning and doffing of respirators. If you find anything wrong with a respirator, do not use it; instead, have it repaired or replaced immediately.

Lesson 3: Head, Hand, Face, and Foot Protection

Lesson Focus

This lesson focuses on the following topics:

- Why Head Protection Is Important
- Potential Hazards
- Occupational Noise
- Why Hand Protection Is Important
- Foot Protection Is Important

Why Head Protection Is Important

Your head is a very delicate part of the human body.

Injuries to the head may be very serious. For this reason, head protection and safety are very important.

Potential Hazards

Impact to the head

Falling or flying objects are a common cause of head injuries. Also, falling or walking into hard objects can cause head injuries. These injuries include neck sprains, concussions, and skull fractures.

Electrical Shocks

Accidents involving electricity result in electrical shocks and burns.

How Hard Hats Protect an Employee's Head

Hard hats protect an employee's head by providing the following features:

- A rigid shell that resists and deflects blows to the head and a suspension system inside the hat that acts as a shock absorber.
- Some hats serve as insulators against electrical shocks.
- Hard hats may shield the scalp, face, neck, and shoulders against splashes, spills, and drips.
- Some hard hats are constructed so that face shields, goggles, hoods, or hearing protection can be added.

When necessary, employers must provide hard hats that meet the appropriate ANSI standards.

Note: OSHA does not specify when an employer must provide hard hats. It's up to each employer to make that determination based on the hazards of the workplace and through the required completion of a proper hazard assessment.

Occupational Noise

Noise is a common problem found in many workplaces. Research has shown that high levels of noise will damage your hearing. Losing your hearing is typically a gradual process, and is less noticeable than other types of workplace injuries. It is, however, a permanent handicap for those who are affected.

Remember the four "P"s of hearing loss: It's Painless, Permanent, Progressive and usually, Preventable.

When an employer determines the "Action Level" of 85 dBs has been reached, they must create a formal written hearing conservation program.

Types of Hearing Protection Devices

Many types of hearing protection devices are available. Popular types of hearing protection devices are the following:

1. Foam Earplugs
2. PVC Earplugs
3. Earmuffs

Caring for Hearing Protection Devices

Foam Earplugs

- When not using your foam earplugs, store them in a clean, cool, dry place. If your foam earplugs become soiled, torn or stiff, discard them and ask your supervisor or safety manager for a new pair.

PVC Earplugs

- When not using your PVC earplugs, store them in a clean, cool, dry place. If your PVC earplugs become soiled, you can clean them according to the manufacturer's guidelines. If your PVC earplugs become torn or brittle, discard them and ask your supervisor or safety manager for a new pair.

Earmuffs

- When not using your earmuffs, store them in a clean, cool, dry place. Always inspect your earmuffs for cracks around the foam cups. If your earmuffs are damaged, have them repaired immediately or ask your supervisor or safety manager for a new pair.

Why Hand Protection Is Important

Take a moment to hold your hands out in front of you. Look at them. They are the only two hands you will ever have.

A number of disabling accidents on the job involve the hands. Without your fingers or hands, your ability to work would be greatly reduced.

Human hands are unique. No other creature in the world has hands that can grasp, hold, move, and manipulate objects like human hands. They are one of your greatest assets and, as such, must be protected and cared for.

Potential Hazards to the Hand

Traumatic Injuries

An employee can suffer a traumatic injury to his or her hands in many ways:

- Tools and machines with sharp edges can cut hands.
- Staples, screwdrivers, nails, chisels and stiff wire can puncture hands.
- Getting your hands caught in machinery can sprain, crush or remove your hands and fingers.

Contact Injuries

Coming into contact with caustic or toxic chemicals, biological substances, electrical sources or extremely cold or hot objects can irritate or burn one's hands.

Toxic substances are poisonous substances, some of which can be absorbed through one's skin and enter the body.

Repetitive Motion Injuries

Whenever you repeat the same hand movement over a long period of time, you run the risk of repetitive motion problems. Repetitive motion problems can appear as a numb or tingling sensation, chronic or acute pain, loss of gripping power in your hands, or in many other ways.

Preventative Measures

Poorly maintained machinery, tools, sloppy work areas, and cluttered aisles all contribute to hand injuries.

Good hygiene includes hand washing. Proper washing helps remove germs and dirt from your hands. Clean hands are less susceptible to infection and other skin problems such as contact dermatitis.

Instructions for the Safe Removal of Contaminated Gloves

When removing contaminated gloves, remember to:

1. Pull one glove near your wrist towards your fingertips until the glove folds over.
2. Carefully grab the fold and pull towards your fingertips. As you pull you are turning the inside of the glove outwards.
3. Pull the fold until the glove is almost off.
4. To avoid contamination of your environment, continue to hold the removed glove. Completely remove your hand from the glove.
5. Slide your finger from your glove-free hand under the remaining glove. Continue to slide your finger towards your fingertips until almost half of your finger is under the glove.
6. Turn you finger 180 degrees and pull the glove outwards and towards your fingertips. As you do this, the first glove will be encased in the second glove. The inside of the second glove will also be turned outwards.
7. Grab the gloves firmly, by the uncontaminated surface (the side that was originally touching your hand). Release your grasp of the first glove you removed. Pull your second hand free from its glove. Dispose of the gloves properly.

Foot Protection Is Important

Scientists and engineers for centuries have marveled at the design and structure of the human foot.

The human foot is rigid enough to support the weight of your entire body, and yet flexible enough to allow you to run, dance, play sports, and take you anywhere you want to go. Without your feet and toes, your ability to work at your job would be greatly reduced.

Potential Hazards to the Foot

Impact Injuries

If you have ever stubbed your toe, you know that impact injuries can hurt. At work, heavy objects can fall on your feet. If you work around sharp objects, you might step on something sharp and puncture your foot.

Injuries from Spills and Splashes

Liquids such as acids, caustics, and molten metals can spill onto your shoes and boots. These hazardous materials can cause chemical and heat burns.

Compression Injuries

Heavy machinery, equipment, and other objects can roll over your feet. The result of these types of accidents is often broken or crushed bones.

Electrical Shocks

Accidents involving electricity can cause severe shocks and burns.

Extremes in Cold, Heat, and Moisture

If not protected, your feet can suffer from frostbite if you must work in an extremely cold environment. Extreme heat, on the other hand, can blister and burn your feet. Finally, extreme moisture in your shoes or boots can lead to fungal infections.

Slipping

Oil, water, soaps, wax, and other chemicals can cause you to slip and fall.

Preventative Measures for Foot Safety

Housekeeping

Poorly maintained machinery, tools, sloppy work areas, and cluttered aisles all contribute to foot injuries.

Wearing and Using Safety Footwear

Select and use the right kind of footwear for the job you are going to be performing. Footwear should meet or exceed the standards set by ANSI (ANSI Z41-1991) or the newer ASTM F 2413 Specifications for Performance Requirements for Protective Footwear.

Avoid footwear made of leather or cloth if you work around acids or caustics. These chemicals quickly eat through the leather or cloth and can injure your feet.

Remember to:

- Select footwear that fits properly.
- Inspect your footwear before you use it. Look for holes and cracks that might leak.
- Replace footwear that is worn or torn.
- After working with chemicals, cleanse your footwear appropriately to rinse away any chemicals or dirt before removing footwear.
- Avoid borrowing footwear; footwear is personal protective equipment.
- Store footwear in a clean, cool, dry, ventilated area.

Lesson Summary

Head injuries may include neck sprains, concussions, and skull fractures, or electrical shocks and burns.

OSHA does not dictate specifically when an employer must provide hard hats; rather it is up to each employer to make that determination based on the hazards of the workplace.

When employers determine that noise levels in excess of the "Action Level" of 85 dBS has been reached, they must create a formal written hearing conservation program.

Liquids such as acids, caustics, and molten metals can spill onto your shoes and boots. These hazardous materials can cause chemical and heat burns, and extreme moisture in your shoes or boots can lead to fungal infections.

Because poorly maintained machinery, tools, sloppy work areas, and cluttered aisles all contribute to foot injuries, footwear should meet or exceed the standards formerly set by ANSI and now set by ASTM F2413. Avoid borrowing footwear; footwear is personal protective equipment.

Module 7: Materials Handling

Module Description

Many injuries in the industry occur while employees move materials. In every day operations, workers handle, transport, and store materials. They may do so by hand, or with manually operated or power operated equipment.

This module specifies the materials handling and storage procedures developed by the Occupational Safety and Health Administration (OSHA) to reduce injuries resulting from mishandling or improper storage.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Name general material handling concepts.
- Service multi-piece and single-piece rim wheels.
- Identify powered industrial trucks.
- Maintain overhead and gantry cranes.
- Discuss the safety requirements for crawler, locomotive, and truck cranes.
- Identify different types of slings.
- Describe the requirements for inspecting derricks.
- List the safety requirements for helicopter operations.

Lesson 1: Material Handling Essentials

Lesson Focus

This lesson focuses on the following topics:

- Introduction to Material Handling
- Potential Hazards and Precautions When Moving Materials
- Use of Mechanical Equipment
- Servicing Multi-Piece and Single Piece Rim Wheels
- Safe Operating Procedures

Introduction to Material Handling

Handling and storing materials involve diverse operations such as hoisting tons of steel with a crane; driving a truck loaded with concrete blocks; carrying bags or materials manually; and stacking palletized bricks or other materials such as drums, barrels, kegs, and lumber.

The efficient handling and storing of materials are vital to industry. In addition to raw materials, these operations provide a continuous flow of parts and assemblies through the workplace and ensure that materials are available when needed. Unfortunately, the improper handling and storing of materials often result in costly injuries.

What should employees know before moving, handling, and storing materials?

In addition to training and education, applying general safety principles—such as proper work practices, equipment, and controls—can help reduce workplace accidents involving the moving, handling, and storing of materials. Whether moving materials manually or mechanically, employees should know and understand the potential hazards associated with the task at hand and how to control their workplaces to minimize the danger.

Because numerous injuries can result from improperly handling and storing materials, workers should also be aware of accidents that may result from the unsafe or improper handling of equipment as well as from improper work practices. In addition, workers should be able to recognize the methods for eliminating—or at least minimizing—the occurrence of such accidents. Employers and employees should examine their workplaces to detect any unsafe or unhealthy conditions, practices, or equipment and take corrective action.

Potential Hazards and Precautions When Moving Materials

What are the potential hazards for workers?

Workers frequently cite the weight and bulkiness of objects that they lift as major contributing factors to their injuries. In 1999, for example, more than 420,000 workplace accidents resulted in back injuries. Bending, followed by twisting and turning, were the more commonly cited movements that caused back injuries.

Other hazards include falling objects, improperly stacked materials, and various types of equipment. Employees should be aware of potential injuries that can occur when manually moving materials, including the following:

- Strains and sprains from lifting loads improperly or from carrying loads that are either too large or too heavy,
- Fractures and bruises caused by being struck by materials or by being caught in pinch points
- Cuts and bruises caused by falling materials that have been improperly stored or by incorrectly cutting ties or other securing devices.
- Loads shifting from loose blocking when managing loads manually.

What precautions should workers take when moving materials manually?

When moving materials manually, workers should attach handles or holders to loads.

In addition, workers should always wear appropriate personal protective equipment and use proper lifting techniques. To prevent injury from oversize loads, workers should seek help in the following:

- When a load is so bulky that employees cannot properly grasp or lift it,
- When employees cannot see around or over a load, or
- When employees cannot safely handle a load.

Using the following personal protective equipment prevents needless injuries when manually moving materials:

- Hand and forearm protection, such as gloves, for loads with sharp or rough edges.
- Eye protection.
- Steel-toed safety shoes or boots.
- Metal, fiber, or plastic metatarsal guards to protect the instep area from impact or compression.

Use of Mechanical Equipment

Where mechanical handling equipment is used, sufficient safe clearances shall be allowed for aisles, at loading docks, through doorways, and wherever turns or passage must be made.

Aisles and passageways should be kept clear and in good repair, without any obstruction that could create a hazard. Permanent aisles and passageways should be properly marked.

Secure Storage

Material storage should not create a hazard. Bags, containers, bundles, etc., stored in rows should be stacked, blocked, interlocked, and limited in height so that they do not slide or collapse.

Housekeeping

Storage areas should be kept free from collections of materials that may cause hazards like tripping, fire, explosion, or pest harborage. Vegetation control should be implemented where necessary.

Clearance Limits

Clearance signs should be used to warn employees about clearance limits.

Rolling Railroad Cars

Derail or bumper blocks should be provided on spur railroad tracks where a rolling car could contact other cars being worked on or entering a building, work, or traffic area.

Guarding

Covers or guardrails should be used to protect employees from the hazards of open pits, tanks, vats, and ditches, etc.

Servicing Multi-Piece and Single Piece Rim Wheels

Employee Training

The employer shall train all employees that service rim wheels in the hazards involved and the safety procedures to be followed. The employer shall make sure that no employee services any rim wheel without proper training for the type of wheel being serviced.

The training program must include all the topics in the rim manual. The topics include:

- Demounting of tires (including deflation)
- Inspecting and identifying of the rim wheel components
- Mounting of tires (including inflation with a restraining device or other safeguard)

- Using the restraining device or barrier or other equipment
- Handling of rim wheels
- Inflating the tire when a single-piece rim wheel is mounted on a vehicle
- Understanding why it's necessary to stand outside the trajectory both during inflation of the tire and during inspection of the rim wheel following inflation
- Installing and removing rim wheels

Tire Servicing Equipment

The employer must provide a restraining device for inflating tires on multi-piece wheels and single piece wheels, unless the rim wheel will be bolted onto a vehicle during inflation.

Restraining devices and barriers must fulfill the following requirements:

- The restraining device or barrier should be strong enough to withstand the maximum force that could be transferred to it during a rim wheel separation, occurring at 150 percent of the maximum tire pressure specification for the type of rim wheel being serviced.
- Restraining devices and barriers should be capable of preventing the rim wheel components from being thrown outside or beyond the barrier.

Restraining devices and barriers shall be visually inspected before being put to use. The restraining device or barrier must be removed immediately from service if any of the following defects are noticed:

- Cracks at welds.
- Cracked or broken components.
- Bent or sprung components caused by mishandling, abuse, tire explosion, or rim wheel separation.
- Pitting of components due to corrosion.
- Structural damage, which would decrease its effectiveness.

Restraining devices or barriers removed from service shall not be returned to service until they are repaired and re-inspected.

The employer must ensure that an air line assembly consisting of the following components is being used for inflating tires:

- A clip-on chuck
- An in-line valve with a pressure gauge or a preset regulator; and
- A sufficient length of hose between the clip-on chuck and the in-line valve if one is used to allow the employee to stand outside the trajectory.

Current charts or rim manuals containing instructions for the type of wheels being serviced must be available in the service area.

The employer should also furnish and ensure that only tools recommended by the rim manual for the type of wheel being serviced are used.

Wheel Component Acceptability

Multi-piece wheel components should only be interchanged as recommended by the appropriate rim manual.

Multi-piece wheel components and single piece wheels should be inspected before assembling. Any wheel or wheel component that is bent out of shape, pitted from corrosion, broken, or cracked should not be used, marked or tagged unserviceable, and removed from the service area.

Damaged or leaky valves should be replaced.

Make sure that rim flanges, rim gutters, rings, bead seating surfaces, and the bead areas of tires are free from dirt, surface rust, scale, or loose or flaked rubber build-up before mounting and inflation.

More Information: Make sure the tire and the wheel are compatible before assembling.

Safe Operating Procedures

Multi-Piece Rim Wheels

The employer is required to establish a safe operating procedure for servicing multi-piece rim wheels. The procedure must include at least the following elements:

- Tires must be completely deflated before demounting by removing the valve core, including in the following situations:
 - The tire has been driven at 80% or less of its recommended pressure.
 - There is obvious or suspected damage to the tire or wheel components.
- If a tire on a vehicle is under-inflated but has more than 80% of the recommended pressure, the tire may be inflated while the rim wheel is on the vehicle, provided remote control inflation equipment is used, and no employees remain in the trajectory during inflation.
- Rubber lubricant should be applied to bead and rim mating surfaces during assembly of the wheel and inflation of the tire, unless the tire or wheel manufacturer recommends against it.
- Tires should only be inflated outside a restraining device to a pressure sufficient enough to force the tire bead onto the rim ledge and create an airtight seal with the tire and bead.
- No attempt must be made to correct the seating of side and lock rings by hammering, striking, or forcing the components while the tire is pressurized.
- Whenever a rim wheel is in a restraining device, the employee must not rest or lean any body part or equipment on or against the restraining device.
- After tire inflation, the tire and wheel components should be inspected while still within the restraining device to make sure that they are properly seated and locked. If further adjustment to the tire or wheel components is necessary, the tire should be deflated by removing the valve core before the adjustment is made.

Single-piece Rim Wheels

The employer must establish a safe operating procedure for servicing single-piece rim wheels. The procedure will include at least the following elements.

1. Tires must be completely deflated before demounting by removing the valve core.
2. Mount and demount the tires from only the narrow ledge side of the wheel. Avoid damaging the tire beads while mounting tires on wheels.
3. Tires should only be mounted on compatible wheels of matching bead diameter and width.
4. Nonflammable rubber lubricant can be applied to bead and wheel mating surfaces before assembly of the rim wheel unless the tire or wheel manufacturer recommends against the use of rubber lubricants.
5. If a tire-changing machine is used, the tire can only be inflated to the minimum pressure necessary to force the tire bead onto the rim ledge.
6. Tires must be inflated within a restraining device, positioned behind a barrier or bolted on the vehicle with the lug nuts fully tightened.
7. Tires must not be inflated when any flat, solid surface is in the trajectory and within one foot of the sidewall.
8. Employees must stay out of the trajectory when inflating a tire.
9. Tires shall not be inflated to more than the inflation pressure stamped in the sidewall to seat tire bead firmly against the rim flange unless a higher pressure is recommended by the manufacturer.
10. Cracked, broken, bent, or otherwise damaged wheels should not be reworked, welded, brazed, or otherwise heated.

Lesson Summary

The employer must provide a restraining device for inflating tires on multi-piece wheels and single piece wheels. The employer should make sure that no employee services any rim wheel without proper training for the type of wheel being serviced.

The employer should train all employees that service rim wheels in the hazards involved and the safety procedures to be followed, including information about material storage, and what clearance signs should be used to warn employees about clearance limits.

Lesson 2: Commercial and Industrial Vehicles (a)

Lesson Focus

This lesson focuses on the following topics:

- Powered Industrial Trucks
- Overhead and Gantry Cranes

Powered Industrial Trucks

All new powered industrial trucks acquired and used by an employer after February 15, 1972 must meet the design and construction requirements for powered industrial trucks established in the American National Standard for Powered Industrial Trucks, except for vehicles intended primarily for earth moving or over-the-road hauling.

Approved trucks must bear a label or some other identifying mark indicating approval by the testing laboratory. No modifications or additions that affect capacity and safe operation shall be made without the manufacturer's written approval. Nameplates and markings must be in place and in legible condition.

Vehicles Covered

Vehicles covered include:

- High- lift trucks
- Counter-balanced trucks
- Cantilevered trucks
- Rider trucks
- Forklift trucks
- High- lift platform trucks
- Low- lift trucks
- Low- lift platform trucks

- Motorized hand trucks
- Pallet trucks
- Narrow- aisle rider trucks
- Straddle trucks
- Reach rider trucks
- Single- side- loader rider trucks
- High- lift order- picker rider trucks
- Motorized hand/rider trucks
- Rough- terrain trucks

Designation of Industrial Trucks

There are 11 different designations of industrial trucks.

D

The D designated units are diesel-powered units having minimum acceptable safeguards against inherent fire hazards.



DS

The DS designated units are diesel-powered units that are provided with additional safeguards to the exhaust, fuel, and electrical systems. They may be used in some locations where a D unit may not be considered suitable.



DY

The DY designated units are diesel-powered units that have all the safeguards of the DS units; in addition, they do not have any electrical equipment, including the ignition, and are equipped with temperature limitation features.



E

The E designated units are electrically powered units that have minimum acceptable safeguards against inherent fire hazards.



ES

The ES designated units are electrically powered units that, in addition to all of the requirements for the E units, are provided with additional safeguards to the electrical system to prevent emission of hazardous sparks and to limit surface temperatures. They may be used in some locations where the use of an E unit may not be considered suitable.



EE

The EE designated units are electrically powered units that have, in addition to all of the requirements for the E and ES units, the electric motors and all other electrical equipment completely enclosed. In certain locations the EE unit may be used where the use of an E and ES unit may not be considered suitable.



EX

The EX designated units are electrically powered units that differ from the E, ES, or EE units in that the electrical fittings and equipment are so designed, constructed and assembled that the units may be used in certain atmospheres containing flammable vapors or dusts.

**G**

The G designated units are gasoline-powered units having minimum acceptable safeguards against inherent fire hazards.

**GS**

The GS designated units are gasoline-powered units that are provided with additional safeguards to the exhaust, fuel, and electrical systems. They may be used in some locations where the use of a G unit may not be considered suitable.

**LP**

The LP designated unit uses liquefied petroleum gas as fuel having minimum acceptable safeguards against inherent fire hazards.

**LPS**

The LPS designated units are liquefied petroleum gas powered units that are provided with additional safeguards to the exhaust, fuel, and electrical systems. They may be used in some locations where the use of an LP unit may not be considered suitable.



Designated Locations

Power-operated industrial trucks must not be used in atmospheres containing hazardous concentration of acetylene, butadiene, ethylene oxide, hydrogen, propylene oxide, acetaldehyde, cyclopropane, diethyl ether, ethylene, isoprene, or unsymmetrical dimethyl hydrazine (UDMH) or other flammable substances.

Power-operated industrial trucks must not be used in atmospheres containing: hazardous concentrations of metal dust, including aluminum, magnesium, and their commercial alloys; other metals of similarly hazardous characteristics; or in atmospheres containing carbon black, coal, or coke dust.

Only approved power-operated industrial trucks designated as EX may be used in atmospheres containing acetone, acrylonitrile, alcohol, ammonia, benzene, benzol, butane, ethylene dichloride, gasoline, hexane, lacquer solvent vapors, naphtha, natural gas, propane, propylene, styrene, vinyl acetate, vinyl chloride, or xylenes in quantities sufficient to produce explosive or ignitable mixtures and where such concentrations of these gases or vapors exist continuously, intermittently or periodically under normal

operating conditions or may exist frequently because of repair, maintenance operations, leakage, breakdown or faulty operation of equipment.

Power-operated industrial trucks designated as DY, EE, or EX may be used in locations where volatile flammable liquids or flammable gases are handled, processed or used, but in which the hazardous liquids, vapors or gases will normally be confined within closed containers or closed systems from which they can escape only in case of accidental rupture or breakdown of such containers or systems, or in the case of abnormal operation of equipment; also in locations in which hazardous concentrations of gases or vapors are normally prevented by positive mechanical ventilation but which might become hazardous through failure or abnormal operation of the ventilating equipment; or in locations which are adjacent to Class I, Division 1 locations, and to which hazardous concentrations of gases or vapors might occasionally be communicated unless such communication is prevented by adequate positive-pressure ventilation from a source of clear air, and effective safeguards against ventilation failure are provided.

In locations used for the storage of hazardous liquids in sealed containers or liquefied or compressed gases in containers, approved power-operated industrial trucks designated as DS, ES, GS, or LPS may be used. This classification includes locations where volatile flammable liquids or flammable gases or vapors are used, but which, would become hazardous only in case of an accident or of some unusual operating condition. The quantity of hazardous material that might escape in case of accident, the adequacy of ventilating equipment, the total area involved, and the record of the industry or business with respect to explosions or fires are all factors that should receive consideration in determining whether or not the DS or DY, ES, EE, GS, LPS designated truck possesses sufficient safeguards for the location. Piping without valves, checks, meters and similar devices would not ordinarily be deemed to introduce a hazardous condition even though used for hazardous liquids or gases. Locations used for the storage of hazardous liquids or of liquefied or compressed gases in sealed containers would not normally be considered hazardous unless subject to other hazardous conditions also.

Only approved power operated industrial trucks designated as EX shall be used in atmospheres in which combustible dust is or may be in suspension continuously, intermittently, or periodically under normal operating conditions, in quantities sufficient to produce explosive or ignitable mixtures, or where mechanical failure or abnormal operation of machinery or equipment might cause such mixtures to be produced.

The EX classification usually includes the working areas of grain handling and storage plants, room containing grinders or pulverizers, cleaners, graders, scalpers, open conveyors or spouts, open bins or hoppers, mixers, or blenders, automatic or hopper scales, packing machinery, elevator heads and boots, stock distributors, dust and stock collectors (except all-metal collectors vented to the outside), and all similar dust producing machinery and equipment in grain processing plants, starch plants, sugar pulverizing plants, malting plants, hay grinding plants, and other occupancies of similar nature; coal pulverizing plants (except where the pulverizing equipment is essentially dust tight); all working areas where metal dusts and powders are produced, processed, handled, packed, or stored (except in tight containers); and other similar locations where combustible dust may, under normal operating conditions, be present in the air in quantities sufficient to produce explosive or ignitable mixtures.

Only approved power-operated industrial trucks designated as DY, EE, or EX shall be used in atmospheres in which combustible dust will not normally be in suspension in the air or will not likely to be thrown into suspension by the normal operation of equipment or apparatus in quantities sufficient to produce explosive or ignitable mixtures but where deposits or accumulations of such dust may be ignited by arcs or sparks originating in the truck.

Only approved power-operated industrial trucks designated as DY, EE, or EX shall be used in locations that are hazardous because of the presence of easily ignitable fibers or flyings but in which such fibers or flyings are not likely to be in suspension in the air in quantities sufficient to produce ignitable mixtures.

Only approved power-operated industrial trucks designated as DS, DY, ES, EE, EX, GS, or LPS shall be used in locations where easily ignitable fibers are stored or handled, including outside storage, but are not being processed or manufactured. Industrial trucks designated as E that have been previously used in these locations may be continued in use.

On piers and wharves handling general cargo, any approved power-operated industrial truck designated as Type D, E, G, or LP may be used.

Only approved power-operated industrial trucks designated as DS, DY, ES, EE, EX, GS, or LPS shall be used in locations where easily ignitable fibers are stored or handled but not being processed or manufactured. Industrial trucks designated as E that have been previously used in these locations may be continued in use.

Power-operated industrial trucks that have been originally approved for the use of gasoline for fuel, when converted to the use of liquefied petroleum gas fuel, may be used in those locations where G, GS or LP, and LPS designated trucks have been specified.

Safety guards: High-lift Rider trucks shall be fitted with an overhead guard manufactured in accordance with paragraph ANSI B56.1-1969, unless operating conditions do not permit.

Employers must ensure that operators are trained, competent, and certified.

Changing and Charging Storage Batteries

Guidelines for changing and charging storage batteries include:

- Battery charging installations shall be located in areas designated for that purpose. Facilities shall be provided for flushing and neutralizing spilled electrolyte, for fire protection, for protecting charging apparatus from damage by trucks, and for adequate ventilation for dispersal of fumes from gassing batteries.

- A conveyor, overhead hoist, or equivalent material handling equipment shall be provided for handling batteries. Reinstalled batteries shall be properly positioned and secured in the truck. A carboy tilter or siphon shall be provided for handling electrolyte.
- When charging batteries, acid shall be poured into water; water shall not be poured into acid.
- Trucks shall be properly positioned, and the brake applied before attempting to change or charge batteries. Care shall be taken to assure that vent caps are functioning. The battery (or compartment) cover(s) shall be open to dissipate heat.
- Smoking shall be prohibited in the charging area.
- Precautions shall be taken to prevent open flames, sparks, or electric arcs in battery charging areas. Tools and other metallic objects shall be kept away from the top of uncovered batteries.

Operator Training

Guidelines for powered industrial truck training include:

- Before permitting an employee to operate a powered industrial truck (except for training purposes), the employer must ensure that the operator has successfully completed the required training.
- Trainees may operate a powered industrial truck only under direct supervision of a person who has the knowledge, training, and experience to train operators and evaluate their competence, in an area where such operation does not endanger the trainee or other employees.
- Training should be based on formal instruction (lecture, discussion, interactive computer learning etc.), practical training (demonstrations and exercises performed by the trainee), and evaluation of the operator's performance in the workplace.
- Each type of powered industrial truck has its own unique characteristics and inherent hazards. To be effective, training must address the unique characteristics of the type of vehicle the employee is being trained to operate.

General Requirements

An evaluation of each powered industrial truck operator's performance shall be conducted at least once every three years.

If an operator has previously received training in a topic specified, and such training is appropriate to the truck and working conditions encountered, additional training in that topic is not required if the operator has been evaluated and found competent to operate the truck safely.

Certification

Certification should include the following:

- Name of operator
- Date of training
- Date of evaluation
- Name of trainer performing the evaluation

Overhead and Gantry Cranes

In many industries, overhead cranes are used to move heavy and oversized objects. These cranes have a railed support structure called a bridge, and a wheeled trolley that travels across the bridge horizontally. Several varieties of overhead cranes exist including gantry, semi-gantry, cantilever gantry, storage bridge, and wall cranes.

General Requirements

The general requirements for overhead cranes include:

- All overhead cranes installed after August 31, 1971, must meet the specifications of the American National Standard Safety Code for Overhead and Gantry Cranes.

- Cranes can be modified to increase their load capacity as long as the modifications and associated structure are approved by a qualified engineer or the equipment manufacturer.
 - The rated load of the crane must be clearly marked on each side of the crane. If more than one hoist is present, each hoist will have its rating displayed. The crane must not have any obstructions on either side.
- The clearance area must not be used as a walkway, as it compromises employee safety during crane operation.
- The employer must ensure that only a qualified and trained employee operates a crane.

Initial Inspection

Before initial use, all new and altered cranes shall be inspected to insure compliance with:

- Hoisting and lowering mechanism
- Trolley travel
- Bridge travel
- Limit switches, locking, and safety devices function
- Conducting a load test of not more than 125% of rated load

Routine inspections should be conducted to ensure continued operation of the crane and the safety of the employees around it. Frequent inspections done on a daily to monthly basis should be conducted.

Daily to monitor:

- All functional operating mechanisms for maladjustment interfering with proper operation.
- Deterioration or leakage in lines, tanks, valves, drain pumps, and other parts of air or hydraulic systems.

- Hooks with deformation or cracks (visual inspection).
- Hoist chains, including end connections, for excessive wear, twist, distorted links interfering with proper function, or stretch beyond manufacturer's recommendations (visual inspection).
- All functional operating mechanisms for excessive wear of components.
- Rope reeving for noncompliance with manufacturer's recommendations.

Monthly to monitor:

- Hooks with deformation or cracks with a certification record, which includes the date of inspection, the signature of the person who performed the inspection, and the serial number, or other identifier, of the hook inspected.
- Hoist chains, including end connections, for excessive wear, twist, distorted links interfering with proper function, or stretch beyond manufacturer's. Inspection with a certification record, which includes the date of inspection, the signature of the person who performed the inspection, and an identifier of the chain which was inspected.
- All functional operating mechanisms for excessive wear of components.
- Rope reeving for noncompliance with manufacturer's recommendations.

Periodic Inspection

The purpose of periodic inspection (those conducted in 1 to 12 month intervals) is to determine the extent of wear, deterioration, or malfunction of critical components.

The purpose of the inspection is to identify and maintain the following:

- Deformed, cracked, or corroded members.
- Loose bolts or rivets.
- Cracked or worn sheaves and drums.
- Worn, cracked, or distorted parts such as bearings, gears, rollers, etc.
- Excessive wear on brake system parts.

- Inaccuracies in load, wind, and other indicators.
- Electric or fossil-fuel motors.
- Excessive wear of chain drive sprockets and chain.
- Deteriorated electrical components such as pushbuttons, limit switches, or contactors.

Maintenance Requirements

A preventive maintenance program based on the crane manufacturer's recommendation must be followed to make sure that the crane is safe to operate. If repairs or maintenance is required, the crane must not be operated until the necessary modifications have been made by a qualified and trained technician in accordance with the manufacturer's specified guidelines.

More Information: Make sure that the crane does not accidentally energize itself during maintenance by controlling the release of hazardous energy through lockout/tagout. The equipment must be de-energized before performing the maintenance and repair operation.

The main, or emergency switch shall be open and locked in the open position. Warning or "out of order" signs must be placed on the crane, on the floor beneath, or on the hook where visible from the floor.

Employee Training

To minimize injury and ensure proper functioning of the equipment, employees should be appropriately trained on the following topics:

- How to operate cranes
- Hazards associated with crane operations

- Repair and maintenance requirements
- The purpose and functioning of energy control procedures
- Limitations of lockout/tagout devices
- Standard hand signals for controlling overhead and gantry cranes

Retraining employees is necessary whenever replacement or a major modification is made to the crane that brings about a new hazard or when operator issues indicate a need for retraining.

Operation

Follow the manufacturer's instruction when operating the crane to ensure employee safety and proper functioning of the equipment. Attach the load to the crane, following all manufacturer's guidelines and making sure there are no obstacles that may pose a potential hazard. Once the load is secured and balanced, slowly raise the load. To prevent the load from swinging, complete all movements as instructed by the equipment manufacturer.

Carrying loads over personnel is never allowed under any circumstances. A load should not be left suspended.

Standard Hand Signals for Controlling Overhead and Gantry Cranes

Standard hand signals must be used in a workplace to make sure that there is visual communication where audible and discernable voice communication is not possible. Signals must be understood and agreed upon by all individuals using the crane. Some of the signals being used in the industry for visual communication are as follows.

Hoist:

With forearm vertical, forefinger pointing up, move hand in small horizontal circle.



Lower:

With arm extended downward, forefinger pointing down, move hand in small horizontal circles.



Bridge Travel:

Arm extended forward, hand open and slightly raised, make pushing motion in direction of travel.



Trolley Travel:

Palm up, fingers closed, thumb pointing in direction of motion, jerk hand horizontally.



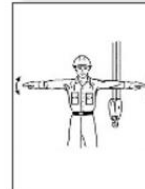
Stop:

Arm extended, palm down, move arm back and forth horizontally.



Emergency Stop:

Both arms extended, palms down, move arms back and forth horizontally.

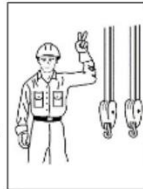


Standard Hand Signals for Controlling Overhead and Gantry Cranes

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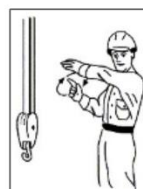
Multiple Trolleys:

Hold up one finger for block marked "1" and two fingers for block marked "2." Regular signals follow.



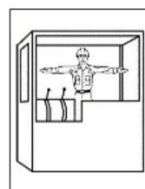
Move Slowly:

Use one hand to give any motion signal and place other hand motionless in front of hand giving the motion signal.



Magnet is Disconnected:

Crane operator spreads both hands apart--palms up.



Lesson Summary

Each type of powered industrial truck has its own unique characteristics and some inherent hazards. To be effective, training must address the unique characteristics of the type of vehicle the employee is being trained to operate. Thus, employers must ensure that operators are trained, competent, and documented.

With cranes, a preventive maintenance program based on the crane manufacturer's recommendation must be followed to make sure that the crane is safe to operate. If repairs or maintenance is required, the crane must not be operated until the necessary modifications have been made by a qualified and trained technician in accordance with the manufacturer's specified guidelines.

Lesson 3: Commercial and Industrial Vehicles (b)

Lesson Focus

This lesson focuses on the following topics:

- Crawlers, Locomotives, and Truck Cranes
- Slings
- Derricks
- Helicopters

Crawlers, Locomotives, and Truck Cranes

Crawlers, locomotives, and truck cranes are powered by internal combustion engines or electrical motors and utilize drums and ropes.

Only a designated employee who is familiar with these types of cranes and their usage should be permitted to operate them. For the operator's convenience, a durable rating chart with clearly legible letters and figures should be securely fixed to the crane cab in a location where it is easily visible to the operator while seated at the control station.

Clearances

Before starting operations near energized electrical wires, contact the authorized representative and notify them about the operation with relevant information.

Where electrical power lines have been de-energized or insulated barriers have been put up, the minimum clearance between the wires and any part of the crane or load should be as follows:

- For lines rated 50 KV or below—10 feet
- For lines rated over 50 KV—10 feet plus 0.4 inch for every KV over 50 KV, or twice the length of the wire insulator but never less than 10 feet
- In transit with no load and boom lowered—4 feet minimum

More Information: Any overhead wire must be considered energized unless the electrical utility verifies that the wires are not energized.

All moving parts should be guarded so that a person may not come into contact with these parts and cause a hazard in any way. Moving parts include but are not limited to: belts, gears, shafts, pulleys, sprockets, spindles, drums, fly wheels, and chains.

If the area inside the swing radius of the rear of the crane is accessible, this area should be barricaded to prevent employees from being struck or crushed by the crane accidentally.

Cab Safety Requirements

All cab windows should be made of safety glass. Ensure that there are no distortions on the windscreen that may hinder visibility.

For rigging, a ladder or steps should be made available to provide easy access to the cab roof when necessary.

The platforms should have anti-skid surfaces to prevent an employee from accidentally slipping and getting injured.

Frequent Inspections

Inspection requirements for crawlers, locomotives, and truck cranes depend on their exposure to wear, deterioration, or malfunction. Frequent inspections, in accordance with the manufacturer's guidelines, are required to monitor control mechanism adjustments interfering with proper operations, contamination by lubricants, and deterioration or leakage in hydraulic systems.

Concerns that require frequent inspection are as follows:

- All control mechanisms for maladjustment interfering with proper operation
- All control mechanisms for excessive wear of components and contamination by lubricants or other foreign matter
- All safety devices for malfunction
- Deterioration or leakage in air or hydraulic systems, which need to be checked daily
- Hooks with deformation or cracks on more than 15% of the normal throat opening or with more than a 10-degree twist from the plane of the unbent hook
- Rope reeving for noncompliance with manufacturer's recommendations
- Electrical apparatus for malfunctioning, signs of excessive deterioration, dirt, and moisture accumulation

Periodic Inspections

Complete inspection of the cranes must be carried out on a periodic basis. The interval between such inspections depends upon the activity, severity of service, and environment. The items being inspected include:

- Deformed, cracked, or corroded members in the crane structure and boom
- Loose bolts or rivets; worn, cracked, or distorted parts such as pins, bearings, shafts, gears, rollers, and locking devices
- Cracked or worn sheaves and drums
- Excessive wear on brake and clutch system parts, linings, pawls, and ratchets
- Load, boom angle, and other indicators over their full range for any significant inaccuracies

Inspect:

- Gasoline, diesel, electric, or other power plants for improper performance or noncompliance with safety requirements
- Excessive wear of chain-drive sprockets and excessive chain stretch

- Travel steering, braking, and locking devices, for malfunction
- Excessively worn or damaged tires

Cranes Not in Regular Use

A crane that has been idle for a period of time should be inspected thoroughly according to the manufacturer's guidelines before putting it back in service.

Slings

Slings, used in tandem with other material handling equipment are employed in hoisting various materials. Slings are made of wire rope, metal mesh, alloy steel chains, natural or synthetic fiber, or a synthetic web of nylon polyester and polypropylene.

For safety purposes, follow these protective guidelines:

- Do not use damaged or defective slings.
- Do not shorten the length of slings by tying knots or by using bolts or other makeshift devices.
- Do not twist sling legs for any purpose.
- Do not load any material that is in excess of the rated capacity of the sling.
- Secure the sling by carefully attaching it to the load.
- Use appropriate padding material to protect the sling from the sharp edges of loads

To maintain employee safety:

- Keep away from loads about to be lifted or suspended.
- Do not place hands or fingers between the sling and its load while tightening the sling around the load.
- Shock loading is prohibited.
- Do not pull the sling from under the load while it is resting.

Inspections

A competent person designated by the employer must inspect the sling and all fasteners to make sure there are no defects or damages on it before starting work. Damaged or defective slings shall be immediately removed from service.

Removal

If any of the following conditions are found during inspection, the sling should be removed from service:

- Ten random broken wires in one rope lay.
- Five broken wires in one strand.
- Wear or scraping of 1/3 original diameter of the outside wires.
- Kinking, crushing, bird caging.
- Heat related damages.
- Damage to end attachments.
- Hook throat is opened more than 15 percent of normal.
- Hook is twisted more than 10 degrees from normal.
- Corrosion is detected.

Alloy Steel Chain Slings

A permanent identification label must be attached to alloy steel chain slings. The label must include the following information:

- Size
- Grade
- Rated Capacity
- Reach

All attachments must have the same rated capacity as that of the alloy steel chain sling. If this is not the case, the minimum rating capacity must be identified from the weakest link.

Minimum sling lengths include:

- Cable laid and 6x19 or 6x37 slings shall have a minimum clear length of wire rope 10 times the component rope diameter between splices, sleeves, or end fittings.
- Braided slings shall have a minimum clear length of wire rope 40 times the component rope diameter between the loops or end fittings.
- Cable laid grommets, strand laid grommets, and endless slings should have a minimum circumferential length of 96 times their body diameter.

Metal Mesh Slings

Metal mesh slings must be labeled with durable markings stating the rated capacity of the vertical basket hitch and choker hitch loadings. Other guidelines include:

- Make sure that handles have at least the same rated capacity as that of the metal fabric. Handles must be placed so that they distribute the load evenly across the sling and do not diminish its rated capacity.
- Use appropriate padding wherever it is necessary to protect the sling from the sharp edges of its load.
- Metal mesh slings must be removed from service if an inspection reveals reduction in wire diameter of 25 percent due to abrasion or 15 percent due to corrosion.

Natural and Synthetic Fiber Rope Slings

Natural and synthetic fiber rope slings must never be used over their rated capacities.

This type of sling can be used within the temperature range designated by the manufacturer without decreasing the working load limit.

Clamps not designed specifically for fiber ropes shall not be used for splicing.

Removal

If any of the following conditions are met, fiber rope slings must be immediately removed from service:

- Abnormal wear.
- Powdered fiber between strands.
- Broken or cut fibers.
- Variations in the size or roundness of strands.
- Discoloration or rotting.
- Distortion of hardware in the sling.

Synthetic Web Slings

Synthetic web slings must be marked, labeled, or coded to identify their rated capacity for the type of hitch and synthetic web material used. Webbing must be made of uniform thickness and width, and all fittings should have a rated capacity at least equal to the rated capacity of the web material.

Precautions

When using synthetic web slings, observe the following precautions:

- Do not use nylon web slings in a workplace where fumes, vapors, and mists or liquids of acid and phenolic are present.
- Do not use polyester and polypropylene web slings where mists or liquids of caustics are present. If the fittings are made of aluminum, the slings should also not be used in a similar environment.
- Do not use synthetic web slings of nylon and polyester where temperature is expected to rise over 180 degrees Fahrenheit. If the sling is made of polypropylene, the temperature should not exceed 200 degrees Fahrenheit.

Removal

Remove the synthetic web slings from service if any of the following conditions occur:

- Tears and cuts on the surface of web slings
- Acid or caustic burns
- Snags and punctures
- Broken or worn stitches
- Distortion of fittings

Derricks

Derricks are capable of handling loads at various reaches and are powered by hoists through a system of rope reeving. Only a designated employee should be allowed to operate derricks.

For permanently installed derricks with fixed lengths of boom guy and mast, a rating chart must be provided for easy reference. The rating chart must include the load ratings for different boom angles and operating radii, lengths of different components and parts required for hoist reeving. This chart must be available on the derrick and in the jobsite office.

For non-permanent derricks, sufficient information must be available to prepare customized rating charts.

Inspection

The following items require frequent inspection for proper functioning.

- Control mechanism
- Chords and lacing
- Plumb of the mast
- Hydraulic systems

- Tension in guys
- Hooks
- Rope reeving
- Hoist brakes, clutches, and operating levers
- Electrical apparatus

Complete inspections shall be conducted to maintain or repair the following:

- Cracks, corrosion, or deformation of structural members
- Pins, bearings, shafts, and clamping devices
- Bolts and rivets and gudgeon pins
- Performance of power plants
- Foundation and support
- Cracked hooks

A derrick that has been idle for more than six months must be inspected thoroughly before putting it back into operation.

Operational Tests

Before putting up a new derrick in operation, certain tests need to be performed to evaluate the performance and safety issues. These tests must be conducted to analyze the following operations.

- Operation of clutches and brakes of hoist
- Load hoisting and lowering
- Boom up and down
- Swing

Rock and hairpin anchorage require special testing; a competent person is required to perform the tests and evaluate the performance.

Rope Inspection

Ropes are the strength of a derrick. Any deterioration resulting in appreciable loss of original strength must be carefully observed. Some of the conditions that could result in an appreciable loss of strength are the following:

- Reduction of rope diameter below nominal diameter due to the loss of core support, internal or external corrosion, or wear of outside wires
- A number of broken outside wires and the degree of distribution or concentration of such broken wires
- Worn outside wires
- Corroded or broken wires at end connections
- Corroded, cracked, bent, worn, or improperly applied end connections
- Severe kinking, crushing, cutting, or unstranding

Holding the Load

- The operator must not leave a suspended load unattended.
- No employee should ever be allowed to pass under a suspended load.
- If the load needs to be suspended for a considerable amount of time, a dog, pawl, and/or ratchet should be used in conjunction with the brake to hold the load.

Helicopters

Before starting work, the pilot should be briefed about the day's work along with all related employees who are part of the operation. Proper care must be taken to load slings.

Pressed sleeve, swaged eyes, or equivalent equipment should be used to secure the suspended load. Cargo hooks must have emergency mechanical controls for releasing the load. As a primary safety precaution, all mechanical devices and controls shall be inspected before starting the operation.

Proper communication must be maintained between the pilot and the ground crew, who acts as a signalman during the loading/depositing operation. The signalman must be distinctively identifiable from the rest of the crew.

Safety Measures

Personal safety equipment, such as hard hats secured by chin straps and complete eye protection, must be made available to the employees receiving the load. Employees should not be allowed to wear loose fitting clothes that may flap because of the rotor downwash. Such clothing can easily be snagged on the hoist line and cause a safety hazard.

The landing area must be clear of objects that may fly because of the rotor downwash. Loose gear and other such objects should be removed to a position of at least 100 feet from the lifting/depositing area.

A lift cannot be made if the helicopter pilot feels that the load is too heavy or out of the rated proportions.

No employees can be allowed within 50 feet of the helicopter's radius while the rotor blades are turning. Only authorized employees can be allowed if their presence is strictly necessary.

Lesson Summary

Proper communication must be maintained between helicopter pilots and the ground crew that act as signalmen during loading/depositing operations. The signalman must be distinctively identifiable from the rest of the crew.

A derrick that has been idle for more than six months must be inspected thoroughly before putting it back into operation. Only designated employees should be allowed to operate derricks.

Natural and synthetic fiber rope slings must never be used over their rated capacities. Clamps not designed specifically for fiber ropes are not to be used for splicing.

A permanent identification label must be attached to alloy steel chain slings. Damaged or defective slings are to be immediately removed from service.

Complete inspection of cranes must be carried out on a periodic basis. The interval between such inspections depends upon the activity, severity of service, and environment. Inspection requirements for crawlers, locomotives, and truck cranes depend on their exposure to wear, deterioration, or malfunction. Frequent inspections are required to monitor control mechanism adjustments interfering with proper operations, contamination by lubricants, and deterioration or leakage in hydraulic systems. All moving parts should be guarded so that a person may not come in contact with them and cause a hazard in any way. Moving parts include but are not limited to: belts, gears, shafts, pulleys, sprockets, spindles, drums, fly wheels, and chains.

Module 8: Hazard Communication

Module Description

The Hazard Communication Standard (HCS) provides information to workers and employers about various chemical hazards that exist in the workplace, and what protective measures they can take in order to prevent the adverse effects of such hazards.

This module gives you a basic understanding of how to deal with hazardous chemicals and how workers can prevent and protect themselves from chemical hazards.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Describe the purpose of the Hazard Communication Standard (HCS).
- Summarize the Global Harmonization System overview and changes
- Discuss labels and Safety Data Sheets.
- Differentiate between physical and health hazards associated with hazardous chemicals.
- Distinguish between symbols used to identify hazards.
- State how to prepare and implement a written hazard communication program.
- Defend training and its importance.

Lesson 1: Introduction to Hazard Communication

Lesson Focus

This lesson focuses on the following topics:

- The Hazard Communication Standard (HCS) Aligned with GHS.
- Hazard Communication/GHS Pictograms
- Hazardous Materials
- Important Definitions

The Hazard Communication Standard (HCS) Aligned with GHS

The Need of a Hazard Communication Standard (HCS)

According to OSHA, over 650,000 hazardous chemical products exist and hundreds of new ones are being introduced annually. More than 32 million workers are potentially exposed to one or more chemical hazards in more than 3 million American workplaces.

The Concept of Hazard Communication Standard (HCS)

The simple idea behind the Hazard Communication Standard (HCS) is that workers have both a need and a right to know about the hazards and identities of the chemicals they are exposed to when performing their tasks and duties.

The Hazard Communication Standard Coverage

Implementation of HCS for all those companies who import, produce, distribute, or use hazardous chemicals in the United States is mandatory. They must provide proper information and training to all of their affected employees.

The Hazard Communication Standard covers both physical (such as explosive, flammable) and health (acute and chronic) hazards. Being a worker or an employer, it is a need and a "right to know" how you can perform job responsibilities safely.

The Hazard Communication Standard (HCS) is now aligned with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS). This update to the Hazard Communication Standard (HCS) will provide a common and coherent approach to classifying chemicals and communicating hazard information on labels and safety data sheets.

This update will also help reduce trade barriers and result in productivity improvements for American businesses that regularly handle, store, and use hazardous chemicals while providing cost savings for American businesses that periodically update safety data sheets and labels for chemicals covered under the hazard communication standard.

OSHA is modifying its Hazard Communication Standard (HCS) to conform to the United Nations' Globally Harmonized System of Classification and Labelling of Chemicals (GHS). OSHA has determined that the modifications will significantly reduce costs and burdens while also improving the quality and consistency of information provided to employers and employees regarding chemical hazards and associated protective measures.

OSHA has concluded this improved information will enhance the effectiveness of the HCS in ensuring that employees are apprised of the chemical hazards to which they may be exposed, and in reducing the incidence of chemical-related occupational illnesses and injuries.

The modifications to the standard include revised criteria for classification of chemical hazards; revised labeling provisions that include requirements for use of standardized signal words, pictograms, hazard statements, and precautionary statements; a specified format for safety data sheets; and related revisions to definitions of terms used in the standard, and requirements for employee training on labels and safety data sheets.

OSHA is also modifying provisions of other standards, including standards for flammable and combustible liquids, process safety management, and most substance-specific health standards, to ensure consistency with the modified HCS requirements.

The consequences of these modifications will be to improve safety, to facilitate global harmonization of standards, and to produce hundreds of millions of dollars in annual savings.

Hazard Communication Standard Changes

Major changes to the Hazard Communication Standard include:

- **Hazard classification:** Provides specific criteria for classification of health and physical hazards, as well as classification of mixtures.
- **Labels:** Chemical manufacturers and importers will be required to provide a label that includes a harmonized signal word, pictogram, and hazard statement for each hazard class and category. Precautionary statements must also be provided.
- **Safety Data Sheets:** Will now have a specified 16-section format.
- **Information and training:** Employers are required to train workers by December 1, 2013 on the new labels elements and safety data sheets format to facilitate recognition and understanding.

Hazard Communication/ GHS Pictograms

As of June 1, 2015, the Hazard Communication Standard (HCS) will require pictograms on labels to alert users of the chemical hazards to which they may be exposed. Each pictogram consists of a symbol on a white background framed within a red border and represents a distinct hazard(s). The pictogram on the label is determined by the chemical hazard classification.

Hazardous Materials

Hazardous and toxic materials are those chemicals that may be present in a workplace that have a capacity to cause harm. Mixtures, fuels, solvents, paints, and dusts are all considered hazardous substances or materials.

Recognition of Hazardous Chemicals

Before working with or using chemicals, it is important to recognize those that may be physically hazardous or are capable of posing serious health problems to you. Recognition of hazardous chemicals prior to work can reduce the risk of chemical accidents.

Important Definitions

Personal Protective Equipment

The devices or clothing used by workers to protect against hazards in the environment are called Personal Protective Equipment (PPE).

Examples

Some common examples of PPE are:

- Respirators
- Gloves
- Chemical splash goggles

Toxicity

The term toxicity is used to describe the ability of a substance to cause a harmful effect or damage to an organism, cell, or organ. Toxicity also refers to the basic biological property of a material reflecting its inherent capacity to produce injury—adverse effects resulting from overexposure to a material, generally via the mouth, skin, eyes, or respiratory tract. Almost everything is toxic at some dose or concentration.

Example: When you inhale a toxic chemical, the dose you receive depends on four factors: (1) the level (concentration) of chemical in the air; (2) how hard (fast and deep) you are breathing, which depends on your degree of physical exertion; (3) how much of the chemical that is inhaled stays in your lungs and is absorbed into your bloodstream; and (4) how long the exposure lasts.

Flashpoint

The factor that determines whether a liquid is flammable or not is its flashpoint.

The flashpoint is a minimum temperature at which a liquid produces enough vapor within a test vessel in adequate concentration to form a flammable mixture with air near the surface of the liquid.

Combustible Liquids

Combustible liquids are those that have a flashpoint at or above 100 °F (37.8 °C).

Flammable Liquids

Flammable liquids are those that have a flashpoint below 100 °F (37.8 °C). Flammable materials require more care than combustible materials because they ignite at lower temperatures.

Container

Container means any bag, barrel, bottle, box, can, cylinder, drum, reaction vessel, storage tank, or the like that contains a hazardous chemical. For the purposes of this section, pipes or piping systems, and engines, fuel tanks, or other operating systems in a vehicle are not considered to be containers.

Exposure or Exposed

Exposure (or exposed) means that an employee is subjected, as a condition of employment, to a chemical that is a physical or health hazard, including potential (accidental or possible) exposure. "Subjected" in terms of health hazards includes any route of entry (such as inhalation, ingestion, skin contact, absorption, or injection).

Hazard Warning

Hazard warning means any pictograms, words, pictures, symbols, or combination thereof appearing on a label or other appropriate form of warning that conveys the specific physical and health hazard(s), including target organ effects of the chemical(s) in the container(s).

Immediate Use

Immediate use means that the hazardous chemical will be under the control of and used only by the person who transfers it from a labeled container and only within the work shift in which it is transferred.

Any hazardous material decanted—transferred from a primary to a secondary container—should have the labeling information transferred to the secondary container also.

Organic Peroxide

Organic peroxide is any carbon-containing compound with 2 oxygen atoms joined together. Organic peroxides can be severe fire and explosion hazards.

Oxidizer

An oxidizer is a chemical other than a blasting or explosive agent that initiates or promotes combustion in other materials, thereby causing fire either by itself or through the release of oxygen or other gases.

Pyrophoric

Pyrophoric means a chemical will ignite spontaneously in air at a temperature of 130 °F (54.4 °C) or below.

Unstable (Reactive)

Unstable (reactive) means a chemical that in the pure state, or as produced or transported, will vigorously polymerize, decompose, condense, or become self-reactive under conditions of shocks, pressure, or temperature.

Water-Reactive

Water-reactive means a chemical that reacts with water to release a gas that is either flammable or presents a health hazard.

Lesson Summary

The simple idea behind the Hazard Communication Standard (HCS) is that workers have both a need and a right to know about the hazards and identities of the chemicals they are exposed to when performing their tasks and duties. This standard covers both physical and health hazards.

As a worker or an employer there is a need and a right to know how you can perform your job responsibilities safely. Hazard warning means any pictograms, words, pictures, symbols, or combination thereof appearing on a label or other appropriate form of warning that conveys the specific physical and health hazards of the chemicals contained.

Lesson 2: Labels, SDSs, Symbols and Hazards

Lesson Focus

This lesson focuses on the following topics:

- Labels
- Safety Data Sheet (SDS)
- Symbols
- Hazards
- Controlling Physical and Health Hazards
- Hazard Communication Program
- Training

Labels

Any written, printed, or graphic materials displayed on or affixed to containers of hazardous chemicals are called labels. Labels are considered the most immediate source of information about chemicals and their hazard potential. It is obligatory that all hazardous chemicals containers must be labeled.

The following information must be included on all labels:

- Complete chemical name or names, (no abbreviations); formula may be used as an option.
- A warning statement, (symbol or message) conveying what the hazards are for the hazardous chemical.
- The manufacturer's name and address and some way to reach the manufacturer.

Labels must appear on each container. Labels must be legible and must be written in English. Other languages can be used additionally if desired.

Note: Many manufacturers of chemicals include safe handling procedures on labels.

Colors on Labels

This is a label created by the Paint and Coatings Association. Its use is voluntary and is not part of OSHA's HazCom Standard, but many manufacturers use it on containers they use, or ship to help further identify hazards of the material in the container.

The following colors are used on this voluntary label to represent different types of hazards:

- Blue is used to indicate the degree of health hazards.
- Red indicates how flammable the chemical is.
- Yellow represents the reactivity characteristics of the material.
- The White field is used to display any other special symbols, such as one to indicate the material is an oxidizing agent or is water reactive.

Safety Data Sheet (SDS)

The Hazard Communication Standard (HCS) requires chemical manufacturers, distributors, or importers to provide Safety Data Sheets (SDSs) (formerly known as Material Safety Data Sheets or MSDSs) to communicate the hazards of hazardous chemical products. As of June 1, 2015, the HCS will require new SDSs to be in a uniform format.

The Safety Data Sheet (SDS) provides detailed information about a specific hazardous material. Although labels are a good way to provide information about hazardous chemicals, sometimes you need more information about the chemical, but it may not be possible to provide all the information on a label.

SDS must be maintained in the facility for use by personnel while the material is in the facility. It is the recommended best practice to retain Safety Data Sheets for a period of at least 30 years. Employers who choose not to retain Safety Data Sheets are required to retain a record of the identity of the substance or agent, where it was used, and when it was used, for at least 30 years.

SDS Presentation

The following information must be included in an SDS:

- Identification of the substance or mixture and of the supplier
- Hazard identification
- Composition/information on ingredients
- First aid measures
- Fire-fighting measures
- Accidental release measures
- Handling and storage
- Exposure controls/personal protection
- Physical and chemical properties
- Stability and reactivity
- Toxicological information

The following information may also be included on an SDS if desired:

- Ecological information
- Disposal information
- Transport information
- Regulatory information
- Other information

Before an employee is exposed, it is the employer's responsibility to convey information about the hazards of working with materials in the facility.

Although it can also be stored electronically, an SDS must always be immediately available for review for all affected employees.

More information: [Sample SDS](#)

Chemical Manufacturer and SDSs

It is obligatory that chemical manufacturers include all the hazard information on SDSs and distribute it to their customers/clients at the time of first shipment of the hazardous chemicals or materials.

Note: It is necessary that manufacturers or importers of hazardous chemicals must update an SDS within three months of discovering new and important information about chemical hazards.

Availability of SDSs

Every employer must train every effected employee in the information derived from the SDS for each hazardous material in the work area, before exposing the employee to the hazard. An employee should ideally only need to consult SDS on an infrequent or emergency basis.

Note: Remember, it is the employer's responsibility to ensure that each employee who handles or uses any hazardous material knows where SDSs are located and how to read and understand them.

Symbols

OSHA created no symbols or specific color designation as part of the HAZCOM Standard. However, OSHA does recognize the use of U.S. Dept. of Transportation's hazard class symbol and color system in any employer's training programs.

Whenever you are working with materials that have a DOT hazard class shipping label, you should be aware that this represents a specific hazard. Information regarding specific hazards denoted by DOT labels is contained on the material's SDS.

Explosive symbols (DOT hazard class labels) are used with those materials that release a great amount of energy in the form of light, expanding pressure, and heat within a short passage of time. Water reactive materials react with water and can explode. Furthermore, unstable reactive materials can react or become self-reactive subject to pressure, temperature, or shock.

Compressed gas cylinder symbols were created by the Compressed Gas Association and adopted by OSHA and should appear on every cylinder of compressed hazardous gas.

Gases are used in various manufacturing processes. Because these gases are bottled under great pressure, misuse or unsafe handling could lead to an accident.

Note: Do not expose flammable or combustible materials to fire, heat, sparks, flames or any other source of heat or ignition.

Hazards

There are frequently two types of hazards (physical and health) present in a workplace where hazardous chemicals or materials are present. Physical hazards are one of two major definitions of hazards covered by OSHA's Hazard Communication Standard. These hazards are responsible for hundreds of deaths and injuries each year in the United States.

Health Hazards

Hazardous chemicals can affect our health in different ways. Generally, two terms—acute and chronic—are commonly used in order to understand the nature of the health hazards.

Determination of Health Hazards

Determining whether a hazard is affecting the health of a worker or not is very difficult. Often, the signs and symptoms of acute or chronic health effects are the same as those possessed by a non-occupationally exposed person. For example, lung cancer, kidney failures, and nervous system breakdowns are some common health problems that occur in occupationally exposed persons but can also be found in non-occupationally exposed persons as well.

Acute effects indicate that symptoms have occurred rapidly as compared to chronic effects, which means symptoms have manifested themselves over a period of time. The former are of short duration and/or the result of short-term exposures.

Example: If you accidentally spill a strong acid on your hand and the acid begins to burn your hand it is immediately considered to be acute and would be an accident as opposed to an illness effect.

Chronic Health Hazards

Chronic effects develop as a result of long-term exposures. Some manifestation examples of chronic effects follow:

- Silicosis
- Contact dermatitis
- Lung cancer from occupational exposure to carcinogens

Note: Asbestos is a good example of a chronic health hazard. Those people who are exposed to asbestos may take many years to develop serious lung diseases.

Health Hazard Symbols

The following symbols are used to identify various kinds of health hazards:

- The skull and crossbones symbol represents a poisonous material.
- The hand and test tubes symbols are used to identify corrosive materials.
- The radiating fan symbol is used to represent radioactive materials.

Routes of Exposure-Health Hazards

Health hazards can affect a body through four routes of entry:

1. Absorption
2. Inhalation
3. Ingestion
4. Injection

Note: Be sure you understand the information contained on labels and SDSs before working with hazardous chemicals.

Skin Absorption

Some chemicals enter the body by absorption through the skin. Always use proper personal protective equipment or clothing to protect your body from skin contact with hazardous material.

Inhalation

Inhalation is the most common route of exposure into the body. It normally occurs when you inhale fumes, vapors from solvents, hazardous gasses, or dust. It is the employer's responsibility to be aware of such hazards in the workplace and to protect employees from inhalation hazards.

Ingestion

The primary means of ingestion is by eating or smoking after handling hazardous material. When working with hazardous materials, wear the appropriate PPE, and always wash your hands with soap and water before eating, drinking, or smoking.

There are a number of ways through which you can control physical and health hazards associated with chemicals in a workplace. The following measures can protect you from physical and health hazards:

- Safe Work Practices
- Product Substitution
- Engineering Controls
- Training and Communication
- Environmental Monitoring
- Personal Monitoring
- Personal Protective Equipment

Controlling Physical and Health Hazards

Safe Work Practices

Safe work practices help assure that you are using chemicals safely and correctly.

Product Substitution

There are many chemicals that perform similar jobs. One of the many responsibilities an employer has is to attempt to find a chemical that is less toxic but able to accomplish the same job.

Engineering Controls

An orderly and well-designed workplace can minimize exposure to hazardous chemicals. Some engineering controls, like exhaust systems and wetting systems used to control dust, are good examples of hazard control.

Training and Communication

Training and communication play an essential role in every field of life. It is vital to know how to work safely with hazardous chemicals.

Environmental Monitoring

Environmental monitoring is a component in keeping an environment free from a buildup of hazardous chemicals, which could lead to an unsafe working environment.

Personal Monitoring

Monitor yourself and coworkers for symptoms (such as dizziness, eye or throat irritation, skin rashes) that would indicate that you or your coworkers have been exposed to a hazardous material or chemical. If these or other symptoms appear, report them to your supervisor immediately.

Personal Protective Equipment

Always use gloves, aprons, masks, or other PPE whenever called for on a label or SDS.

Hazard Communication Program

It is required that all chemical manufacturers, importers, and distributors convey complete information about a chemical and its hazards in the form of labels and SDSs. It is also mandatory that employers conduct hazard communication training programs in order to

provide complete information to their employees through SDSs, labels, and training sessions.

Note: Employers are responsible for informing and training their employees about the hazards that exist in their workplaces.

Written Hazard Communication Program

A written program must be established in all workplaces where employees are exposed to hazardous chemicals. It should include a list of all hazardous chemicals that are present in the workplace and where employees can get copies of written information about safe chemical handling procedures.

A written program also indicates the person in the facility who is responsible for the various aspects of the program. The written program must also describe requirements and information about labels, SDSs, and employee training.

Note: Written programs may not be required in laboratories and those workplaces where employees are dealing with sealed containers.

Training

Training must be provided to all employees exposed to hazardous chemicals. It must include information on how to handle chemicals safely, how to read and understand labels, SDSs, and other warning information, and what PPE is required before handling or using the hazardous material. It is required, and critical, that employees be trained before working with materials that represent a hazard.

Note: Remember, it is not sufficient to just provide SDSs to read.

Lesson Summary

Labels are considered the most immediate source of information about chemicals and their hazard potential. It is obligatory that all hazardous chemical containers be labeled.

It is the employer's responsibility to translate the information contained on the SDS, into an understandable format, and convey that information about the hazards associated with working with any of the hazardous materials in the facility, before an employee is ever exposed to a hazard.

Module 9: Hazardous Materials (Flammable and Combustible Liquids, Spray Finishing, Compressed Gases, Dipping and Coating Operations)

Module Description

Hazardous materials are considered any substance or compound that has a capability of producing adverse effects on the health and safety of humans. Every year, thousands of workers are injured or killed because of chemical or other toxic hazards. OSHA and other authorities have provided many rules and regulations to prevent hazards and to enable employees to perform safe operations at the workplace.

It is very important to know how we can save lives at a workplace. This module gives you a basic understanding of how to deal with hazardous materials and how we can protect ourselves from their potentially lethal hazards to prevent injury, illness, and/or death.

Module Learning Objectives

At the conclusion of this module, you will be able to:

- Discuss general requirements of handling/working with hazardous materials.
- Explore how to store hazardous materials.
- Explain the use and disposal of hazardous materials.
- Name the steps necessary to take during an accident.
- Illustrate training, first aid, and other essential factors associated with hazardous materials.

Lesson 1: Introduction to Hazardous Materials

Lesson Focus

This lesson focuses on the following topics:

- Hazardous Materials
- Flammable and Combustible Liquids
- Flammable and Combustible Liquid Classifications
- Important Definitions

Hazardous Materials

Hazardous materials are those chemicals that may be present in a workplace and have a capacity to cause harm if improperly managed. Pure chemicals, formulations, fuels, solvents and paints may all be considered hazardous materials.

Flammable and Combustible Liquids

The factor that determines whether or not a liquid is flammable is its flashpoint.

Flashpoint

The flashpoint is a minimum temperature at which a liquid produces enough vapor within a test vessel in adequate concentration to form a flammable mixture with air near the surface of the liquid.

Flammable Liquids

Flammable liquids are those that have a flashpoint below 100°F (37.8°C). Flammable materials require more care than combustible materials because they can ignite at lower temperature.

Combustible Liquids

The combustible liquids are those that have a flashpoint at or above 100°F (37.8°C).

More Information: It is important to remember that it is vapor and not the liquid itself that is more easily ignited.

Flammable and Combustible Liquid Classifications

Flammable Liquids

Flammable liquids are also known as Class I liquids. Class I liquids are further divided into three subclasses, as follows:

NFPA Class IA, IB, and IC

Class IA—includes liquids having flashpoint below 73°F (22.8°C) and having a boiling point below 100°F (37.8°C).

Class IB—includes liquids having flashpoint below 73°F (22.8°C) and having a boiling point at or above 100°F (37.8°C).

Class IC—includes liquids having flashpoint at or above 73°F (22.8°C) and below 100°F (37.8°C).

Combustible Liquids

Combustible liquids are divided into the two following classes:

NFPA Classes II and III

Class II—includes liquids with flashpoints at or above 100° F (37.8°C) and below 140°F (60°C).

Class III—includes liquids with flashpoints at or above 140°F (60°C). This class is further divided into two subclasses:

Class IIIA—includes liquids with flashpoints at or above 140°F (60°C) and below 200°F (93.3°C).

Class IIIB—includes liquids with flashpoints at or above 200°F (93.3°C).

Flammable Liquids and the Flashpoints

Some common flammable liquids and their approximate flashpoints are:

- Triethalemine: 17°F
- Barsol A-2316: 54°F
- Acetone Solvent: 0°F
- Glycol Ether: 94°F
- Methanol: 45°F
- Xylene: 79°F
- Gasoline: -45°F

Note: Flammable or combustible liquids must be stored in a properly designed closed container or tank, that is appropriately vented.

Important Definitions

Personal Protective Equipment (PPE)

The equipment used by workers to protect against hazards in the environment is called Personal Protective Equipment (PPE).

Example:

Some common examples of PPE are the following:

- Respirators
- Gloves
- Chemical splash goggles

Explosive Limits

A material's explosive limit is the concentration (in air) that is needed for a gas or vapor to explode. There is a lower explosive limit (LEL) and an upper explosive limit (UEL). At concentration below the LEL, there is not enough fuel to initiate or continue an explosion. At concentrations above the UEL, the fuel has displaced sufficient air that an inadequate level of oxygen is present to initiate a reaction. Concentrations of explosive gases are often given in terms of percent of lower explosive limit (%LEL) and percent of the upper explosive limit (%UEL).

Lower Explosive Limit (LEL)/Lower Flammable Limit (LFL)

The lower explosive limit is the lowest concentration of a substance that will produce a fire or flash when an ignition source (flame, spark, etc.) is present. It is expressed in percentage of vapor or gas in the air by volume. Below the LEL, the air/contaminant mixture is theoretically too "lean" to burn.

Upper Explosive Limit (UEL)/Upper Flammable Limit (UFL)

The upper explosive limit is the highest concentration (expressed in percentage of vapor or gas in the air by volume) of a substance that will burn or explode when an ignition source is present. Theoretically, above this limit, the mixture is said to be too "rich" to support combustion.

Grounding

A grounding wire should always be used to ground the static charge that might exist in an employee's body or the containers of materials being worked with. Proper grounding discharges the static charge, which minimizes the possibility of a spark igniting the flammable vapor of a material.

Bonding

Proper bonding of two containers together minimizes the possibility of static charge/ spark near any flammable vapor. This procedure is referred to as bonding.

Safety Can

A safety can for flammable liquids is an approved red-colored container that has a capacity of not more than five gallons and a spring-closing lid and spout cover to allow for the discharge of vapor safely without building excessive pressure in the container.

Oily Waste Can

An approved container with a self-closing metal lid must be used to store oil-soaked rags.

Vapor

Vapor is the gas phase of material that would otherwise be in a liquid or solid state. The temperature at which this change in physical state begins is referred to as the material's boiling point.

Evaporation Rate

A material's evaporation rate is the speed at which a material is converted into vapor (evaporates) at a given temperature and pressure. A material's evaporation rate must be taken into consideration when addressing storage, handling, and disposal issues regarding hazardous materials.

Spraying Area

Any work operation where flammable materials are sprayed will result in the release of vapor, mist, or dust. This situation generally occurs in a paint spraying operation.

Spray Booth

A spray booth is an enclosure that may contain a flammable or combustible vapor as a result of the spraying operation. These work environments require special attention be paid to ventilation and control of the hazardous material being released into the air, as well as safety issues involving the employees who work in such an environment. Special attention must be paid to PPE and respiratory protection issues to keep employees safe.

Lesson Summary

The factor that determines whether or not a liquid is flammable is known as its flashpoint. The flashpoint is a minimum temperature at which a liquid produces enough vapor within a test vessel and in adequate concentration to form a flammable mixture with air near the surface of the liquid.

Flammable or combustible liquids must be stored in properly designed and closed containers or tanks that are provided with appropriate venting. It may be necessary to wear PPE when working with such materials.

A material's explosive limit is the concentration (in air) that is needed for a gas or vapor to explode. There is a lower explosive limit, (LEL) and an upper explosive limit (UEL). At concentrations below the LEL, there is not enough fuel to initiate or continue an explosion, while at concentrations above the UEL, the fuel has displaced so much air that there is inadequate oxygen to initiate a reaction.

Lesson 2: Handling, First Aid, Precautions, and Training

Lesson Focus

This lesson focuses on the following topics:

- Storage
- Transferring Flammable Liquids
- The Use and Disposal of Hazardous Materials
- First Aid
- The Use of Labels
- Training

Storage

Safe Storage

The following points are essential to consider when discussing the storage of hazardous materials:

- Smoking, unprotected welding, and open flames are not allowed in hazardous materials storage areas.
- Do not place containers or tanks in direct sunlight—always provide for proper storage.
- Flammable or combustible liquids must be stored in a proper area.
- Flammable liquid drums must be grounded and bonded when materials are to be transferred from one container to another.
- Suitable fire control devices, such as portable fire extinguishers, must be available at locations where flammable or combustible liquids are stored.

More Information: The warmer a flammable liquid becomes, the greater the potential for evaporation and the generation of vapor.

Storage Inside Building

Do not store containers of combustible or flammable liquids in stairways, near exits, or in areas used for safe exit of employees. Flammable or combustible liquids containers/portable tanks are strictly prohibited in office premises, except when building or equipment maintenance operations are in progress. Contents should still only be contained in approved containers or cabinets.

Leaking tanks or containers must be removed immediately from the storage area where practical, but in such an event, the contents must be transferred to a stable container or the entire container placed in a DOT-approved over pack.

Bulk Oxygen

Oxygen cylinders must be stored in properly ventilated buildings or areas and must be 20 feet from any highly combustible materials or outdoors.

Case Study

An Explosion and Fire Erupted in the Storage Tanks

In a product refinery, newly hired and experienced workers were performing their regular tasks. No warning signs and symbols were present on the hazardous materials storage tanks.

Two workers who recently joined the refinery were welding pipes near the storage tanks when an explosion occurred, and fire spread to the tanks, which contained flammable hydrocarbons and wastewater.

Three workers were killed on the spot and seven were seriously wounded. Three workers died later as a result of their injuries.

Reasons

- Warning signs/labels were not present on the tanks' storage area.
- Workers did not know that the tanks contained hazardous materials.
- The employer or other workers did not tell them not to weld near hazardous materials.
- The workers were new and had not received essential safety training.

Transferring Flammable Liquids

The following points are essential to consider while transferring flammable liquids:

- Liquids must be kept in covered and closed containers when they are not in use.
- Smoking and open flames are strictly prohibited in product transfer areas.
- The bulk transfer of liquids must take place through a pump drawing through the top of a drum.
- Those containers from which flammable liquids are transferred must be grounded and bonded to minimize the chances of fire due to potential static electricity sparks.
- Flammable liquids can only be transferred into approved, properly labeled containers.

More Information: The use of air pressure for transfer of hazardous materials is prohibited.

The Use and Disposal of Hazardous Materials

For proper use and disposal of flammable liquids, the following points are important to consider:

- Unprotected welding, open flames, or smoking are strictly prohibited in places where flammable liquids are transferred, used, disposed of or stored.
- Always keep containers, tanks, and drums closed when not in use.
- Immediately clean up spills.
- Ventilation is very important to keep a storage place free of vapor build-up.
- Used rags must be properly disposed of or stored in a self-closing oil rag waste can.

First Aid

The guidelines for immediate first aid are as follows:

- Call 911 or any other local emergency number immediately for medical care.
- Do not begin treating an accident victim until the hazardous substance has been identified and the respective authorities give approvals/signs to go near the victims; otherwise, it could be very harmful. After getting permission, you may move the victim to fresh air.
- Remove any contaminated clothing and shoes from the victim if safe and appropriate to do so and place them in a clean container.
- If it is safe to do so, clean the victim affected by chemicals by pouring cold running water on their skin and eyes for at least 15 minutes, unless the respective authorities direct you not to use water for that particular chemical.
- Do not inhale gas, fumes, and smoke in the hazardous accident area. If necessary, use an appropriate respirator while within and/or leaving the area.

More Information: If you do not know what to do in case of an emergency or are in doubt, contact your supervisor or call 911.

Safety Procedures

There is always the potential that liquids or gases may escape from containers or tanks, which can lead to a hazardous situation. It is very important that everyone who works with or around hazardous materials observes the highest level of safety precaution.

Significant points to remember to perform safe operations are:

- Always read the labels on containers/tanks so that you know what type of chemical and hazard with which you are dealing.
- Do not forget to read the SDS when you are handling or using the container or its contents.
- Always follow the symbols and signs present inside or outside the workplace.
- Do not enter areas that you are not authorized to enter.

Some common signs used in the workplace include/indicate:

- Do not smoke
- Highly flammable
- Do not use sparking tools
- Exit
- To Exit
- Hazardous
- Fire Extinguisher

Fire Related Emergencies

If you discover a fire in your workplace:

- Activate your Emergency Evacuation Plan.
- Immediately activate the fire alarms in your building.
- Shutdown equipment and appliances in the immediate area of the fire if possible and appropriate to do so safely.

- Evacuate the work area immediately.
- Try to isolate the fire area if safe to do so.
- Use portable fire extinguishers if you have been trained how to use them and if the conditions are appropriate for the use of this type of equipment.
- Help other workers evacuate the workplace if it is safe.
- Provide complete details of the incident to police and/or other rescue teams.
- Account for your team members by immediately reporting anyone missing if you are a supervisor or a team leader.
- Do not enter the building until you are instructed that you may do so.
- Observe any other procedures contained in your facility's EAP.

The Use of Labels

It is extremely important that every container or tank containing chemicals has a label describing exactly what it contains and what hazards are associated with those chemicals.

The characteristics of labels:

- A label (for our purposes) is any printed, written, or graphic material displayed on containers of hazardous materials.
- Labels on containers of hazardous chemicals must not be defaced or removed.
- Never use an unlabeled container.
- Labels are considered forms of hazard warnings—Placards and tags also provide warnings of potential danger.
- The manufacturer must also provide Safety Data Sheets.

Labels Must Contain

The following information is mandatory for the label of every chemical container:

- The contents of the container

- The name and address of the manufacturer and some way to contact them
- Health and physical hazards
- Recommended PPE

Note: To make sure containers' labels are in good condition, one should perform visual checks as frequently as necessary to ensure full compliance.

Training

It is mandatory to provide training to all affected individuals, including those workers who are working with and/or handling hazardous materials. This includes:

- Interpreting the SDS and labels
- The procedures for safe handling
- How and when to use PPE
- Awareness of the location of hazardous materials
- How to dispose of waste
- Chronic and acute effects of exposure to hazardous chemicals
- Cleanup procedure
- What to do in the event of exposure

Lesson Summary

Containers of combustible or flammable liquids should not be stored in stairways, near exits, or in areas used for safe egress of employees. For this reason, and many others, it is extremely important that every container or tank containing chemicals have a label describing exactly what it contains and what hazards are associated with those chemicals.

In the event that the worst-case scenario occurs, remember what you've been instructed to do and follow your company's EAP.

Module 10: Permit-Required Confined Space

Module Description

This module is intended for all employees who are required to enter confined or enclosed spaces. This module deals with the safety issues concerned with entering permit-required confined spaces.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Identify confined spaces.
- Identify confined space hazards.
- Discuss OSHA requirements regarding confined spaces.
- Explore the functions and duties of confined space workers.
- Name the conditions for and the steps of permit-required confined space entry procedure.
- List the basic qualifications and requirements for employee training and rescue services.

Lesson 1: Introduction to Confined Spaces

Lesson Focus

This lesson focuses on the following topics:

- Confined Spaces
- Confined Space Hazards

Confined Spaces

Introduction

Many employers require their employees to work within spaces that are considered "confined." These spaces normally hinder the activities of any employee who enters and works in them. In many instances, employees who work in confined spaces face an increased risk of exposure to serious physical injury from hazards such as entrapment, engulfment, and hazardous atmospheric conditions.

OSHA estimates that almost 2.1 million workers enter confined spaces annually. In order to reduce the workers' chances of becoming victims in confined spaces, OSHA has specified certain regulations contained in 29 CFR 1910.146 that must be followed by all employees.

Identifying Confined Spaces

The first step toward conducting a safe confined-space entry is to identify the space as potentially dangerous. As defined by OSHA, a confined space is large enough for an employee to enter and perform the assigned task but has limited or restricted means of entry or exit and is not designed for continuous occupancy by employees. Examples of confined spaces include underground vaults, storage bins, tanks, trenches, tunnels, pits, vessels, silos and manholes.

Non-Permit Confined Spaces

Non-permit confined spaces cannot contain any hazard that may cause death or any serious physical harm to the personnel entering them. Some examples of non-permit confined spaces may include walk-in freezers or refrigerators, certain air plenums, and certain pipe chases.

Permit-required Confined Spaces

A permit-required confined space is one that meets the definition of a confined space and has one or more of these characteristics: (1) contains or might have the potential of containing a hazardous atmosphere; (2) contains a material that can engulf an entrant; (3) has an internal configuration that may cause an entrant to be trapped or suffocated by inwardly converging walls or by a floor that slopes downward and becomes narrower; and/or (4) contains any other known serious safety or health hazards.

Confined Space Hazards

There are a number of hazards associated with confined spaces. Confined spaces pose a threat of serious injuries or even death to workers entering them. Entering confined spaces without proper equipment and training increases the inherent danger. It has also been noticed that confined spaces can cause multiple fatalities as would-be rescuers often jump right in and become victims themselves.

Case Study

Fatal Rescue Try

On March 6, 1998, four men were given orders to work in an empty sewage tank. The first worker who entered the tank had a safety harness, but no safety line. As soon as he detached a hose that was used to drain off the tank's contents, sewage and a mixture of methane and hydrogen sulfide from the line flowed back into the tank. The gases immediately overcame the man, and he lost consciousness.

His co-workers, who were watching from outside the tank, entered the tank in an attempt to rescue him. They were all overcome within a minute and drowned in sewage.

The company was fined \$125,000 by the Occupational Safety and Health Administration (OSHA) and was ordered to provide hands-on training to all employees about working in confined spaces.

The tragic event illustrates the danger involved with confined space entry. It is obvious that hazards associated with confined spaces can result in the death of the employees entering them. This danger is increased if the workers are not properly equipped or trained to deal with the situation. Another important consequence of a lack of training is the occurrence of multiple fatalities. This happens because often co-workers or rescuers leap right in to help a victim and become victims themselves.

Classification of Confined Space Hazards

In order to evaluate the potential hazards of a confined space, many factors must be taken into consideration. It is not always enough to identify the inherent hazards of the space; the hazards present in the immediate vicinity of the space that may affect entry operations must also be taken under consideration. Also, it is good practice to consider the range of work done in the space.

Hazards in a confined space may be determined by its atmospheric conditions, its current and previous contents, configuration of the space, nature of the work, external hazards, and mechanical and physical hazards.

Atmospheric Conditions

Before workers enter a confined space, the employer must check the atmospheric conditions of the space for potential hazards. OSHA has specified four categories of hazardous atmospheres. These are:

1. Oxygen-deficient/asphyxiating,
2. Flammable,

3. Toxic,
4. Irritant and/or corrosive atmospheres.

OSHA's accident data reveals that most of the confined space deaths and injuries are caused by asphyxiates, flammables, and toxics.

The OSHA standard specifies the following order of testing:

1. Oxygen content
2. Flammable gases and vapors
3. Potential toxic air contaminants

Oxygen-deficient/asphyxiating Atmospheres

Normal atmosphere consists of approximately 21% oxygen, 78% nitrogen, 1% argon, and other trace substances, including carbon dioxide. An atmosphere that contains less than 19.5% oxygen is considered to be oxygen deficient.

In a confined space, the lack of oxygen may occur due to consumption or displacement. One cause of oxygen consumption is the combustion process of flammable materials during many tasks such as welding, cutting, or brazing.

Consumption of oxygen can also occur during bacterial actions such as decay, and during chemical reactions such as the formation of rust on the exposed surfaces of a confined space. Also, the consumption of oxygen can be influenced by the number of people working in a confined space and the amount of physical activity involved. Sometimes, the oxygen level in a confined space can be reduced due to the displacement of oxygen by other gases.

The Health Effects of Different Levels of Oxygen	
Level of Oxygen	Results
21%	Average atmospheric oxygen level
19.5%	Absolute minimum level of oxygen present to enter a confined space without air-supplied respirator
16%	Tire rapidly; cannot think clearly
14%	Not enough oxygen to breath
<10%	Inability to move, fainting, loss of consciousness, death

Flammable Atmospheres

Flammable atmospheres are caused by the presence of flammable gases, vapors, combustible dust, and oxygen-enriched atmosphere. According to OSHA, an atmosphere is classified as oxygen-enriched if it contains an oxygen concentration greater than 23.5%. An oxygen-enriched atmosphere may cause flammable materials such as clothing and hair to burn violently when ignited.

There are three components that cause the atmosphere to become flammable: fuel and oxygen, in proper proportion, and a source of ignition. The proper proportion of fuel and oxygen is different for each gas and is referred to as the difference between the lower flammability limit (LFL) and upper flammability limit (UFL). OSHA specifies that employees should not enter a confined space if flammable levels are in excess of 10% of a substance's LFL.

Confined spaces often don't have adequate ventilation, so flammable and combustible gases or vapors may accumulate within these spaces. It is very important to conduct atmospheric tests at multiple levels of all confined spaces since gases that are heavier than air normally accumulate in the lower levels of a confined space and others may

accumulate at upper levels. Work such as spray painting, coating, or the use of flammable solvents for cleaning can create a flammable atmosphere. Also, it should be noted that in closed top tanks, gases that are lighter than air might rise and develop a flammable concentration under the lid of the tank.

Atmospheric testing should begin near the point of confined space entry. When monitoring for entries involving a descent into atmospheres that may be stratified, the atmospheric envelope should be tested at an appropriate distance in the direction of travel.

Toxic Atmospheres

Toxic materials in a confined space may cover a wide variety of substances including gases, vapors, and finely divided airborne dust. Toxic substances may form by one or more of the following means: product stored in a confined space, work being conducted in a confined space, or in areas adjacent to a confined space.

Product Stored in a Confined Space

A product that is stored in a confined space may contain a toxic component, which could be absorbed into the walls. This can then be released later depending on the use of the space. Moreover, the product itself may release the toxic substance.

Work Being Conducted in a Confined Space

The nature of the work being conducted within a confined space may also produce toxic atmospheres. Some examples of such work are scraping, painting, sanding, and welding or brazing with metals capable of producing toxic vapors. There are many solvents used for cleaning or degreasing that produce highly toxic vapors.

Areas Adjacent to a Confined Space

Toxic fumes produced by work near a confined space may enter and accumulate in the confined space. There are certain gases that are commonly found in various vessels; one such gas is carbon monoxide (CO). This is an odorless and colorless gas with almost the same density as air. It is usually formed due to the incomplete combustion of certain materials such as wood, oil, gas, etc.

Another gas that is commonly found is hydrogen sulfide (H₂S). Hydrogen sulfide may be formed in several ways; however, it is usually formed when hydrochloric acid is combined with iron sulfide. Microbial breakdown of organic material, such as garbage, sewage and manure, may also contribute to the formation of hydrogen sulfide.

Irritant or Corrosive Atmospheres

Irritants or corrosive substances can be classified as primary or secondary. Primary irritants do not have a toxic effect on the entire body. Examples of primary irritants include ozone, chlorine, hydrofluoric acid, hydrochloric acid, sulfuric acid, ammonia, nitrogen dioxide, and sulfur dioxide.

Secondary irritants may produce systemic toxic effects along with surface irritation. Examples of secondary irritants are benzene, ethyl chloride, carbon tetrachloride, trichloroethylene, trichloroethane, and chloropropene.

Continued exposure to irritant or corrosive concentrations in a confined space may cause immediate, acute impact on a worker or little or no evidence of irritation. However, exposure does result in a general weakening of the defense reflexes from changes in sensitivity. An associated danger in this situation is that the worker may not be aware of any increase in the exposure to toxic substances.

Physical Hazards

Apart from atmospheric hazards, certain physical hazards must also be taken under consideration. These can be caused due to:

- **Extreme Temperatures:** Extremely hot or cold temperatures can make working in a confined space very hazardous. For example, steam tunnels and large ovens may have very high temperatures that can increase the risk of a heat stroke.
- **Engulfment:** Engulfment by loose granular material in a confined space is one of the most common physical hazards. Engulfment can occur in storage bins and hoppers that are used to store materials such as grain, sand, or coal.
- **Noise:** In a confined space, noise can be amplified due to the design and acoustic properties of the space. Apart from damaging hearing, too much noise can also adversely affect communication.
- **Wet Surfaces:** Wet surfaces inside confined spaces can cause workers to slip and fall. This can lead to injury or even death. This hazard of slipping increases if the space has inwardly converging walls, or a floor that slopes downward and tapers to a smaller cross-section.
- **Falling Objects:** Falling objects can pose a threat to workers in confined spaces, especially the ones that have topside openings for entry and where there is some work being done above the worker.

Mechanical Hazards

In a confined space, hazards could be caused due to the movement or rotation of a mechanical part or an energy source. It is important to identify and lock out all rotating or moving equipment within a confined space.

Lesson Summary

In many instances, employees who work in confined spaces face an increased risk of exposure to serious physical injury from hazards such as entrapment, engulfment, and hazardous atmospheric conditions.

Before workers enter a confined space, the employer must check the atmospheric conditions of the space for potential hazards.

Toxic substances may form by one or more of the following means: product stored in a confined space, work being conducted in a confined space, or in areas adjacent to a confined space. Toxic materials in a confined space may cover a wide variety of substances including gases, vapors, and finely divided airborne dust.

Lesson 2: OSHA Standard

Lesson Focus

This lesson focuses on the following topics:

- OSHA Requirements
- Permit-required Confined Space Program
- Training
- Rescue and Emergency Services

OSHA Requirements

OSHA requires all organizations to comply with its Confined Space Entry Standard. The standard specifies that all employers must evaluate their workplace in order to find out if spaces are permit-required confined spaces. If any spaces are found to be permit-required spaces, all employees should be informed of their existence, location, and the danger they pose. OSHA requires posting danger signs (or other equally effective means) at the entrance of permit-required confined spaces in order to do this. The sign must be clearly labeled with an appropriate warning.

If the permit-required spaces are too dangerous to enter, employers must take effective measures to prevent their employees from entering them. If, however, it is necessary for the employees to enter permit-required spaces, employers are required to develop a written permit space program, which must be available to employees.

Permit-Required Confined Space Program

After evaluating their workplaces, employers are required to prevent unauthorized entry. Usually, this is accomplished by labeling all confined spaces. The next step is to make a list of all the confined spaces and to provide appropriate training and oversight, including a policy strictly forbidding employees from entering.

OSHA standards lay all the responsibilities of identifying, labeling, and listing the permit-required confined spaces on the employer. OSHA also requires the employer to identify and assess the hazards associated with each confined space. All the hazards that have been identified and evaluated must be clearly stated in the organization's written confined space entry program.

The next step involves the development of written procedures and practices for those workers who are required to enter the permit-required confined spaces. All the procedures and practices required for permit-required confined space entry must be in writing, and must include:

- Conditions that are acceptable for entry
- Information for isolating the permit space
- Information for inserting, purging, flushing, or ventilating the permit-required space as necessary to protect entrants from external hazards
- Information about pedestrian, vehicle, or other barriers as necessary to protect entrants from hazards
- Verification of the conditions in the permit space that are acceptable for entry throughout the duration of an authorized entry
- Name of at least one attendant who is to be stationed outside the permit-required space throughout the duration of an authorized entry

During the entry, the employees must be made to leave the space immediately if any hazardous conditions are detected. The employers are then required to identify the cause of the hazardous atmospheres and take appropriate corrective actions before anyone returns to the confined space. The entry conditions would then be reverified before entry.

In cases where there are changes in their use or configuration, OSHA requires employers to reevaluate all non-permit and permit-required confined spaces and to reclassify them as necessary.

Permit System

Before a permit-required space is entered, a confined space entry permit must be completed and signed by the entry supervisor. The permit must verify that all pre-entry preparations have been carried out and that the space is safe to enter. The permit must then be posted at entrances or otherwise made available to entrants before they enter a permit space.

The supervisor must make sure that the duration of the entry permit does not exceed the time required to complete the assignment. However, if such a situation arises, it is the responsibility of the entry supervisor to restrict entry and cancel permits as soon as an assignment has been completed. In case any new conditions arise, they must be specified on the canceled permit and used in revising the permit space program.

The OSHA standard also requires the employer to keep all canceled entry permits for at least one year to facilitate a review of the effectiveness of the confined space entry program.

Entry Permits

Entry permits must include the following information:

- Name of permit-required space to be entered
- Acceptable entry conditions
- Test results
- Initials or signature of the tester
- Name and signature of the supervisor who authorizes entry
- Name(s) of the authorized entrant(s)
- Name(s) of eligible attendant(s)
- Name(s) of entry supervisor(s)
- Purpose of entry
- Hazards of the permit space to be entered
- Date and authorized duration of entry

- Measures that should be taken to isolate permit-required spaces in order to eliminate or control space hazards, i.e., locking out of equipment and procedures for purging, making inert, ventilating, and flushing permit spaces
- Rescue and emergency services that can be summoned and the means for summoning those services
- Methods and equipment of communication that would be used to maintain contact during entry
- Special equipment and procedures, including personal protective equipment, rescue equipment and alarm systems
- Additional permit(s), such as for hot work, that have been issued to authorize work in the permit-required space
- Any other information needed to ensure employee safety

Training

OSHA is very particular about its requirements to train confined space entry personnel. Before work is started, the employer is required to provide proper training to all employees who are supposed to work in permit-required spaces. After the completion of training, the employees must be able to recognize the hazards associated with confined spaces and must have the necessary skills for safe performance of their job.

Sometimes, employers are required to provide additional training. This is required when:

- Job duties change.
- Some changes have been made in the permit space program.
- The permit-required space presents a new hazard.
- An employee's job performance demonstrates inadequacies in knowledge or use of prescribed procedures.

OSHA also requires employers to train all employer-based rescue team members. They must be provided with cardiopulmonary resuscitation and first-aid training. At the end of the training, employers must certify in writing that an employee has been trained.

The certificate of training should include the employee's name, signature or initials of trainer(s), and dates of training. Employees are required to make their certificates available for inspection by authorized representatives whenever required.

Some employers decide to use outside resources such as the local fire department to provide rescue services. The employer must verify that agency's ability to provide confined space rescue.

Duties of Authorized Entrants

The OSHA standard specifies duties that must be performed by all workers who have been authorized to enter a confined space. These include:

- Knowing the hazards that may be faced during entry. This includes the information about signs, symptoms, or consequences of exposure.
- Knowing how to use the equipment required for entry, including safety, monitoring, ventilation, and lighting equipment. The entrants must also know about the barriers/shields and the equipment required during an emergency.
- Maintaining effective communications with the attendant in order to allow him or her to monitor the authorized entrant.
- Alerting the attendant if any warning signs of exposure are detected, or if prohibited conditions are presented.
- Exiting the space if the attendant orders evacuation, if warning signs or symptoms of exposure are detected, if a prohibited condition occurs or an order to evacuate is given by the attendant or entry supervisor, or if an evacuation alarm is activated.

Duties of Attendants

Attendants are those individuals who are stationed outside permit-required confined spaces and who monitor authorized entrants along with performing duties as specified by OSHA. Attendants are responsible for:

- Knowing the hazards that may be faced during entry. This includes the information about signs, symptoms, or consequences of exposure.

- Knowing the behavioral effects of exposure to hazards. Examples of these effects include deterioration in performance and slurred speech.
- Maintaining an accurate count of entrants and ensuring that the permit correctly identifies all the personnel present in the space.
- Remaining outside the permit space at all times until another attendant takes his or her place.
- Maintaining effective communications with entrants in order to monitor their status and alert them if the need to evacuate the space arises.
- Monitoring activities inside and outside the space to identify potential hazards to the entrants and call for evacuation if unsafe conditions are detected.
- Summoning rescue and other emergency services as soon as it is determined that the entrants may need assistance to escape.
- Removing or forbidding unauthorized personnel from entering the permit space.
- Performing non-entry rescues.
- Performs no duties that might interfere with the primary duty of monitoring and protecting the authorized entrants.

Duties of Entry Supervisors

Entry supervisors are mainly responsible for ordering entries into permit-required spaces. Entry supervisors are responsible for:

- Knowing the hazards that may be faced during entry. This includes the information about signs, symptoms, or consequences of exposure.
- Making sure that all entry-permit requirements have been satisfied.
- Authorizing entry by signing the entry permit.
- Making sure to terminate the entry permit properly.
- Canceling the permit if unacceptable conditions arise during entry.
- Making sure that rescue services are available and can be summoned whenever required.
- Removing unauthorized individuals from the entry area.

- Verifying that the terms of the entry permit are followed and that acceptable entry conditions are maintained.
- Conducting pre-planning meetings with all employees who may be involved in permit-required confined space entries.

Rescue and Emergency Services

The OSHA standard requires all employers to ensure that rescue service personnel are provided with proper training and rescue equipment. The rescue personnel must also be provided with the authorized entrant's training. OSHA also states that the rescuers must be trained and certified in first aid and CPR. OSHA advises employers to conduct practice rescue exercises on at least an annual basis and allow rescue personnel to enter permit-required spaces so that they can practice rescue operations. Rescuers should also be made aware of all the hazards of the permit space.

The employers are also required to provide all rescuers with personal protective equipment needed to perform confined space rescues safely. Rescuers required to enter a confined space should be provided appropriate equipment such as a full body harness that has a retrieval line attached to the center of their backs near shoulder level or above their heads.

OSHA allows the use of wristlets, but only if employers can demonstrate that the use of a chest or full body harness is not feasible or creates a greater hazard. The standard also requires employers to ensure that the other end of the retrieval line is attached to a fixed point or a mechanical device outside the permit-required space. If a permit-required space is more than five feet deep, a mechanical device must be available to retrieve the personnel.

OSHA also requires employers to keep a Safety Data Sheet (SDS) or other similar written information in case an injured entrant is exposed to a substance. This document must be made available to the medical facility treating the exposed entrant.

Lesson Summary

Before a permit-required space is entered, a confined space entry permit must be completed and signed by the entry supervisor. The permit must verify that all pre-entry preparations have been carried out and that the space is safe to enter. The permit must then be posted at entrances or otherwise made available to entrants before they enter a permit space.

Before work is started, the employer is required to provide proper training to all employees who are supposed to work in permit-required spaces. After the completion of training, the employees must be able to recognize the hazards associated with confined spaces and must have the necessary skills for safe performance of their jobs.

Module 11: Lockout / Tagout

Module Description

This module is designed to inform employees about the requirements for lockout/tagout during servicing and maintenance of machines or equipment. Also covered are the requirements of the OSHA Control of Hazardous Energy, or Lockout/tagout Standard, procedures for the application of locks and tags, and a discussion of the types of energy these procedures are designed to control.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Identify key components of the lockout/tagout standard.
- Name the purpose and elements of an energy control program.
- Discuss the responsibilities of people involved with lockout/tagout.
- List elements that must be included in energy control procedures.
- Summarize the training and inspection requirements.
- Illustrate steps to safely apply energy shutdown.
- Describe the basic requirements for lockout/tagout devices.

Lesson 1: Introduction to Lockout/Tagout

Lesson Focus

This lesson focuses on the following topics:

- Lockout
- Tagout
- Conditions for Lockout/Tagout
- Purpose of Lockout/Tagout

Lockout

What is Lockout?

Lockout is the process of blocking the flow of energy from an energy source to a piece of equipment, and keeping it blocked out. Lockout is accomplished by the placement of a lockout device on an energy isolating device, in accordance with an established procedure, ensuring that the energy isolating device and the equipment being controlled cannot be operated until the lockout device is removed. A lockout device is a lock that holds an energy isolating device in a safe position.

What Are Lockout Devices?

Lockout devices—such as locks, blank flanges, and bolted slip blinds—are devices that use positive means to hold an energy-isolating device in a safe position. These devices prevent the accidental or inadvertent energizing of machinery or equipment.

Tagout

What is Tagout?

Tagout is the placement of a tagout device on an energy isolating device, in accordance with an established procedure, to indicate that the energy isolating device and the

equipment being controlled may not be operated until the tagout device is removed. The tag acts as a warning to not restore energy, though it is not a physical restraint.

Tags must clearly display a "Do Not Operate" sign and must be applied by hand. In order to allow the use of a "Tagout" system instead of a "Lockout" system, the employer must be able to demonstrate that the tag system provides at least the level of protection that would be provided with physical locks.

What Are Tagout Devices?

Tagout devices are warning devices (such as tags) that can be attached to energy-isolating devices to indicate that the equipment to which they are fastened may not be operated until such time that the tagout device has been removed.

Conditions for Lockout/Tagout

What Must Be Locked or Tagged Out?

Any machines or equipment whose unexpected energization, release of energy, or start-up could harm people when that equipment is under repair, replacement, renovation, modification, or adjustment must be locked or tagged out.

Who are the authorized employees?

Those employees who physically service and maintain equipment and lock out or tag out equipment for servicing or maintenance are authorized employees.

Note: These individuals are not necessarily the people who normally operate the equipment.

Who are the affected employees?

Those employees who are required to use machines or equipment on which servicing is performed or who perform other job responsibilities in an area where such servicing is performed are called affected employees.

Which energy sources are potentially hazardous?

Most people immediately think of electricity as the only potentially hazardous energy source, but there are other sources of energy that can be just as hazardous. These energy sources include thermal, chemical, pneumatic, hydraulic, mechanical, and gravity. It is important to remember that all sources of energy that have the potential to start-up, energize, or release unexpectedly must be identified and controlled before service or maintenance is performed.

Purpose of Lockout/Tagout

The purpose of lockout/tagout is to prevent any injuries to employees during the maintenance, repair, inspection, and adjustment of machines or equipment. All energy sources that are used, stored, or produced by the equipment must be controlled by lockout/tagout.

What are energy isolation devices?

The mechanical devices that can physically prevent the transmission or release of energy are called energy isolation devices and include:

- A manually operated electrical circuit breaker
- A disconnect switch
- A manually operated switch by which the conductors of a circuit can be disconnected from all underground supply conductors and, in addition, no pole can be operated independently

- A line valve
- A block

What are the general requirements for lockout and tagout devices?

The general requirements for lockout/ tagout devices are as follows:

- So as to withstand environmental conditions for the required period of time, lockout/tagout devices must be manufactured from durable materials.
- Each lockout/ tagout device must be clearly identifiable.
- When controlling energy, only lockout/tagout devices are to be used.
- Lockout/tagout devices are not to be used for any purposes other than controlling energy.
- Lockout/tagout devices must be standardized and classified using all or any of the following criteria: color, shape, and/or size.
- All lockout/tagout devices must have a tag clearly stating the identity of the employee(s) who installed the device.

What protective materials and hardware must the employer provide for isolating, securing, or blocking machines or equipment from energy sources?

The following protective materials and hardware are examples of what must be provided by the employer for isolating, securing, or blocking machines or equipment from energy sources:

- Locks
- Tags
- Chains
- Wedges
- Key blocks
- Adapter pins
- Self-locking fasteners

The equipment used for lockout must be strong enough so that it can only be removed by applying excessive force or using tools used to cut metals, such as bolt cutters. In other words, it must be strong enough to prevent involuntary or accidental removal.



The lockout equipment must not deteriorate in areas where corrosive substances, such as acid and alkali chemicals, are present.



The means of attachment for a tagout device must not be reusable. They should be attachable by hand and self-locking, with a minimum unlocking strength of 50 pounds. OSHA recommends that the attachment means should possess the general design and basic qualities of a one-piece nylon cable tie that can endure all environmental conditions.



The tag must be constructed in a way that the message printed on it does not get affected when exposed to harsh weather. Also, the message on the tag should be printed using standard print and format, and it should not fade away and become illegible when placed in wet or humid locations. Depending on the requirement, the messages printed on the tags can include legends such as "Do Not Open", "Do Not Start", "Do Not Energize", "Do Not Close", and "Do Not Operate". If any hazardous conditions are present due to the energization of any machine or equipment, a warning message must be included.



Lesson Summary

The purpose of lockout/tagout is to prevent any injuries to employees during the maintenance, repair, inspection, and adjustment of machines or equipment. All energy sources that are used, stored, or produced by the equipment must be controlled by lockout/tagout.

Tagout devices are warning devices that can be attached to energy-isolating devices to indicate that the equipment to which they are fastened may not be operated until such time that the tagout device has been removed.

Lesson 2: Energy Control and Lockout/Tagout

Lesson Focus

This lesson focuses on the following topics:

- Energy Control Application
- Release from Lockout/Tagout
- Energy Control Program

Energy Control Application

Application of Energy Control

Authorized employees are required to perform energy control procedures in order to safely de-energize machines or other equipment. The following steps highlight a series of measures.

Preparation for Shutdown

Before an authorized or affected employee turns off a machine or equipment, the authorized employee must have knowledge of the type and magnitude of the energy, the hazards of the energy to be controlled, and the method or means to control the energy.

Machine or Equipment Shutdown

Notify all affected employees that servicing or maintenance is required on a machine or equipment and that the machine or equipment must be shut down and locked out to perform the servicing or maintenance. The machine or equipment must be turned off or shut down using the procedures established for it to avoid any additional or increased hazards to employees as a result of the machine or equipment stoppage.

Machine or Equipment Isolation

The machine or equipment shall be turned off or shut down using the procedures established for the machine or equipment. An orderly shutdown must be utilized to avoid any additional or increased hazard(s) to employees as a result of the equipment stoppage. Engage the energy isolating device(s) so that the machine or equipment is isolated from the energy source(s). All energy isolating devices that are needed to control the machine's energy source must be located. These devices must then be used to isolate the machine or equipment from its energy source.

Lockout or Tagout Device Application

Lockout devices must be installed in such a manner that they will keep any energy isolating devices in the "safe" or "off" setting. While installing tagout devices, employees must ensure that the device clearly states the operations and/or movement of any energy-isolating devices. Furthermore, the tag must be installed in a location that is as close to a device as possible and is clear to employees operating the device. Both lockout and tagout devices can only be installed to the energy isolating device by an authorized employee. Lock out the energy isolating device(s) with assigned individual lock(s).

Stored Energy

After the energy isolating device has been locked out or tagged out, all potentially hazardous stored or residual energy must be relieved, disconnected, restrained, or otherwise rendered safe.

Verification

As a final precautionary measure, before allowing other employees to perform the necessary maintenance operations, an authorized employee must check the equipment and make sure that:

- The equipment is properly shut down.
- The energy source has been isolated.

- Lock-out and tag-out devices are in place.
- Stored energy is released and the machine has been completely de-energized.
- Ensure that the equipment is disconnected from the energy source(s) by first checking that no personnel are exposed, then verify the isolation of the equipment by operating the push button or other normal operating control(s) or by testing to make certain the equipment will not operate.

Release from Lockout/Tagout

There are lockout/tagout requirements for releasing machines or equipment that have been locked out or tagged out prior to restoring energy to the equipment and using it. Before lockout or tagout devices are removed, and energy restored, the procedures that follow are a must for authorized employees to follow.

1. **Machine/Equipment Inspection:** The work area must be inspected to ensure that nonessential items (tools, spare parts, etc.) have been removed and that all of the machine or equipment components are operationally intact.
2. **Positioning of Employees:** The work area must be checked to ensure that all employees have been safely positioned or have cleared the area. In addition, all affected employees must be notified that the lockout or tagout devices have been removed before the equipment is started. Verify that the controls are in neutral.
3. **Lockout or Tagout Device Removal:** Each lockout or tagout device must be removed from the energy-isolating device by the employee(s) who applied the device. Notify affected employees that the servicing or maintenance is completed, and the machine or equipment is ready for use.

What is the unique circumstance that allows an employee other than the one(s) who applied the lockout/tagout device to remove the device?

When the authorized employee(s) who applied the lockout or tagout device is not available to remove it, that device may be removed under the direction of the employer,

provided that specific procedures and training for such removal have been developed, documented, and incorporated into the employer's energy control program.

What steps must the employer take if an employee, other than the one(s) who applied the lockout/tagout device, removes the device?

- The employer must verify that the authorized employee who applied the device is not at the facility.
- The employer must make all reasonable efforts to contact the authorized employee to inform him/her that his/her lockout or tagout device has been removed.
- The employer must ensure that the authorized employee knows that the lockout device has been removed before he/she resumes work at the facility.

Testing of Machines

When may lockout or tagout devices be temporarily removed?

In some circumstances, employees need to temporarily restore energy to a machine or piece of equipment during servicing or maintenance to test and/or reposition the machine or piece of equipment. Working under controlled conditions in accordance with company procedures, lockout or tagout devices may be removed temporarily in order to perform these tasks.

What sequence of actions must occur in the temporary removal of the lockout/tagout devices?

1. The machine or equipment must be cleared of tools and materials.
2. Employees must be removed from the machine or equipment area.
3. All lockout or tagout devices may then be removed.
4. Authorized employees may then proceed to energize and test or position the equipment or machinery.

5. Following testing or positioning, all systems must be deenergized and energy control measures reapplied to continue the servicing and/or maintenance.

Outside Personnel (Contractors)

What are the obligations of the outside contractor and the on-site employer?

The on-site employer and outside employer are required to notify each other of their lockout/tagout procedures in situations when outside personnel, for example contractors, are present on the site and involved in activities that need to fulfill lockout/tagout requirements. The on-site employer is responsible for making his or her employees understand and follow the lockout/tagout methods of the outside employer.

Shift and Personnel Changes

How is the continuity of lockout or tagout protection maintained during shift or personnel changes?

A high percentage of accidents historically occur shortly after a shift change and are often due to a lack of communication. During a shift change, exiting personnel should meet oncoming personnel at the lockout/tagout device.

Specific procedures shall be utilized during shift or personnel changes to ensure the continuity of lockout or tagout protection, including provision for the orderly transfer of lockout or tagout device protection between off-going and oncoming employees, to minimize exposure to hazards from the unexpected energization.

Example: The oncoming authorized employee should place his/her lock or tag on the energy isolating device before the exiting authorized employee removes his /her lock or tag.

If this is not possible, the oncoming authorized employee should place his/her lock or tag on the energy- isolating device immediately after the exiting authorized employee removes his/her lock or tag. Exiting employees must inform oncoming employees of any problems or concerns regarding the service and maintenance of the machinery or equipment.

Group Lockout/Tagout Requirements

What procedures must be followed that will offer group employees the same protection that is provided to individual employees?

1. A group lockout/tagout must afford each employee a level of protection equivalent to that provided by the implementation of a personal lockout or tagout device.
2. The primary responsibility for a set number of employees working under the protection of a group lockout or tagout device must be vested in a single authorized employee.
3. The single authorized employee must determine the exposure status of individual group members.
4. If there will be more than one crew, department, or group involved in the activity, a single authorized employee must be designated to coordinate affected workforces and to ensure continuity of protection.
5. Each authorized employee must affix a personal lockout or tagout device to the machine or equipment when work begins and remove it when work is completed.

Energy Control Program

There exist three integral steps in any energy control program. They are as follows:

1. **Energy Control Procedures:** Procedures must point out the appropriate steps that an authorized employee would have to undergo to accomplish a successful lockout/ tagout. This would include the rules, regulations, and various techniques that can be employed in the control of hazardous energy. This includes:

- Specific procedural steps for shutting down, isolating, blocking, and securing machines or equipment to control hazardous energy.
 - Specific procedural steps for the placement, removal, and transfer of lockout devices or tagout devices and the responsibility for them.
 - Specific requirements for testing a machine or equipment to determine and verify the effectiveness of lockout devices, tagout devices, and other energy control measures.
- 2. Inspections:** Regular, periodic inspections must be carried out no less than annually to ensure that employees are following energy control procedures.
- 3. Training:** All employees expected to work under a lockout or tagout system must undergo extensive training and retraining programs to ensure that all operational energy control procedures and regulations are properly understood. The employer shall provide training to ensure that the purpose and function of the energy control program are understood by employees and that the knowledge and skills required for the safe application, usage, and removal of the energy controls are acquired by employees. The training shall include the following:
- Authorized employee shall receive training in the recognition of:
 - Applicable hazardous energy sources
 - The type and magnitude of the energy present in the workplace
 - Methods and means necessary for energy isolation and control
 - Affected employee shall be instructed in the purpose and use of the energy control procedure.
 - All other employees whose work operations are or may be in an area where energy control procedures may be utilized, shall be instructed about the procedure, and about the prohibition relating to attempts to restart or reenergize machines or equipment that are locked out or tagged out.
 - When tagout systems are used, employees shall also be trained in the limitations of tags.
 - Retraining shall be provided for all authorized and affected employees whenever there is a change in:
 - Their job assignments

- Machines, equipment, or processes that present a new hazard
 - The energy control procedures
- Retraining shall also be conducted whenever a periodic inspection reveals, or whenever the employer has reason to believe, that there are deviations from or inadequacies in the employee's knowledge or use of the energy control procedures.
- The retraining shall reestablish employee proficiency and introduce new or revised control methods and procedures, as necessary.
- The employer shall certify that employee training has been accomplished and is being kept up to date. The certification shall contain each employee's name and dates of training. 1910.147(c)(8)

What is the purpose of the energy control program?

The purpose of the energy control program is to guarantee that before any employee services equipment—if the potential exists for unexpected energization or startup or the release of stored energy, the machine or equipment first is isolated from all energy sources and rendered inoperative.

Energy Control Procedures-Documentation

What is the employers' obligation in establishing energy control procedures?

Employers must develop, document, and use specific procedures to control potentially hazardous energy when employees are servicing equipment or machinery.

What specific elements must be documented in the employer's energy control procedures?

The employer's energy control procedures must be documented properly to include the scope, purpose, authorization, rules, and techniques that must be used in order to prevent hazards related to energy sources. Employers are also required to specify the means that

are to be adopted in order to ensure strict adherence to the specified procedures.

These include, but are not limited to:

- Step-by-step instructions to isolate, shut down, block, and secure equipment and machines that may emit hazardous energy.
- Detailed instructions to place, remove, and transfer lockout or tagout devices along with specifying names of the personnel who are responsible for these devices.
- Exact requirements that would allow the personnel to test a machine or equipment to ensure lockout/tagout procedures are working effectively.

Periodic Inspection

What is the intent of the requirement for the employer to conduct periodic inspections?

The intent is to ensure that:

- The energy control procedures continue to be implemented properly.
- The employees are familiar with their responsibilities.
- Any deviations or procedural inadequacies observed are corrected.

Who performs the periodic inspection and what does it entail?

An authorized employee who is not involved in the energy control procedure being inspected performs the periodic inspection.

The periodic inspection must first identify any deficiencies or deviations and then correct them. Where lockout is used, the inspector must review each authorized employee's responsibilities under the procedure with that employee (although group meetings are acceptable).

Where tagout is used, the inspector must review both the authorized and affected employee's responsibilities with those employees for the energy control procedure being inspected.

Employee Training and Communication

Why must employees be trained?

Employees must be trained so that they can understand the purpose and function of the energy control program and acquire the knowledge and skills necessary for the safe application, usage, and removal of the energy controls.

What are the differences in the levels of training for the three categories of employees—"authorized employees," "affected employees," and "other employees"?

1. **Authorized employees** must receive training on the recognition of applicable hazardous energy sources, the type and magnitude of the energy available in the workplace, and the methods and means necessary for energy isolation and control.
2. **Affected employees** must receive training on the purpose and use of the energy control procedure.
3. **Other employees**—those whose work activities are or may be in an area where energy control procedures may be utilized must be instructed about the procedure and the prohibition relating to attempts to restart or reenergize machines or equipment that are locked out or tagged out.

Additional Training

What additional training is required when tagout systems are used?

Employers must train employees in the following limitations of tags:

- Tags are essentially warning devices affixed to energy isolating devices. They do not provide the physical restraint on those devices as is provided by a lock.
- When a tag is attached to an energy isolating device, it is not to be removed without authorization, and it is never to be bypassed, ignored, or otherwise defeated.
- Tags must be legible and understandable by all employees.
- Tags and their means of attachment must be made of materials that will withstand the environmental conditions encountered in the workplace.
- Tags may evoke a false sense of security, and their meaning needs to be understood as part of the overall energy control program.
- Tags must be securely attached to energy isolating devices so that they cannot be inadvertently or accidentally detached during use.

Employee Retraining

What triggers the retraining requirements for employees?

- A change in job assignments
- A change in machines, equipment, or processes that present a new hazard
- A change in the energy control procedures
- Periodic inspections reveal that there are deviations in the energy control procedure
- The employer believes that there are deviations from, or inadequacies in, the employee's knowledge or use of the energy control procedures

Case Study

This accident occurred in a semi-conductor manufacturing plant where 4Mbit memory and other products are produced.

The victim in this accident worked for a subcontractor of the parent company, which owned the plant where the victim and his colleagues were engaged in silicon wafer polyimide processing and back face processing.

On the day of the accident, two maintenance workers, including one leader, started repair work when a sputtering unit used in the processing wafer surface failed.

As the first step, workers tried to replace a power amplifier and an intermediate amplifier with no success in restoring the functions of the unit. They then checked the voltage inside the high-frequency power source unit and detected abnormal figures in the voltages of four inspection terminals.

After taking a rest break, the leader made a telephone call at about 2:30 to the maintenance company responsible for the unit to report their failure in repairing the sputtering unit. He came back to the repair site about one hour later, and found the victim lying unconscious on the floor between the sputtering unit and the power source rack.

The victim's body showed evidence of electric current passing through from the first and second fingers of the right hand to both knees and legs. In addition, some rubber-like material was adhering to a burned spot near the bottom of the metal cap on the top of the power amplifier in the high-frequency power unit.

What do you think were some of the causes?

- The power source was turned on.
- Although the leader had turned off the power switch of the high-frequency power unit by himself before taking a break, the victim turned on the switch while the leader was talking to the maintenance company by phone to locate the failure and perform other repair work.
- Personal protective equipment to prevent electric shock was not used.
- The victim was wearing polyester work clothes for the clean room, with a hood and anti-static rubber shoes, but no anti-shock protective equipment.
- Wrong operational procedures.
- The sputtering unit handling manual indicates that the unit can generate electrical discharges and requires adequate precautions for high voltages. However, the worker started the operation without reconfirming the contents of the manual.

Lesson Summary

When the authorized employee who applied the lockout or tagout device is not available to remove it, that device may be removed under the direction of the employer, provided that specific procedures and training for such removal have been developed, documented, and incorporated into the employer's energy control program.

All employees expected to work under a tagout system must undergo extensive training and retraining programs to ensure that all operational energy control procedures and regulations are properly understood.

Module 12: Machine Guarding

Module Description

There seem to be as many hazards created by moving machine parts as there are types of machines. Safeguards are essential for protecting workers from injuries. Any machine part, function, or process that may cause injury must be safeguarded. When the operation of a machine or accidental contact with it can injure the operator or others in the vicinity, the hazard must be controlled or eliminated.

This module will help workers to protect themselves from moving machinery. After completion of this module, workers will be able to properly apply the OSHA Machine and Machine Guarding requirements to their day-to-day work activities.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Identify hazards associated in working with and around machinery and the guards used to protect the operator.
- Identify hazards and machine guards for various types of specialized machinery.

Lesson 1: Introduction to Machines and Machine Guarding

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- Basics of Machine Safeguarding
- Training

Introduction

The intent of machine guarding is to minimize the risk of accidents from machine-operator contact. This includes: making contact with the machine; making contact with parts via flying chips, sparks, chemical and hot metal splashes, saw kickbacks, etc.; or making contact with the machine parts as a result of machine malfunctions.

Basics of Machine Safeguarding

The list of possible machinery-related injuries is as long as it is horrifying. Crushed hands and arms, severed fingers, blindness—there seems to be as many hazards created by moving machine parts as there are types of machines. Safeguards are essential for protecting workers from needless and preventable injuries.

Where Mechanical Hazards Occur

There are three basic areas that need safeguarding from dangerous moving parts:

1. **The point of operation:** The point where work is performed on the material, such as cutting, shaping, boring, or forming of stock.

2. **Power transmission apparatus:** All components of the mechanical system that transmit energy to the part of the machine performing the work. These components include flywheels, pulleys, belts, connecting rods, couplings, cams, spindles, chains, cranks, and gears.
3. **Other moving parts:** All parts of the machines that move while the machine is working. These can include reciprocating, rotating and transverse moving parts, as well as feed mechanisms and auxiliary parts of the machine.

Hazardous Mechanical Motions and Actions

A wide variety of mechanical motions and actions may present hazards to the worker. These can include the movement of rotating members, reciprocating arms, moving belts, meshing gears, cutting teeth, and any parts that impact or shear. These different types of hazardous mechanical motions and actions are basic to nearly all machines, and recognizing them is the first step toward protecting workers from the danger they present.

The basic types of hazardous mechanical motions and actions are as follows:

Motions

- Rotating (including in-running nip points)
- Reciprocating
- Transverse

Actions

- Cutting
- Punching
- Shearing
- Bending

Motions

Rotating motion can be dangerous; even smooth, slowly rotating shafts can grip clothing, and even the slightest skin contact can force an arm or hand into a dangerous position. Injuries due to contact with rotating parts can be severe.

Collars, couplings, cams, clutches, flywheels, shaft ends, spindles, and horizontal or vertical shafting are some examples of common rotating mechanisms that may be hazardous. There is added danger when bolts, nicks, abrasions, and projecting keys, or set screws are exposed on rotating parts of machinery.

Actions

Cutting action involves rotating, reciprocating, or transversing motion. The danger of cutting action exists at the point of operation where finger, head, and arm injuries can occur, and where flying chips or scrap material can strike the eyes or face. Such hazards are present at the point of operation in cutting wood, metal, or other materials. Typical examples of machinery that may involve cutting hazards include bandsaws, circular saws, boring or drilling machines, turning machines (lathes), or milling machines.

The Need for Safeguards

Safeguards must meet these minimum general requirements:

1. **Prevent contact:** Safeguards must be in places to prevent hands, arms, or any part of a worker's body or clothing from making contact with dangerous moving parts. A good safeguarding system eliminates the possibility of the operator or other workers placing parts of their bodies near hazardous moving parts.
2. **Secure:** Workers should not be able to easily remove or tamper with the safeguard, because a safeguard that can easily be removed or made ineffective is no safeguard at all. Guards and safety devices should be made of durable material that will withstand the demands of normal use. They must be firmly secured to the machine.

3. **Protect from falling objects:** The safeguard should ensure that no objects can fall into moving parts. A small tool dropped into a cycling machine could easily become a projectile that could strike and injure someone.
4. **Create no new hazards:** A safeguard defeats its own purpose if it creates a hazard of its own, such as a shear point, a jagged edge, or an unfinished surface that can cause a laceration. The edges of guards, for instance, should be rolled or bolted in such a way that they eliminate sharp edges.
5. **Create no interference:** Any safeguard that prevents a worker from performing the job quickly and comfortably might soon be overridden or disregarded. Proper safeguarding can actually enhance efficiency since it can relieve the worker's apprehensions about injury.
6. **Allow safe lubrication:** If possible, one should be able to lubricate the machine without removing the safeguards. Locating oil reservoirs outside the guard, with a line leading to the lubrication point, will reduce the need for the operator or maintenance worker to enter the hazardous area.

Training

Even the most elaborate safeguarding system cannot offer effective protection unless the worker knows how to use it and why. Specific and detailed training is therefore a crucial part of any effort to provide safeguarding against machine-related hazards. Thorough operator training should involve instruction or hands-on training in the following:

- A description and identification of the hazards associated with particular machines.
- The safeguards themselves, how they provide protection, and the hazards for which they are intended.
- How and why to use the safeguards.
- How and under what circumstances safeguards can be removed, as well as by whom (in most cases, by repair or maintenance personnel only).
- What to do (e.g., contact the supervisor) if a safeguard is damaged, missing, or unable to provide adequate protection.

This kind of safety training is necessary for new operators and maintenance or setup personnel when any new or altered safeguards are put into service or when workers are assigned to a new machine or operation.

Case Study

This accident occurred at a plant that manufactures machines to develop photographic film.

As the victim was engaged in processing work using a turret punch press, and all by himself, the accident apparently occurred in this way:

A turret punch press is a numerically controlled machine that can continue automatic processing once the operator has put raw materials in place. The machine the victim was using is believed to have suddenly stopped during its operation due to some apparent trouble. After judging that some abnormality occurred to the crankshaft, the victim climbed onto the table and inserted a handcrank (a manual tool) to turn the crankshaft. When the crankshaft suddenly started to move again, the handcrank was also rotated, striking a hard blow to the victim's head.

The handcrank was used roughly once a month as a manual tool to adjust any gaps that occurred when the crankshaft stopped at the top, dead center.

What do you think some of the causes were?

1. When the turret punch press stops operations in the automatic mode, it must be changed to a manual mode to turn the crankshaft from the outside. Immediately after the accident, however, the machine's display indicated that it was left in automatic mode.
2. The grip portion of the handcrank the victim was using has a length of some three feet, posing a risk of striking anyone within a wide reach. The victim was working within that range.

3. Although press machines are legally required to undergo specified inspections on a regular basis, no inspection had been carried out on the relevant press.

Lesson Summary

Different types of hazardous mechanical motions and actions are basic to nearly all machines, and recognizing them is the first step toward protecting workers from the danger they present. Even the slightest skin contact can force an arm or hand into a dangerous position, and injuries due to contact with rotating parts can be severe.

A good safeguard system may eliminate the possibility of the operator or other workers placing parts of their bodies near hazardous moving parts. But because even a small tool dropped into a cycling machine could easily become a deadly projectile, specific and detailed training is a crucial part of any effort to prevent machine-related hazards.

Lesson 2: OSHA Requirements

Lesson Focus

This lesson focuses on the following topics:

- General Requirements
- Woodworking Machinery Requirements
- Abrasive Wheel Machinery Requirements
- Mechanical Power Presses
- Mechanical Power-Transmission Apparatus

General Requirements

One or more methods of machine guarding should be provided to protect employees in the machine area from hazards such as those created by point of operation, ingoing nip points, rotating parts, flying chips, and sparks. The following are some general guidelines that apply to all machines:

- Guards should be securely affixed to the machine; a guard should not create an accident hazard.
- The point of operation for machines that expose an employee to injury must be guarded.
- Revolving drums, barrels, and containers should be guarded by an enclosure that is interlocked with the drive mechanism.
- When the periphery of the blades of a fan is less than seven feet above the floor or working level, the blades should be guarded with a guard having openings no larger than 1/2 inch.
- Machines designed for a fixed location should be securely anchored to prevent walking or moving.
- In operations where power failures are a possibility, provisions shall be made to prevent machines from automatically starting upon restoration of power. This will

reduce the incidence of injury from motors restarting unexpectedly. If a machine was not manufactured with a restart switch, one can be added on "inline" requiring it to be reset upon restoration of power.

Woodworking Machinery Requirements

All woodworking machinery such as table saws, swing saws, radial arm saws, band saws, jointers, tenoning machines, boring and mortising machines, shapers, planers, lathes, sanders, veneer cutters, and other miscellaneous woodworking machinery must be effectively guarded to protect the operators and other employees from hazards inherent to their operation.

Radial Saws

Requirements for radial saw guarding include:

- The upper hood shall completely enclose the upper portion of the blade down to a point that includes the end of the saw arbor.
- The sides of a blade's lower exposed portion shall be guarded to the full diameter of the blade by a device that will automatically adjust itself to the thickness of the stock and remain in contact with the stock being cut.
- Radial arm saws used for ripping shall have non-kickback fingers or dogs.
- An adjustable stop shall be provided to prevent the forward travel of the blade beyond the position necessary to complete the cut in repetitive operations.
- Installation shall be completed so that the cutting head to return gently to the starting position when released.

Table saws

Requirements for table saw guarding include:

- Circular table saws shall have a guard over the portion of the blade that is above and below the table. The guard must automatically adjust itself to the thickness of the material being cut and remain in contact with it.
- Circular table saws shall have a spreader aligned with the blade and mounted in the saw. The requirement of having a spreader installed does not apply while grooving, dadoing, or rabbeting.
- Circular table saws used for ripping shall have non-kickback fingers or dogs.
- Feed rolls and blades of self-feed circular saws shall be protected by a hood or guard to prevent the hand of the operator from coming into contact with the in-running rolls at any point.

Swing or Sliding Cut-Off Saws

Requirements for swing or sliding cut-off saw guarding include:

- All swing or sliding cut-off saws shall be provided with a hood that completely encloses the upper half of the saw.
- Limit stops shall be provided to prevent swing or sliding type cut-off saws from extending beyond the front or back edges of the table.
- Each swing or sliding cut-off saw shall be provided with an effective device to return the saw automatically to the back of the table when released at any point of travel.
- Inverted sawing or swing cut-off saws shall be provided with a hood that covers the part of the saw protruding above the top of the table or material being cut.

Bandsaws and Band Resaws

Requirements for bandsaw guarding include:

- All bandsaws and band resaws require safety devices.

- All portions of the saw blade shall be enclosed or guarded, except for the working portion of the blade between the bottom of the guide rolls and the table.
- Contact also can occur during machine repair or cleaning if care is not taken to de-energize the machine—that is, if lockout/tagout procedures are not followed.

Bandsaw wheels shall be fully encased; the outside periphery of the enclosure shall be solid, and the front and back shall be either solid or made of wire mesh or perforated metal.

Jointers

Each hand-fed jointer, with a horizontal cutting head, shall be equipped with an automatic guard that covers the entire section of the head on the working side of the fence or gage.

Abrasive Wheel Machinery Requirements

Introduction

An abrasive wheel is made up of individual particles that are bonded together to form it. The potential hazard exists where an improperly mounted and used wheel can explode, causing serious or fatal injury to the operator.

Abrasive Wheel Machinery—Mounting

Requirements for abrasive wheel machinery safety include:

- Immediately before mounting, all wheels shall be closely inspected and sounded (ring tested) by a trained and qualified individual. This is done by gently striking the wheel and listening for any abnormal sounds to ensure that the wheel has not been damaged.

- The spindle speed of the machine shall be checked before mounting the wheel to be certain that it does not exceed the maximum operating speed marked on the wheel.
- Always ensure that the manufacturer's instructions are followed to properly mount the wheel.
- The operator should also always ensure that the machine guards are in place.

Abrasive Wheel Machinery—Machine Guards

Abrasive wheels can only be used on machines that have safety guards that normally cover the spindle end, nut, and flange projections.

Exceptions to the above machine guarding requirement include machines with:

- Wheels used for internal work being ground while within the work
- Mounted wheels, used in portable operations, two inches and smaller in diameter
- Type 16, 17, 18, 18R, and 19 cones, plugs, and threaded hole pot balls where the work offers protection
- Safety guards may be constructed so that the spindle end, nut, and outer flange are exposed on all operations where the work provides a suitable measure of protection to the operator
- Work practices that entirely cover the side of the wheel. In that case, the side covers of the guard may be omitted

Abrasive Wheel Machinery—Work Rests

For offhand grinding machines, there must not only be machine guards, but adjustable work rests of rigid construction to support the work.

Work rests shall be adjusted closely to the wheel with a maximum opening of 1/8 inch to prevent the work from being jammed between the wheel and the rest. Breakage could result if jamming were to occur.

Abrasive Wheel Machinery—Angular Exposure and Exposure Adjustment

Abrasive wheel safety guards for bench stands, floor stands, and cylindrical grinders should not expose the grinding wheel periphery for more than 65 degrees above the horizontal plane of the wheel spindle.

The protecting member of the abrasive wheel safety guard should be adjustable for variations in wheel size so that the distance between the wheel periphery and the adjustable tongue or the end of the peripheral member at the top must not exceed 1/4 inch.

Mechanical Power Presses

The employer should provide and ensure the usage of point-of-operation guards or properly applied and adjusted point-of-operation devices to prevent entry of hands or fingers into the point of operation by reaching through, over, under and around the guard for every operation performed on a mechanical power press. This requirement does not apply when the point-of-operation opening is 1/4 inch or less.

A substantial guard should be placed over the treadle on foot-operated presses.

Pedal counterweights, if provided on foot-operated presses, should have the travel path of the weight enclosed.

Machines using full revolution clutches must all incorporate a single stroke mechanism.

Employers shall establish a program for regular inspections of their power presses to ensure safe operating conditions and must maintain a record of inspections and maintenance work.

All point-of-operation injuries to operators must be reported within 30 days to OSHA or the state agency administering a plan approved by OSHA.

Mechanical Power-Transmission Apparatus

Mechanical power-transmission apparatus refers to all components of the mechanical system that transmit energy from the prime mover (power source) to the part of the machine performing the work. These components include flywheels, pulleys, belts, connecting rods, shafting, couplings, cams, spindles, chains, cranks, and gears.

Requirements for mechanical power-transmission apparatus safety include:

- It must be ensured that employees are safeguarded from injuries caused by rotating members, in-running nip points, sprockets or pulleys, and the like.
- Guards for mechanical power-transmission equipment shall be made of metal or other suitable material. Wood guards may be used in the woodworking and chemical industries, in industries where atmospheric conditions would rapidly deteriorate metal guards, or where temperature extremes make metal guards undesirable.
- All pulleys, belts, sprockets and chains, flywheels, shafting and shaft projections, gears, and couplings, or other rotating or reciprocating parts, or any portion thereof within seven feet of the floor or working platform shall be effectively guarded.
- All guards for inclined belts shall conform to the standards for construction of horizontal belts and should be arranged in such a manner that a minimum clearance of seven feet is maintained between the belt and floor at any point outside the guard.
- Flywheels protruding through a working floor shall be guarded.
- The guard shall extend at least 15 inches above the belt where both runs of horizontal belts are seven feet or less from the floor or working surface.
- Gear guards, where required, shall extend six inches above the mesh point by a band guard covering the face, or otherwise be completely enclosed.
- Couplings with bolts, nuts, or set screws extending beyond the flange of the coupling shall be guarded by a safety sleeve.

Exceptions

Belts, pulleys, and shafting located in rooms used exclusively for power-transmission apparatus need not be guarded when all of the following requirements are met:

- Basements, towers, or rooms occupied by transmission equipment are locked against unauthorized entrance.
- Vertical clearance in passageways between the floor and power-transmission beams, ceiling, or any other objects is not less than five feet six inches.
- The intensity of illumination conforms to the requirements of ANSI A11.1-1965(R1970).
- Route followed by the oiler is protected in such a manner as to prevent accidents or injury.

Lesson Summary

Because an abrasive wheel is made up of individual particles bonded together, the potential hazard exists for an improperly mounted and used wheel to explode, causing serious or fatal injury to the operator. Employers should establish a program for regular inspections to ensure safe operating condition and maintain a record of inspections and maintenance work.

Module 13: Welding and Cutting

Module Description

This module is intended to provide an overview of the hazards associated with welding, cutting, and brazing and the protective measures necessary to assure the work is performed safely. The module is specifically designed to help workers meet OSHA compliance regarding welding, cutting, and brazing.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Discuss OSHA's general requirements concerning welding, cutting, and brazing.
- Identify the general safety requirements for all types of welding.
- Describe the general characteristics for various types of welding.
- Name the specific safety requirements for oxygen-fuel gas welding and cutting, arc welding and cutting, and resistance welding.

Lesson 1: General Requirements

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- Fire Prevention and Protection
- Personnel Protection
- Health Protection and Ventilation

Introduction

Welding, cutting, and brazing are some of the most common industrial processes. However, if proper preventive measures are not taken, these processes can be extremely hazardous. Burns to the skin, flash burns to the eyes and fire are some of the more immediate and serious hazards associated with welding, cutting, and brazing.

The use of compressed gases to create extremely hot flames can expose workers to highly dangerous conditions. There are many welding techniques that have different types of hazards associated with them, including the risk of fire, explosion, gas, and fume hazards.

The OSHA standard mandates all employers adopt and follow good work practices in all welding, cutting, and brazing techniques to prevent injuries, fires, and explosions.

Supervisor's Responsibility

Supervisor's duties include:

- Safe handling of the cutting or welding equipment, and the safe use of the cutting or welding process.
- Determining the combustible materials and hazardous areas present or likely to be present in the work location

Protecting combustibles from ignition by implementing the following measures:

- Have the work moved to a location free from dangerous combustibles.
- If the work cannot be moved, have the combustibles moved to a safe distance from the work or have the combustibles properly shielded against ignition.
- See that cutting and welding are so scheduled that plant operations that might expose combustibles to ignition are not started during cutting or welding.
- Secure authorization for the cutting or welding operations from the designated management representative.
- Determine that the cutter or welder secures approval that conditions are safe before going ahead.
- Determine that fire protection and extinguishing equipment are properly located at the site.

Fire Prevention and Protection

In order to protect workers from fire-related hazards, OSHA has specified basic preventive measures along with certain special preventive measures for exceptionally dangerous tasks, like welding and cutting containers and welding in confined spaces.

The OSHA standard's basic preventive measures for fire in welding, cutting, and brazing focus on eliminating fire hazards. These include:

- Before an object is welded or cut, it must be moved to a safe place.
- If the object to be welded or cut cannot be moved, then the area must be cleared of all moveable fire hazards.
- If certain fire hazards cannot be removed from the area, then workers are required to use guards to protect the immovable fire hazards from heat, sparks, and slag.
- If, however, the requirements mentioned above cannot be met, then employees must not perform any welding and cutting tasks.

Apart from these basic preventive measures for fires in welding and cutting, OSHA requires all workers to take certain special precautions. The OSHA standard requires workers to have proper fire extinguishing equipment available. All fire extinguishing equipment must be in proper working condition and must be kept close by for instant use. Preventive equipment may consist of buckets of sand, pails of water, fire extinguishers, or hoses. However, these may vary according to the nature of the work and the quantity of combustible material present in the vicinity.

Where there is a considerable amount of combustible material present closer than 35 feet from the work location, OSHA requires employers to station fire watchers whenever welding or cutting is performed. OSHA also requires employers to station fire watchers if considerable amounts of combustible materials are present that can be easily ignited by sparks, even if they are more than 35 feet away. The OSHA standard requires the fire watch to be maintained for at least a half hour after the welding or cutting task has been completed, and for a longer period of time if necessary.

Case Study

Welding Hazard

In a product refinery, two newly recruited workers were required to weld pipes near some storage tanks. These tanks contained highly flammable hydrocarbons. Some time after work commenced, the combustible material caught fire and resulted in a big explosion. Three workers who were working in the area died on the spot and seven were seriously wounded. Three of those workers died later as a result of their injuries.

This accident occurred because no preventive measures were taken before welding operations were started. No one made sure that there were no hazardous materials present in the vicinity of the work area, as required by the OSHA standard. Also, the workers were new and had not received any safety training and education regarding hazards associated with tasks like welding, cutting and brazing.

When Welding is Prohibited

The OSHA standard has specified certain situations in which welding, cutting, or brazing is not allowed. Cutting or welding is not allowed in:

- Buildings that have damaged sprinkler systems.
- Areas that are not authorized by management.
- Locations that have explosive atmospheres due to the presence of mixtures of flammable gases, liquids or dusts in the air.
- Areas where there is a risk of explosive atmospheres. These include areas inside unclean or improperly prepared tanks and areas with an accumulation of flammable gasses, vapors, liquids, or combustible dusts.

There are many containers that require additional safety precautions apart from the basic preventive measures. The OSHA standard specifies that welding, cutting, or other hot work should not be performed on barrels, used drums, tanks, or other containers until they are properly cleaned. Cleaning should remove all materials or substances like tars, acids, greases, or other flammable or toxic materials. It is also important to remove or blank any pipeline or connections to the container being welded or cut.

Welding in Confined Spaces

Personnel who are required to work in confined spaces must follow additional safety guidelines. OSHA defines a confined space as a relatively restricted space that has limited means of entry or exit. Examples of confined spaces include tanks, boilers, and pressure vessels. Before working in a confined space, all appropriate safety measures must be in place. The space must be ventilated thoroughly to be rid of any flammable or toxic gasses.

It is important for all employees who are assigned the task of welding or cutting inside a confined space to leave their gas cylinders and welding machines outside the space.

If arc welding is suspended for a long period of time, such as during lunch or overnight, all electrodes must be removed from the holders. The holders must be placed carefully so that accidental contact does not occur. Also, the machine must be disconnected from the power source when not in use.

When an operation is suspended, all torch valves must be closed so that no gas escapes through the nozzle. Whenever the torch is not to be used for a long period of time, the gas supply to the torch must be properly shut off at a point outside the confined area. If possible, the hose and torch must also be removed from the confined space.

If a welder is to enter a confined space, employers must station an attendant outside the space to monitor the welder at all times. The attendant must be capable of putting rescue operations into effect.

Personnel Protection

OSHA has specified guidelines for personnel involved in welding and cutting operations to take protective measures against various other hazards apart from fire hazards. These measures provide guidelines to personnel for fall protection, eye protection, and protective clothing. If welders are required to work at a height, for example on a platform, a runway, or a scaffold, OSHA mandates the use of railings, lifelines, or some equally effective safeguards.

Performing operations like welding, cutting or brazing without proper eye protection may cause serious eye injuries. OSHA requires all personnel to wear helmets and hand shields during all arc welding or arc cutting operations in order to protect themselves from direct radiant energy from the arc. However, OSHA does not require personnel to wear helmets and hand shields while involved in submerged arc welding. OSHA also mandates all helpers and attendants to have proper eye protection.

The material that is used to make the helmets and hand shields must be an insulator for heat and electricity.

It is very important to ensure that the helmets, goggles, and shields are not readily flammable, and they must be strong enough to withstand sterilization. If the work permitted, welding should be performed behind a non-combustible screen with a low reflectivity surface, such as zinc oxide.

This helps in absorbing ultra-violet radiation that may harm a welder; however, all booths and screens used must allow circulation of air at floor level. People in the vicinity of the area where welding or cutting operations are being carried out must be provided with non-combustible screens, shields, or goggles if necessary.

All personnel involved in welding, cutting, or brazing operations must be provided with personal protective equipment to protect themselves from burns and fires.

The type of protective clothing may vary according to the nature of the welding operation and the location where work has to be performed. Employers must provide personal protective equipment materials that are designed to provide maximum protection from hot metal and sparks.

Health Protection and Ventilation

Some materials that are used in welding operations can be extremely hazardous even if precautionary measures have been taken. These materials must be removed by proper ventilation. The materials that are considered extremely hazardous include fluorine compounds, zinc, lead, beryllium, cadmium, mercury, certain cleaning compounds, and stainless steel.

Lesson Summary

All fire extinguishing equipment must be in proper working condition and must be kept close by for instant use. However, these devices may vary according to the nature of the work and the quantity of combustible material present in the vicinity.

OSHA requires employers to station fire watchers if considerable amounts of combustible materials are present that can be easily ignited by sparks and requires the fire watch to be maintained for at least a half hour after the welding or cutting task has been completed or longer if necessary.

OSHA additionally requires all personnel to wear helmets and hand shields during all arc welding or arc cutting operations in order to protect themselves from direct radiant energy from the arc. It mandates all helpers and attendants to have proper eye protection. Also, if a welder is to enter a confined space, employers must station an attendant outside the space to monitor the welder at all times, and this attendant must be capable of putting rescue operations into effect.

Lesson 2: Oxygen-fuel Gas Welding and Cutting

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- Cylinders and Containers
- Manifolding of Cylinders
- Service Piping Systems
- Protection of Service Pipe Systems

Introduction

In this lesson, we will discuss the preventive and protective measures that focus on hazards associated with oxygen-fuel gas welding and cutting. Welders must guard against mixtures of fuel gasses and air or oxygen, as such mixtures may be explosive. No devices are to be used that mix air or oxygen with flammable gasses before they have to be consumed, unless they have been approved by an authorized person.

The OSHA standard requires all welders to take special care when dealing with acetylene. Acetylene should never be generated or stored except in approved cylinders. It should not be utilized at a pressure exceeding 15 pounds per square inch gauge (psig) or 30 pounds per square inch absolute (psia). Using acetylene at pressures exceeding 15 psig (30 psia) can be extremely hazardous.

Gaseous acetylene is highly unstable at pressures above 15 psig and it may decompose with an explosion. The decomposition of acetylene must be avoided by storing it in cylinders that have been specially designed for storing such gasses.

Cylinders and Containers

The OSHA standard requires all portable cylinders that are used for storage and shipment of compressed gasses to be constructed and maintained according to the regulations set forth by the U.S. Department of Transportation, 49 CFR Parts 171-179.

OSHA has provided specific guidelines for using and handling cylinders and containers containing compressed gasses. All personnel must know about marking of cylinders, storage of cylinders, and their operating procedures.

Approval and Marking

OSHA mandates that all compressed gas distributors clearly mark all compressed gas cylinders. Markings must include either the chemical or trade name of the gas stored in the cylinder. It is very important to ensure that all stamping, stenciling, or labeling used for marking the cylinder is not easily removable. The cylinder must be marked on the shoulder where it does not come into contact with other surfaces. Marking cylinders not only ensures proper use but also helps personnel to store them properly. An employer or employee must not attempt to use the contents of an unmarked cylinder.

Storage of Cylinders

Proper storage of cylinders is also stressed by the OSHA standard. All cylinders must be stored away from radiators and other devices that produce heat. If cylinders are stored inside buildings, steps must be taken to protect and ventilate the location thoroughly. It also must be ensured that the storage area is dry and at least 20 feet (6.1 m) away from highly combustible materials. It is additionally important to store cylinders away from stairs, elevators, exit routes, or other areas where they can be toppled or damaged by passing people or falling objects, or where they could otherwise be tampered with.

Personnel responsible for storing cylinders must make sure to close valves of all empty cylinders. Some cylinders are designed with valve protection caps.

When the cylinders are not in use, these protection caps must be placed on the valves and appropriately tightened. The purpose of the valve protection cap is to protect the valve and to buffer the shock received if the cylinder falls.

Fuel Gas Cylinders

All fuel gas cylinders stored inside a building must be limited to a total gas capacity of 2,000 cubic feet (56 m³) or 300 pounds (135.9 kg) of liquefied petroleum gas. All fuel gas cylinders with a gas capacity greater than 2,000 cubic feet (56 m³) or 300 pounds of liquefied petroleum gas must be stored in a separate room or a special building. These buildings or rooms must not have any heating or lighting and should be kept properly ventilated at all times. However, this limitation is not applicable for cylinders that are being used or are ready to be used.

Acetylene cylinders are required to be stored with the valve end up. If these cylinders are stored on their side, acetone may leak out and create a hazardous condition.

Oxygen Cylinders

Oxygen cylinders must be stored away from fuel-gas cylinders or other combustible materials, such as oil or grease. These cylinders must be stored at a minimum distance of 20 feet (6.1 m) from highly combustible material.

Oxygen cylinders outside generator houses are required to be separated from the generator by using a partition that has a fire-resistance rating of at least one hour. The partition must not have any openings and must not allow any gasses to accumulate. OSHA has specified this preventive measure in order to protect personnel from fire hazards.

Operating Procedures

Due to the potential hazards associated with cylinders containing compressed gasses, OSHA requires all personnel to store and operate these cylinders properly.

Care must be taken to eliminate all traces of oil and grease from cylinders, cylinder valves, regulators, couplings, hoses, and other apparatus when welding operations are being carried out. Welders must not operate or handle cylinders with oily hands or gloves. Also, it must be ensured that no oily surface or greasy clothing is exposed to a jet of oxygen.

Compressed gas cylinders have two critical components that must be handled carefully: valve outlet and the regulator. Valve protection caps are used to protect valves from getting damaged and from coming into contact with oil or grease. The OSHA regulation specifies that workers should never use valve protection caps for lifting the cylinders as the caps may accidentally come loose and fall, possibly causing a sudden release of pressure.

Apart from protecting the valve outlet, measures must be taken to protect the regulator. Regulators must be removed and valve-protection caps placed properly before cylinders are moved.

Cylinders that are not equipped with fixed hand wheels must have keys, handles, or non-adjustable wrenches on valve stems while they are being used.

A regulator must always be attached to a compressed gas cylinder before it is used, except when connected to a manifold. It is important to ensure that the regulator being used is compatible with the gas in the cylinder and its service pressure. The regulator must be cleaned with a clean filter installed.

Before a regulator is attached, the protective cap must be removed from the cylinder and the valve opened slightly for an instant, and then closed immediately.

This practice ensures that the cylinder valve is cleared of any dust or dirt that may have accumulated during storage. Dirt can damage vital parts of a regulator and may cause a fire or an explosion.

The valve of a cylinder containing acetylene should not be opened more than one and a half turns of the spindle and preferably no more than three-fourths of a turn. This allows adequate flow of the gas and permits the welder to close the valve quickly in an emergency situation.

While the welding or cutting operations are being carried out, all cylinders must be kept far away from the work location so that no sparks, hot slag, or flame can reach them. If they cannot be removed, they must be protected with fire-resistant shields. The OSHA standard also specifies that cylinders must not be placed in an area where they can become a part of an electric circuit.

Manifolding of Cylinders

Advantages of using manifolds:

- Because gas cylinders have not been scattered throughout the work area but have been kept centrally:
 - The possibility of accidents is reduced.
 - More space is available at each workplace.
- In case of fire, one can easily reach the cylinders.
- There is more effective use of gases.
- Cylinders are transported by less distance.
- There is no replacement of cylinders inside the workshop.

Welders do not always use portable cylinders as a source of gas. Sometimes a service pipe system is used to provide a manifolding effect. Manifolding is the process of using multiple-line fluid inputs into a single intake chamber for the purpose of combining gasses when they are needed for welding.

Portable outlet headers are used in order to control the flow of a particular gas. Each portable outlet header consists of a nozzle and a hose that can be connected to a portable cylinder or a service pipe. The OSHA standard specifies that, except for temporary service where conditions preclude a direct supply, portable outlet headers must never be used indoors.

All service piping outlets used to withdraw and supply oxygen or fuel gas to a portable outlet header must consist of a shut-off valve that is easily accessible.

The service outlet on the portable outlet header must be equipped with a valve assembly that must consist of a detachable outlet seal cap attached to the body of the valve.

The use of a seal cap ensures that the outlet pipe thread will remain free from oil or grease and not be damaged. Damaged pipe threads can result in leaky connections.

Service Piping Systems

All service piping systems being utilized must be designed and installed according to the safety requirements specified by OSHA. The pipes to be used should be at least Schedule 40, and all fittings must be of standard weight in sizes up to and including 6-inch nominal. Schedule 40 pipe has a working pressure of up to 125 psi, and it should always be tested before use.

A close inspection is also required because problems may arise if line extensions are made with pipe type other than Schedule 40. OSHA has specified some special requirements when personnel are using service pipe systems with oxygen or acetylene.

The piping system is required to have a minimum pressure of 250 psig if oxygen is supplied to a service piping system from a low-pressure oxygen manifold without any pressure-regulating device. When the connected equipment is used at pressures less than 250 psig, a pressure-regulating device must be used for each station outlet.

For acetylene or acetylenic compounds, the piping must be made of steel or wrought iron. Under certain conditions, acetylene forms explosive compounds with copper, silver, and mercury; therefore, employers must ensure that unalloyed copper is not used for acetylene or acetylenic compounds except when using certain listed equipment.

All piping must be installed in a way that it runs as directly as possible. Some space should be allowed for expansion, contraction, jarring, and vibration. This will help protect the pipes from damage. Pipes that have to be laid underground must be located below the frost line, and protective measures must be taken to prevent or protect against corrosion.

After installation, the piping has to be blown out with air, nitrogen, or carbon dioxide to remove any foreign materials. For oxygen piping, the air, nitrogen, or carbon dioxide that is used must be completely oil-free. When air or gas is released from combustible gas lines, care must be taken to ensure that no source of ignition is allowed near uncapped openings.

Underground piping must be protected against corrosion by covering or painting it with a suitable material that would allow maximum protection against corrosion.

After installing the piping systems, they must be tested to ensure that they are gas-tight at 1.5 times the maximum operating pressure. Oxygen must be tested by only using oil-free and non-combustible materials. Under no circumstances are personnel allowed to use flames to detect leaks.

Protection of Service Pipe Systems

The OSHA standard requires employers to ensure that the service pipe system is always protected against the build-up of excessive pressure and leaks. Protection may be provided using pressure relief devices, protective equipment, regulators, and hoses.

Pressure Relief Devices

Pressure relief devices are used to protect service piping from excessive pressure build-up. Pressure relief devices must be set to function at not more than the pressure specified for the system. The pressure relief device must be made to discharge upwards to a safe location. OSHA mandates the use of pressure relief valves for preventing excessive build-up in fuel-gas piping systems.

Piping Protective Equipment

The OSHA standard mandates the use of approved protective equipment for fuel-gas piping in order to prevent:

- The backflow of oxygen into the fuel-gas system.
- The passage of a flash-back into the fuel-gas supply system.
- Excessive back-pressure of oxygen in the fuel-gas supply system.

Personnel can use one device that performs all three functions of the protective equipment or separate devices to achieve each task separately. In a fuel-gas piping system, the protective equipment is required either at the main supply line, at the head of each branch line, or at each location where fuel-gas is withdrawn. However, it is preferable to place the protective equipment at the main supply as this would provide the best protection.

Regulators

Pressure-reducing regulators can be used to control pressure in piping systems. OSHA has specified that pressure-reducing regulators shall only be used for the gas and pressures for which they are made.

If any part of these regulators is out of order and needs repair, only skilled mechanics that have had proper training should fix it. The regulators for oxygen piping systems must be labeled with a warning advising the user to: "USE NO OIL."

Hoses

Apart from using release devices, the piping system must be designed using proper hoses to protect against leaks. The OSHA standard specifies that the hose and hose connections must be clamped or securely fastened so that they can withstand twice the pressure to which they are normally subjected. This pressure cannot be less than 300 psi. To test the hoses, oil-free air or oil-free inert gas must be used.

The oxygen hose is green and has a right-hand threaded nut for connecting to the torch. The acetylene fuel-gas hose is usually red (sometimes black) and has a left-hand threaded nut for connecting to the torch.

Hoses that are burnt, worn, or have other defects must be replaced or repaired before any operations are performed. OSHA also prohibits personnel from using leaky hoses.

When inspecting hoses, look for charred sections close to the torch. These may have been caused by flash-back. Also check that hoses are not taped up to cover leaks.

Lesson Summary

All personnel who weld or supervise those individuals who do must know about marking of cylinders, storage of cylinders, and their operating procedures. OSHA mandates that all compressed gas distributors, for example, clearly mark all compressed gas cylinders. Markings must include either the chemical or trade name of the gas stored in the cylinder.

It is additionally important to store cylinders away from stairs, elevators, exit routes, or other areas where they can be toppled or damaged by passing people or falling objects, or where they could otherwise be tampered with. Due to the potential hazards associated with cylinders containing compressed gases, OSHA requires all personnel to store and operate these cylinders properly.

The OSHA standard requires employers to ensure that the service pipe system is always protected against the build-up of excessive pressure and leaks. Protection may be provided using pressure relief devices, protective equipment, regulators, and hoses. OSHA mandates the use of pressure relief valves for preventing excessive build-up in fuel-gas piping systems.

Lesson 3: ARC Welding and Resistance Welding

Lesson Focus

This lesson focuses on the following topics:

- Arc Welding and Cutting
- Resistance Welding

ARC Welding and Cutting

The information presented in this lesson will focus on preventing accidents and promoting work safety when performing arc welding or cutting operations. In the arc welding process, an electric current is passed through the welding rod and is forced to jump or arc across a gap. The heat produced through this process is intense enough to perform welding and cutting operations. Most of the precautions and safe practices specified by OSHA are common to oxy-fuel gas welding, but there are certain requirements that are unique to arc welding.

One of the unique requirements of arc welding is shielding. It has been noticed that if air is kept away from the weld puddle, the welds produced have better physical and chemical properties. Some gasses like oxygen, hydrogen, and nitrogen, when mixed with moisture, can reduce the quality of the weld. Therefore, shielding is used to preserve the integrity of the weld joint.

Application

The OSHA standard requires construction of standard machines for arc welding service. These must be designed to carry their rated load with rated temperature rises. The maximum temperature of the cooling air should be 40 degrees Celsius (104 degrees Fahrenheit), and the maximum altitude is 3,300 feet (1,005.8 m).

The machines should be suitable for operation in atmospheres containing gases, dust, and light rays produced by the welding arc.

To carry out the arc welding process, a welder can either use an alternating current (AC) or a direct current (DC). For alternating current machines, the voltage should not exceed the limit of 80 volts for manual and 100 volts for automatic arc welding and cutting. For direct current machines, the voltage must not exceed the limit of 100 volts for both manual and automatic arc welding and cutting.

If certain special welding and cutting processes are carried out that require values of the voltages higher than the specified limits, OSHA mandates employers provide all personnel with adequate insulation or other means that would ensure their safety.

If arc welding and cutting operations are being carried out at a location where the surroundings are warm and humid, or where perspiration is a factor, OSHA recommends the use of reliable automatic controls for reducing no-load voltage to reduce the shock hazard. Some of the older AC machines do not have an automatic control and are on load all the time.

Design

The OSHA standard requires all types of arc welding machines and control apparatus to be enclosed except for the operating wheels, levers or handles. It is also very important to enclose input power terminals, tap change devices, and live metal parts so that they can only be accessed by means of tools.

While carrying out welding operations, all personnel are required to protect the terminals for welding leads from accidental contact by workers or by metal objects such as hooks, vehicles, cranes, etc. To achieve this, OSHA specifies the use of:

- Dead-front receptacles for plug connections.
- Recessed openings with non-removable hinged covers.
- Heavy insulating sleeving or taping.

The connections for portable control devices must never be connected to an AC circuit of higher than 120 volts. On circuits above 50 volts, all exposed metal parts of portable control devices should be grounded by a grounding conductor in the control cable.

Personnel must never use AC reactors or auto transformers to obtain welding current directly from an AC power source with a voltage exceeding 80 volts.

Installation

Proper installation of equipment is very important to ensure safety during arc welding and cutting operations. One of the most critical requirements is proper grounding. OSHA requires personnel to ensure that the frame or case of the welding machine (except with engine-driven machines) is properly grounded before it is used.

The OSHA standard specifies that conduits that contain electrical conductors must never be used for completing a work-lead circuit. Pipelines must not be used as a permanent part of a work-lead circuit. However, they may be used during extension, construction, or repair, but only when current is not being carried through threaded joints, flanged bolted joints, or caulked joints. Special precautions must be taken to avoid sparking at the connection of the work-lead cable.

Supply Connections and Conductors

If a welding machine does not have a disconnecting switch or a controller as an integral part, one must be provided at or near the machine. A disconnect switch with overload protection is required for each outlet that has to be connected to a portable welding machine.

The rated current-carrying capacity of the supply conductors for individual welding machines must not be less than the rated primary current of the welding machines.

Operations that involve many welders working on one structure may require both polarities in the DC welding process. Supply circuit limitations for AC welding may require allocation of machines among the phases of the supply circuit.

In a situation like that, no-load voltages between electrode holders should be twice as normal in DC or 1, 1.41, 1.73, or twice as normal on AC machines.

Operation and Maintenance

OSHA requires personnel to check all connections of the machines before initiating the operations to ensure that all requirements have been fulfilled. Work clamps that are magnetic must be cleared of all metal particles that might be sticking to them. Welders must spread out all coiled welding cables before use to prevent overheating and damage to insulation.

While the welding operations are being carried out, cables with splices within 10 feet (3 meters) of the holder must not be used. It is important that the welding electrode cables are never coiled around parts of the welders' bodies.

OSHA mandates the replacement of all cables that have damaged insulation or exposed bare conductors. Work and electrode cables must only be joined by using means that are particularly designed for that purpose.

Resistance Welding

The third type of welding process is resistance welding. Resistance welding utilizes pressure and heat that is generated in the pieces to be welded by resistance to an electric current.

When conducting resistance welding, all equipment should be installed by a qualified electrician. It must be ensured that all machines have disconnecting switches or circuit breakers that are located at or near the machine. Also, thermal protection switches must be provided for all ignitron tubes used in resistance welding equipment.

All personnel who are required to perform resistance welding must be properly trained and judged to be competent before they perform any tasks.

For all non-portable spot and seam welding machines, all external circuits must not operate on a voltage higher than 120 volts. All resistance welding equipment and control panels that involve voltages of over 550 volts must be properly insulated and shielded. OSHA requires all doors and access panels for the resistance welding machines and control panels to be kept locked and interlocked in order to prevent unauthorized persons from coming in contact with the live portions of the equipment.

Where there is a possibility of the operator's fingers being under the point of operation, all press welding machine operations shall be effectively guarded by the use of a device such as an electronic eye safety circuit, two hand controls, or protection similar to that prescribed for punch press operation. In all press welding operations, if the operator is required to have his or her fingers under the point of operation, he or she must be provided with effective protective equipment.

Wherever practical, a shield guard of safety glass or suitable fire-resistant plastic must be installed at the point of operation to avoid the hazard of flying sparks. Also, protective shields should be installed to prevent flying sparks from harming passing persons.

All foot switches that may be present on the machine must be guarded so that the machine does not get started accidentally. On special multi-spot welding machines, including 2-post and 4-post weld presses, there must be at least two safety emergency stop buttons.

OSHA requires all portable welding guns, transformers, and related equipment suspended from overhead structures to be outfitted with safety chains or cables. In case of failure of any component of the supporting system, these safety chains and cables must be capable of enduring the shock load.

All initiating switches, including retraction and dual schedule switches, located on the portable welding gun shall be equipped with suitable guards capable of preventing accidental initiation.

The outer casing of all portable welding transformers must be grounded. The OSHA standard requires all flash welding machines to comprise a protective hood to control flying flash.

Lesson Summary

For direct current machines, the voltage must not exceed the limit of 100 volts for both manual and automatic arc welding and cutting. OSHA requires all doors and access panels for the resistance welding machines and control panels to be kept locked and interlocked in order to prevent unauthorized persons from coming in contact with the live portions of the equipment.

Wherever practical, a shield guard of safety glass or suitable fire-resistant plastic must be installed at the point of operation to avoid the hazard of flying sparks. Also, protective shields should be installed to prevent flying sparks from harming passing persons.

Module 14: Hazardous Substances & Industrial Hygiene

Module Description

This module will introduce the contents of 29 CFR 1910 Subpart Z, which deals with toxic and hazardous substances.

This module will focus on the aspects of Subpart Z, which deal with keeping hazardous chemicals and materials from negatively affecting your health. This includes the use of personal protective equipment, decontamination procedures in the event of exposure to hazardous materials, and the procedures in place for conducting first aid and recording an incident of exposure in the workplace.

Module Learning Objectives

At the completion of this module, you should be able to:

- Describe how the body can be exposed to toxic substances.
- Discuss the methods by which you can detect a toxic substance.
- Name effects that toxic substances can have on the body.
- Illustrate how to reduce exposure to toxic substances.

Lesson 1: Exposure to Toxic Substances

Lesson Focus

This lesson focuses on the following topics:

- How Can the Body Be Exposed to Toxic Substances?
- Clues of Exposure to Toxic Substances
- Why is the Safety Data Sheet Needed?

How Can the Body Be Exposed to Toxic Substances?

Inhalation

For a toxic substance, inhalation is the most common form of exposure to the body. Inhaled materials can be deposited in the lungs and can have numerous harmful effects on the body. They may block the air sacs (alveoli) or may be further absorbed into the bloodstream, where they can cause harm elsewhere in the body. If the inhaled substance is an irritant, it may lead to nose and throat irritation, or may also lead to coughing and chest pains if it comes in contact with the bronchi.

Skin Contact

Some substances are caustic, meaning that they can chemically burn the skin; examples include strong acids and alkalis. Some substances are irritants, dissolving the oils produced by the skin, leaving it dry and cracked. This makes the skin more susceptible to absorption of toxic materials; if the skin is cracked or cut, substances can be absorbed and passed into the bloodstream more easily.

Eye Contact

Some substances can irritate or burn the eye. In some cases, the substance in contact with the eye may be absorbed into the bloodstream. Eyes are highly susceptible to damage and any contact or irritation should be taken seriously.

Ingestion

Conscious ingestion of a toxic substance is not common, though it may occur if the substance has been left on your hands or has contaminated food or drink.

Injection

While the unintentional injection of toxic substances into the body is uncommon in most workplaces, it can occur when a sharp object (e.g., needle) punctures the skin and injects a chemical directly into the bloodstream or allows the chemical to enter the body. In some workplaces, such as the medical care and veterinary industries, biological or chemical substances can be injected into the body by the accidental puncturing of the skin with a contaminated needle.

Odor

As a general rule, if you can smell the substance, then you are inhaling it. However, many substances are odorless, so odor cannot be used as a steadfast criterion in determining whether or not a chemical is being inhaled. An individual should not depend on smell to provide sufficient warning as to whether they have been exposed to a chemical. One individual's sense of smell will differ from another's—what one person may be able to smell, another may not.

More Information: Another issue to consider is olfactory fatigue. If you are exposed to a chemical that emits a strong odor for a long period of time, your nose may simply "shut-down," which effectively, reduces your ability to smell that particular odor.

Taste

Many chemicals will leave a taste in your mouth if they are ingested.

Clues of Exposure to Toxic Substances

If coworkers in your work area begin showing symptoms known to be from exposure to a toxic substance, then it is likely that you have been exposed yourself.

Settled Dust or Mist

If you find any dust that has settled on clothes or in your general surroundings, it is also likely that you have inhaled it.

Particles in Nose or Mucous

If after blowing your nose or sneezing, you find particles in your handkerchief, this indicates that you have inhaled minute particles of a substance. Unfortunately, many particles that are inhaled are too small to be seen.

Why is the Safety Data Sheet Needed?

An employer has the responsibility to inform, and employees have the right to be informed, of any hazardous substances that they may be exposed to in the workplace, as well as the damage their bodies may suffer as a consequence of that exposure. Employers are required to inform employees of the hazards of the chemicals they use and how to protect themselves in the workplace, including through the use of PPE. Employers are required to provide information concerning the effects to workers' bodies as a result of exposure to a hazardous chemical.

The Safety Data Sheet is one document that can assist employers in accomplishing this responsibility. Another is the inventory of hazardous chemicals, which provides a list of hazardous substances that are physically present in the given workplace.

The Safety Data Sheet (SDS) may provide the permissible exposure limit to a material an individual may experience without suffering any permanent damage; this is known as the "PEL." The SDS also states basic first-aid procedures that should be followed should exposure/overexposure occur, and common precautions in the handling of these dangerous materials.

In most circumstances, an SDS will provide emergency response procedures in the event of chemical spills.

More information: [Sample SDS](#)

Lesson Summary

The body can be exposed to toxic substances through inhalation, skin or eye contact, ingestion or injection. Clues to toxic exposure may include the presence of settled dust or mist, or odor or taste. But because exposure may take place without one's knowledge, employers are responsible for informing employees of what chemicals they may be exposed to before they are exposed to them, what the consequences of such exposure may be, as well as how to protect themselves from unnecessary exposure.

Lesson 2: Effects and Exposure

Lesson Focus

This lesson focuses on the following topics:

- The Effects of a Toxic Substance on the Body
- How to Reduce Exposure
- Exposure Measurement and Monitoring

The Effects of a Toxic Substance on the Body

Local Effect

When and where a toxic substance comes in contact with and causes damage to the body is known as its local effect. For example, a toxic substance comes into contact with skin and causes inflammation or irritation.

Systemic Effect

In the case of a systemic effect, the substance is absorbed by the body and enters the bloodstream and may eventually cause damage to internal organs. For example, if taken through inhalation, the substance enters the bloodstream by way of the lungs and eventually damages the liver.

More Information: A toxic substance may have local effects, systemic effects, or both. For example, a substance that irritates the respiratory tract (nose, throat, lungs) when inhaled, subsequently passes into the bloodstream and damages the kidneys. In some cases, the local effects may provide a warning that exposure to a toxic substance has occurred. If timely action is taken, it may prevent systemic damage from taking place.

Cancer

Some chemicals are known to induce cancer (the uncontrolled growth and spread of abnormal cells) in the body. Chemicals that cause cancer are known as carcinogens. There are many different proven carcinogens for humans and even more that cause cancer in test animals— those are considered to be potentially threatening to humans.

Trying to determine what substances cause cancer is difficult due to temporal constraints. Cancer is a chronic disease—meaning that its effects are only apparent after a body has been continually exposed to a substance over a period of time. The existence of cancer may only become apparent after a very long time, which makes it hard to determine which chemical, if any, was the cause, especially if the individual has been exposed to a number of chemicals over a certain period.

Although the exposing of a chemical to laboratory animals is considered to be a time-cost-efficient method of predicting the potential of a chemical to cause cancer in humans over a long period of time, the risk factor and precise correlation are difficult to determine.

Mutation

It has been discovered that some toxic substances can lead to changes in the genetic makeup of a person, even to the point of damaging the genes. This kind of genetic change is commonly known as mutation, and the substance that induces the genetic change is known as a mutagen. Mutation of genetic material will ultimately mean that the cells affected are unlikely to function or reproduce properly. Furthermore, cells reproduced from the mutated cell are also not likely to function properly.

More Information: Further tests may be carried out to determine whether a mutagen can cause cancer in the body.

The Reproductive System

Exposure to some chemicals can lead to the inability to bear children or may adversely affect unborn children.

Toxic substances can lead to infertility, disruption in the menstrual cycle, fetal damage, stillbirths, spontaneous abortions, and defects in born children - both at birth and in the future.

Substances that cause defects in children, stillbirths, or otherwise damage the fetus in a mother's womb are known as teratogens. Other chemicals that generally damage the fetus are known as fetotoxins. Furthermore, as a general rule, any chemicals that lead to health complication in a pregnant woman are likely to have the potential to damage the fetus. In the case of males, some chemicals can reduce sperm count, cause infertility, sterility, and a reduction in sex drive.

Reaction and Interaction

The existence of more than one chemical in a work environment can lead to a more complicated form of exposure. Chemicals can react together when they are exposed to each other, depending on the types of chemicals, temperatures, and pressure. These reactions may lead to the creation of other chemicals that can affect the body in a different manner than individual chemicals would. As such, employees will need to be aware of the chemicals that they are exposed to and the possible reactions and interactions that may exist in the case of two or more chemicals.

The individual chemicals in a reaction may be harmless to the individual, but when they come into contact with one or more chemicals, the result may be toxic. Though common reactions can be accounted for and documented on the SDS, some cannot be anticipated and as such, will not be included. The interaction of more than one chemical may lead to synergism. That is, the interaction of two or more chemicals will lead to an effect that is greater than the sum of their individual effects.

More Information: Another effect of interaction is called potentiation. This means that the existence of one chemical may not lead to any harmful effects, but the existence of a second chemical can enhance its potential to harm the individual.

Sensitivity

Due to the diverse nature of the human population, particularly in the case of physical and biological traits, the reaction of one individual to a chemical as opposed to another may differ. Some can and will tend to be more sensitive to the toxic effects of a substance than others. As a result, they may be susceptible to the toxic effects of a substance at lower levels of exposure, while others will be more tolerant to the toxicity of a substance, meaning that they will be able to endure longer exposure periods or larger amounts of the substance than others.

How to Reduce Exposure

Knowledge

All employees expected to work with toxic materials must be sufficiently informed about the toxicity levels of the substance, the duration of time that they may be safely exposed to it, and what would constitute a harmful dose.

Therefore, it remains the responsibility of the employer to fully train every effected employee as to the hazards they will encounter and how to protect themselves.

Engineering Controls

These procedures constitute limiting the level of toxicity and level of hazard at the source. They include the following measures:

- **Substitution:** Using a less hazardous substance is a good measure for limiting exposure to a toxic substance. This, however, may not be a solution if no substitution exists, or if the substitution results in lower or undesirable performance.
- **Process or equipment enclosure:** This may isolate the toxic substance, limiting its exposure to the employee and/or ensuring that those who do have access have the necessary clearance and, as such, are well informed of the effects and necessary precautions.
- **Local exhaust ventilation:** This is a ventilation that is placed over the substance so as to capture contaminated air and/or particles before they are able to disperse.

- **General ventilation:** This means continuous ventilation of an area to ensure that the toxicity levels of a substance are always kept below a hazardous level. However, if the ventilation is not equalized over the whole enclosed area, there may be pockets of high concentration that will be hazardous.

Personal Protective Equipment

Personal protective equipment (PPE) may be an acceptable alternative when engineering controls and/or administrative controls are insufficient. PPE can include the use of protective goggles to protect the eyes, as well as gloves and aprons to protect the hands and body. In some cases, the nature of the substance may warrant the use of a respirator to prevent the inhalation of toxic materials.

In some cases, the use of creams to coat the face and skin may be used to reduce exposure, particularly where the nature of the work prevents the use of gloves or facemasks.

Exposure Measurement and Monitoring

Exposure Limits

Exposure limits are set by OSHA to control the permissible levels of exposure to a substance without being adversely affected. Exposure limits are usually expressed in terms of maximum concentration of a substance in the air over a prescribed period of time.

- Although substances that have a hazardous effect in the short term may be easy to observe, measure, and document, that may not be the case for substances that have an effect on the body in the long run. Chronic exposure to a toxic substance can sometimes be observed only after the individual has been exposed to it for a long period of time (often measured in years, sometimes decades).
- While exposure limits are set to protect individuals, there may be a number of individuals who are particularly sensitive to a certain substance and may have

reactions to certain hazards at levels below what is normally acceptable. Similarly, the nature of the work may lead to increased exposure to a substance than is the case in normal circumstances. For example, heavy physical work may lead workers to breathe in more air at a greater frequency, meaning that they may inhale more of a substance than is usual.

- Exposure limits may not take into account the possible interactions of two or more chemicals. The existence of one substance alone may pose no threat, but when mixed with another, the resultant combination may be harmful to the body.
- As previously mentioned, exposure limits are often expressed in terms of concentration in the air over prescribed periods of time; this does not take into account the possibility of exposure through skin or ingestion.

The level of inhalation exposure that an employee is subject to in the workplace can often be measured by what is called air or environmental monitoring or sampling, which may be carried out by an industrial hygienist or safety professional. Samples are collected from within an individual's breathing zone (around the nose and mouth) so as to provide more realistic results. Unfortunately, if exposure to the toxic substance did not occur through inhalation but through the skin or ingestion, air sampling will provide poor and inaccurate results that do not reflect the true level of toxicity in the workplace.

The use of biological sampling is another alternative. Biological sampling is carried out by way of stool, blood, and urine tests. When the existence of certain toxic substances in the air rises above a certain level, by law biological sampling is required (i.e., Cadmium).

Lesson Summary

Though common reactions can be accounted for and documented on the SDS, some are more difficult to anticipate and as such, may not be included.

All employees expected to work with toxic materials must be sufficiently informed about the toxicity levels of the substance, the duration of time that they may be safely exposed to it, and what would constitute a harmful dose.

Module 15: Bloodborne Pathogens

Module Description

This training module is designed to provide a basic understanding of bloodborne pathogens, common modes of their transmission, methods of prevention, and other pertinent information such as what requirements of the OSHA standard should be met in the handling and proper disposal of such materials. Employers are to implement written Exposure Control Programs to address what duties employees are to perform on a daily basis to keep their work areas and their persons safe.

Module Learning Objectives

At the completion of this module, you should be able to:

- Summarize what is meant by bloodborne pathogens.
- Identify the types of diseases that can be transmitted through the bloodborne pathogens.
- Name the symptoms of bloodborne diseases.
- Explain how bloodborne pathogens may be transmitted in the workplace.
- Discuss prevention measures and what to do if exposed to bloodborne pathogens.

Lesson 1: Introduction to Bloodborne Pathogens

Lesson Focus

This lesson focuses on the following topics:

- Bloodborne Pathogens
- Hepatitis B
- Human Immunodeficiency Virus
- Modes of Transmission
- Vehicles of Transmission

Bloodborne Pathogens

Bloodborne pathogens are pathogenic microorganisms (e.g. viruses, bacteria, or parasites) that may be present in human blood, tissue, or organs and which can cause disease in humans. There are several bloodborne diseases that you could be exposed to on the job. The most significant include human immunodeficiency virus (HIV) and the hepatitis B virus (HBV).

Hepatitis B

Hepatitis means "inflammation of the liver." HBV is the most common infectious bloodborne hazard facing healthcare workers. It infects approximately 3,000 healthcare workers a year.

Blood and blood components are the main vehicles for occupational transmission of HBV.

According to the CDC, when signs and symptoms are present, they can include:

- Fever
- Fatigue
- Loss of appetite
- Nausea

- Vomiting
- Abdominal pain
- Dark urine
- Clay-colored bowel movements
- Joint pain
- Jaundice

Persons with chronic HBV infection might be asymptomatic, have no evidence of liver disease, or have a spectrum of diseases ranging from chronic hepatitis to cirrhosis or hepatocellular carcinoma (a type of liver cancer).

Although the infected person may not feel any symptoms at all, Hepatitis B can cause severe damage to the liver, ultimately resulting in cirrhosis (scarring of the liver) and possibly death. HBV may also spread to others via blood and body fluids, but fortunately there is a vaccine available to prevent Hepatitis B.

Human Immunodeficiency Virus

The human immunodeficiency virus (HIV) attacks the body's immune system, and may cause the disease known as AIDS, or Acquired Immune Deficiency Syndrome.

Currently, there is no vaccine or cure for HIV.

Needle-sticks are a cause of occupational HIV transmission.

A person with HIV may:

- Carry the virus without developing symptoms for several years.
- Develop AIDS or AIDS-related symptoms, including neurological problems, cancer, and other opportunistic infections.
- Suffer from flu-like symptoms, fever, diarrhea, and fatigue.
- Transmit the disease to others via intimate contact with blood or body fluids.

Modes of Transmission

Bloodborne pathogens such as HBV and HIV can be transmitted through contact with infected human blood and other potentially infectious body fluids such as those that follow:

- Semen
- Vaginal secretions
- Cerebrospinal fluid
- Synovial fluid
- Pleural fluid
- Peritoneal fluid
- Amniotic fluid
- Saliva (in dental procedures)
- Anybody fluid that is visibly contaminated with blood

It is important to know the ways exposure and transmission are most likely to occur in your particular situation, be it through providing first aid to a fellow student, handling blood samples in the laboratory, or cleaning up blood spilled in a hallway.

Vehicles of Transmission

Bloodborne pathogens may enter the body and infect a person through a variety of means including those that follow:

- Needles
- Scalpels
- Broken glass, or anything that can pierce skin
- Open cuts, acne, and mucous membranes
- Indirect transmission such as touching a contaminated object with your mouth, eyes, nose, or open skin

Accidental Puncture from Contaminated Needles or other Sharps

In most healthcare or laboratory situations, transmission is most likely to occur because of:

- Accidental puncture from contaminated needles, broken glass, or other sharps;
- Contact between broken or damaged skin and infected body fluids; or
- Contact between mucous membranes and infected body fluids.

Example: If someone infected with HBV cuts his or her finger on a piece of glass, and then you cut yourself on the now-infected piece, it is possible that you could contract the disease. Anytime there is blood-to-blood contact with infected blood or body fluids, there is the potential for transmission.

Unbroken skin forms an impervious barrier against bloodborne pathogens. However, infected blood can enter your system through any of the following means:

- Open sores
- Cuts
- Abrasions
- Acne
- Any sort of damaged or broken skin, such as sunburn or blisters

Bloodborne pathogens may also be transmitted through the mucous membranes of the eyes, nose, and mouth. For example, a splash of contaminated blood to your eye, nose, or mouth could result in transmission.

Lesson Summary

Bloodborne pathogens such as HBV and HIV can be transmitted through contact with infected human blood and other potentially infectious body fluids.

HBV is the most common infectious bloodborne hazard facing health care workers today, and blood and blood components are the main vehicles for occupational transmission of HBV.

Anytime there is blood-to-blood contact with infected blood or body fluids, there is the potential for transmission. For this reason, it is important to know the ways exposure and transmission are most likely to occur in your particular situation.

Lesson 2: Prevention

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- Universal Precautions
- Reducing the Risk
- Signs, Labels, and Color Coding
- Exposure Incident Procedures
- Emergency Procedures

Introduction

Employers having employees with reasonably anticipated skin, eye, mucous membrane, or parenteral contact with blood or other potentially infectious materials that may result from the performance of an employee's duties shall have a written Exposure Control Program. Examples of workers who are at risk include:

- Physicians, nurses, and emergency room personnel
- Orderlies, housekeeping personnel, and laundry workers
- Dentists and other dental workers
- Laboratory and blood bank technologists and technicians
- Medical examiners
- Morticians
- Law enforcement personnel
- Firefighters
- Paramedics and emergency medical technicians
- Anyone providing first-response medical care
- Medical waste treatment employees
- Home healthcare workers

An Exposure Control Program shall be updated annually and whenever necessary to reflect new/modified tasks, etc. Employers are required to solicit employee input to the plan.

Exposure determination includes:

- A list of all job classifications in which all employees in those job classifications have occupational exposure.
- A list of job classifications in which some employees have occupational exposure.
- A list of all tasks and procedures or groups of closely related task and procedures in which occupational exposure may occur and that are performed by employees in job classifications.

The exposure determination shall be made without regard to the use of Personal Protective Equipment (PPE).

The employer must develop a schedule and method of implementation for:

- Methods of compliance
- HIV and HBV research laboratories and production facilities
- Hepatitis B vaccination and post-exposure evaluation and follow-up
- Communication of hazards to employees
- Recordkeeping, of this standard procedure for the evaluation of circumstances surrounding exposure incidents

As well, the employer shall ensure that a copy of the Exposure Control Program is accessible to employees.

Universal Precautions

HIV and HBV infect people of all ages, socioeconomic classes, races, and backgrounds. Therefore, you cannot easily identify every patient who may transmit infection.

Universal precautions resolve this uncertainty by requiring you to treat all human blood and bodily fluids as if they were known to be infected.

Reducing the Risk

There are five major tactics to help reduce your risk of exposure and infection. They are as follows:

1. Work Practices and Personal Hygiene
2. Personal Protective Equipment
3. Engineering Controls
4. Housekeeping
5. Hepatitis B Vaccine

All of these are encompassed in an employer's responsibility to create and maintain an Exposure Control Plan for any facility where there exists a reasonable expectation that one or more employee will experience an occupational exposure incident by virtue of their work responsibilities.

Work Practices

Needles and sharps potentially contaminated should not be bent, recapped, or removed unless it can be demonstrated that no alternative is feasible or that these actions are required for a specific procedure.

More guidelines for handling sharps include:

- Do not overfill the sharps containers. Never fill beyond indicated line on the container.
- Do not ever reach into any container used for sharps.
- Place leaking or punctured sharps containers inside a secondary container to prevent leakage. Also, notify the personnel responsible for handling the containers that the container has been punctured.
- Always use a utensil (tongs, dustpan, etc.) to pick up contaminated broken glass, needles or other sharps.
- Assume all used needles and other sharps are contaminated.

More Work Practices

If you come in contact with infectious materials, wash the exposed area immediately in order to lessen the chance of becoming infected.

In laboratory and patient care areas you should wash your hands whenever you do the following:

- Remove your gloves
- Come in contact with blood or other bodily fluids
- Change workstations
- Enter a "clean" area

Additional controls should be followed to protect yourself from infection, such as:

- Always minimize splashing, spraying, and spattering of any material that may contain a human body fluid.
- Do not eat, drink, smoke, apply cosmetics, or handle contact lenses where you may potentially be exposed to infectious materials.

- Avoid lubricants that are not compatible with the gloves and other PPE in use. Improper lubricants can severely degrade the protection intended to be provided by the equipment. Appropriate lubricants and moisturizers should only be used in an appropriate manner.
- Never pipette or suction by mouth!
- Don't store food and drinks where blood and other infectious materials may be present.

Personal Protective Equipment

Personal Protective Equipment (PPE) helps protect you from contact with potentially infectious materials.

Gloves

Always use gloves before touching or if likely to come in contact with blood, bodily fluids, non-intact skin, mucous membranes, or if performing venipuncture.

Masks/Eye Protection

Always use masks/eye protection equipment when you perform procedures likely to generate splashes of blood or other bodily fluids.

Gowns/Coats

Use gowns/coats for procedures likely to generate splashes of blood or other bodily fluids.

Gloves Removal

As previously mentioned, gloves are very important when it comes to protecting yourself. The CDC has developed guidelines on the accepted method for the removal of contaminated and potentially contaminated gloves.

These guidelines can be found [here](#).

Removed gloves should be properly disposed of immediately and the hands thoroughly washed.

Engineering Controls

To help eliminate hazards, employers provide engineering controls. Some examples are as follows:

- Self-sheathing needles
- Sharps disposal containers
- Biohazard waste bags
- Biological safety hoods
- Autoclaves

Good Housekeeping

Good housekeeping means following established protocol and applying knowledge gained during related training to keep your work areas clean. This will help protect you and your colleagues.

General rules include:

- Clean and decontaminate your workspace at the end of each shift or more frequently as necessary, and as dictated in the exposure control plan.
- Replace protective coverings on equipment.
- Do not pick up broken glass with your hands.
- Place contaminated sharps into the proper leak-proof containers.

- Read and respect all biohazard labels.
- Always dispose of hazardous materials in the proper RED (NOT orange) container or bag.

Note: All of these activities must be included in an employer's Exposure Control Plan (ECP).

HBV Vaccination

If your occupation has the potential for HBV exposure, your employer will offer the hepatitis B vaccine at no cost to you.

The vaccine is administered in three injections: one at employment orientation, one at three months, and the final at six months—or as directed by your physician. A post vaccination serology test should be scheduled to check antibody levels. All three injections should be received for the vaccine to be effective. Today's vaccines are safe and are 85–97% effective.

Participation in a prescreening program is not a prerequisite for receiving the hepatitis B vaccination.

If the employee initially declines the hepatitis B vaccination but at a later date while still covered under the standard decides to accept the vaccination, the employer shall make available the hepatitis B vaccination at that time.

The employer shall assure that employees who decline to accept the hepatitis B vaccination offered by the employer sign the statement declining the vaccination. There is a specific form for this required by OSHA.

Signs, Labels, and Color Coding

Warning labels must be affixed to containers of regulated waste, refrigerators, and freezers containing blood or other potentially infectious material and other containers used to store, transport, or ship blood or other potentially infectious materials. These labels are fluorescent orange, or orange-red with lettering and symbols in contrasting colors.

Bags used to dispose of regulated waste must be red and employees properly trained on the significance of all color coding used in dealing with bloodborne pathogens.

Note: Labels should display the universal biohazard symbol.

Exposure Incident Procedures

An exposure incident is a spill, splash, needlestick, ingestion, or accident, in which you have direct and unprotected contact with human blood, fluids, or tissues.

In the event of an exposure incident, an employee should:

- Contact emergency medical personnel as necessary.
- Immediately initiate first aid at the work site if trained to do so.
- Notify his or her supervisor or someone else who can notify his or her supervisor.
- Seek further medical treatment as necessary.
- Ensure the incident is reported to the employer.

Emergency Procedures

In the case of an emergency, each employer will have as part of their Emergency Action Plan, emergency numbers, locations, and procedures for all personnel.

Post-exposure follow-up:

- Document routes of exposure and how exposure occurred.

- Record injuries from contaminated sharps in a sharps injury log, if required.
- Obtain consent from the source individual and the exposed employee, and test blood as soon as possible after the exposure incident.
- Provide risk counseling and offer post-exposure protective treatment for the disease when medically indicated in accordance with current U.S. Public Health Service guidelines.
- Provide a written opinion of findings to the employer and copy to the employee within 15 days of the evaluation.

Lesson Summary

Employers have a responsibility to create and maintain Exposure Control Plans for any facility where there exists a reasonable expectation that one or more employees will experience an occupational exposure incident by virtue of their work responsibilities. If you should come in contact with infectious materials, wash the exposed area immediately in order to lessen the chance of becoming infected.

Module 16: Ergonomics

Module Description

Employees who work in non-office environments are routinely required to carry out tasks that involve movement and physical exertion. These forceful exertions associated with such tasks may lead to fatigue, musculoskeletal disorders, and other serious injuries.

This module is designed to help employees identify work-related problems and learn to apply the principles of ergonomics in order to make their jobs less physically demanding, thereby increasing their overall efficiency.

Module Learning Objectives

At the completion of this module, you should be able to:

- Discuss what is meant by ergonomics.
- Name the factors that may contribute to problems in a non-office work environment.
- Identify the causes of musculoskeletal disorders.
- Apply effective ergonomic principles at your workplace.

Lesson 1: Ergonomics in the Workplace

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- Musculoskeletal Disorders (MSDs)
- Risk Factors

Introduction

Many workers across the United States have to carry out physically demanding tasks each day while on the job. If these tasks are not carried out in proper postures, they may cause fatigue or discomfort.

Musculoskeletal Disorders (MSDs)

Carrying out such tasks for prolonged periods may cause severe damage to muscles, ligaments, tendons, blood vessels, and nerves. Such injuries include musculoskeletal disorders (MSDs). MSDs not only affect individual workers but also increase the cost of business in terms of higher workers' compensation premiums, increased employee turnover, absenteeism, and decreased efficiency. Overall productivity can also be greatly affected.

In order to reduce the costs associated with MSDs and avoid the problem of work-related injuries, employers must implement the principles of ergonomics in their workplaces.

Why Are MSDs a Problem?

Many reasons exist for considering MSDs a problem, including the following:

- MSDs are among the most prevalent lost-time injuries and illnesses in almost every industry [Bureau of Labor Statistics 1995, 1996; National Safety Council 1995; Tanaka et al., 1995].
- MSDs, specifically those involving the back, are among the most costly occupational problems [National Safety Council 1995; Webster and Snook 1994; Guo et al., 1995; Frymoyer and Cats-Baril 1991].
- Job activities that may cause MSDs span diverse workplaces and job operations ([see Table 1; see also Tray 1-A of the Toolbox](#)).
- MSDs may cause a great deal of pain and suffering among afflicted workers.
- MSDs may decrease productivity and the quality of products and services. Workers experiencing aches and pains on the job may not be able to do quality work.
- Because musculoskeletal disorders have been associated with non-work activities (e.g., sports) and medical conditions (e.g., renal disease, rheumatoid arthritis), it is difficult to determine the proportion due solely to occupation. For example, in the general population, non-occupational causes of low back pain may be more common than workplace causes [Liira et al., 1996]. However, even in these cases, the musculoskeletal disorders may be aggravated by workplace factors.

Medical Conditions Associated with Musculoskeletal Symptoms

Conditions associated with MSDs include:

- Muscle strain or fatigue
- Tendonitis
- Epicondylitis or "tennis elbow"
- Carpal tunnel syndrome
- Other "pinched" nerve entrapment syndrome

- Nerve injury/irritation from external compression
- Arthritis and other rheumatological disorders

Ergonomics, also known as human engineering, is the practice of designing machines, products, and places to better accommodate people. The principles of ergonomics are geared towards adapting the design and engineering of products and workplaces to people's sizes and shapes, physical strengths and limitations, biological needs, ability to handle information and make decisions, as well as their capacities for dealing with such psychological factors as isolation and stress.

Risk Factors

There are certain aspects of tasks that can increase the risk of fatigue, musculoskeletal disorder (MSD) symptoms and injuries, or other types of problems. These risk factors can be found in a variety of tasks that an employee is required to carry out and can be divided into two categories: physical factors and environmental factors.

Physical factors

These are fundamentally the interactions between the work area and the worker, and include:

- Awkward postures
- Forceful exertions
- Repetitive tasks or motions
- Pressure points
- Recovery time
- Vibration

Awkward Postures

The position of the body while performing a task is known as posture. The muscle groups used while performing a task are affected by the worker's posture. An awkward position can make the task more physically demanding by overexerting small muscle groups and not using larger muscle groups. This can increase the likelihood of poor blood-flow, which can lead to fatigue and injury.

Some examples of awkward postures include repeated or prolonged reaching, bending, twisting, kneeling, holding fixed positions, or squatting. Several areas of the body can be affected due to these postures including shoulders, arms, wrists, hands, knees, neck, and back. Awkward postures are often caused by poorly designed work areas, equipment, and tools along with poor work practices.

Postures to Avoid:

- Prolonged or repetitive flexion or extension of the wrist
- Prolonged or repetitive bending at the waist
- Prolonged standing or sitting without shifting your position
- Suspending an outstretched arm for extended periods of time
- Holding or turning your head consistently to one side
- Any posture that is held repeatedly or for a prolonged time

Motions to Avoid:

- Repeated motion without periods of rest
- Repeated motion with little or no variation
- Repeated motions done with force
- Resting or compressing a body part on or against a surface
- Lifting heavy objects away from the body
- Frequent reaching or working above shoulder height

The following types of employee behavior may indicate the presence of ergonomics-related problems:

- Employees shaking arms and hands or rolling shoulders due to discomfort.
- Employees voluntarily modifying workstations and equipment to increase comfort.
- Employees bringing in ergonomic products to the worksite (such as wrist braces).

Forceful Exertions

The amount of muscular effort required to perform a task is called force. Exerting more force than a body can sustain can cause severe damage to muscles and ligaments. The amount of force required for tools or machinery depends upon various factors, including those that follow:

- Load weight, shape, and bulkiness
- Grip type
- Amount of pressure required to accelerate, or decelerate, the load

The degree of risk generally increases with increasing force. Various parts of the body can be affected due to high force, including shoulders, neck, lower back, forearm, wrist, and hands.

Note: The risk of injury due to forceful exertions may increase if other risk factors such as awkward posture, repetitive tasks, and extended duration are also present.

Repetitive Tasks or Motions

Repetitive tasks require workers to perform the same task over and over again, using the same muscles, ligaments, and tendons. Repetitive motion may cause severe damage, injury, and discomfort to the worker. The risk of injury increases if the worker fails to take any breaks to relax his or her muscles.

Pressure Points

Exerting pressure on different parts of the body by pressing them against hard or sharp surfaces may cause injury. Some body parts are at a greater risk as nerves, blood vessels, and tendons are present just under the skin in certain areas. Fingers, wrists, palms, elbows, forearms, and knees are examples of such body parts.

Recovery Time

Recovery time is the amount of time allocated to rest the muscles and tendons in any strained part of the body. It is very important for workers to take pauses between tasks that require forceful exertions. These breaks not only provide relief to workers but also enhance their performance. Employers must assess the duration of breaks according to the workload along with the risk factors present.

Vibration

Exposure to continuous vibration can cause damage if uncontrolled. Exposure to vibration can occur with the use of vibrating tools such as sanders, chippers, chain saws, drills, grinders, routers, and impact guns. Vibrations can cause fatigue, numbness and pain in the exposed area. It may also cause decreased sensitivity to touch and increased sensitivity to cold.

Environmental Factors

Heat Stress

Heat stress is the amount of heat that a body is exposed to while performing a task. Heat stress can be attributed to the worker's environment and also his or her own internal metabolism.

Exposure to excessive heat can cause various disorders, including heat exhaustion, heat cramps and heat stroke. The symptoms of heat stress may include headaches, thirst, nausea, muscle cramps, dizziness, and weakness.

Due to the severity of the consequences of heat stress, employees must regularly monitor the workplace and take appropriate preventive measures.

Cold Stress

Cold stress occurs when a worker is exposed to cold temperatures. Cold stress results in the decrease of the worker's body temperature and may cause shivering, unconsciousness, pain, and inadequate circulation of the blood. Cold stress may also cause the worker to lose the ability to grasp due to the decrease in body strength. Cold temperatures combined with the risk factors above may increase the risk of musculoskeletal disorders.

Noise

Continuous sound at levels above 80-85 dB in the workplace can cause severe damage to a worker's hearing. Continued exposure to high noise levels may also result in impaired hearing or permanent deafness, tinnitus, or speech misperception. Furthermore, high levels of noise may also affect the worker's ability to concentrate on his or her work.

Lighting

Improper lighting in the workplace may cause eye fatigue and may result in headaches and a loss of focus. A worker's ability to perform tasks efficiently depends greatly on the proper lighting of the work area.

Lesson Summary

Repetitive tasks require workers to perform the same task over and over again, using the same muscles, ligaments, and tendons. Repetitive motion may cause severe damage, injury, and discomfort to the worker. The risk of injury increases if the worker fails to take breaks to relax his or her muscles.

Recovery time is the amount of time allocated to rest the muscles and tendons in any strained body part. It is very important for workers to take adequate pauses between tasks that require exertions. These breaks not only provide relief to workers but also enhance their performance.

High or continuous vibrations can cause fatigue, numbness, and pain in the exposed area. It may also cause decreased sensitivity to touch and increased sensitivity to cold. To prevent the consequences of heat stress, employers must regularly monitor the workplace and take the appropriate measures to ensure that all employees are adequately protected. In short, in order to reduce the costs associated with musculoskeletal disorders and to avoid the problem of work-related injuries, employers must implement principles of ergonomics in their workplaces.

Lesson 2: Improving the Workplace

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- Engineering Improvements
- Administrative Improvements
- Use of Protective Equipment
- Training

Introduction

Employers can adopt various ergonomic improvements in their workplaces in order to improve the working environment for their employees. Ergonomic improvements can be divided into three categories:

- Engineering Controls
- Administrative Controls
- Use of Protective Equipment

Engineering Controls

Employees can make engineering controls in their workplaces by redesigning, rearranging, modifying or replacing tools, equipment, workstations, products, or actions that increase the risks of injury. Implementing effective engineering controls can greatly reduce the risk factors. The engineering controls that follow are recommended.

Install work tables with work surfaces that can be raised or lowered according to the employee's body size and position. This can reduce bending, reaching, and awkward postures that can contribute to body damage.

Cutout work surfaces may be appropriate to allow employees to adjust their distance from the work table. This can help reduce awkward postures.

By allowing employees to reposition their work, their bending and reaching efforts can be reduced.

Modifying the work surface according to the task can reduce the effort needed to complete the task. For example, to deliver packages from one area to another, workers can slide and roll the packages on the surface rather than lifting and carrying them manually.

Ladders, scaffolds, steps or work platforms must be provided to employees who are required to reach a surface high off the ground to retrieve packages or containers.

If employees are required to place packages and containers on surfaces high off the ground, certain mechanical lifting devices can be used to lift them. This can reduce stress to shoulders, neck, back, and other body parts. This can also reduce force, repetition, and awkward postures in tasks related to lifting or handling heavy objects.

Certain jobs, such as loading and unloading packages from delivery trucks, may require workers to bend their torsos repeatedly. This can again increase the risk of damage to a variety of body parts. This type of operation can be improved significantly by implementing lifting devices, altering the work site, and adopting a wide range of ergonomic solutions.

Employers should provide adjustable equipment that allows employees to accomplish tasks in a comfortable, upright working posture.

All materials, products, and tools that have to be used frequently must be stored in a place that can be accessed easily without requiring the worker to reach high or adopt awkward postures.

Materials should be transported around the workplace using mechanical aids when possible and appropriate. This may reduce the force required, repetition of motions, and awkward postures, while increasing workers' efficiency.

Unnecessary repetitive reaching, twisting, bending, and forceful exertions can be avoided by properly organizing the equipment and materials stored by grouping stored items by container size or shape.

Installing proper lighting systems in the workplace, including all storage facilities, can help reduce eye strains and headaches.

By utilizing good design and carrying out proper maintenance of all machinery and equipment, employees can ensure pressure points on the hands and wrists, awkward postures, and forceful exertions are minimized or avoided.

Exposure to vibration may be reduced by routine maintenance of vibrating equipment, covering handles with vibration-dampening wraps, operating the tool from a distance when possible, using vibration-dampening gloves, using alternate tools that produce less vibration, utilizing vibration isolators for workers who are seated, employing cushioned floor mats for tasks that have to be carried out while standing, or using vibrating tools at low speeds.

Administrative Controls

Administrative controls involve developing work practices and methods that best protect the worker. These are often focused towards devising and implementing new practices and policies in order to allow employees to carry out their jobs effectively and efficiently and avoid any on-the-job injuries, illnesses, and accidents. Administrative controls rely on communication and training, as well as feedback from management and employees on the effectiveness of the controls. Administrative controls may include:

- Job Rotation
- Adjusting Work Schedules and Work Pace
- Allowing More Frequent Breaks
- Modifying Work Practices
- Regular Housekeeping and Maintenance
- Encouraging Regular Exercise

Job Rotation

Adopting a job rotation system may be one effective measure to reduce damage caused to employees by using the same muscle groups every day. With a properly designed and implemented system, employees are rotated through different jobs, thus increasing job and muscle-use variety. Another system through which employees can increase job variety is through job enlargement. Through this system, employers combine two or more jobs or add different tasks to an existing job.

These systems aim to prevent overuse and overexertion of muscles and body parts, by reducing the amount of repetition, altering the pace of work, reducing the physical exertion required, and controlling visual and mental demands.

Adjusting Work Schedules and Work Pace

Employers must be careful not to assign too heavy of a workload to employees. They also must limit the amount of time that an employee spends performing a particularly challenging job in awkward positions even when physical improvements have been incorporated.

Allowing More Frequent Breaks

Breaking work into smaller tasks allows employees to take adequate breaks between them. These breaks may help employees relax their muscles, thus preventing fatigue and injury.

Modifying Work Practices

Supervisors and managers should regularly observe how workers perform their jobs. When employees perform all jobs while in a neutral posture, the body is less susceptible to injury. Employees may be able to adopt this posture by sitting or standing upright and not bending any joints into extreme positions; they should keep their necks, backs, arms,

and wrists in a neutral position. Supervisors should encourage employees to work in a comfortable position and shift their positions or stretch often.

Other work practices:

- Minimize distances for carrying, pushing, and pulling
- Manage an equal amount of weight in each hand
- Avoid unnecessarily twisting of the body
- Use smooth and even motions, avoiding jerking
- Utilize legs to accomplish tasks rather than using the upper body or back
- Ensure that all paths are free from obstacles and even-surfaced
- Organize tasks to provide a gradual increase in the amount of force required
- Ensure shoes worn are slip-resistant.

Regular Housekeeping and Maintenance

Employers must devise a system to carry out regular housekeeping and maintenance of workspaces, equipment, machinery, and tools. There should be no cluttering in the workspace, as clutter can force employees to reach, bend, or twist their bodies while handling different objects. Additionally, employers must ensure that workspaces comply with the following points:

- All floor surfaces must be kept dry and free of any obstacles when possible. This can minimize hazards associated with slipping or tripping in the work area. Problems related to overexertion can often be minimized by carrying out regular maintenance of all tools and equipment.
- Ensure that handles and padding on vibrating tools are well maintained to help reduce vibration and awkward postures while tasks are being performed.
- All moving or mechanical parts on carts and pulleys are properly lubricated and maintained so as to reduce the amount of force required to move them.

Encouraging Regular Exercise

Regular exercise is very important to one's well-being. It not only keeps the body fit, it reduces the risk of injury. Individuals who are in good physical condition are more productive and sustain fewer injuries.

Employers may encourage their workers to increase their energy levels, coordination, and alertness by exercising regularly. Regular exercise can also increase the efficiency of their joints and improve blood circulation.

Some organizations allow and encourage employees to warm up and engage in proper stretching before beginning work and while taking a break from work.

Use of Protective Equipment

Personal Protective Equipment (PPE) includes all protective equipment, such as gloves, footwear, knee and elbow pads, eye protection and other equipment that employees wear according to the type of task they are involved in.

Gloves

Properly selected gloves help to protect hands from sustaining injuries, improve grip, and avoid contact with chemicals. However, if gloves do not fit properly or are not made of the proper materials, they can restrict hand movement and make it harder for employees to grip things.

Footwear

Choosing the proper footwear according to the nature of the job can greatly reduce the risk of slipping. Some soles are designed to reduce fatigue for employees who are required to stand for long hours while performing a task.

Knee and Elbow Pads

Knee and elbow pads can protect body parts that are pressed against hard or sharp surfaces. These aim to minimize the risk of negatively affecting pressure points until proper engineering improvements can be made.

Back Belts

OSHA does not recognize back belts as effective engineering controls to prevent back injury. While they may be accepted by individual workers because they feel as if they provide additional support, the effectiveness of back belts in the prevention of low back injuries has not been proven in the work environment.

OSHA's preferred approach to prevention of injuries and illnesses, including back injuries, is to eliminate the hazardous conditions in the workplace, primarily through engineering controls.

Training

Training is also an important element of the ergonomics process. Training ensures that employees are informed about ergonomic concerns in the workplace and ways to minimize the risk of injury. Training is best provided by individuals who have experience with ergonomic issues in their workplace. Training should be provided annually in a manner and language that all employees can understand.

Training prepares employees for active participation in the ergonomics process, including identifying potential problems, implementing solutions, and evaluating the process. Effective training includes:

- Proper use of equipment, tools, and machine controls
- Good work practices, including proper lifting techniques
- Awareness of work tasks that may lead to pain or injury
- Recognition of MSDs and their early indications
- Addressing early indications of MSDs before serious injury develops

- Procedures for reporting work-related injuries and illnesses

Employees will benefit from orientation and hands-on training received before starting tasks with potential ergonomic risk factors. Employees should also be notified of workplace changes, instructed on using new equipment, and notified of new work procedures.

An effective ergonomics program involves training and education. In order to thoroughly implement the principles of ergonomics, all employees must be trained and provided with guidance about how to use new tools, machinery, equipment, and proper work procedures. An effective training program includes a mix of theoretical and practical ways in which employees can develop their skills to work safely.

An ergonomics training program must include:

- All employees who are exposed to different risk factors
- Supervisors
- Managers
- Appropriate engineers and maintenance personnel

Qualified persons who have a thorough knowledge about the principles of ergonomics greatly improve the quality of training. Additionally, trainers must thoroughly familiarize themselves with the workplace before devising a program.

All supervisors who have to administer an ergonomics program should be provided with special training focusing on how to effectively incorporate sound ergonomic principles and practices into the workplace.

Trainers must consider communication levels before devising the training program. They should adopt a language and style that is easy for all employees to understand and relate to.

Apart from ergonomic principles, the training must also include the risk potential of damage to the body and injuries that can result from the failure to adopt sound ergonomic principles and practices.

Employee training must consist of both general and specific job-related skill sets.

General Training

Employers must provide formal training to all employees who could be exposed to ergonomic hazards to inform them about the hazards associated with their jobs and the tools, machinery, and equipment they use. Information that must be included in the training includes specific risk factors, their causes, recognizing and reporting symptoms, and the prevention of these occurrences.

Job Specific Training

All new employees and those assigned new tasks must be made aware of the specific risks associated with a particular job before they start their work. A practical demonstration should be arranged in order to show the employees how to use all the tools and equipment properly and how to carry out all procedures efficiently. The initial training program should incorporate the following:

- How to use, handle, and maintain all tools, machinery, and equipment that have to be used as a part of the job
- How to use the special tools, if any, associated with a particular job
- How to use safety equipment and guards along with personal protective equipment to ensure safety
- How to properly lift and the proper procedures to follow when an object is too heavy to lift safely without assistance

Training for Supervisors

It is the responsibility of supervisors to ensure that all employees are properly trained to follow safe work practices and that these practices are followed on a consistent basis. In addition to the training received by employees, supervisors should receive additional training that enables them to recognize early risk signs and symptoms, hazardous work practices, how to correct those practices, and how to reinforce the ergonomic program.

Training for Managers

Apart from employees and supervisors, managers must also be made aware of their responsibilities to implement ergonomic principles that ensure the safety and health of all employees. They must also be familiar with the problems and risks associated with all tasks.

Training for Engineers and Maintenance Personnel

On-site engineers and maintenance personnel must also be trained so that they can recommend and implement the best possible machinery, equipment, designs, work practices, and tools to reduce the risk of injury and bodily damage to employees.

Lesson Summary

All supervisors required to administer an ergonomics program should be provided with special training focusing on how to effectively make the workplace safe by adopting sound ergonomic principles and practices. Apart from ergonomic principles, the training must also include the risk potential of damage to the body and injuries that can result from the failure to adopt proper ergonomic practices.

Employers must provide formal training to all employees who could be exposed to hazards to inform them about the hazards associated with their jobs and the tools, machinery, and equipment they use.

New employees and those assigned new tasks must be made aware of the specific risks associated with a particular job before they start their work.

An ergonomics training program must include:

- All employees who are exposed to different risk factors
- Supervisors
- Managers
- All engineers and maintenance personnel

An effective training program includes a mix of both theoretical and practical ways in which employees can develop their skills to work safely. They must manage the amount of time that an employee spends performing a particularly challenging job.

Module 17: Fall Protection

Module Description

This module gives you a basic understanding about OSHA's role in the prevention and elimination of work-related illnesses and injuries. The OSHA standard identifies areas or activities where fall protection is needed.

It clarifies what an employer must do to provide fall protection for employees, such as identifying and evaluating fall hazards and providing training. Under the standard, employers are able to select fall protection measures compatible with the type of work being performed.

OSHA places its rules for fall protection in several different subparts in the standards, depending primarily on the nature of the work. The standard covers most workers, except those inspecting, investigating, or assessing workplace conditions prior to the actual start of work or after all work has been completed.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Discuss falls in the workplace.
- Identify preventative measures to avoid falls.
- Discuss the duty to have fall protection.
- Identify and describe fall protection systems.
- Develop a fall protection plan.
- Inspect fall protection equipment.

Lesson 1: Fall Protection

Lesson Focus

This lesson focuses on the following topics:

- Case Study
- Falls
- Protection from Falling Objects
- Types of Fall Protection—Passive Systems
- Types of Fall Protection—Active Systems

Case Study

Worker Falls from Scaffolding

This accident occurred during the construction of a new two-story wood frame house. The work procedures on the day of the accident involved working on the ground to insert pillars into the floor joists, which were then lifted by a mobile crane. Three workers were engaged in this assembly work on the ground, and the victim and two coworkers carried out the elevated assembly work. The framing for the second-floor roof was completed in the morning.

After the lunch break, work was resumed with the same assignments as those in the morning. The victim was on a scaffolding board that was stretched over a second-floor ceiling beam, and suddenly fell onto the first-floor concrete foundation.

What do you think were some of the causes of the accident?

- The scaffolding board was not fixed.
- No guardrail system or personal fall protection system was used by employees while working at heights greater than six feet.
- No competent person for fall protection was onsite during the construction of this project.

- While the victim had been newly employed the day of the accident, he was not given new hire safety orientation prior to starting work.

Falls

Falls are the leading cause of fatalities in the construction industry with 349 total deaths related to falls in the construction industry reported in 2014.

The Physics of a Fall

A body in motion can cover vast distances in a short period of time. Consider this:

- A body in free fall can travel 4 feet in 0.5 seconds.
- A body in free fall can travel 16 feet in 1 second.
- A body in free fall can travel 64 feet in 2 seconds.
- A body in free fall can travel 144 feet in just 3 seconds.

Fall Prevention Measures

Fall prevention measures include:

- Select fall protection systems appropriate for given situations.
- Use proper construction and installation of safety systems.
- Supervise employees properly.
- Use safe work procedures.
- Train workers in the proper selection, use, and maintenance of fall protection systems.

Areas Required to Have Fall Protection

Depending on the circumstances, the following areas are required to have fall protection:

- Unprotected sides and edges
- Leading edges
- Hoist areas
- Holes
- Formwork and reinforcing steel

- Ramps, runways, and other walkways
- Excavations
- Dangerous equipment
- Overhand bricklaying and related work
- Roofing work on low-slope roofs
- Roofs
- Pre-cast concrete erection
- Residential construction
- Wall openings
- Walking/working surfaces not otherwise addressed

Duty to Have Fall Protection

Fall protection duties include:

- Fall protection is generally required when one or more employees have exposure to falls of six feet or greater to the lower level.
- Surfaces must be inspected before the work begins.
- Employees are only permitted to be on surfaces that are strong enough to support them.

Employers are required to assess the workplace to determine if the walking/working surfaces on which employees are to work have the strength and structural integrity to safely support workers. Employees are not permitted to work on those surfaces until it has been determined that the surfaces have the requisite strength and structural integrity to support workers. Once employers have determined that the surface is safe for employees to work on, the employer must select one of the available options for the work operation if a fall hazard is present.

Example: For example, if an employee is exposed to falling 6 feet (1.8 meters) or more from an unprotected side or edge, the employer must provide a guardrail system, safety net system, or personal fall arrest system to protect the worker. Similar requirements are prescribed for other fall hazards as follows.

Leading Edge Work

Each employee who is constructing a leading edge six feet (1.8 meters) or more above lower levels shall be properly protected. Suitable protection may be provided by guardrail systems, safety net systems, or personal fall arrest systems.

Hoist Areas

Each employee in a hoist area shall be protected from falling 6 feet (1.8 meters) or more by guardrail systems, personal fall arrest systems, or other appropriate means. If guardrail systems (or chain gate or guardrail) or portions thereof must be removed to facilitate hoisting operations, as during the landing of materials, and a worker must lean through the access opening or out over the edge of the access opening (to receive or guide equipment and materials, for example), that employee must be protected by one of the appropriate means.

Formwork and Re-Bar

During formwork or re-bar assembly employees shall be protected from falls of six feet or more by personal fall arrest systems, safety net systems, or positioning device systems.

Ramps, Runways, and Walkways

Each employee using ramps, runways, and other walkways shall be protected from falling 6 feet (1.8 meters) or more.

Excavations

Each employee at the edge of an excavation 6 feet (1.8 meters) or deeper shall be protected from falling by guardrail systems, fences, barricades, or covers when the excavations are not readily seen because of plant growth or other visual barrier.

Where walkways are provided to permit employees to cross over excavations, guardrails are required on the walkway if it is 6 feet (1.8 meters) or more above the excavation.

Dangerous Equipment

Each employee working above dangerous equipment must be protected from falling into or onto the dangerous equipment by guardrails systems or by equipment guards even in those cases where the fall distance is less than 6 feet (1.8m).

Overhand Bricklaying

Except as otherwise provided in the OSHA Fall Protection Standards, each employee performing overhand bricklaying and related work 6 feet (1.8 m) or more above lower levels, shall be protected from falling by guardrail systems, safety net systems, personal fall arrest systems, or shall work in a controlled access zone.

Note: Bricklaying operations performed on scaffolds are regulated by subpart L of OSHA 1926 – Scaffolds.

Low-Sloped Roof Work

Each employee engaged in roofing activities on low-slope roofs, with unprotected sides and edges six feet or more above lower levels, shall be protected from falling by guardrail systems, safety net systems, and personal fall arrest systems, or a combination of a warning line system and guardrail system, warning line system and safety net system, warning line system and personal fall arrest system, or warning line system and safety monitoring system.

- **Safety Monitoring System:** a safety system in which a competent person is responsible for recognizing and warning employees of fall hazards.
- **Warning Line System:** a barrier erected on a roof to warn employees that they are approaching an unprotected roof side or edge, and which designates an area in which roofing work may take place without the use of guardrails, body belts, or safety net systems to protect employees in the area.

Steep Roofs

Each employee on a steep roof with unprotected sides and edges 6 feet (1.8 meters) or more above lower levels shall be protected by guardrail systems with toe-boards, safety net systems, personal fall arrest systems, or by other appropriate means.

Pre-Cast Concrete

Each employee who is 6 feet (1.8 meters) or more above lower levels while erecting pre-cast concrete members and related operations such as grouting of pre-cast concrete members shall be protected by guardrail systems, safety net systems, or personal fall arrest systems.

Wall Openings

Each employee working on, at, above, or near wall openings (including those with chutes attached) where the outside bottom edge of the wall opening is 4 feet (1.2 meters) or more above lower levels and the inside bottom edge of the wall opening is less than 39 inches (1.0 meter) above the walking/working surface must be protected from falling by the use of a guardrail system, a safety net system, or a personal fall arrest system.

Protection from Falling Objects

When employees are exposed to falling objects, the employer must have employees wear hardhats and implement one of the following measures:

- Erect toe-boards, screens, or guardrail systems to prevent objects from falling from higher levels.

OR

- Erect a canopy structure and keep potential fall objects far enough from the edge so that those objects will not go over the edge if they are accidentally displaced.

OR

- Barricade the area to which objects could fall, prohibit employees from entering the barricaded area, and keep objects that may fall far enough away from the edge of a higher level so that those objects would not go over the edge if they were accidentally displaced.

Types of Fall Protection—Passive Systems

Passive systems are protective systems that do not involve the actions of employees. An example of a passive system is a catch platform extending around the perimeter of the work area.

Guardrails

Guardrails are one the most common forms of fall protection. They can be constructed of wood, pipe, structural steel, or wire rope. Flags must be provided on wire rope to increase visibility. Guardrails must have top rail, mid rail, posts, and when necessary, a toe board.

- Guardrail systems shall be capable of withstanding, without failure, a force of at least 200 pounds (890 N) applied within 2 inches (5.1 cm) of the top edge, in any outward or downward direction, at any point along the top edge.
- Steel or plastic bands must not be used as top rails or midrails.
- Manila, plastic or synthetic rope being used for top rails or midrails shall be inspected as frequently as necessary to ensure that it continues to meet the mandated strength requirements.

Note: When the 200-pound test is applied in a downward direction, the top edge of the guardrail shall not deflect to a height less than 39 inches (1.0 m) above the walking/working level. Guardrail system components selected and constructed in accordance with the Appendix B to subpart M of OSHA 1926 will be deemed to meet this requirement.

Guardrails: Design Criteria

Requirements for the design of guardrails include:

- Top edge height of top rails, or equivalent guardrail system members, shall be 42 inches (1.1 m) plus or minus 3 inches (8 cm) above the walking/working level. When conditions warrant, the height of the top edge may exceed the 45-inch height, provided the guardrail system meets all other criteria of this paragraph.
- Midrails, screens, mesh, intermediate vertical members, or equivalent intermediate structural members shall be installed between the top edge of the guardrail system

and the walking/working surface when there is no wall or parapet wall at least 21 inches (53 cm) high.

- Midrails, when used, shall be installed at a height midway between the top edge of the guardrail system and the walking/working level.
- Top rails and midrails shall be at least one-quarter inch (0.6 cm) nominal diameter or thickness to prevent cuts and lacerations. If wire rope is used for top rails, it shall be flagged at not more than 6-foot intervals with high-visibility material.
- For pipe railings: posts, top rails, and intermediate railings shall be at least one and one-half inches nominal diameter (schedule 40 pipe) with posts spaced not more than 8 feet (2.4 m) apart on centers.
- For structural steel railings: posts, top rails, and intermediate rails shall be at least 2-inch by 2-inch (5 cm x 10 cm) by 3/8-inch (1.1 cm) angles, with posts spaced not more than 8 feet (2.4 m) apart on centers.
- Screens and mesh, when used, shall extend from the top rail to the walking/working level and along the entire opening between top rail supports.
- Intermediate members (such as balusters), when used between posts, shall not be more than 19 inches (48 cm) apart.
- Other structural members (such as additional midrails and architectural panels) shall be installed such that there are no openings in the guardrail system that are more than 19 inches (.5m) wide.

Safety Net Systems

Safety net systems must comply with the following provisions:

- They must be installed as close as practicable under the walking or working surface on which employees are working, but in no case more than 30 feet below the surface.
- If the net is not vertically more than 5 feet from the working level, the safety net must extend outward from the outermost projection of the work by 8 feet.
- If the net is not vertically more than between 5 feet and 10 feet from the working level, the safety net must extend outward from the outermost projection of the work by 10 feet.

- If the net is vertically more than 10 feet from the working level, the safety net must extend outward from the outermost projection of the work by 13 feet.
- Safety nets must be drop-tested at the jobsite after they are installed and before use, whenever relocated, after major repair, and at 6-month intervals after installation if left in one place.
- Drop-tests must consist of a 400-pound bag of sand 28-32 inches in diameter being dropped into the net from the highest working or walking surface, but not from less than 42 inches above that level.
- Safety nets must have enough clearance beneath them to prevent contact with the surface or structures below when a load equal to the drop-test weight is dropped on them.
- Safety nets must be capable of absorbing an impact force that is equal to the drop test weight.
- Defective nets cannot be used.
- All materials, scraps, equipment, and tools that have fallen in the net must be removed as soon as possible and at least before the next work shift.
- The maximum size of each safety net mesh opening shall not exceed 36 square inches (230 cm²) nor be longer than 6 inches (15 cm) on any side, and the opening, measured center-to-center of mesh ropes or webbing, shall not be longer than 6 inches (15 cm).
- The safety net must have a border rope with a minimum breaking strength of at least 5,000 pounds.
- If safety nets are connected together, the connection must be as strong as the individual nets and not more than 6 inches apart.

Types of Fall Protection—Active Systems

Active fall protection systems require workers to be engaged in ensuring that proper protection is in use. This may include activities such as donning a full-body harness with an attached lanyard and attaching the lanyard to appropriate anchorage point.

Active systems are designed to operate in free fall situations. Active systems must be connected to other systems/components or activated to provide protection. Active systems are designed to protect employees from the following:

- Falls
- Forces that can cause injury

An example of an active system is a personal fall arrest system (PFAS).

Personal Fall Arrest Systems (PFAS)

Personal Fall Arrest Systems (PFAS) shall not be attached to a guardrail system or hoists.

All components of a fall arrest system must be inspected before each use and after impact. Defective components must be removed from service. Personal fall arrest systems and components subjected to impact loading shall be immediately removed from service and shall not be used again for employee protection until inspected and determined by a competent person to be undamaged and suitable for reuse.

Action must be taken promptly to rescue fallen employees or be assured they can rescue themselves.

When stopping a fall, a PFAS must:

- Limit maximum arresting force on an employee to 1,800 pounds (8 kN) when used with a body harness.
- Be rigged such that an employee can neither free fall more than 6 feet (1.8 m), nor contact any lower level.
- Be attached to an anchor point capable of withstanding 5000 pounds of force or shall be designed, installed, and used as part of a complete personal fall arrest system that maintains a safety factor of at least two and is used under the supervision of a qualified person.
- Bring an employee to a complete stop and limit maximum deceleration distance an employee travels to 3.5 feet (1.07 m).

- Have sufficient strength to withstand twice the potential impact energy of an employee free falling a distance of 6 feet (1.8 m), or the free fall distance permitted by the system, whichever is less.

Lesson Summary

Steel or plastic bands must not be used as top rails or midrails. Midrails, when used, shall be installed at a height midway between the top edge of the guardrail system and the walking/working level.

When employees might be exposed to falling objects, the employer must have employees wear hardhats and erect toe-boards, screens, or guardrail systems to prevent objects from falling from higher levels. This means that employers must either erect a canopy structure or ensure that potential fall objects are far enough from the edge so that those objects will not go over the edge if they are accidentally displaced, or barricade the area to which objects could fall thereby prohibiting employees from entering the barricaded area.

Each employee engaged in roofing activities on low-slope roofs with unprotected sides and edges six feet or more above lower levels shall be protected from falling by guardrail systems, safety net systems, and personal fall arrest systems, or a combination of a warning line system and guardrail system, warning line system and safety net system, warning line system and personal fall arrest system, or warning line system and safety monitoring system.

During formwork or re-bar assembly employees shall be protected from falls of six feet or more by personal fall arrest systems, safety net systems, or positioning device systems. Passive systems are protective systems that do not involve the actions of employees.

Lesson 2: Inspection and Safety Monitoring Systems

Lesson Focus

This lesson focuses on the following topics:

- Inspecting Fall Protection Equipment
- Positioning Device Systems
- Safety Monitoring System
- Fall Protection Plan
- Training
- Case Study

Inspecting Fall Protection Equipment

Equipment must be inspected before each use for:

- Tears, cuts, burns and abrasions
- Distorted hooks, damaged springs, and non-functioning parts
- Manufacturer labels
- Deformed eyelets, D-rings and other metal parts
- Dirt, grease, oil, corrosives, and acids

PFAS—Harnesses

- Harness systems are constructed of synthetic fibers. Only systems that encompass the entire body (full body harness) are permitted.
- Body belts cannot be used for fall arrest.
- A full body harness will distribute weight across the waist, pelvis, and thighs.

PFAS—Lanyards

Lanyards are flexible synthetic fiber lines or wire rope that have a connector at each end for connecting the body belt or body harness to a deceleration device, lifeline, or anchorage.

- Lanyards and vertical lifelines must have a minimum breaking strength of 5000 pounds.
- Lanyards should be attached to a D ring between the shoulder blades above the employee.
- There are several types of lanyards including: synthetic webbing, synthetic rope, and shock absorbing.

Types of Lanyards

- **Self-retracting:** Eliminates excess slack in the lanyard (cable, rope, or web)
- **Shock absorbing:** Device slows and eventually stops descent and absorbs the forces (i.e., rip stitch controlled tearing)
- **Synthetic rope:** Absorbs some of the force by stretching
- **Synthetic webbing:** Strong but not flexible (absorbs little force)

PFAS—Life Lines

- Life lines consist of flexible material connected at one or both ends to an anchorage point.
- There are two types of life lines:
 - Vertical: hangs vertically (5000 pound minimum breaking strength).
 - Horizontal: connects at both points to stretch horizontally (serves as connection point for other components of PFAS- total system must have safety factor of two and be capable of locking in both directions on the lifeline).

Lifelines, Safety Belts, and Lanyard (PPE)

Lifelines, safety belts, and lanyards shall be used only for employee safeguarding. Any lifeline, safety belt, or lanyard actually subjected to in-service loading, as distinguished from static-load testing, shall be immediately removed from service and shall not be used again for employee protection until inspected and determined by a competent person to be undamaged and suitable for reuse.

Vertical lifelines shall have a minimum breaking strength of 5,000 pounds (22.2 kN). Self-retracting lifelines and lanyards that automatically limit free fall distance to 2 feet (0.61 m) or less shall be capable of sustaining a minimum tensile load of 3,000 pounds (13.3 kN) applied to the device with the lifeline or lanyard in the fully extended position.

More Information: All safety belt and lanyard connectors shall be drop forged, pressed or formed steel, or made of equivalent materials. Each connector shall have a corrosion-resistant finish. The surface shall be smooth and free of sharp edges.

PFAS—Snaphooks

- Snaphooks are used to connect lanyards to D-rings on a body harness.
- D rings must be compatible and connected to harness or anchorage point only.
- Snaphooks and D rings must have tensile strength of 5000 pounds and be proof tested to a minimal tensile load of 3600 pounds.
- When using snaphooks:
 - All snap hooks must have a locking mechanism.

Snaphooks

Locking Snaphooks

Locking snaphooks have a self-closing, self-locking keeper, which remains closed and locked until unlocked and pressed open for connection or disconnection.

PFAS—Anchorage Points

- The anchorage point is most effective when it is above the employee's head; located as to not allow an employee to fall more than 6 feet.
- Anchorages used for the attachment of personal fall arrest equipment shall be independent of any anchorage being used to support or suspend platforms and capable of supporting at least 5,000 pounds per employee attached, or shall be designed, installed, and used as follows:
 - As part of a complete personal fall arrest system that maintains a safety factor of at least two, and

- Under the supervision of a qualified person.

Positioning Device Systems

A positioning device system is a body belt or body harness system rigged to allow an employee to be supported on an elevated vertical surface, such as a wall, and work with both hands free while leaning.

- Positioning device systems must be inspected before each use for defects, and defective components must be removed from service.
- Positioning devices shall be rigged such that an employee cannot free fall more than 2 feet (0.6 m).
- Positioning devices shall be secured to an anchorage capable of supporting at least twice the potential impact load of an employee's fall or 3,000 pounds, whichever is greater.
- A positioning device system is not a fall arrest system!

Warning Line System

A warning line system is an awareness device erected on a roof to warn employees that they are approaching an unprotected roof side or edge, and which designates an area in which roofing work may take place without the use of guardrail, body belt, or safety net systems to protect employees in the area.

Warning line systems and their use shall comply with the following provisions:

- The warning line shall be erected around all sides of the roof work area.
- Warning lines shall consist of rope, wire, chains, or supporting stanchions, which are used to warn employees of an unprotected edge, and must be erected as follows:
 - It must be flagged at not more than 6-foot intervals with high-visibility materials.
 - The rope, wire, or chain must be rigged and supported such that:

- Its lowest point (including sag) is no less than 34 inches from walking/working surface.
- Its highest point no more than 39 inches from surface.
- Stanchions, with rope, chain, or wire attached, must be able to withstand, without tipping over, a force of 16 pounds applied horizontally against the stanchion, 30 inches (.8 m) above the walking/working surface, perpendicular to the warning line, and in the direction of the floor, roof, or platform edge.
- The rope, wire, or chain used must have a minimum tensile strength of 500 pounds.
- No employee is permitted between roof's edge and a warning line unless the employee is performing roof work in that area.

Note: A warning line system is used mainly on roofs, where the use of PFAS is impractical.

Controlled Access Zone (CAZ)

- When used to control access to areas where overhand bricklaying or related work are taking place, only qualified personnel involved in overhand bricklaying or related work are permitted in the controlled access zone.
- Ropes, wires, tapes, or chains with supporting stanchions are used to designate the area.
- Must be erected between 6 and 25 feet away from unprotected edge.
- The control line shall be connected on each side to a guardrail system or a wall.
- CAZ must be defined by a control line erected 10-15 feet from the edge.
- Lines must be flagged at 6-foot intervals and have a minimum breaking strength of 200 pounds.

Safety Monitoring System

The employer must designate a competent person to monitor the safety of other employees, and the employer has the duty to ensure that the safety monitor complies with the following requirements:

- He/she must be competent to recognize fall hazards.
- He/she must warn the employee when it appears that the employee is unaware of a fall hazard or is acting in an unsafe manner.
- He/she must be on the same walking/working surface and within visual sighting distance of employee being monitored.
- He/she must be close enough to communicate orally with the employee.
- He/she must not have other responsibilities that could take attention from monitoring functions.

Note: Each employee working in a controlled access zone must be directed to comply with all instructions from the monitor.

It is recommended that you have a written plan for using the safety monitoring system to address:

- Identification of the monitor
- Roles of employees in monitoring system
- Training for using the monitoring system

Covers

Covers are used to protect personnel from falling through holes in walking surfaces. Covers for holes in floors, roofs, and other walking/working surfaces shall meet the following requirements:

- All covers shall be secured when installed so as to prevent accidental displacement by the wind, equipment, or employees.
- All covers shall be color coded or they shall be marked with the word "HOLE" or "COVER" to provide warning of the hazard.
- Covers located in roadways and vehicular aisles shall be capable of supporting, without failure, at least twice the maximum axle load of the largest vehicle expected to cross over the cover.

- All other covers shall be capable of supporting, without failure, at least twice the weight of employees, equipment, and materials that may be imposed on the cover at any one time.

Note: This provision does not apply to cast iron manhole covers or steel grates used on streets or roadways.

Falling Objects

Employers are required to protect their employees from falling objects. Some methods that might have to be used (when necessary) consist of:

- Installation of toe boards (at least 3.5 inches wide) erected along the edges of the overhead walking/working surfaces for a distance sufficient to protect persons working below.
 - Toe boards shall be capable of withstanding, without failure, a force of at least 50 pounds applied in any downward or outward direction at any point along the toe board.
 - Where tools, equipment, or materials are piled higher than the top edge of a toe board, paneling or screening shall be erected from the walking/working surface or toe board to the top of a guardrail system's top rail or midrail, for a distance sufficient to protect employees below.
- Building barricades and restricting entrance.

Fall Protection Plan

The fall protection plan option is available only to employees engaged in leading edge work, precast concrete erection work, or residential construction work who can demonstrate that it is unfeasible, or it creates a greater hazard to use conventional fall protection equipment.

If used, the plan should be strictly enforced.

- A Fall Protection Plan must be prepared by a qualified person and developed specifically for each site.
- The Fall Protection Plan must be maintained up to date.
- Any changes to the plan must be approved by a qualified person.
- A copy of the plan with all approved changes must be maintained at the site.
- The fall protection plan shall document the reasons why the use of conventional fall protection systems (guardrail systems, personal fall arrest systems, or safety nets systems) is infeasible or why their use would create a greater hazard.

Elements of a Fall Protection Plan include:

- Statement of Policy
- Fall Protection Systems to be Used
- Implementation of Plan
- Enforcement
- Accident Investigation
- Changes to the Plan

Training

All employees exposed to fall hazards must receive training by a competent person who addresses:

- The nature of fall hazards in the work area.
- Procedures for erecting, maintaining, disassembling and inspecting fall protection systems to be used.
- Use and operation of fall arrest equipment.

Training elements include:

- Role of employees in a safety monitoring system (when used)
- Limitations on the use of mechanical equipment for low-slope roofs
- Role of employees in fall protection plans
- Standards contained in 29 CFR 1926.500-503

- Procedure for handling and storage of equipment

Case Study

Fall During the Assembly of a Suspended Scaffold for Bridge Painting

The following is a case study of an accident involving falls and fall protection.

This accident occurred while a suspended scaffold was being installed for painting bridge girders. The suspended scaffold was comprised of the main pipes supported by chains hung from the bridge girders and single tubes that extended perpendicular to the main pipes. The plan called for installing two layers of scaffold.

Measures to prevent a fall when this scaffold was completed included safety netting that was stretched below the bottom of the lower scaffold platform and scaffolding boards beneath, to the right and left of bridge girders to make personnel movement easy.

Two lift trucks, each with a maximum work height of 15 meters, were being used for this work. Each truck was moved after each scaffold section was complete.

On the day of the accident, three workers exited onto the ground from one of the lift trucks in order to move the vehicle. However, the truck could not be moved forward because the truck tires were stuck in river sand. Five other workers, who had their safety belts on and attached to the hanging chains, were on scaffold boards watching and waiting for the truck to be moved.

After several minutes, some of these workers who were tied off on the scaffold heard a loud sound and turned in time to see a fellow worker attempting to catch one of the chains as he fell beneath the single tubes. The worker had been attempting to adjust the chains by himself. He fell while attempting to adjust the hanging chain or during his movement from the scaffold board to a flange below.

What do you think were some of the causes of the accident?

1. Although the victim had been wearing a safety harness while waiting on the scaffold, when he moved to adjust the hanging chains he was no longer wearing it and it wasn't tied off.
2. Workers should have waited on the ground while the truck was being moved, and not at an elevated site that posed a danger of falling.
3. The operations chief for scaffolding erection did not provide proper supervision regarding the proper use of personal fall arrest systems.

Lesson Summary

A positioning device system is a body belt or body harness system rigged to allow an employee to be supported on an elevated vertical surface, such as a wall, and work with both hands free while leaning.

A warning line system is an awareness device erected on a roof to warn employees that they are approaching an unprotected roof side or edge, and which designates an area in which roofing work may take place without the use of guardrail, body belt, or safety net systems to protect employees in the area.

A Fall Protection Plan must be prepared by a qualified person and developed specifically for each site. The Fall Protection Plan must be maintained up to date. Any changes to the plan must be approved by a qualified person. A copy of the plan with all approved changes must be maintained at the site. The fall protection plan shall document the reasons why the use of conventional fall protection systems (guardrail systems, personal fall arrest systems, or safety nets systems) is infeasible or why their use would create a greater hazard.

Module 18: Safety and Health Programs

Module Description

Have you ever been injured on the job? Do you know what steps to take in the event of sickness, injury, or death due to your workplace environment? More importantly, do you know how to protect yourself, as well as others, and help promote healthy working conditions? Every year, more than 50,000 workers die from exposure to various hazards in the workplace. The Occupational Safety and Health Administration (OSHA) is committed to saving lives, preventing injuries, and protecting the health of workers all across America. This module will show you how to identify workplace hazards and become involved with ensuring healthy and safe working environments.

Module Learning Objectives

At the completion of this module, you should be able to:

- Describe the importance of effective safety and health programs.
- Summarize the common characteristics of exemplary workplaces.
- Explain the General Guidelines of an effective safety and health program.
- Discuss the major elements of an effective safety and health program.
- Name the state programs.
- List consultation services.
- Describe the Voluntary Protection Program (VPP).
- Discuss the Safety Health Achievement Recognition Program (SHARP).
- Describe the Strategic Partnership Program.
- Apply training and education.
- Utilize electronic information.

Lesson 1: Effective Program Elements

Lesson Focus

This lesson focuses on the following topics:

- Importance of Effective Safety and Health Programs
- Common Characteristics of Exemplary Workplaces
- The Guidelines—General
- Major Elements

Importance of Effective Safety and Health Programs

It has been found that effective management of worker safety and health programs:

- Reduces the extent and severity of work-related injuries and illnesses.
- Improves employee morale and productivity.
- Reduces workers' compensation costs.

Common Characteristics of Exemplary Workplaces

Organized and systematic methods are used to:

- Assign responsibility to managers, supervisors, and employees.
- Regularly inspect for hazards and control the hazards.
- Regularly re-evaluate the effectiveness of existing programs.
- Orient and train all employees to eliminate or avoid hazards.

The Guidelines—General

An effective program:

- Includes provisions for systematic identification, evaluation, and prevention or control of hazards.

- Goes beyond specific requirements of the law to address all hazards.

Written program:

- As the size and complexity of the worksite or process increases, so does the need for written guidance.
- Some individual OSHA standards require a written program be created. Taken in whole, these individual written programs will often form the basis of the facility safety program.

Major Elements

An effective occupational safety and health program will include the following four elements:

1. Management commitment and employee involvement
2. Worksite hazard analysis
3. Hazard prevention and control programs
4. Safety and health training

1. Management commitment and employee involvement

- Management commitment and employee involvement are complementary.
- Management commitment provides the motivating force and resources for organizing and controlling activities within an organization.
- Employee involvement provides the means through which workers develop and express their own commitment to safety and health protection.

Recommended Actions

- Clearly state a worksite safety and health policy.
- Establish and communicate a clear goal and objective for the safety and health program.

- Provide visible top management involvement when implementing the program.
- Encourage employee involvement in the program and in decisions that affect safety and health (e.g., inspection or hazard analysis teams; developing or revising safe work rules; training new hires or co-workers; assisting in accident investigations).
- Assign and communicate responsibility for all aspects of the program.
- Provide adequate authority and resources to responsible parties.
- Hold managers, supervisors, and employees accountable for meeting their responsibilities.
- Review program operations at least annually to evaluate, identify deficiencies, and revise as needed.

2. Worksite Analysis

- Worksite hazard analysis involves a variety of worksite examinations to identify not only existing hazards, but also conditions and operations where changes might occur to create hazards, anticipating hazards before they become obvious.
- Effective management actively analyzes the work and the worksite to anticipate and prevent harmful occurrences.

Recommended Actions

In order to identify all the hazards:

- Conduct comprehensive baseline and periodic surveys for safety and health.
- Analyze planned and new facilities, processes, materials, and equipment.
- Perform routine job hazard analyses supported by checklists, Job Safety Analysis, or Job Hazard Analysis forms.
- Provide for regular site safety and health inspections.
- Involve the facility's Safety Committee in periodic, but regular, site inspections.

- Provide a reliable system for employees, without fear of reprisal, to notify management about apparent hazardous conditions and to receive timely and appropriate responses.
- Provide for investigation of accidents and "near miss" incidents, so that their causes and means for prevention are identified.
- Analyze injury and illness trends over time, so that patterns with common causes can be identified and addressed with an eye towards prevention.

3. Hazard Prevention and Control

- Where feasible, prevent hazards through effective design of the job or job site, which are referred to as Engineering Controls.
- Where elimination of the hazard is not feasible, control hazards to prevent unsafe and unhealthful exposure.
- Elimination or control must be accomplished in a timely manner.

Recommended Actions

Establish procedures for timely prevention, correction, or control of hazards, including:

- Engineering controls, where feasible and appropriate.
- Administrative controls such as procedures for safe work practices, which are understood and followed as a result of training, positive reinforcement, correction of unsafe performance, and enforcement.
- Provision for personal protective equipment.
- Provide for facility and equipment maintenance.
- Plan and prepare for emergencies by conducting training and drills, not less than annually.
- Establish a medical program consisting of first aid on site and the structure to access nearby physician and emergency medical care.

4. Safety and Health Training

- Safety and health training must address the safety and health responsibilities of all personnel, whether salaried or hourly.
- Safety and health training is most effective when incorporated into other training about performance requirements and job practices. After all, safety is everyone's responsibility and part of everyone's job.
- Complexity of training depends on size, complexity of the worksite, and the nature of hazards.

Recommended Actions

Conduct safety and health orientations before employees or contractors start work to ensure that all employees understand the hazards to which they may be exposed and how to prevent harm to themselves and others because of exposure to these hazards.

Ensure that supervisors carry out their safety and health responsibilities, including:

- Analyzing the work under their supervision to identify previously unrecognized potential hazards.
- Maintaining physical protections in work areas.
- Reinforcing employee training through continual performance feedback and enforcement of safe work practices.
- Ensure that managers understand their safety and health responsibilities, as described under the management commitment and employee involvement element of the guidelines.

Lesson Summary

As the size and complexity of the worksite or process increases, so does a need for written guidance. Individual written programs often form the basis of a facility safety program.

An effective program includes provisions for systematic identification, evaluation, and prevention or control of hazards and goes beyond specific requirements of the law to address all hazards. It has been found that effective management of worker safety and health programs reduces the extent and severity of work-related injuries and illnesses, improves employee morale and productivity, and reduces workers' compensation costs.

Lesson 2: OSHA Safety and Health Programs

Lesson Focus

This lesson focuses on the following topics:

- State Programs
- Consultation Services
- Voluntary Protection Programs (VPPs)
- Strategic Partnership Program
- Training and Education
- Electronic Information

State Programs

The Occupational Safety and Health Act of 1970 (OSH Act) encourages states to develop and operate their own job safety and health plans, which OSHA then approves and monitors. There are currently 28 state/territory plans: 22 cover both private and public (state and local government) employment; 4 states/territories—Connecticut, Illinois, Maine, New Jersey, New York, and the Virgin Islands—cover the public sector only.

More Information: Full details and a current [map](#) of state/territory plans.

States and territories with their own OSHA-approved occupational safety and health plans must adopt and enforce standards identical to, or at least as effective as, the federal standards, and provide extensive programs of voluntary compliance and technical assistance, including consultation services.

Consultation Services

Consultation assistance is available upon request to employers who want help in establishing and maintaining a safe and healthful workplace. Funded equally by OSHA and each state, the service is provided at no cost to the employer.

Each state, whether a federal plan state, or a state plan state, operates the consultation program for the state. Primarily developed for smaller employers with more hazardous operations, the consultation service is delivered by state governments employing professional safety and health consultants.

Comprehensive assistance includes a hazard survey of the worksite and appraisal of all aspects of the employer's existing safety and health management system. In addition, the service offers assistance to employers in developing and implementing an effective safety and health management system. No penalties are proposed or citations issued for hazards identified by the consultant.

The employer's only obligation is to correct all serious hazards identified by the consultant within the agreed upon correction timeframe. OSHA provides consultation assistance to the employer with the assurance that his or her name, firm, and any information about the workplace will not be reported to OSHA enforcement staff.

Under the consultation program, certain exemplary employers may request participation in OSHA's Safety and Health Achievement Recognition Program (SHARP). Eligibility for participation in SHARP includes, but is not limited to, receiving a full-service, comprehensive consultation visit, correcting all identified hazards, and developing an effective safety and health program management system.

Employers accepted into SHARP may receive an exemption from programmed inspections (not complaint or accident investigation inspections) for a period of two years initially, or up to three years upon renewal.

Voluntary Protection Programs (VPPs)

The Voluntary Protection Program and onsite consultation services, when coupled with an effective enforcement program, expand worker protection to help meet the goals of the OSH Act. The three levels of VPP—Star, Merit, and Demonstration—are designed to recognize outstanding achievements by companies that have developed and implemented effective safety and health management systems.

The VPPs motivate others to achieve excellent safety and health results in the same outstanding way as they establish a cooperative relationship between employers, employees, and OSHA.

Strategic Partnership Program

OSHA's Strategic Partnership Program, the newest member of OSHA's cooperative programs, helps encourage, assist, and recognize the efforts of partners to eliminate serious workplace hazards and achieve a high level of worker safety and health. Whereas OSHA's Consultation Program and VPP entail one-on-one relationships between OSHA and individual worksites, most strategic partnerships seek to have a broader impact by building cooperative relationships with groups of employers and employees.

These partnerships are voluntary, cooperative relationships between OSHA, employers, employee representatives, and others such as trade unions, trade and professional associations, universities, and other government agencies.

Training and Education

OSHA's area offices offer a variety of information services, such as compliance assistance, technical advice, publications, audiovisual aids, and speakers for special engagements. OSHA's Training Institute in Arlington Heights, IL, provides basic and advanced courses in safety and health for federal and state compliance officers, state consultants, federal agency personnel, and private sector employers, employees, and their representatives.

The OSHA Training Institute also has established OSHA Training Institute Education Centers to address the increased demand for its courses from the private sector and from other federal agencies. These centers are nonprofit colleges, universities, and other organizations that have been selected after a competition for participation in the program. There is a Federal Education Center in each of the 10 Federal Regions.

OSHA awards grants to nonprofit organizations through its Susan Harwood Training Grant Program in order to provide safety and health training and education to employers and workers in the workplace.

The grants focus on programs that will educate workers and employers in small businesses (fewer than 250 employees) about new OSHA standards or about high-risk activities or hazards. Grants are awarded for one year and may be renewed for an additional 12 to 24 month period, depending on whether or not the grantee has performed satisfactorily.

OSHA expects each organization awarded a grant to develop a training and/or education program that addresses a safety and health topic named by OSHA, to recruit workers and employers for the training, and to conduct the training. Grantees are also expected to follow up with people who have been trained to find out what changes were made in order to reduce the hazards in their workplaces as a result of the training.

Electronic Information

OSHA has a variety of materials and tools available on its [Website](#). These include e-Tools, Expert Advisors, Electronic Compliance Assistance Tools (e-CATs), Technical Links, regulations, directives, publications, videos, and other information for employers and employees. OSHA's software programs and compliance assistance tools walk you through challenging safety and health issues and common problems to find the best solutions for your workplace.

Lesson Summary

OSHA awards grants to nonprofit organizations to provide safety and health training and education to employers and workers in the workplace. It expects each organization awarded a grant to develop a training and/or education program that addresses a safety and health topic, to recruit workers and employers for the training, and to conduct the training. Grantees are also expected to follow up with people who have been trained to find out what changes were made to reduce hazards in their workplaces as a result of the training.

Module 19: OSHA Recordkeeping

Module Description

This module is designed to assist employers in identifying and fulfilling the requirement to post certain records, maintain records of illnesses and injuries, and report specific cases to OSHA. Students who successfully complete this module will be able to identify the OSHA requirements for recordkeeping.

Module Learning Objectives

At the completion of this module, you should be able to:

- Summarize the importance of OSHA recordkeeping and reporting.
- Identify workplace actions that are in compliance with OSHA recordkeeping and reporting requirements.
- Distinguish between OSHA recordable and non-recordable cases.
- Apply the appropriate procedures for determining if a case is OSHA recordable.
- Discuss the appropriate category for a recordable case.

Lesson 1: Recordkeeping

Lesson Focus

This lesson focuses on the following topics:

- Importance of Recordkeeping
- Purpose (of the Rule)
- Partial Exemption
- Size Exemption
- Industry Exemption
- Recording Criteria

Importance of Recordkeeping

What's So Important about Recordkeeping?

- Keeping track of work-related injuries and illnesses can help you prevent them in the future.
- Using injury and illness data helps identify problem areas. The more you know, the better you can identify and correct hazardous workplace conditions.
- You can better administer company safety and health programs with accurate records.
- As employee awareness about injuries, illnesses, and hazards in the workplace improves, workers are more likely to follow safe work practices and report workplace hazards.

Purpose (of the Rule)

This rule requires employers to record and report work-related fatalities, injuries, and illnesses.

Recording/reporting a work-related injury, illness, or fatality does not mean the employer or employee was at fault, an OSHA rule has been violated, or that the employee is eligible for workers' compensation or other benefits.

Note: OSHA injury and illness recordkeeping and workers' compensation are independent of each other.

Partial Exemption

All employers covered by OSHA are covered by the [29 CFR 1904 regulations](#). However, some employers with 10 or fewer employees or in identified low hazard industries do not have to maintain OSHA injury and illness records unless notified by OSHA or the Bureau of Labor Statistics.

Example: Employers with 10 or fewer employees (at all times during the calendar year) and business establishments in certain industry classifications (e.g., specific low hazard retail, service, finance, insurance or real estate, etc.) are partially exempt from keeping OSHA injury and illness records (except in certain states which may have more stringent requirements—see state plans on www.OSHA.gov).

Employers that are partially exempt from recordkeeping requirements because of their size or industry must continue to comply with the following requirements:

- Reporting fatalities, inpatient hospitalization of an injured or ill employee, amputations, and loss of an eye.
- Annual OSHA injury and illness survey (if specifically requested to do so by OSHA).

Size Exemption

The following are some size exemptions:

- If a company had 10 or fewer employees at all times during the last calendar year, it does not need to keep the injury and illness records unless surveyed by OSHA or BLS.
- The size exemption for the company is based on the number of employees in the entire company.
- Temporary employees, who are supervised on day-to-day basis in the count, also should be included.

Industry Exemption

Please review the following industry exemptions:

- The partial industry classification exemption applies only to business establishments in certain retail, services, finance, insurance, or real estate industries. However, certain state plans do not allow exemption due to industry classification.
- Businesses classified in industries including agriculture, mining, construction, manufacturing, transportation, communication, electric, gas, and sanitary services or wholesale trade are not eligible for the partial industry classification exemption in any state.

More Information: The partial industry classification exemption applies to individual business establishments. If a company has several business establishments engaged in different classes of business activities, some of the company's establishments might be required to keep records, while others may be exempt.

Recording Criteria

Covered employers must record each fatality, injury, or illness that is work-related, is a new case, and meets the recording criteria as being recordable.

Work Relatedness

A case is considered work-related if an event or exposure in the work environment either caused or contributed to the resulting condition, or if an event or exposure in the work environment significantly aggravated a pre-existing injury or illness. However, there must be a discernable cause for it to be work-related.

Work relatedness is presumed for injuries and illnesses resulting from events or exposures occurring in the work environment.

Work Environment

The work environment is defined as the establishment and other locations where one or more employees are working or are present as a condition of employment. The work environment includes not only physical locations, but also the equipment or materials used by employees during the course of their work.

Significant Aggravation

A pre-existing injury or illness is significantly aggravated when an event or exposure in the work environment results in any of the following (which otherwise would not have occurred):

- Death
- Loss of consciousness
- Days away, days restricted, or job transfer
- Medical treatment beyond first aid or which causes a change in medical treatment
(What is considered medical treatment will be discussed in more detail later.)

Work Relatedness Exceptions

Work-relatedness exceptions occur in the following cases:

- A worker is present in the work environment as a member of the general public
- Injury or illness involves signs or symptoms arising in the work environment that are solely due to a non-work-related event or exposure

Work-relatedness exemptions also occur in cases where injury or illness results from voluntary participation in a wellness program or in a medical, fitness, or recreational activity such as:

- Blood donation
- Physical examination
- Flu shot
- Exercise class
- Racquetball
- Baseball

Such exemptions also include injury or illness that is solely the result of eating, drinking, or preparing food or drink for personal consumption.

Example: If the employee is injured by choking on a sandwich while in the employer's cafeteria, the case would not be considered work-related.

Note: If the employee is made ill by ingesting food contaminated by workplace contaminants (such as lead) or gets food poisoning from food supplied by the employer, then the case would be considered work-related.

Work relatedness exemptions *may* also occur when:

- Injury or illness is solely the result of employees doing personal tasks ***outside assigned working hours.***
- Injury or illness is caused by a motor vehicle accident in a parking lot/access road during a commute.
- Injury or illness is solely the result of personal grooming, self-medication for non-work-related conditions, or intentionally self-inflicted.
- An employee is afflicted by the common cold or flu.
- An employee is afflicted by mental illness, unless the employee voluntarily provides an opinion from a Health Care Professional stating the employee has a mental illness that is work-related.

Note: Contagious diseases such as tuberculosis, brucellosis, hepatitis A, or plague are considered work-related if the employee is infected at work.

Travel Status

An injury or illness that occurs while an employee is on travel status is work-related if it occurred while the employee was engaged in work activities in the interest of the employer.

An injury or illness is not recorded if:

- The employee checks into a hotel, motel, or other temporary residence for one or more days and is injured in this property, as this becomes his or her "home away from home." Evaluate the employee's activities after he or she checks into the hotel, motel, or other temporary residence for their work-relatedness in the same manner as you evaluate the activities of a non-traveling employee.
- The employee is injured while taking a detour for personal reasons.

Examples of activities in the "interest of the employer" include:

- Travel to and from customer contacts.

- Conducting job tasks.
- Entertaining or being entertained to transact, discuss, or promote business (work-related entertainment includes only entertainment activities being engaged in at the direction of the employer).

Work at Home

Injuries and illnesses that occur while an employee is working at home are work-related if they:

- Occur while the employee is performing work for pay or compensation in the home.
- Are directly related to the performance of work rather than the general home environment.

Examples:

Examples of work at home, work related injuries/illnesses.

If an employee drops a box of work documents and injures his or her foot, the case is considered work-related.

If an employee's fingernail is punctured by a needle from a sewing machine used to perform garment work at home, becomes infected, and requires medical treatment, the injury is considered work-related.

Examples of work at home injuries/illnesses that are not work related.

If an employee is injured because she or he trips on the family dog while rushing to answer a work phone call, the case is likely not considered work-related.

If an employee working at home is electrocuted because of faulty home wiring, the injury is likely not considered work-related.

New Case

A case is new if:

- The employee has not previously experienced a recordable injury or illness of the same type that affects the same part of the body.
- The employee previously experienced a recordable injury or illness of the same type that affected the same part of the body, but after complete recovery, an event or exposure in the work environment caused the signs and symptoms to reappear.

If an exposure triggers a recurrence, it is a new case (e.g., asthma, rashes).

If signs and symptoms recur even in the absence of exposure, it is not a new case (such as silicosis, tuberculosis, and asbestosis).

Days Away Cases

The following information applies to "days away" cases:

- Record if the case involves one or more days away from work.
- Check the box for "days away" cases and count the number of days.
- Do not include the day of injury or illness.
- Check the "days away" box and enter calendar days away from work.
- Count the number of calendar days the employee was unable to work (include weekends, holidays, vacation days, etc.).
- Cap the day count at 180 days.

If a medical opinion exists, the employer must follow that opinion whether or not the employee follows that recommendation.

Restricted Work

Restricted work activity occurs when:

- An employee is unable to perform one or more routine functions (work activities the employee regularly performs at least once per week) of his or her job.
- An employee is unable to work a full workday.
- A Physician or Other Licensed Health Care Professional (PLHCP) recommends either of the above.

Transfer to Another Job

If you assign an injured or ill employee to a job other than his or her regular job for part of the day, the case will involve transfer to another job. In such situations, the following information should be taken into account:

- A transfer to another job counts the same as days away from work or restricted duty.
- If the employer makes permanent changes to the job, you should stop counting "days away", but always count at least one such day.

Medical Treatment

Medical treatment is the management and care of a patient to combat disease, injury or disorder.

Medical treatment does not include the following occurrences:

- Visits to a PLHCP solely for observation or counseling
- Diagnostic procedures including
 - X-rays
 - Blood tests
 - The administration of prescription medications used solely for diagnostic purposes
- First aid

First Aid

The following can be referred to as first aid treatment:

- Using nonprescription medication at nonprescription strength
- Tetanus immunizations
- Cleaning, flushing, or soaking surface wounds
- Wound coverings (bandages), butterfly bandages, or Steri-Strips
- Hot or cold therapy
- Non-rigid means of support
- Temporary immobilization device used to transport accident victims
- Drilling of fingernail or toenail, draining fluid from blister
- Eye patches
- Removing foreign bodies from an eye using irrigation or cotton swab
- Removing splinters or foreign material from areas other than the eye by irrigation, tweezers, cotton swabs, or other simple means
- Finger guards
- Massages
- Drinking fluids for relief of heat stress

Note: Other wound closing devices such as sutures, staples, etc., are considered medical treatment.

Loss of Consciousness

All work-related cases involving loss of consciousness must be recorded.

Needlesticks and Sharps Injuries

An employer has to record all work-related needlesticks and cuts from sharp objects that are contaminated with another person's blood or other potentially infectious material—including human bodily fluids, tissues, and organs; other materials infected with HIV or Hepatitis B (HBV), such as laboratory cultures.

You must record splashes or other exposures to blood or other potentially infectious materials not caused by a cut or scratch where they result in a diagnosis of a bloodborne disease or meet the general recording criteria.

Hearing Loss

The OSHA standard requires the employer to record all of the work-related hearing loss cases where:

- The employee has experienced a Standard Threshold Shift (STS).
- The employee's total hearing level is 25 decibels (dB) or more above audiometric zero [averaged at 2000, 3000, and 4000 hertz (Hz)] in the same ear(s) as the STS.

Medical Removal Cases

- If an employee is medically removed under the medical surveillance requirements of an OSHA standard, an employer must record the case (i.e., exposure to lead, cadmium, methylene chloride, formaldehyde, benzene, etc).
- The case is recorded as either one involving days away from work or days of restricted work activity.
- If the case involves voluntary removal below the removal levels required by the standard, the case need not be recorded.

Tuberculosis

An employer must record a case where an employee is occupationally exposed to someone with a known case of active tuberculosis and subsequently develops a TB infection (as evidenced by a positive skin test or diagnosis by a physician or other licensed healthcare professional).

A case is not recordable when:

- The worker is living in a household with a person who is diagnosed with active TB.
- The public health department has identified the worker as a contact of an individual with active TB.
- A medical investigation shows the employee's infection was caused by exposure away from work.

Lesson Summary

Although some employers with 10 or fewer employees and business establishments in certain industry classifications are partially exempt from keeping OSHA injury and illness records, an employer has to record all work-related needlesticks and cuts from sharp objects that are contaminated with another person's blood or other potentially infectious material. All work-related cases involving loss of consciousness must also be recorded.

An injury or illness that occurs while an employee is on travel status is work-related if it occurred while the employee was engaged in work activities in the interest of the employer. The work environment includes not only physical locations, but also the equipment or materials used by employees during the course of their work. Additionally, a case is considered work-related if an event or exposure in the work environment either caused or contributed to the resulting condition, or if an event or exposure in the work environment significantly aggravated a pre-existing injury or illness.

Lesson 2: Reporting

Lesson Focus

This lesson focuses on the following topics:

- Forms
- Covered Employees
- Annual Summary
- Retention and Updating
- Change of Ownership
- Employee Involvement
- Prohibition Against Discrimination
- State Plans
- Fatality/Catastrophe Reporting
- Providing Records to Government Representatives

Forms

- Employers must enter each recordable case on reporting forms within seven calendar days of receiving information that a recordable case has occurred.
- An equivalent form may be used if it has the same information, is as readable and understandable, and uses the same instructions as the OSHA form it replaces.
- Forms can be kept on a computer as long as they can be produced when they are needed; you may keep your records using the computer system.

OSHA 300 Privacy Protections

- Do not enter the name of an employee on OSHA Form 300 for "privacy concern cases."
- Enter "privacy case" in the employee name column.

- You must keep a separate, confidential list of the case numbers and employee names for your privacy concern cases, so you can update the cases and provide the information to the government if asked to do so.

Privacy Concern Cases

You must consider the following injuries or illnesses to be privacy concern cases:

- An injury or illness to an intimate body part or to the reproductive system
- An injury or illness resulting from sexual assault
- Mental illnesses
- HIV infection, hepatitis, or tuberculosis
- Needlestick or sharps injuries that are contaminated with another person's blood or other potentially infectious material
- Other illness cases where an employee voluntarily requests his/her name be kept off

Note: These are the only injuries and illnesses considered privacy concern cases for Part 1904 purposes.

OSHA 300 Disclosures

- An employer may use discretion in describing the case if the employee can be identified.
- If employers give reporting forms to people not authorized by the rule, they must remove the names first. There are exceptions for:
 - Auditors/consultants hired by the employer to evaluate the health and safety program.
 - Processing a claim for workers' compensation or other insurance benefits.
 - Public health authority or law enforcement agency uses and disclosures not requiring consent or authorization.

Multiple Business Establishments

- An employer must keep a separate OSHA Form 300 for each establishment that is expected to be in operation for more than a year.
- An employer may keep one OSHA Form 300 for all short-term establishments.

Covered Employees

The following employee cases are recordable:

- All employees on payroll including:
 - Labor
 - Executive
 - Hourly
 - Salary
 - Part-time
 - Seasonal
 - Migrant
- Employees not on payroll who are supervised by your supervisors or managers on a day-to-day basis.
- The self-employed and partners should be excluded.
- Temporary help agencies should not record cases experienced by temporary workers who are supervised by the hiring firm.

Annual Summary

At the end of each calendar year an employer must:

- Review the OSHA Form 300 for completeness and accuracy and correct any deficiencies.
- Create an annual summary of injuries and illnesses recorded on the OSHA Form 300A.
- Certify the summary.

- Post the summary in a conspicuous place or places where notices to employees are customarily posted no later than February 1 of the year following the year covered by the records and keep the posting in place until April.

Retention and Updating

Retention

An employer must save all of the following forms for five years (from the end of the calendar year covered by the reports):

- The OSHA 300 Form
- Any privacy case list(s)
- The Annual Summary
- OSHA 301 Incident Report forms

Updating

An employer must update stored OSHA 300 Logs:

- To include newly discovered recordable injuries or illnesses.
- To reflect any changes to the classification of previously recordable injuries and illnesses.

However, employers do not need to update the Annual Summary or OSHA Form 301 Incident Reports but can do so if they desire.

Change of Ownership

The following information applies to change of ownership:

- Each employer is responsible for recording and reporting injuries and illnesses only for that period of the year during which he or she owned the establishment.
- The old owner must transfer records to the new owner.

- The new owner must retain prior records but is not required to update or correct records of the prior owner.

Employee Involvement

Employees and their representatives must be involved in the recordkeeping system in several ways:

- The employer must inform each employee of how to report an injury or illness.
- The employer must set up a way for employees to promptly report work-related injuries and illnesses.
- The employer must tell each employee how to report work-related injuries and illnesses to the employer.

An employer must provide limited access to injury and illness records of current employees, former employees, and personal/authorized representatives.

This requires that the employer:

- Provide a copy of OSHA Form 300, for an establishment the employee or former employee has worked in, by end of the next business day.
- Provide a copy of OSHA Form 301, describing an injury or illness to that employee or former employee, or personal representative by end of the next business day.
- Provide copies of OSHA Form 301 to authorized representatives within seven calendar days and provide only "Information about the case" section of the form.

More Information: This requirement applies when an authorized employee representative asks for copies of OSHA 301 Incident Reports for an establishment where the agent represents employees under a collective bargaining agreement.

Prohibition against Discrimination

An employer must avoid discriminating against an employee for reporting a work-related fatality, injury, or illness.

The employee who files a safety and health complaint, who asks for access to Part 1904 records, or who otherwise exercises any rights afforded by the OSH Act, is protected by law.

State Plans

State-Plan states must have the same requirements as Federal OSHA for determining which injuries and illnesses are recordable and how they are recorded.

For other Part 1904 requirements, State-Plan requirements may be more stringent, or may deny partial exemptions due to industry classification.

Examples of other Part 1904 requirements:

- Industry exemptions
- Reporting of fatalities and hospitalizations
- Record retention
- Employee involvement

Fatality/Catastrophe Reporting

Employers must report orally, within eight hours, any work-related fatality to the area office of OSHA, U.S. Department of Labor, nearest to the site of the incident by telephone or in person. Inpatient hospitalizations, amputations, or the loss of an eye must be reported within 24 hours.

The following also apply to fatality/catastrophe reporting:

- Employers must report fatal heart attacks.
- Employers do not need to report highway or public street motor vehicle accidents (outside of a construction work zone).
- Employers do not need to report commercial airplane, train, subway, or bus accidents.

You must give OSHA the following information for each fatality and 24-hour notification events:

- The establishment name
- The location of the incident
- The time of the incident
- The number of fatalities or hospitalized employees
- The names of any injured employees
- Your contact person and his or her phone number
- A brief description of the incident

Providing Records to Government Representatives

An employer must provide copies of records within four business hours when requested by an authorized government representative such as:

- A representative of the Secretary of Labor conducting an inspection or investigation under the Act.
- A representative of the Secretary of Health and Human Services conducting an investigation under section 20(b) of the Act.
- A representative of a State agency responsible for administering a State plan approved under section 18 of the Act.

An employer should provide access during the business hours of the establishment where the records are located, if records are kept in a location within a different time zone.

Lesson Summary

Employers must report orally, within eight hours, any work-related fatality and must report within 24 hours any inpatient hospitalization, amputation, or loss of an eye to the area office of OSHA, U.S. Department of Labor, nearest to the site of the incident by telephone or in person.

State-Plan states must have the same requirements as Federal OSHA for determining which injuries and illnesses are recordable and how they are recorded. For other Part 1904 requirements, State-Plan requirements may be more stringent or may deny partial exemptions due to industry classification.

Module 20: Process Safety Management

Module Description

In industries that produce and use highly hazardous chemicals, the potential for an accidental release if these substances are not properly controlled creates the possibility of a disaster. OSHA's Process Safety Management of Highly Hazardous Chemicals standard contains requirements for the proper management of the hazards associated with processes that use these chemicals.

Process safety management is the proactive identification, evaluation, and mitigation or prevention of chemical releases that could occur as a result of failures in process, procedures, or equipment. This module gives you a basic understanding of OSHA standards regarding process safety management (PSM) of highly hazardous chemicals.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Discuss OSHA standards related to process safety management (PSM).
- Examine the role of employers and employees.
- Discuss contractors.
- Study training, emergency planning, mechanical integrity, and other essential factors associated with process safety management (PSM).

Lesson 1: Introduction to Process Safety Management

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- OSHA Standards
- About the Process
- About Procedures
- Training

Introduction

The Problem

Unexpected releases of toxic, reactive, or flammable liquids and gases in processes involving highly hazardous chemicals have been reported for many years. Incidents continue to occur in various industries that use highly hazardous chemicals that may be toxic, reactive, flammable, or explosive, or may exhibit a combination of these properties. Regardless of the industry that uses these highly hazardous chemicals, there is a potential for an accidental release any time they are not properly controlled. This, in turn, creates the possibility of disaster.

Major disasters include the 1984 Bhopal, India, incident resulting in more than 2,000 deaths; the October 1989 Phillips Petroleum Company, Pasadena, TX, incident resulting in 23 deaths and 132 injuries; the July 1990 BASF, Cincinnati, OH, incident resulting in 2 deaths, and the May 1991 IMC, Sterlington, LA, incident resulting in 8 deaths and 128 injuries.

More Information: Although these major disasters involving highly hazardous chemicals drew national attention to the potential for major catastrophes, the public record is replete with information concerning many other less notable releases of highly hazardous chemicals.

Hazardous chemical releases continue to pose a significant threat to employees and provide impetus, internationally and nationally, for authorities to develop or consider developing legislation and regulations to eliminate or minimize the potential for such events.

OSHA Standards

The Need for Process Safety Management (PSM)

Incidents resulting from the unexpected release of toxic, reactive, or flammable liquids and gases in processes involving the use of highly hazardous chemicals occur in various industries each year. To eliminate or minimize these incidents, OSHA sets standards that are called *Process Safety Management (PSM)* standards.

OSHA Standards Application

The standard mainly applies to manufacturing industries, particularly those pertaining to chemicals, transportation equipment, and fabricated metal products. Other affected sectors include natural gas, liquids, farm product warehousing, electric, gas, sanitary services, and wholesale trade. The standard also applies to both pyrotechnics and to explosives manufacturers covered under other OSHA rules and has special provisions for contractors working in covered facilities. Furthermore, PSM standards apply to all those companies of each industry that deal with any of more than 130 specific toxic and reactive chemicals in listed quantities; they also include flammable liquids and gases in quantities of 10,000 pounds (4,535.9 Kg) or more except for:

- Hydrocarbon fuels used solely for workplace consumption as a fuel when not used within a process containing another highly hazardous chemical.
- Flammable liquids stored in atmospheric tanks or transferred and kept below their normal boiling point without benefit of chilling or refrigeration.

About the Process

What is process?

A process is any activity involving a highly hazardous chemical, including any use, storage, manufacturing, handling, or the on-site movement of such chemicals, or a combination of these activities. Therefore, any activities involving a group of vessels that are interconnected, or activities involving separate vessels that are located in areas where activities could result in the potential release of a highly hazardous chemical, will be considered a single process.

Process Safety Information

According to the OSHA standards, employers are responsible for the mandatory compilation of written process safety information before conducting any process hazard analysis. Written process safety information will help employers and employees recognize the various hazards posed by those processes involving highly hazardous chemicals. Furthermore, the process safety information must include information about the dangers of the highly hazardous chemicals used by and produced in the process, information pertaining to the technology of the process, and information on the equipment used in the process.

Note: 1910.119(C) and 1926.64(c) require that employees have a role in the development of, and that they or their representatives will have access to, any process hazard analysis or other elements of process safety management required under OSHA standards.

According to OSHA 1910.119 and 1926.64(d)(1), "Information pertaining to the hazards of the highly hazardous chemicals in the process" must include at least the:

- Toxicity.
- Permissible exposure limits.

- Physical data.
- Reactivity data.
- Corrosivity data.
- Thermal and chemical stability data.
- Hazardous effects of the inadvertent mixing of different materials.

OSHA 1910.119 and 1926.64(d)(2) spells out the rules regarding information on the technology of the process. This information must include at a minimum:

- A block flow diagram or simplified process flow diagram.
- Process chemistry.
- Maximum intended inventory.
- Safe upper and lower limits for such items as temperatures, pressures, flows, or compositions.
- An evaluation of the consequences of deviations, including those affecting the safety and health of employees.

If the original technical information is not available, then it may be developed in combination with the process hazard analysis in sufficient detail to support the analysis.

Information on the equipment in the process, as spelled out in 1910.119 and 1926.64(d)(3), must include:

- Materials of construction.
- Piping and instrument diagrams.
- Electrical classification.
- Relief system design and design basis.
- Ventilation system design.
- Design codes and standards employed.
- Material and energy balances for processes built after May 26, 1992.

- Safety systems (such as interlocks, detection, or suppression systems).

The employer shall document that equipment complies with recognized and generally accepted good engineering practices.

Employers also have to determine and document that any existing equipment that may have been designed and constructed in accordance with codes or practices that are no longer in standard use were designed and are used, maintained, inspected, tested, and operated in a safe manner.

Process Hazard Analysis

Process hazard analysis is a systematic approach for identifying and evaluating the hazards of processes involving hazardous chemicals. The employer is required to perform an initial process analysis on all processes covered by this standard. The process hazard analysis must be appropriate to the complexity of the process and should identify, evaluate, and control the hazards involved in the process.

The employer is required to determine and document the priority order for conducting the analysis. This must be based on considerations such as:

- The extent of the process hazard.
- The number of potential effected employees.
- The age of the process.
- The operating history of the process.

Initial process hazard analysis must meet the PSM standards requirement and must be completed as soon as possible. The analysis must be updated and revalidated regularly, at least every five years.

The employer must use one or more of the following methods, as appropriate, to determine and evaluate the hazards of the process being analyzed:

- What-if
- Checklist
- What-If/checklist
- Hazard and operability study (HAZOP)
- Failure mode and effects analysis (FMEA)
- Fault tree analysis
- An appropriate equivalent methodology

Whichever methods are used; the process hazard analysis must address the following:

- The hazards of the process.
- The identification of any previous incident that had a potential for catastrophic consequences in the workplace.
- Engineering and administrative controls applicable to the hazards and their interrelationships, such as appropriate application of detection methodologies to provide early warning of releases. Acceptable detection methods might include process monitoring and control instrumentation with alarms, and detection hardware such as hydrocarbon sensors.

Whichever methods are used, the process hazard analysis must address the following:

- Consequences of failure of engineering and administrative controls.
- Facility siting.
- Human factors.
- A qualitative evaluation of a range of the possible safety and health effects on employees in the workplace if there is a failure of controls.

A team with expertise in engineering and process operations will perform the best process hazard analysis. An inspection team must include at least one employee with experience and knowledge about the process being evaluated and another having knowledge about the specific process hazard analysis methodology used.

There must be a system in place that addresses the team's findings and recommendations. It is important the employer assure that:

- The recommendations are resolved in a timely manner.
- The actions are taken as soon as possible.
- The resolutions and actions to be taken are documented.
- A schedule for when these actions are to be completed is drawn up.
- The actions are communicated to maintenance and other employees who may be affected by the recommended actions.

The analysis must be updated and revalidated at least every five years by a team meeting the standard's requirements to assure that the hazard analysis is consistent with the current process.

Furthermore, employers must keep a copy of the analysis and make it available to OSHA on request. The files must include the process hazard analyses, updates, and revalidation for each process covered by the PSM, along with the documented resolution of recommendations.

About Procedures

Operating Procedures

Employers must develop and implement written operating procedures that take the process safety information into consideration and clearly communicate this information to employees, so that workers are as safe as possible when engaged in processes covered by the procedures. According to OSHA, it is vital that tasks and procedures related to the covered process be appropriate, clear, consistent, and most importantly, well communicated to the employees.

The procedures must address at least the following elements:

Steps for each operating phase:

- Initial startup
- Normal operations
- Temporary operations
- Emergency shutdown, including the conditions under which emergency shutdown is required, and the assignment of shutdown responsibility to qualified operators to ensure that emergency shutdown is executed in a safe and timely manner
- Emergency operations
- Normal shutdown
- Startup following a turnaround, or after an emergency shutdown

Operating limits:

- Consequences of deviation
- Steps required to correct or avoid deviation

Safety and health considerations:

- Properties of, and hazards presented by, the chemicals used in the process
- Precautions necessary to prevent exposure, including engineering controls, administrative controls, and personal protective equipment
- Control measures to be taken if physical contact or airborne exposure occurs
- Quality control for raw materials and control of hazardous chemical inventory levels
- Any special or unique hazards
- Safety systems (e.g., interlocks, detection or suppression systems) and their functions

Note: Any type of change that could affect the process must be reported and updated in the written operating procedures.

The availability of ready and up-to-date references is important for employees engaged in a process. The operating procedures must be reviewed as often as necessary to assure employees are correctly following the most recent recommendations.

The employer shall certify annually that these operating procedures are current and accurate.

Employers are responsible for the development and implementation of safe work practices for the control of hazards during work activities. Some of the operations requiring safe work practices include:

- Lockout/tagout
- Confined space entry
- Opening process equipment or piping
- Control over entrance into a facility by maintenance, contractor, laboratory, or other support personnel

These safe work practices apply to both employees and contract employees.

Training

Initial Training

Implementing an effective training program for the employees who are engaged in a process is essential to enhancing the work quality and safety of those employees. Process safety management requires that each employee who is engaged in operating a process be trained to work safely and that, before being assigned to a new process, employees be trained in an overview and the operating procedures of that process. The training must include:

- Specific safety and health hazards of the process
- All emergency operations, including shutdown
- Other safe work practices related to their duties

Those employees already trained and previously engaged in operating a process may not need to complete initial training before resuming work on that process. Instead of providing initial training for those experienced employees, the employer may certify in writing that these employees possess all the required knowledge, skills, and abilities to perform their assigned tasks safely during the operating process.

Refresher Training

To assure that employees understand and adhere to the current procedure of the process, refresher training is required at least once every three years, or more frequently if necessary, for each employee engaged in operating a process.

More Information: Employees engaged in operating the process and the employer must mutually decide the appropriate frequency of refresher training.

Training Documentation

Documenting all training is also an important task. It is the responsibility of the employer to assure each employee involved in the operation of a process has received and understands their training, and to prepare and maintain records to that effect. These records must contain:

- Identity of the employee
- Date of training
- Procedures or tests used to verify that the employee adequately understood the training

Lesson Summary

A process hazard analysis for identifying and evaluating the hazard of processes involving specific highly hazardous chemicals must be updated and revalidated at least every five years by a team meeting the standard's requirements, to assure that the hazard analysis is consistent with the current process.

Furthermore, employers must keep a copy of the analysis and make it available to OSHA on request. The files must include the process hazard analyses, updates, and revalidation for each process covered by the process safety management (PSM), along with the documented resolution of recommendations. According to OSHA, it is vital that tasks and procedures related to the covered process be appropriate, clear, consistent, and most important, well communicated to employees.

Lesson 2: Contractors and Mechanical Integrity

Lesson Focus

This lesson focuses on the following topics:

- Contractors
- Mechanical Integrity

Contractors

Application

There are different categories of contract labor. This includes those with specialized knowledge or skills who are hired to operate the facility or perform a particular aspect of a job, those who are hired for a short time period to perform turnaround operations or maintenance, and those who provide incidental service, such as janitorial, food, or laundry services. PSM provides special provisions for contractors and all of their employees, emphasizing the importance of protecting those employees who may work for another employer. However, PSM standards need not specifically apply to contractors providing incidental services whose work does not influence process safety.

Exemption

The PSM standard does not apply to contractors providing minor services that do not affect process safety, such as janitorial, food and drink, laundry, delivery, or other supply services. It does apply to contractors performing maintenance or repair, turnaround, major renovation, or specialty work on or adjacent to a covered process.

Employer Responsibilities

When selecting a contractor, the employer operating the facility to which PSM applies is required to obtain and evaluate information regarding the prospective contract employer's safety performance and programs.

It is the operating employer's responsibility to:

- Inform the contract employers about potential fire, explosion, or toxic release hazards related to the contractors work or process
- Explain the provisions of the emergency action plan to the contract employer
- Develop and implement safe work practices to control the presence, entrance, and exit of contract employers and contract employees in covered process areas
- Periodically evaluate the performance of contract employers in fulfilling their obligations
- Maintain a contract employee injury and illness log related to the contractor's work in the process areas

Contract Employer Responsibilities

The contract employer must:

- Assure that contract employees are trained in the work practices necessary to perform their job safely
- Assure that contract employees are instructed in the known potential fire, explosion, or toxic release hazards related to their job and the process, and in the applicable provisions of the emergency action plan
- Document that each contract employee has individually received and understood the training required under the standard by preparing a record that contains the identity of the contract employee, the date of training, and the means used to verify that the employee understood the training

- Assure that each contract employee follows the safety rules of the facility, including the safe work practices required in the operating procedures section of the standard
- Advise the operating employer of any unique hazards presented by or found during the contract employer's work

Pre-Startup Safety Review

It is mandatory that a safety review take place before the introduction of any highly hazardous chemical into the process. The PSM requires the employer to perform a safety review for new facilities and for modified facilities. Before the introduction of a highly hazardous chemical to a process, the pre-startup safety review must confirm the following:

- Construction and equipment are in accordance with design specifications
- Safety, operating, maintenance, and emergency procedures are in place and are adequate
- A process hazard analysis has been performed for new and modified facilities and recommendations have been resolved or implemented before startup, and modified facilities meet the management of change requirements delineated in this lesson
- Training of each employee involved in operating a process has been completed

Mechanical Integrity

OSHA believes that the mechanical integrity of the critical process equipment is assured through proper installation, design, and operation. PSM mechanical integrity requirements apply to the following equipment:

- Pressure vessels and storage tanks
- Piping systems (including piping components such as valves)
- Relief and vent systems and devices
- Emergency shutdown systems

- Controls (including monitoring devices and sensors, alarms, and interlocks)
- Pumps

To maintain the ongoing integrity of the process equipment, the operating employer must establish and implement written procedures. Employees responsible for maintaining the integrity of process equipment should be trained in an overview of the process and the procedures applicable to the employee's job tasks.

Note: Employees must be well aware of the hazards involved.

Good Engineering Practices

Procedures that follow recognized and accepted good engineering practices should be used for the inspection and testing of process equipment. Inspections and tests must be scheduled in accordance with the manufacturer's recommendations, good engineering practices, and prior operating experience. Each test and inspection must be documented. The documentation should include:

- The date on which the inspection or test was conducted
- The inspector's or tester's name
- The serial number or any other identifier of the equipment on which the inspection or test was conducted
- A description of the inspection or test
- The results of the inspection or test

Equipment Deficiencies

Equipment deficiencies outside the acceptable limits defined by the process safety information must be corrected before further use. As long as the deficiencies are corrected in a safe and timely manner, they may not necessarily need to be corrected before further use as long as other sufficient and necessary safety measures are in place.

When constructing new plants and equipment, the operating employer must assure that the equipment as it is fabricated is suitable for the desired process. Checks and inspections must be performed to assure proper installation and consistency with both design specifications and manufacturer's instructions.

Note: The suitability of maintenance material, spare parts, and the equipment for which they will be used must be ensured by the employer.

Hot Work Permit

Hot work permits are essential for hot work operations conducted on or near a covered process. Before starting hot work operations, the implementation of fire prevention and protection requirements must be documented on the hot work permit. The date(s) authorized for hot work and the object on which the hot work is to be performed must also be included on the permit and the permit must be kept on file until hot work is completed.

Case Study

An Explosion and Fire Erupted in the Storage Tanks

In a product refinery, newly hired and old workers were performing their regular tasks. No warning signs or symbols were present on the highly hazardous chemical's storage tanks. Two workers who recently joined the refinery were welding pipes near the storage tanks. An explosion and fire erupted in storage tanks containing flammable hydrocarbons and wastewater.

Three workers were killed on the spot and seven were wounded seriously. Three workers died later as a result of their injuries.

Reasons

- Warning signs/labels were not present on the tanks' storage area
- Workers did not know that tanks contained hazardous materials
- Workers did not tell them (the victims) not to weld in the hazardous chemical storage area
- The workers were new and did not receive any safety training or education

Management of Change

There is nothing permanent except change; therefore, written procedures to manage changes, except for "replacements in kind" (e.g., replacement of defective equipment with same type which does not affect the operation or safety of the process) to process chemicals, equipment, and technology, or any other changes that could affect a covered process, must be established and implemented.

These written procedures must assure that the following considerations are addressed prior to any change:

- The technical basis for the proposed change
- Impact of the change on employee safety and health
- Modifications to operating procedures
- Necessary time period for the change
- Authorization requirements for the proposed change

Those employees who are involved in an operating process, or those whose tasks will be affected by a change in the process, need to be informed and trained before implementing any change in the process. In addition, process safety information and operating procedures or practices must be updated in accordance with the process change and OSHA requirements.

Emergency Planning and Response

Emergency pre-planning and training are vital to avoid catastrophes in a workplace; they enable employees to execute proper actions in case of emergencies. An emergency action plan must be developed and implemented in accordance with the OSHA standards for the entire plant. Furthermore, the emergency action plan must include the procedure for handling small releases of hazardous chemicals.

Incident Investigation

Incident investigation standards require the employer to investigate as soon as possible (but no later than 48 hours after) incidents that did result or could have resulted in catastrophic releases of covered chemicals. The investigation team must include at least one person who is knowledgeable in the process (including a contract employee, if appropriate) to investigate, analyze, and develop a written report on the incident.

An investigation report must include the following points:

- Date of incident
- Date investigation began
- Description of the incident
- Factors that contributed to the incident
- Recommendations resulting from the investigation

Employers must address and document their responses to report findings and recommendations and review findings with affected employees and contractor employees.

Note: Incident investigation reports must be retained for five years.

Compliance Audit

In order to assess whether process safety management standards are adequate and being followed, employers must certify at least every three years that they have carefully applied all the applicable provisions of the PSM standards. The compliance audit must be conducted by at least one person with knowledge of the process and standards requirement. Furthermore, it is the auditor's responsibility to prepare a report based on the findings, noting any deficiencies, and the employer's responsibility to determine and document any responses to the findings and to document any deficiencies that have been rectified.

More Information: Remember! The last two compliance audit reports must be kept on record.

Trade Secrets

It is important that employers provide all information necessary for compliance to the persons responsible for:

- Compiling the process safety information
- Developing the operating procedures
- Developing the process hazard analysis
- Performing incident investigations
- Emergency planning and response
- Compliance audits

Note: Employers must provide all the information without regard to possible trade secret status of such information. However, they may require employees to sign confidentiality agreements to protect those trade secrets.

Lesson Summary

In order to assess whether process safety management standards are adequate and being followed, employers must certify at least every three years that they have carefully applied all the applicable provisions of the PSM standards.

The compliance audit must be conducted by at least one person with knowledge of the process and standards requirement. Furthermore, it is the auditor's responsibility to prepare a report based on the findings, noting any deficiencies, and the employer's responsibility to determine and document any responses to the findings and any deficiencies that have been rectified. Bear in mind that PSM applies to contractors performing maintenance or repair, turnaround, major renovation, or specialty work on or adjacent to a covered process.

Module 21: Asbestos in Workplace

Module Description

Asbestos is a substance that has been used for centuries. Its heat-resistant properties make it almost indestructible; due to this property, asbestos has been widely used in the construction industry, including pipe and boiler insulation, flooring and ceiling tiles, drywall, adhesives, and much more. Asbestos has also been widely used in products such as vehicle brakes, wire insulation, dryers, and much more. Before 1973, asbestos was sprayed onto different surfaces for fire protection purposes, but this practice was banned due to its hazardous nature. Furthermore, it is no longer legal to be used for insulating pipes and boilers in most countries, including the United States.

However, asbestos can still be found in many buildings. Those who work in construction, repair, demolition, and renovations are at a greater risk of contracting asbestos-related diseases such as asbestosis, lung cancer, and mesothelioma. Very stringent exposure guidelines have been established for asbestos.

This module introduces students to the hazards of asbestos in the workplace and provides information about the measures that must be taken in order to minimize the effects of exposure to asbestos.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Describe the hazards of asbestos in the workplace.
- Discuss the health effects of exposure to asbestos.
- Identify and assess asbestos-related work activities.
- Apply control measures for low-risk, moderate-risk, and high-risk activities.
- Relate the proper disposal of asbestos containing material.
- Select appropriate protective equipment.

Lesson 1: Asbestos in the Workplace

Lesson Topics

This lesson focuses on the following topics:

- What is Asbestos?
- Uses of Asbestos
- Health Hazards of Exposure to Asbestos
- Asbestos-related Diseases

What is Asbestos?

Asbestos is a group of naturally occurring mineral silicates whose crystals form long, thin fibers. The three types of asbestos that were commonly used in building materials are Chrysotile, Amosite, and Crocidolite.

Chrysotile

Chrysotile is the most commonly used type of asbestos. This type of asbestos is commonly called "white asbestos."

Amosite

Amosite, commonly referred to as "brown asbestos," is the second most common type of asbestos and accounts for approximately four to six percent of the asbestos used in building materials in the United States.

Crocidolite

Crocidolite is the least common type of asbestos and accounts for only about one to four percent of the asbestos materials used in the United States. Crocidolite is commonly known as "blue asbestos."

Why is Asbestos a Hazard?

Asbestos has been determined to be a hazardous substance because its fiber masses break easily into tiny particles that can easily become airborne. These fibers, when inhaled or swallowed, can cause serious health problems.

Uses of Asbestos

Asbestos has many properties that make it ideal for use in a variety of products. It is strong, fire-resistant, a poor conductor of heat and electricity, corrosion-resistant, and found in abundance. Those products that are made with asbestos are called asbestos-containing materials (ACM).

By the end of the 19th century, and until around 1980, asbestos was widely used in the construction industry for the manufacture of building material. Building materials that contain asbestos are known as asbestos-containing building material (ACBM).

The most common uses of ACBM include:

- **Insulation material:** This was usually applied with spray, trowel, or manually installed on many surfaces to insulate them from heat and cold.
- **Fireproofing material:** This material was applied with a spray to steel beams that were used in construction of multi-story buildings. This helped to protect the structural members from damage caused by heat in case there was a fire.
- **Acoustical or soundproofing material:** Apart from soundproofing, this material was also used for decoration. A mixture of asbestos and some other materials was sprayed onto ceilings and walls to create a textured look.
- **Miscellaneous materials:** Asbestos has been used in the making of many products, including roofing felts, roofing shingles, floor tiles, vehicle brakes, combustion vents, exterior siding and wallboard, and flues for waste gases and heat.

The asbestos fibers in these products are tightly woven into the materials and is easily released if the material is damaged (by a mechanical process such as drilling, grinding, cutting, or sanding). The asbestos present in roofing shingles and siding that is exposed to weathering can deteriorate gradually and release asbestos fibers in the air.

General industry employees may be exposed to asbestos during the manufacture of asbestos-containing products or when performing brake and clutch repairs.

The potential for a product containing asbestos to release breathable fibers depends, in part, on its degree of friability. Friable means that the material can be crumbled or reduced to dust-size particles likely to emit fibers.

The fibrous or fluffy sprayed-on materials used for fireproofing, insulation, or sound proofing are considered to be friable, and they may readily release airborne fibers if disturbed.

Materials such as vinyl-asbestos floor tile or roofing felts are considered nonfriable and generally do not emit airborne fibers unless subjected to damage, polishing, grinding, sanding, and similar operations.

Asbestos-cement pipe or sheet can emit airborne fibers if the materials are cut or sawed, or if they are broken.

Health Hazards of Exposure to Asbestos

Many studies have proven that any prolonged or excessive exposure to asbestos is likely to have adverse health effects. Inhalation of asbestos fibers may cause several lung diseases. It has been demonstrated that the greater the exposure to asbestos fiber, the greater the risk of developing an illness.

If someone is exposed to brief bursts of asbestos fibers in addition to those present in the background, he or she is at a greater risk of contracting an asbestos-related disease.

Construction workers who work on and disturb asbestos-containing materials in buildings are at a greater risk of coming in contact with additional asbestos fibers.

The Respiratory System

Exposure to asbestos fibers often adversely affects the lungs. There are several mechanisms in the human body that are used to "filter" the air it breathes. In the nose and mouth, many large particles from the air are removed. Then it is passed to the airway tubes that are covered with mucus where smaller particles may get trapped. Tiny hair-like cells present in the airway tubes then push these particles upwards into the nose or the back of the mouth. From there they can either be swallowed or expelled.

However, the effectiveness of these hair-like cells may be greatly impaired when a person smokes. This may render the body susceptible to unwanted dust or fibers that may be present in the air.

Dust particles too small or too numerous to be captured and expelled through the body's normal protective mechanisms may enter the tiny air sacs in the lungs where respiration occurs. A smoker is likely more susceptible to this than a non-smoker due to the compromised filtration system. The human immune system then releases large cells called macrophages that attempt to digest the dust particles. This is another one of the body's defense mechanisms against unwanted particles in the lungs.

Asbestos-related Diseases

The risk of developing an asbestos related disease increases greatly when the body's natural defenses are not able to control or remove asbestos fibers that enter the lungs. The three main factors that determine the likelihood of developing an asbestos related disease include smoking, the amount and duration of the exposure, and age. Asbestos-related diseases include asbestosis, lung cancer, mesothelioma, and some other cancers.

From the state of Colorado:

- Asbestosis is a lung disease that causes scarring of the lungs. Eventually, this scarring may become so severe that the lungs cannot adequately function. The latency period (meaning the time it takes for the disease to become developed) can be 15-40 years.
- Mesothelioma is a cancer of the lining of the lung and chest and/or the lining of the abdominal wall. Asbestos exposure is the leading cause of this type of cancer. The latency period for mesothelioma is often 15-50 years.
- Lung cancer can be caused by asbestos. The effects of lung cancer are often greatly increased by cigarette smoking. Cancer of the gastrointestinal tract can also be caused by asbestos. The latency period for cancer is often 15-30 years.

From EPA:

- Asbestosis is a serious, progressive, long-term non-cancer disease of the lungs. It is caused by inhaling asbestos fibers that irritate lung tissues and cause the tissues to scar. The scarring makes it hard for oxygen to get into the blood. Symptoms of asbestosis include shortness of breath and a dry, crackling sound in the lungs while inhaling. There is no effective treatment for asbestosis.
- Lung cancer causes the largest number of deaths related to asbestos exposure. People who work in the mining, milling, manufacturing of asbestos, and those who use asbestos and its products are more likely to develop lung cancer than the general population. The most common symptoms of lung cancer are coughing and a change in breathing. Other symptoms include shortness of breath, persistent chest pains, hoarseness, and anemia.
- Mesothelioma is a rare form of cancer that is found in the thin lining (membrane) of the lung, chest, abdomen, and heart and almost all cases are linked to exposure to asbestos. This disease may not show up until many years after asbestos exposure. This is one of the reasons that great efforts are being made to prevent school children from being exposed.

Asbestosis

Asbestosis is a serious disease that causes severe scarring of the lungs and reduces lung elasticity. Due to this, breathing becomes very difficult. Workers who have been exposed to asbestos fibers for long periods of time often manifest the symptoms of this disease. This disease can lead to disability or even death.

All types of asbestos can cause asbestosis. Just like all other diseases that are associated with asbestos exposure, it may take many years for the disease to show up. Typically, asbestosis has a latency period of 15 to 40 years.

Mesothelioma

Mesothelioma is a type of cancer that affects the abdominal lining or the chest cavity lining. It causes great damage to the vital functions of the body and always results in death. Mesothelioma occurs in people who have been exposed to asbestos for a limited period of time. This type of cancer does not seem to occur due to cigarette smoking; rather, it depends on the amount of asbestos inhaled and the period of exposure. Mesothelioma has a latency period of 15 to 50 years.

Lung Cancer

Lung cancer occurs due to long periods of exposure to asbestos. The risk of contracting this disease increases if the worker smokes cigarettes. In fact, workers who are cigarette smokers are 50% more likely to develop lung cancer than those who do not smoke. Asbestos-related lung cancer has a latency period of 15 to 30 years.

Lesson Summary

If someone is exposed to brief bursts of asbestos fibers in addition to those present in the air, he or she is at a greater risk of contracting an asbestos-related disease.

It has been demonstrated that the greater the exposure to asbestos fiber, the greater the risk of developing an illness.

Asbestos-related diseases include asbestosis, lung cancer, and mesothelioma. Workers who have been exposed to asbestos fibers for long periods of time often manifest the symptoms of asbestosis; this disease can lead to disability or even death. Mesothelioma, on the other hand, may occur in people who have been exposed to asbestos for a limited period of time.

Lesson 2: Protection Against Asbestos

Lesson Topics

This lesson focuses on the following topics:

- Introduction
- Engineering Controls and Safe Work Procedures
- Personal Protective Equipment

Introduction

Compliance Program

Where the Time-Weighted Average (TWA) and/or excursion limit is exceeded, the employer shall establish and implement a written program to reduce employee exposure to or below the TWA and to or below the excursion limit. This is done by means of engineering and work practice controls and by the use of respiratory protection where required or permitted. Such programs shall be reviewed and updated as necessary to reflect significant changes in the status of the employer's compliance program.

The employer shall inform all employees concerning the availability of self-help smoking cessation program material. The employer shall also institute engineering controls and work practices to reduce and maintain employee exposure to or below the TWA and/or excursion limit except to the extent that such controls are not feasible. When they are not sufficient, they will be supplemented by appropriate respiratory protection. Employers must provide respirator training and medical clearance to use respirators.

For any employee exposed to airborne concentrations of asbestos that exceed a Permissible Exposure Limit (PEL), employers must provide and require the use of protective clothing, such as coveralls or similar full-body clothing, head coverings, gloves, and foot coverings.

As well, employers must provide face shields, appropriate eye and face protection, or other appropriate protective equipment wherever the possibility of eye irritation exists and require workers to wear it.

Employers must establish decontamination areas and hygiene practices for employees exposed above a PEL. In addition, employees may not smoke in work areas that might expose them to asbestos. In general industry, employers must provide medical examinations for workers who are exposed above a PEL.

Sign Specifications

The warning signs should indicate the following information:

- Danger
- Asbestos
- Cancer and Lung Disease
- Hazard
- Authorized Personnel Only

In addition, where the use of respirators and protective clothing is required in the regulated area, the warning signs shall include "Respirators and protective clothing are required in this area."

Warning Labels

Warning labels shall be affixed to all raw materials, mixtures, scrap, waste, debris, and other products containing asbestos fibers, or to their containers.

The labels shall comply with the requirements of OSHA's Hazard Communication standard, and shall include the following information:

- Danger
- Contains Asbestos Fibers

- Avoid Creating Dust
- Cancer and Lung Disease Hazard

All hand-operated and power-operated tools that would produce or release fibers of asbestos, such as but not limited to saws, scorers, abrasive wheels, and drills shall be provided with local exhaust ventilation systems.

Local exhaust ventilation and dust collection systems shall be designed, constructed, installed, and maintained in accordance with good practices of ANSI Z9.2-1979.

Insofar as practicable, asbestos shall be handled, mixed, applied, removed, cut, scored, or otherwise worked in a wet state sufficient to prevent the emission of airborne fibers.

Engineering Controls and Safe Work Procedures

The employer shall not use employee rotation as a means of compliance with the TWA and/or excursion limit.

Permissible Exposure

Exposure to airborne asbestos fibers may not exceed 0.1 fibers per cubic centimeter of air (0.1 f/cc) averaged over the 8-hour workday.

Excursion Limit

The employer shall ensure that no employee is exposed to an airborne concentration of asbestos in excess of 1.0 fiber per cubic centimeter of air (1 f/cc) as averaged over a sampling period of thirty (30) minutes.

In general industry, employers must perform initial monitoring for workers who may be exposed above a PEL or above the excursion limit.

Employers must conduct subsequent monitoring at reasonable intervals, and in no case at intervals greater than six months for employees exposed above a PEL.

More Information: In general industry, employers must establish regulated areas wherever asbestos concentrations may exceed a PEL, controlled zones known as regulated areas that are designed to protect employees where certain work with asbestos is performed and limit access to these areas to authorized persons who are wearing appropriate respiratory protection and who have been properly trained. Employers must also prohibit eating, smoking, drinking, chewing tobacco or gum, and applying cosmetics in these areas, and they must display warning signs at each regulated area.

Low-risk Work Activities

Low-risk work activities are those that do not pose a great health and safety threat to workers. Work that has to be done near friable asbestos-containing material without disturbing it is categorized as low-risk work. Moving asbestos containing material that is in clean and sealed bags is also considered a low-risk task. Workers who have to carry out such tasks must be informed about the hazards of asbestos.

Employers are required to identify and clearly label all areas where asbestos-containing materials are present. Employers must also ensure that all workers are properly trained to handle asbestos-containing materials when workers are required to perform this work.

Moderate-risk Work Activities

There are many asbestos-related work tasks that are categorized as moderate-risk activities. These tasks require workers to follow specific procedures in order to ensure safety.

Activities that carry a moderate risk of exposure to airborne asbestos fibers may include:

- Cutting, drilling, grinding, shaping, or removing non-friable manufactured products that contain asbestos with hard tools.
- Using a coarse disc to buff floors.
- Drilling non-friable asbestos-containing materials.
- Taking out mounting screws from cement products containing asbestos.
- Analyzing asbestos samples in a laboratory.
- Collecting samples of materials that might contain asbestos for laboratory analysis.
- False ceiling removal to get to a work area when there are friable asbestos-containing materials found on the surface.
- Taking apart drywall that contains asbestos in the joint-filling material.
- Carrying out a procedure that does not create any friable asbestos waste material such as removing vinyl-asbestos floor coverings.
- Destroying a block wall that contains asbestos fragments in the cavity.
- After completion of the asbestos removal process, taking apart the treated enclosure.
- Installing or removing a glove-bag apparatus in order to remove pipe insulation when the insulation is in good condition.
- Removing asbestos insulation from piping using prefabricated glove-bags.

Measures for Moderate-risk Activities

Workers who are involved in a moderate-risk task must follow specific work procedures. They must make sure that all those present at or near the work area are not exposed to the asbestos fibers in the air. Workers must place necessary enclosures, barricades, or similar structures around the work area so that everyone knows the exact location of the designated work area. They must warn unauthorized people from entering the work area by placing signs around it.

Workers are required to put on suitable protective equipment and clothing. The material of the clothing must not allow penetration of asbestos fibers. Torn or defective clothing must be replaced immediately. A respirator with an appropriate filter must also be worn.

All dust and waste produced during work must be wiped; compressed air must never be used to clean clothing or work surfaces. By using polyethylene sheets, the spread of asbestos dust to other parts of the work area can be controlled by sealing doorways, windows, or any other openings. If a worker suspects the presence of friable asbestos on any surface in the work area, he or she must wipe it off with a damp cloth and take necessary steps to identify the source and correct any deficiencies that may have allowed the dust to be present.

All sealable containers or bags containing asbestos waste must be labeled properly to identify the contents, the associated hazards, and the precautionary measures required in handling the substance. These bags and containers must not be left in the work area and should be removed after each work shift. Workers must make sure to wipe the surface of each container or bag before it is removed from the work area.

After completion of the tasks, all polyethylene sheets and work area barriers should be sprinkled with water or dust otherwise properly controlled. They should then be folded carefully in order to contain any remaining dust. After that, they should be placed in bags or containers and properly sealed. The bags should be disposed of as asbestos waste.

Before leaving the work area, workers must use proper materials to clean all protective equipment and clothing before taking it outside the work area. All protective clothing must be left inside the work area for cleaning. If there is no designated storage area within the work area, the clothing must be placed in a sealable bag or container and disposed of as

asbestos waste. The encapsulation of asbestos-containing materials requires covering with a sealant or glue to prevent the release of fibers.

After the work has been completed, a document must be presented to the employer stating that the work area is safe to enter by unprotected personnel.

High-risk Work Activities

Those activities that are categorized as high-risk work activities must be carried out very carefully, following certain specific procedures in order to protect workers and other personnel who may be otherwise affected by them.

Some of the tasks that may pose a high risk of exposure to asbestos fibers in the air include:

- Maintaining, cleaning, or removing air-handling equipment in structures and buildings where sprayed asbestos-containing fireproofing material is used.
- Encapsulating, removing, or enclosing friable asbestos-containing materials while repairing, altering, maintaining, or demolishing a building or a structure.
- Repairing, dismantling, demolishing, or altering any structure, building, or device that has asbestos-containing material.
- Any task where large amounts of friable asbestos fibers can become airborne.

Measures for High-risk Activities

High-risk activities require special measures for specific tasks including isolating the work area, decontaminating workers, controlling airborne asbestos fibers, disposing of asbestos containing material, and collecting samples of materials that may contain asbestos.

Isolating the Work Area

All designated work areas must be isolated in order to prevent unauthorized personnel from entering. In order to isolate a work area, signs should be placed around it to warn unauthorized people against entering. The work area must be enclosed with proper polyethylene sheets in order to contain the asbestos fibers. If the work area cannot be enclosed, a notice must be provided to all workers, specifying alternative work procedures that can control the risk of asbestos exposure.

In order for the contaminated air to remain in the enclosed area or filtered prior to release, it must be ensured that the air pressure in the work area is lower than the surrounding area. This low pressure should be maintained until all work has been completed. Stairways and elevators must be sealed off using polyethylene sheets and tape. The air heating and ventilation system in the work area must be shut down and all ducts should be sealed off with polyethylene sheets.

All devices and fixtures that can be removed should be wiped with damp cloths and removed from the contaminated area. All non-removable devices and fixtures should be covered with polyethylene sheets. The floor of the work area must be covered with polyethylene sheets and extended at least 12 inches up the wall. All enclosures must be inspected on a regular basis in order to ensure that there are no breaks, tears, or leaks.

Decontaminating Workers

In order to ensure that workers do not carry any contaminants outside the work area, they must be decontaminated in a designated space inside the work area. Before entering the contaminated work area, workers should remove their street clothing in a clean room and put on clean work clothes. They must make sure that their respirator fits properly. Workers are required to enter the personnel transfer room and put on the required personal protective equipment. They can then proceed towards the contaminated work area to carry out the assigned tasks.

After leaving the contaminated work area, workers should remove the asbestos fibers from personal protective equipment using a proper system. They must remove all protective clothing and equipment in the personnel transfer room. Any disposable clothing must be placed in polyethylene bags and disposed of properly.

When the respirator and its harness are free from all contaminants, these should then be removed. The workers can then enter the clean room to put on their street clothes. The respirator should be cleaned and disinfected and stored in the clean room until it is required again. All tools and equipment must be stored in the equipment holding room.

Controlling Airborne Asbestos Fibers

In order to properly control the asbestos fibers in the air, certain rules must be followed. Materials containing asbestos should only be removed after getting an approval from authorized personnel. Workers must never use dry clothes or compressed air to clean asbestos-containing materials.

After all the asbestos-containing materials have been removed, all exposed surfaces in the work area must be properly cleaned.

Employers must identify all asbestos material that is encapsulated and must ensure that it is firmly bound by the encapsulating material so that there are no loose fibers. Workers must ensure that encapsulating material does not disturb the bond between asbestos-containing material and its supporting surface.

Workers must make sure that they disturb or displace the least amount of asbestos-containing material possible while repairing it. After performing this task, they should seal the exposed asbestos and then remove the work enclosure. All surfaces that are exposed must be washed thoroughly and properly protected.

Disposing of Asbestos Waste Materials

To ensure that asbestos waste is properly disposed of, all waste materials should be placed in leak-proof containers while remaining inside the enclosed area. These containers should then be sealed and labeled. The outside surface of all containers must be decontaminated before they are removed from the storage room and the waste transfer room.

In order to prevent the containers from getting damaged or torn while being transported to the disposal site, they must be properly packaged using double polyethylene bags.

Employers must ensure that the asbestos-containing waste material is disposed of immediately or after each work shift. All arrangements must be made in advance with the concerned authorities to transport the waste to assigned dumpsters. It is very important to inform the transport drivers about the hazards of asbestos and the appropriate measures they must take. Signs must be placed on the transport vehicles displaying the nature of the materials being transported.

By washing and vacuuming, one must then decontaminate the enclosure and all the equipment. Finally, the air inside the enclosure must be decontaminated before taking apart the enclosure.

Collecting Samples of Materials that May Contain Asbestos

Only a qualified person may collect samples of materials that may contain asbestos. These samples must then be sent immediately to a laboratory where their contents can be determined.

In order to collect samples, a qualified person must take measures recommended for moderate-risk activities. Before starting work, it must be ensured that there are no unprotected workers present in the work area. The materials should be prepared to ensure that no asbestos fibers will be released during the collection process, and the workers must disturb the least amount of material possible.

A sample should be taken by deep penetration into a material to ensure that the sample is valid. The sample collector must observe the color and the texture of the various materials present in the work location in order to collect samples of all those present. These should then be placed in leak-proof containers and properly sealed. Labels must also be placed on the containers to identify the contents as laboratory samples.

To collect any loose material that may break off while sampling, the floor of the work area should be covered with polyethylene sheets when necessary. After completion of the work, these sheets must be placed in leak-proof containers and disposed of properly.

All workers present near the sampling area must be equipped with a respirator as sampling can create airborne asbestos fibers that could be injurious to health.

Personal Protective Equipment

Protective Clothing

Personal protective equipment includes all asbestos protective clothing such as coveralls or any other full-body clothing, hard hats, gloves, rubber boots without laces, and eye protection.

All workers who are required to perform asbestos-related tasks must wear the appropriate protective equipment. Protective clothing should be made with a material that does not allow asbestos fibers to penetrate. The protective clothing should cover the whole body, and it should fit comfortably at the neck, wrists, and ankles. Headgear and proper boots that can resist the penetration of asbestos fibers must be worn. If protective clothing gets damaged or torn, it must be replaced immediately.

It must be ensured that workers do not exit the work area while wearing their protective clothing. All contaminated clothing and equipment must be taken off and kept in the equipment room or work area so that the contaminants do not enter the clean area.

Respiratory Protection

Many types of respirators provide protection from asbestos dust and fibers. Usually a respirator is selected according to the nature of the work and the amount of asbestos that may be produced.

Before entering the work area, workers must put on their respirators, make sure that they fit properly, and check that there are no gaps from which asbestos fibers may enter.

Types of Respirators

Respiratory protection against asbestos can be obtained through two basic types of respirators: air-purifying respirators and atmosphere-supplying respirators.

Air-purifying Respirator

A respirator with an air-purifying filter, cartridge, or canister is called an air-purifying respirator. This respirator removes asbestos contaminants from the air by passing it through an air-purifying component, making it acceptable to breathe normally.

Atmosphere-supplying Respirator

An atmosphere-supplying respirator consists of a component that provides the user with breathable air that is not taken from the ambient atmosphere. There are two types of atmosphere-supplying respirators: supplied-air respirators (SARs) and self-contained breathing apparatus (SCBA) units.

- **Supplied-air respirators (SAR):** It uses a hose called an airline to provide clean air from an air tank. There are two types of supplied-air respirators: pressure demand respirators and continuous-flow respirators.

Pressure-demand respirators prevent the contaminated air from entering the face-piece by maintaining positive pressure. Continuous-flow respirators also maintain positive pressure by constantly supplying fresh air to the face-piece.

- **Self-contained Breathing Apparatus (SCBA) Unit:** It consists of a hose that is connected to a cylinder of compressed air. Activities that carry a risk of exposure to asbestos do not usually require the use of SCBAs. However, in some cases, pressure-demand SCBAs are used, which supply air to the face-piece if the inside pressure reduces due to inhalation or leakage.

Employee Notification of Monitoring Results

The employer must, within 15 working days after the receipt of the results of any monitoring performed, notify each affected employee of these results either individually in writing or by posting the results in an appropriate location that is accessible to affected employees.

The written notification shall contain the corrective action being taken by the employer to reduce employee exposure to or below the TWA and/or excursion limit, wherever monitoring results indicated that the TWA and/or excursion limit had been exceeded.

You must keep accurate records of the following:

- All measurements taken to monitor employee exposure to asbestos—30 years.
- Medical records, including physician's written opinions—duration of the employee's employment plus 30 years.
- Training records—1 year beyond the last date of employment.

Lesson Summary

All workers who are required to perform asbestos-related tasks must wear the appropriate protective equipment.

Protective clothing should be made with a material that does not allow asbestos fibers to penetrate. The protective clothing should cover the whole body, and should fit comfortably at the neck, wrists, and ankles. Headgear and boots that resist the penetration of asbestos fibers must also be worn, and if protective clothing gets damaged or torn, it must be replaced immediately.

To collect any loose material that may break off while sampling, the floor of the work area may need to be covered with polyethylene sheets. After completion of the work, these sheets must be placed in leak-proof containers and disposed of properly. Only a qualified person may collect samples of materials that may contain asbestos. These samples must then be sent immediately to a laboratory where their contents can be determined.

Workers must disturb the least amount of material possible. By washing and vacuuming, one must then decontaminate the enclosure and all of the equipment. Next, all exposed surfaces from which asbestos has been removed should be sealed or protected. Finally, the air inside the enclosure must be decontaminated before taking apart the enclosure.

In order to prevent the containers from getting damaged or torn while being transported to the disposal site, they must be properly packaged using double polyethylene bags. All permanent enclosures around asbestos-containing materials must be airtight, and warning signs must be put up inside enclosures to warn workers about the dangers of asbestos and any protective measures that must be taken.

Workers must never use dry clothes or compressed air to clean asbestos-containing materials. Also, in order for the contaminated air to remain in the enclosed area or filtered prior to release, it must be ensured that the air pressure in the work area is lower than the surrounding area. This low pressure should be maintained until all work has been completed. All designated work areas must be isolated in order to prevent unauthorized personnel from entering. In order to isolate a work area, signs should be placed around it to warn unauthorized persons against entering.

Module 22: Scaffolds

Module Description

This module provides a general overview of the safety measures that are required when working on a scaffold. The module begins with an introduction into the various types of scaffolds and goes on to outline the OSHA safety requirements and safety measures that can be taken to ensure that employees working on scaffolds are safe.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Identify the various classifications of scaffolds.
- Discuss the basic OSHA safety requirements.
- Describe the measures that can be taken to ensure safe working conditions on a scaffold.

Lesson 1: Introduction to Scaffolds

Lesson Focus

This lesson focuses on the following topics:

- What is a Scaffold?
- Types of Scaffolds

What is a Scaffold?

A scaffold is a term used to describe any sort of temporary elevated platform that is used to support either workers, materials, or both. They are commonly used in the construction field (usually in the construction of buildings); however, they are also used in other fields such as ship construction and by cleaning services (to clean the outer windows of high rises).

The main purpose behind the use of scaffolds is to provide support and balance to an employee and his or her materials as the employee conducts tasks in otherwise inaccessible or difficult-to-reach areas.

Types of Scaffolds

Suspended Scaffolds

These are types of scaffolds that are suspended from a fixed overhead position (often placed at the top of a building, but it can be any fixed elevated structure). The various types of suspended scaffolds are discussed next.



Two-Point (Swing Stage)

These are one of the most common types of suspended scaffolds. You see them frequently on construction sites and on skyscrapers and other high-rises where they are used by window washers and exterior building workers. The scaffold is hung from the top of a building by two ropes that are secured at either end of the platform, usually with stirrups to provide for vertical movement.

Single-Point Adjustable (Spider Scaffolds)

This scaffold is smaller than the two-point, as it usually accommodates only one person at a time. It is secured to the top of the building using one rope from a fixed overhead support. As with the two-point, it has a mechanism to allow for vertical movement (up and down). Again, it is often used by window washers.

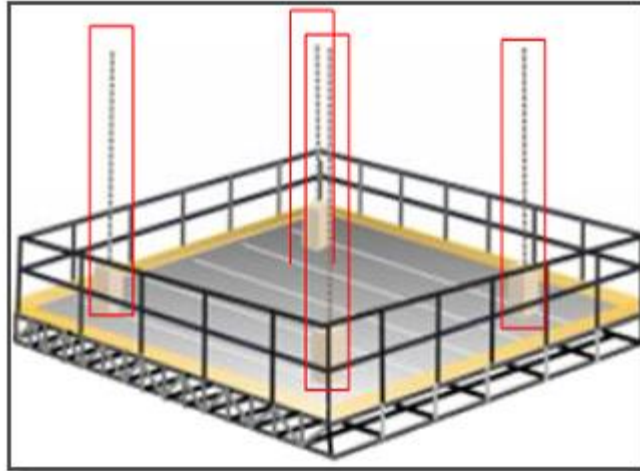


Suspended Platform

This scaffold consists of horizontal and parallel ropes that are used to support a platform on which personnel and/or materials are placed. This is hung from vertical ropes suspended from a structure. These are often used in buildings still under construction.

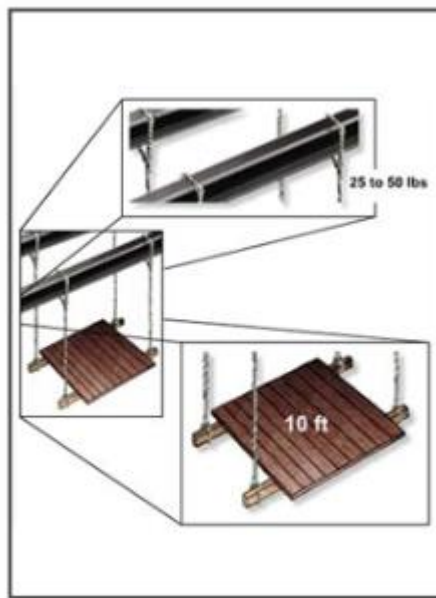
Suspended Scaffold Platform

These consist of at least one platform (they may be more, but they rarely exceed a single platform) that is supported by more than two ropes (usually four ropes are used, one secured at each corner of the platform) from an overhead structure and fitted with a mechanism to allow for vertical movement. They are commonly used as chimney hoists in the cleaning and repair of chimney and/or other ventilation shafts.



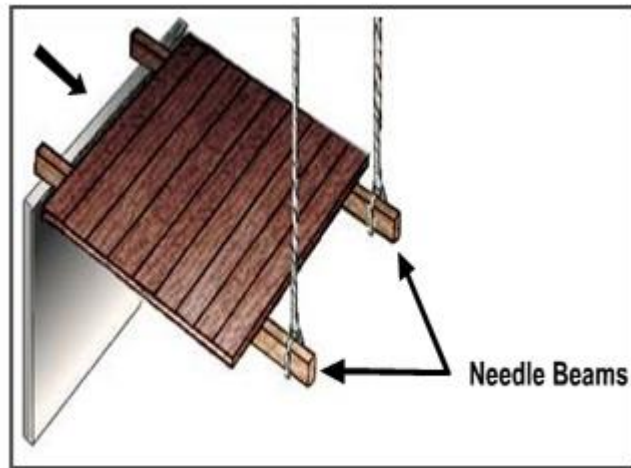
Interior Hung

This scaffold consists of two parallel, horizontal beams upon which a board or plank is placed. It is then suspended by ropes (sometimes up to four, one at each corner of the platform) from a roof or ceiling. This is primarily used indoors, hence the name, because it must be hung from roof structures.



Needle Beam

In this case, the platform (usually a simple wooden board) is supported by two needle beams. The beams are hung from ropes. Usually one end of the platform is supported on the edge of a permanent structure.



Multi-Level

A multi-level scaffold is a series of platforms that are arranged vertically, one on top of the other. Visually it looks like two or more two-point scaffolds (or multi-point scaffolds) on top of one another. The platforms are placed within the same framework, secured by a single stirrup from the same fixed overhead.



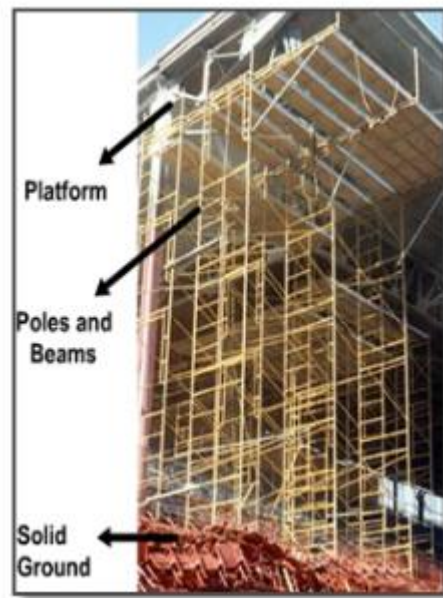
Float (Ship)

The float (or ship) scaffold consists of a platform supported by two bearers and is hung from a fixed overhead support with ropes.



Supported Scaffolds

These are scaffolds that consist of one or more platforms elevated on poles and beams, which are placed upon a solid ground. The various types of supported scaffolds are discussed next.



Frame or Fabricated

This is one of the most common types of scaffolds used. They are easy to build, economical, and practical. They range in size from small set-ups used for one or two story residential houses to large set-ups used on commercial projects; however, it is possible to make them stand several stories high if the need arises. They consist of a single or multiple platforms elevated on tubes.



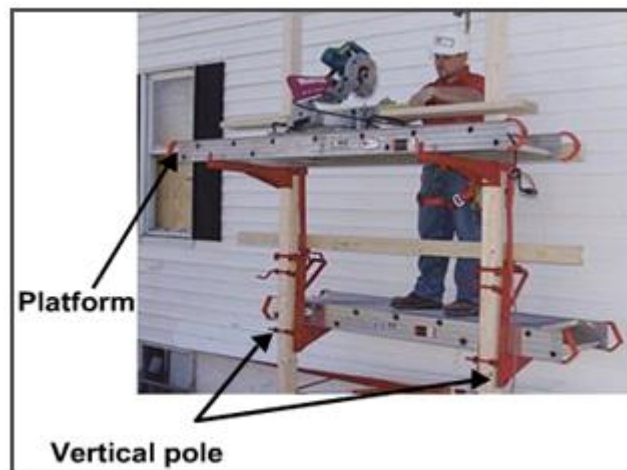
Mobile

Mobile scaffolds are visually similar to frame scaffolds, but with one main difference: they are mounted on wheels or casters. They are commonly used by painters, plasterers, electricians, plumbers, and others who need to change position frequently, hence the wheels.



Pump Jack Scaffold

This scaffold consists of a platform that is elevated on a pair of vertical poles and fitted with moveable brackets. The brackets allow the platform to be moved up and down using a jack (similar to those used to change a car tire). They are attractive because of their ability to be raised and lowered somewhat quickly and easily and because they are relatively inexpensive.



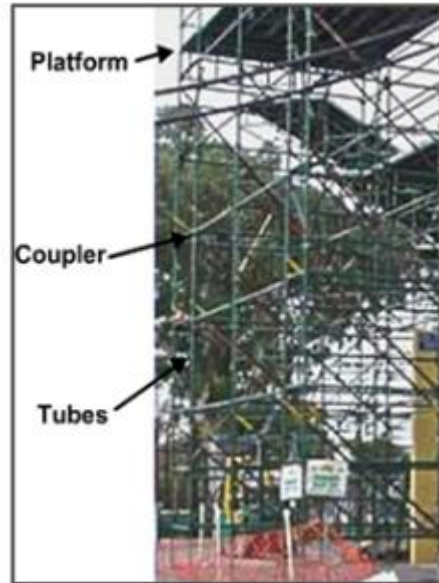
Ladder Jack

This is a very simple scaffold. In this case, the platform is placed in brackets that are attached to a pair of ladders. They are mostly used for light applications. They are popular mostly due to their practicality, ease, and economy.

Tube and Coupler

In this case, tubes (which look like pipes) are placed together in a lattice-like framework using couplers (joints) to hold them together. The lattice-like pattern into which they are arranged adds balance and strength to the framework. Platforms can be placed at the top of the framework and at multiple levels. Tube and coupler scaffolds are often used to carry heavy weights, and they can be built to a height of multiple stories while being able

to be erected to conform to the shape of a building. A prime advantage lies in that they can be built in several directions and combinations that cater to any structure.



Poles

This type of scaffold is also known as Wood Pole, because every part of the scaffold, from the platform on down, is built using wooden poles and boards. There are two types: single pole, which are supported on one side (the interior side) by a fixed structure, such as a building; and two-pole, which are supported by two sets of wooden poles on either side of the platform. They are not commonly used due to the fact that they must be built from scratch, cannot be reused, and are difficult to erect and use in a safe and compliant manner.



Specialty

There are a large number of scaffolds that are not widely used but are employed only in specific jobs. Some of the more common ones include the following:

- Plasterers', Decorators', and Large-Area Scaffolds
- Bricklayers' Square Scaffolds
- Horse Scaffolds
- Form Scaffolds and Carpenters' Bracket Scaffolds
- Roof Bracket Scaffolds
- Outrigger Scaffolds
- Window Jack Scaffolds
- Crawling Boards (Chicken Ladders)
- Step, Platform, and Trestle Ladder Scaffolds



Lesson Summary

A scaffold refers to a temporary, elevated platform that is used to support workers and/or materials. There are several types of scaffolds:

1. **Suspended scaffolds:** the type that are suspended by a wire rope from a fixed overhead position (usually placed at the top of a building). They can be broken down further into the subcategories that include: Two-point (or swing stage) scaffolds; single-point adjustable (spider) scaffolds; catenary (using horizontal and parallel wire ropes); multi-point adjustable; interior-hung; float (ship); and more.
2. **Supported scaffolds:** the type that consist of one or more platforms elevated on poles and beams, which are placed upon a solid ground. They can be broken down into the following subcategories: Frame or fabricated; mobile; pump jack; and more.
3. **Specialty scaffolds:** includes decorators', plasterers', and large-area scaffolds, among others.

Lesson 2: Overview of OSHA Directives for the Construction of Scaffolds

Lesson Focus

This lesson focuses on the following topics:

- Suspension Scaffolds
- Supported Scaffolds

Suspension Scaffolds

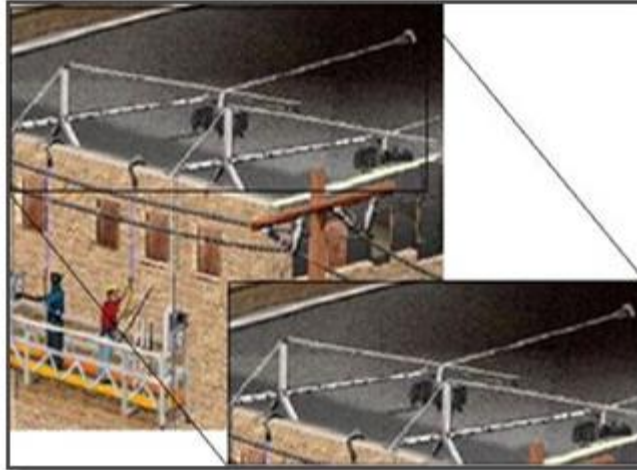
The regulations presented for the two-point scaffold are applicable to all other types of suspension scaffolds, unless stated otherwise.

Two-Point (Swing Stage)

Anchorage

When taking into account the safety of a suspension scaffold, a primary factor that must be considered is the anchorage. Because a suspension scaffold is, in effect, hanging from a structure, its weight must be sufficiently supported and balanced to ensure that it doesn't plummet to the ground. The anchorage that the tieback is secured to must be capable of withstanding four times the intended load of the scaffold.

As such, the tieback should not be secured to objects such as vents or piping but to structural members (foundation beams, for example).



Support

Suspension scaffolds are designed for vertical motion (that is up and down); as such, the scaffold must be capable of supporting its own weight as well as that of the personnel and materials on it while it is either stationary or in motion. The scaffold and its anchorage must be capable of handling up to four times the maximum intended load, and the support ropes must be capable of handling up to six times the intended maximum weight of the scaffold and its load. Scaffold components that have been manufactured by different manufacturers should not be mixed, unless specified. In addition, no modifications should be made to the scaffold components for any reason without approval from the manufacturer.

Access

Though it is possible for suspension scaffolds to be accessed using ladders, it is a preferred practice to access the scaffold from the roof. Safe access methods are discussed in more detail in the next lesson.

Fall Protection

Personnel working from suspension scaffolds may be a great distance above the ground; as such, a common accident is falls by workers from elevated positions.

Both fall arrest systems and guardrails should be provided on any scaffold that is expected to be elevated to more than ten feet above the ground. This will be discussed in more detail in the next lesson.

Platform

The platform is the part of the scaffold where the personnel work from and the materials are placed. It is clearly an important component of the scaffold and must be checked regularly to ensure that it meets the applicable safety standards. Each scaffold platform must be at least 18 inches wide. Also, a platform should be able to support its own weight in addition to four times the maximum allowed weight.

Note: The platform for a two-point scaffold should be no more than 36 inches wide unless designed by a qualified person to prevent unstable conditions.

Stability

The scaffold must be designed in such a manner that it does not sway while it is being raised or lowered. In addition to the engineering methods that are employed to minimize horizontal movement of the scaffold, the employees working on the scaffold must exercise caution at all times.

Electrical Hazards

Due to the fact that scaffolds are typically constructed primarily of metal, it is important that they remain clear of any electrical power sources, as this would increase the risk of electrocution. A minimum clearance to maintain from all electrical lines up to 50 kV is 10 feet. Unless the line is insulated and carrying less than 300 volts, in which case the minimum clearance to maintain is 3 feet. Lines carrying more than 50 kV require greater clearance distances as determined by 1926.451(f)(6).

Personnel Training

One of the most important OSHA requirements is proper training, by competent persons, of the scaffold builders and those who are expected to work on the scaffold.

Single-Point Adjustable

General Requirements

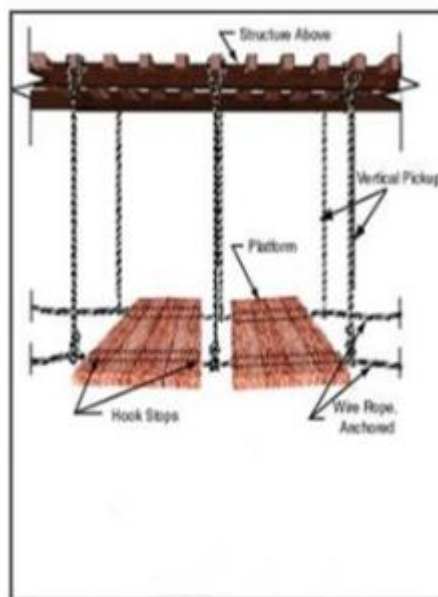
The support rope (the rope that is between the scaffold and the suspension device) must be maintained in a vertical position unless all of the following conditions are met:

- The rigging has been designed by a qualified person.
- The scaffold is accessible to rescuers.
- The supporting rope is protected to ensure that it will not chafe at any point where a change in direction occurs.
- The scaffold is positioned so that swinging cannot bring the scaffold into contact with another surface.

Catenary

General Requirements

Catenary scaffolds should not have more than two interconnected platforms at any one time. Furthermore, because catenary scaffolds do not usually have guardrails, all employees on a scaffold must be protected with personal fall arrest systems if operated at a height in excess of ten feet.



Multi-Point Adjustable

General Requirements

Passage from one platform to another is only allowed when the platforms abut and are at the same height if bridges are not used.

These scaffolds shall be suspended from metal outriggers, brackets, wire rope slings, hooks, or other means that meet the equivalent criteria for strength and durability.

Furthermore, multiple multi-point scaffolds must not be bridged together unless the design of the scaffolds allows them to, the bridge connections are articulated, and the hoists are properly sized.

Interior Hung

General Requirements

Because interior hung scaffolds can only be suspended from the roof structure or other structural members, these structures must be inspected to ensure that they possess sufficient strength.

Needle Beam

General Requirements

Ropes or hangers must be used to support one end of the scaffold. The other end may be supported by ropes, hangers, or a permanent structural member. Needle beams need to be properly secured to ensure that they do not roll or are displaced. In addition, the platform needs to be securely attached to the needle beams with bolts or equivalent means.

Float (Ship)

General Requirements

Platforms must be secured to a minimum of two bearers (such as wooden beams) that extend to at least six inches beyond the platform on each side. Rope connections must be secure so that the platform does not shift or slip. Each employee on a float scaffold

must be protected by a personal fall arrest system if working at an elevation above ten feet.



Supported Scaffolds

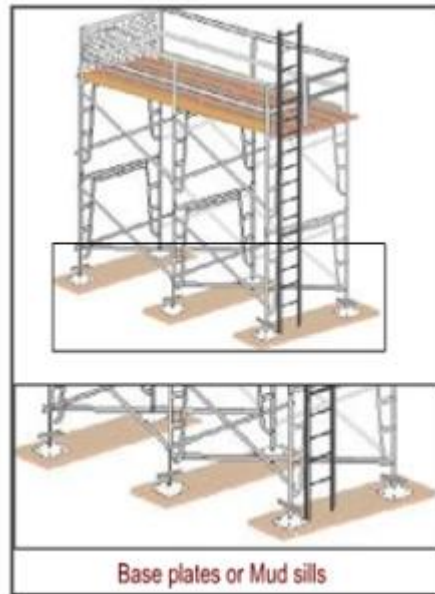
The regulations presented for the frame or fabricated scaffold are applicable to all other types of supported scaffolds, unless stated otherwise.

Fabricated Frame or Tubular Welded Frame

Base Section

- **Firm Foundation:** Supported scaffolds must be set on base plates or mud sills to ensure the stability of the structure. As such, the footings of the scaffold must be capable of withstanding the load weight that is set upon it without movement or displacement.
- Frames and panels shall be braced by cross, horizontal, or diagonal braces, or combination of these. These frames and panels secure vertical members together laterally. The cross braces shall be of the proper length to ensure that vertical members are square and aligned. With this system properly used, the completed

scaffold is always plumb, level, and square. All brace connections shall be secured.



Support Structure

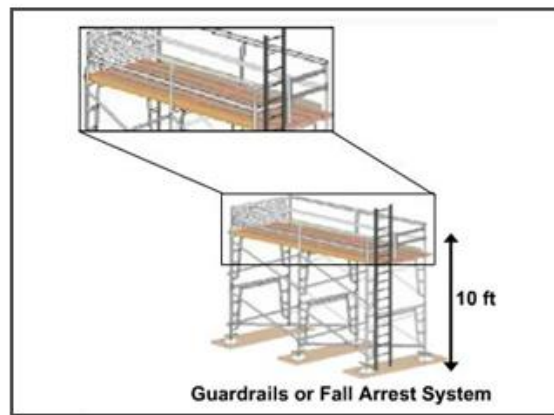
Two important factors in ensuring that the scaffold is stable and is not prone to swaying are the strength and structural integrity of the supports. The supports of the scaffold must be capable of withstanding up to four times their intended maximum weight load as well as the weight of the structure.

Access

Personnel are subject to accidents while they are climbing on or off a scaffold. The mode of accessing the scaffold must be designed in a manner that ensures that employees are not needlessly being endangered, either by accessing the scaffold or putting undue pressure on the structure. Employees are prohibited from climbing the cross-braces as a means of accessing the scaffold. This is extremely dangerous and may undermine the structural integrity of the structure.

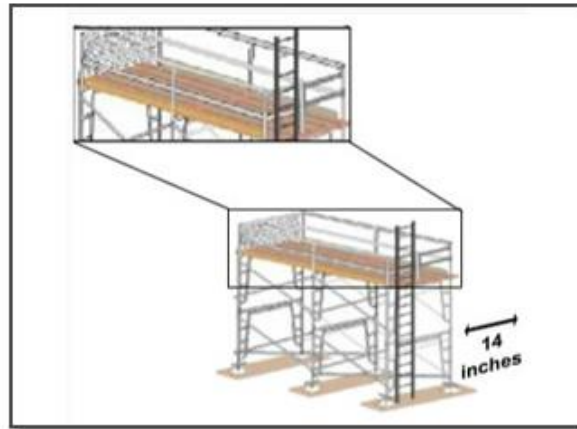
Fall Protection

A common hazard to the employees on scaffolds are falls. The installation of proper fall protection minimizes the risk of such an occurrence. This may include the installation of guardrails on the platform or the use of personal fall arrest systems. Employees on scaffold platforms that are higher than 10 feet above the next level must be adequately protected from falling through the use of proper protective systems and practices.



Platform

The platform is the work area of the scaffold. It supports both the employees and the equipment and materials that they use. It is essential that the platform conform to safety measures to ensure that the employees are not endangered in the course of their work. The platform should be fully planked, and the gaps between the planks should not exceed 1 inch. The platform should not be more than 14 inches from the structure being worked on unless guardrail systems are erected along the front edge and/or personal fall arrest systems are in use for all exposed employees. The maximum distance from the structure for plastering and lathing operations is 18 inches.



Keeping Upright

Scaffolds that exceed a height four times their minimum base dimension must be restrained from tipping with ties, braces, guys, or equivalent means. Unless they have been designed for the specific purpose, scaffolds should not be moved horizontally while employees are on them.

Electrical Hazards

As with suspension scaffolds, the height of supported scaffolds often means that they may be built within close proximity of power lines. In addition, the fact that scaffolds are often built using metal parts makes the risk of electrocution particularly high. As a result, specific measures must be taken to minimize the risk to employees.

Personnel Training

One of the most important OSHA requirements is the proper training by qualified experts of both the scaffold builders and those that are expected to work on the scaffold.

Mobile

Moving

Scaffolds must be stabilized to prevent tipping over while in motion.

When the scaffold is being moved manually, the force must be applied to an area that is as close to the base as possible, and in an area that is not more than five feet above the ground. If the scaffold is moved using a powered system, the system must be designed for that purpose.

Scaffolds shall not be moved horizontally while employees are on them, unless all of the following conditions are met:

- The surface on which the scaffold is being moved is within 3 degrees of level and free of pits, holes, and obstructions;
- The height to base width ratio of the scaffold is two to one or less, unless the scaffold is designed and constructed to meet or exceed nationally recognized stability test requirements;
- Required outrigger frames are installed on both sides of the scaffold;
- When power systems are used, the force is applied directly to the wheels and a speed in excess of 1 foot per second is not generated; and
- No employee is present on any part of the scaffold that extends outward beyond the wheels, casters, or other supports.



Pump Jack

General Requirements

The bracket and braces of the pump jack must be made from metal plates and angles to ensure strength. Each bracket must be fitted with two positive gripping mechanisms to prevent slippage.

Poles must be secured to the structure by rigid triangular bracing at the bottom, top, and other points as needed.

If the pump jack has to pass bracing that is already installed, an additional brace must be installed approximately four feet above the brace to be passed and must be left in place until the pump jack has been moved and the original brace reinstalled.

If wood is used for poles, it must be straight-grained and free of shakes, dead knots, and other defects.

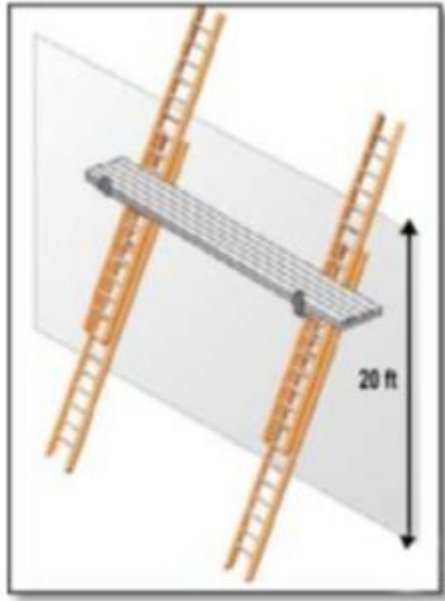
Ladder Jack

Support

Ladders that are used to support ladder jack scaffolds must be placed, fastened, or otherwise equipped with devices in such a manner so as to prevent the scaffold from slipping. They should also be appropriately fastened to ensure overall stability.

Platforms

The maximum height at which a platform can be placed is 20 feet.



Lesson Summary

There are safety requirements applicable to all scaffolds and additional standards applicable to specific types of scaffolds. Unless otherwise stated, the safety regulations for the two-point scaffold apply to all other types of suspension scaffolds. With suspended scaffolding, a primary factor to consider in terms of safety is the anchorage of the scaffold from the building (or other structure). Additional requirements focus on how the scaffold is supported, how it can be accessed (preferably from the roof, rather than a ladder); fall protections, such as arrest systems and guardrails, which are required for any scaffold higher than ten feet in the air; and the platform, which must be at least 18" wide and able to support at least its own weight plus four times the maximum allowed weight. Many of these (not suspension-specific) requirements apply to supported scaffolds, as well.

Further safety requirements (which apply to both supported and suspended scaffolds) address stability (so the scaffold does not sway during its raising or lowering); control of electrical hazards (meeting the minimum 10-foot clearance from all electrical lines up to 50 kV, and other controls); and of vital importance to OSHA and worker safety, proper training to those building and working on scaffolds.

Lesson 3: Scaffold Safety Measures

Lesson Focus

This lesson focuses on the following topics:

- Introduction
- How Do You Minimize Risks?
- Guardrails
- Personal Fall Arrest Systems

Introduction

One out of every three deaths in construction results from fatal falls. This high fatality rate indicates the importance of ensuring that scaffold workers must be well protected against accidental falls. Generally, precautions must be taken for employees expected to work at heights above six feet. Fall protection is not required (but may be an excellent practice) for employees working on scaffolding at a height of ten feet or less.

Working on scaffolds may be more hazardous due to the fact that there is often very little space for a person to maneuver, especially when space is also taken up by the various materials that the employee needs to complete his or her job.

How Do You Minimize the Risks?

There are various ways that employers can minimize the risk to their employees to a sufficiently acceptable level.

Uniformity

- Ensure that the scaffold has been erected in accordance with the instructions of the manufacturer.

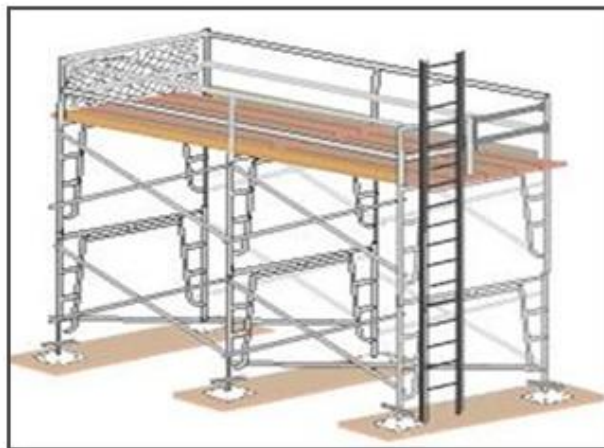
- Do not alter or modify any of the components of the scaffold. If you are faced with a problem, contact the manufacturer.
- If you have scaffolding components from two or more manufacturers, do not under any circumstances mix the components.
- Do not use incompatible metals for the components of the scaffold.

Guardrails

All open and exposed sides of the scaffold at levels above ten feet should be fitted with guardrails to prevent falls or all exposed employees shall be provided with proper fall arrest systems. The guardrails should consist of at least a top-rail, a mid-rail, and when necessary, a toe-board.

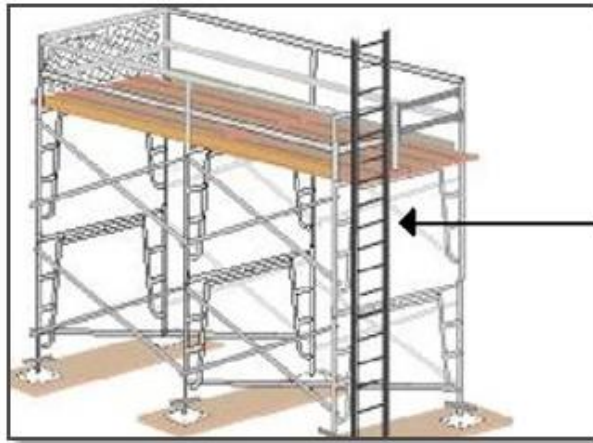
Prohibit Climbing

Prohibit climbing on the cross-braces of the scaffold because it is extremely dangerous and against OSHA regulations to use the cross-braces as a means of accessing the scaffold. Climbing the cross-brace can destabilize the scaffold and cause it to tip over. In addition, employees climbing up or down will not be protected by a personal fall arrest system and would be at an increased risk of falling.



Provide safe ways to access the scaffold.

Provide the employees with personal fall arrest systems when appropriate (this will be discussed in more detail later in the lesson).



Guardrails

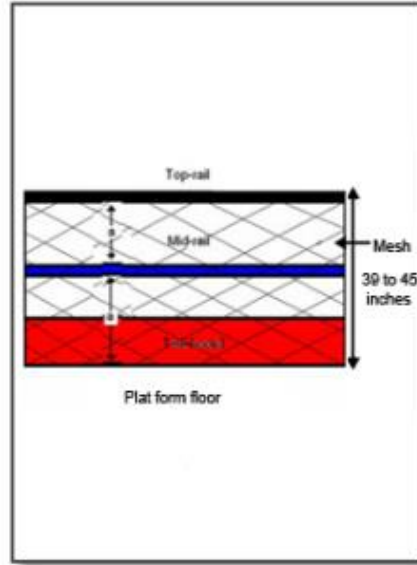
If an employee is expected to work at heights that exceed ten feet, the employer must ensure that the scaffold is fitted with guardrails or that proper personal fall arrest systems are in use. The following provisions must be met when installing guardrails.

Top- and Mid-Rails

The top-rail (that is the topmost part of the guardrail) must be between 39 and 45 inches above the platform or walking level of the scaffold.

A mid-rail is the portion that exists below between the top-rail and the work platform. The mid-rail must be halfway between the top-rail and the surface of the platform. In case there is the need for other structural panels to be installed (such as additional mid-rails), they should be installed so as not to leave openings that are wider than 19 inches.

Toe-boards or mesh screening may also be installed, especially if there is any concern of accidentally falling objects striking workers below.



Strength of Guardrails

Guardrails must be manufactured in such a manner that they would be capable of withstanding up to 200 pounds of force applied at any point and from any direction upon the top-rail. Mid-rails must be capable of withstanding up to 150 pounds of force applied at any point and from any direction.

Smoothness

The guardrails must not have any jagged or protruding surfaces that could cause punctures (either in clothing or in skin). In addition, the top-rail and mid-rails must be constructed in such a manner so as not to overhang the scaffold, thereby causing a projection hazard.

Personal Fall Arrest Systems

Personal fall arrest systems are one of the options that can be used to minimize the risks of injuries related to falls from an elevation. They should be used in the absence of, or instead of, a guardrail system if the employee is expected to work at heights exceeding ten feet.

Types and Components of Personal Fall Arrest Systems

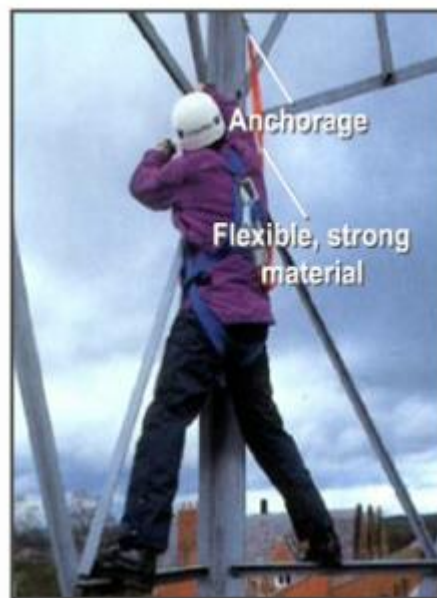
Body Harness

Full body harnesses are designed to minimize the stress on the body that is induced by a fall. Employees must be trained by a competent person in fall protection systems prior to using a full body harness. Harnesses should be inspected prior to each use for mildew, wear, damage, and other deterioration. Defective harnesses shall be removed from service.

More Information: Body harnesses are designed to minimize the stresses on the body that is induced by a fall. Due to high incidences of internal injuries suffered during falls, OSHA has deemed the body belts used for fall protection present a danger to workers and are no longer allowed.

Lanyards/Vertical Lifelines

Vertical lifelines consist of a line, made of a strong material, which is attached to an appropriate anchorage at one end and attached to the full body harness at the other end. Advances in synthetic materials have led to the creation of shock-absorbing lanyards with increased strength.



Self-Retracting Lifelines (SRLs)

SRLs are commonly used as an attractive alternative to fixed-length lanyards. The line is encased in a protective housing that extends and retracts automatically, which increases mobility and may reduce the risk of tripping. SRLs can be anchored to an appropriate anchorage point above an employee's head for increased safety and mobility. These devices are also commonly called "yo-yos."



Horizontal Lifelines

Horizontal lifelines are may be practical in areas where the possible anchorage points are otherwise limited or difficult to access. They are easy to transport and relatively quick to set up. The idea is pretty simple: Two anchorage points are connected at two points and an extended cable is then run between the two anchorages.

PFAS Use Requirements

Requirements for using personal fall arrest systems include:

- When stopping a fall, personal fall arrest systems should limit the maximum arresting force to no more than 900 pounds.

- The line should be rigged to ensure that a worker does not freefall for more than 6 feet and does not come into contact with any surface.
- The fall arrest system must have sufficient strength to withstand twice the potential impact energy of a worker freefalling for six feet.
- Any systems that have been involved in a fall impact (that is, have been involved in arresting a falling worker) should be removed immediately and not used again until they have been inspected by a competent person to ensure that they are undamaged.
- Workers who have been involved in a fall must be assisted immediately, or at least be in a position to rescue themselves—otherwise suspension trauma or other harm may result. This is a condition where pooling of the blood in the extremities may cause loss of consciousness or even death. To avoid this, when suspended, the worker must limit restriction points and move their arms and legs to maintain circulation. If the worker is unconscious when rescued, proper treatment and medical attention should be provided immediately.
- Ensure that regular checks are conducted on the systems for wear and tear and that any defective items are immediately replaced.
- Do not, under any circumstances, attach a fall arrest system to guardrails. They are not designed for that purpose and are unlikely to withstand a fall impact.

Case Study

There are numerous cases of severe injuries and death when personnel working on scaffolds did not take, or were not equipped with, the necessary safety measures. Here are some examples:

- An employee was taking measurements from a scaffold not fitted with guardrails. He tripped and fell 14 feet to the ground. He died instantly.
- A worker was working on a snow and ice covered scaffold with no guardrails or a fall arrest system. He slipped and fell approximately 20 feet. He died instantly.
- An employee was working on a scaffold during stormy weather. It is suspected he lost balance due to a strong gust of wind. He was not wearing a personal fall arrest system and fell 15 feet to his death.

Lesson Summary

A scaffold must not just be anchored properly and be strong and stable; the workers themselves must be trained in how to safely use and maneuver (themselves and their equipment) in the often small space.

Employers can minimize the risk to employees by making sure all scaffolding construction, materials, and other elements are compatible. Scaffolds must contain safe, well-constructed guardrails that consist of three rails: top, mid, and, when necessary, a toe board or employees must be provided with appropriate personal fall arrest systems; guardrails must be sufficiently strong and smooth according to OSHA guidelines. Employers must also prohibit climbing on a scaffold's cross braces, and if there are no guardrails, provide personal fall arrest systems for workers. These systems may include body harnesses, vertical lifelines, self-retracting lifelines and horizontal lifelines, and must meet other OSHA requirements.

Module 23: HAZWOPER

Module Description

This regulation overview module discusses 29 CFR 1910.120, which covers hazardous waste operations and emergency response. It tells the history behind why Congress passed the OSH Act, which created the Occupational Safety and Health Administration, or OSHA, under the U.S. Department of Labor. The module covers information regarding: those industries regulated, how NIOSH assists in rule making, key terms, Hazard Communication Standard (HCS) elements, and the CFR. Finally, there is an important discussion of what to include in a Hazard Communication Program to inform employers about their duties.

Module Learning Objectives

At the conclusion of this module, you should be able to:

- Discuss the creation of OSHA and NIOSH
- Elaborate on the regulations set by OSHA in relation to employee rights and responsibilities
- Describe the Hazard Communication Standards (HCS) and its importance
- Explain various terms in relation to hazards and HAZWOPER requirements

Lesson 1: Regulation Overview

Lesson Focus

This lesson focuses on the following topics:

- About the Occupational Safety and Health Act (OSH Act)
- The Creation of OSHA and NIOSH
- Hazardous Waste Operations and Emergency Response
- Hazard Communication Standard (HCS)
- Understanding OSHA Standards
- Hazardous Waste Legislation (EPA)
- Hazard Definitions

About the Occupational Safety and Health Act (OSH Act)

Before 1971, a variety of state health and safety regulations existed, but they were rarely enforced. As a result, a number of workers were becoming ill, injured, or killed due to the conditions in their workplace. In 1971, the Occupational Safety and Health Act (OSH Act) went into effect. It was the first uniform federal safety and health regulation.

The OSH Act's purpose is *“to assure so far as possible every working man and woman in the Nation safe and healthful working conditions and to preserve our human resources.”*

General Duty Clause 5(a)(1)

A few facts about the OSH Act:

- Employers must ensure that workers have a safe and healthy environment.
- The OSH Act places responsibility on employers and employees.
- The OSH Act resulted in the creation of Occupational Safety and Health Administration, or OSHA.

Employers can be cited for violating the General Duty Clause if there is a recognized hazard and they do not take reasonable steps to prevent or abate the hazard. However, failure to implement these recommendations is not, in itself, a violation of the General Duty Clause. Citations can only be based on standards, regulations, and the General Duty Clause.

The Creation of OSHA and NIOSH

The OSH Act resulted in the creation of the **Occupational Safety and Health Administration, or OSHA**, which is part of the U.S. Department of Labor. OSHA establishes workplace standards for general industry, construction, shipyard employment, and for long shoring and marine terminals.

In order to develop regulations, OSHA needs subject-specific information. Because of this need, the **National Institute for Occupational Safety and Health, or NIOSH**, was created as a research agency within the Department of Health and Human Services. NIOSH conducts research on occupational hazards. Based on the research results, NIOSH makes recommendations to OSHA regarding the creation or revision of OSHA standards. NIOSH also provides education and training in occupational safety and health.

Hazardous Waste Operations and Emergency Response

Hazardous waste operations and emergency response is covered in **29 CFR 1910.120**. Later in the lesson, we will explain what 29 CFR 1910.120 is and what it means. It was created to protect the health and safety of those working at waste sites or as emergency responders dealing with hazardous substances.

People covered under this rule include anyone who is:

- Cleaning up hazardous waste sites

- Working with hazardous waste at TSD (treatment, storage, and disposal sites)
- Responding to emergencies involving hazardous material releases

Because of the seriousness of the safety and health hazards related to hazardous waste operations and emergency response, the Occupational Safety and Health Administration (OSHA) issued its Hazardous Waste Operations and Emergency Response (HAZWOPER) standard, Title 29 Code of Federal Regulations (CFR) Parts 1910.120 and 1926.65 (see 54 Federal Register 9294-9336, March 6, 1989) to protect employees in this environment and to help them handle hazardous substances safely and effectively.

Note: If HAZWOPER conflicts or overlaps with any other OSHA standard, the provision more protective of employee safety and health takes precedence (e.g., with Hazard Communication, 29 CFR 1910.1200 and 1926.59, Bloodborne Pathogens, 29 CFR 1910.1030; Permit-Required Confined Spaces, 29 CFR 1910.146, and other OSHA-expanded health standards under Subpart Z).

MORE ABOUT:

Furthermore, if OSHA considers a worksite response activity a “HAZWOPER Emergency Response,” then employers and employees at the worksite performing emergency response must comply with HAZWOPER paragraph (q) and all other General Industry (1910) or Construction Industry (1926) standards.

For example, when the following conditions, or similar conditions, may develop as a consequence of a release of hazardous substances or threat of release, such situations would be considered emergency situations requiring an emergency response effort:

- High concentrations of toxic substances
- Immediately Dangerous to Life and Health (IDLH) environments
- Situations that present an oxygen-deficient atmosphere
- Conditions that pose a fire or explosion hazard
- Situations that require an evacuation of the area
- Situations that require immediate attention because of the danger posed to employees in the area

Risk identification

Once the presence and concentrations of specific hazardous substances and health hazards have been established, the risks associated with these substances shall be identified. Furthermore, employees who will be working on the site shall be informed of any risks that have been identified. However, in situations covered by the Hazard Communication Standard, 29 CFR 1910.1200, training required by that standard need not be duplicated.

Note:

Risks to consider include, but are not limited to:

[a] Exposures exceeding the permissible exposure limits and published exposure levels

[b] IDLH Concentrations

[c] Potential Skin Absorption and Irritation Sources

[d] Potential Eye Irritation Sources

[e] Explosion Sensitivity and Flammability Ranges

[f] Oxygen deficiency

Employee Rights and Responsibilities

Under OSHA, workers involved in any given work activity are covered by a general set of rights and responsibilities.

Some examples of employee rights are:

- Be informed of your rights and responsibilities.
- Review copies of appropriate OSHA standards.
- Request information on health and safety from employers.
- Receive adequate training and information on workplace safety and health hazards.
- Request an OSHA investigation.
- Be advised of OSHA's actions.

Employee Responsibilities

Along with rights, employees also have responsibilities. Employees must comply with all applicable rules, regulations, and standards pertaining to occupational safety and health.

Some of these responsibilities are to:

- Read the OSHA poster.
- Comply with standards.
- Follow SOPs and wear required PPE.
- Report hazardous conditions to supervisors.
- Cooperate with OSHA.
- Exercise OSH Act rights in responsible manner.

Employer Responsibilities

Employers also have responsibilities under the OSH Act. Employers must ensure that their workers have a safe and healthy environment. If possible, employers should eliminate recognized hazards from the workplace.

The employer must try to eliminate the hazards to which the employees are exposed by sequentially following these methods:

1. Using engineering controls (e.g., ventilation, barriers, tools, equipment, etc.)
2. Following work practices (e.g., methods, training, rules, etc.)
3. Providing personal protective equipment (PPE); e.g., respirators, gloves, goggles, clothing, etc.

(PPE can only be used when it is not feasible to eliminate the hazard using the first two methods.)

Some of these responsibilities include:

- Report to the nearest OSHA office within 8 hours of a fatal accident or hospitalization of three or more employees.
- Post work-related injuries yearly.
- Post OSHA posters in prominent places.
- Cooperate with OSHA compliance officer.

Hazard Communication Standard (HCS)

One of the needs addressed by the OSH Act is the protection of workers from chemicals in the workplace. The Hazard Communication Standard requires employers to evaluate chemical hazards in their facilities and to communicate related hazard information to employees.

The following steps must be included in the employer's hazard communication program:

- Conduct a chemical inventory.
- Obtain and file a Safety Data Sheet (SDS) for each chemical.
- Make available a written document on the hazard communication program to all employees in a readily accessible location.
- Label all containers and storage containing chemicals with the required markings and warnings for the hazard.
- Educate employees about the hazards and proper use of the chemicals.

Understanding OSHA Standards

Many OSHA general industry standards are applicable to cleanup activities at hazardous waste sites. OSHA standards come from three sources:

- **Consensus standards:** which are prepared by organizations that develop industry-wide standards
- **Proprietary standards:** which are prepared by professional experts within particular industries, professional societies, and associations
- **Some federal worker protection laws:** OSHA now enforces the standards issued under earlier Acts in all industries where they apply.

Horizontal and Vertical Standards

When dealing with standards, there are two main categories: horizontal standards and vertical standards.

- **Horizontal:** Most standards are horizontal, or general, and apply to any employer in any industry. Examples include standards applying to fire protection, working surfaces, and first aid.
- **Vertical:** Other standards only apply to a particular industry. These standards are called vertical standards. Examples include standards that apply to the longshoring or construction industries.

Sources for Finding Rules, Regulations and Standards Under OSHA

OSHA regulations are available from several sources:

- The Code of Federal Regulations (CFR) lists the general and permanent rules of federal agencies. Between printings, the Code is kept current by individual issues of the Federal Register.
- The Federal Register is a daily publication that contains meeting notices, proposed new regulations, amendments to existing standards, and final regulations.
- The OSHA CD-ROM is an electronic copy of OSHA regulations, some documents, and technical information. It is made available by the OSHA Computerized Information System.

Keys to Understanding the Code of Federal Regulations

To understand federal regulations, remember:

- The **preamble** explains the history behind the regulation published in the Federal Register. Preambles are not found in CFRs.
- The **scope and application** paragraph explains who is subject to the regulation and under what circumstances. It is usually included in the first paragraph of a CFR standard.
- **Definitions** are commonly found in the next paragraph of a CFR standard. Carefully review definitions before applying the standard.
- A standard may have an **appendix** attached to it. Some appendices are mandatory and others are not.

Titles in the Code of Federal Regulations

CFRs are divided into 50 titles, which represent broad areas that are subject to federal regulation. There are several CFR titles applicable to waste site work. They are:

- **Title 29**, which covers labor and the protection of the worker.

- **Title 40**, which covers the protection of the environment and issues concerning the Environmental Protection Agency, or EPA.
- **Title 49**, which covers protection during transportation of hazardous materials and issues concerning the Department of Transportation, or DOT.

Citations

The correct way to write this particular reference is 29 CFR 1910.120, which is called a **citation**. Notice that the subpart is not written in the citation. The subpart is shown only in the text of the CFR. Each CFR volume is updated once per calendar year. As of July 1 each year, OSHA's regulations (Title 29) are updated. After six months, new volumes are received.

The Code is kept current by individual issues of the Federal Register between printings. This is a daily publication that includes meeting notices, proposed new regulations, amendments to existing standards, and final regulations.

Paragraph Numbering System

The sections of CFR, such as 1910.120, are divided into **paragraphs**. A paragraph in the CFR is not the same as the traditional paragraph in a story or essay. The traditional paragraph is indented at its beginning, is often brief, and focuses on one main idea. On the other hand, a CFR paragraph can run for a number of pages and cover many concepts.

Paragraphs in CFR and their **subparagraphs** are marked with numbers and letters in outline fashion.

Hazardous Waste Legislation (EPA)

In the early 1970s, industries in the U.S. were producing approximately 57 million metric tons of hazardous waste annually. Congress determined it was essential to handle this waste in a manner that would protect the environment.

Hazard Definitions

Important terms include:

- **Hazardous waste:** The EPA defines **hazardous waste** as a hazardous substance that has no commercial value. A hazardous waste is ignitable, corrosive, reactive, or toxic.
- **Hazardous substances:** Substances determined by the EPA to present a danger to the environment.
- **Hazardous chemicals:** Chemicals that are hazardous to people in the workplace or the community if released, as determined by OSHA.
- **Hazardous materials:** Those materials that can present a danger during shipment by truck, rail, pipeline, air, or water, as determined by the Secretary of Transportation.

RCRA and CERCLA

In response to problems posed by industrial hazardous waste, Congress enacted two legislative acts: The Resource Conservation and Recovery Act (RCRA) and The Comprehensive Environmental Response, Compensation and Liability Act of 1980 (CERCLA).

- **RCRA:** Regulates an industry's management of its hazardous wastes from the time the wastes are generated to the time they are disposed of on-site or off-site.
- **CERCLA:** Regulates and enforces the cleanup of abandoned hazardous waste sites and hazardous waste sites created prior to RCRA.

SARA Although RCRA and CERCLA greatly helped in properly managing hazardous wastes, there were no laws that specifically addressed the health and safety of the employee who worked with hazardous wastes and substances.

These issues were addressed in 1986 when the Superfund Amendments and Reauthorization Act (SARA) was passed. SARA addressed the risk of exposure and the need to protect employees exposed to hazardous wastes

State, county, and municipal employees, including hazardous waste treatment, storage, and disposal facility employees, and first responders, such as fire and rescue personnel, police, and medical personnel, are covered by HAZWOPER and other regulations issued by the 26 states and territories operating their own OSHA-approved safety and health programs.

EPA HAZWOPER regulations cover these employees in states without OSHA-approved state plans. The EPA adopted the HAZWOPER standard at 40 CFR Part 311 for public employees (either compensated or non-compensated) who perform operations within the scope of the standard in states that do not have an OSHA-approved state plan.

HAZWOPER Requirements that Apply to Marine Oil Spills

The NCP defines oil as any kind of oil in any form, including petroleum, fuel oil, sludge, oil refuse, and oil mixed with wastes but not dredged spoil (dirt or rock).

Response actions conducted under the NCP must comply with the provisions of HAZWOPER. You'll find this requirement in 40 CFR 300.150. Therefore, if your workers are participating in a response action under the NCP, you must have an occupational safety and health program consistent with HAZWOPER and you must train your workers according to HAZWOPER's training requirements. This applies whether the responsible party or a government agency is directing the cleanup.

For marine oil spill emergency response, the HAZWOPER provisions that most directly apply include:

- Emergency response operations in HAZWOPER paragraph (q), and
- Post-emergency response cleanup operations in paragraph (q)(11).

Example:

For example, drums and containers used during the cleanup must meet the required OSHA, EPA (40 CFR Parts 264-265 and 300), and Department of Transportation (DOT) regulations (49 CFR Parts 171-178) and must be properly inspected and labeled.

More About:

See also emergency response training provisions in paragraph (q)(6), and post-emergency response training requirements in paragraph (q)(11).

When HAZWOPER Does Not Apply

HAZWOPER does not apply to incidental releases that are limited in quantity and pose no safety and health threat to employees working in the immediate vicinity of the spill. These oil spills can be absorbed or controlled at the time of the release by employees in the immediate vicinity. The difference between emergency spills and incidental releases is described in the definition of emergency response in HAZWOPER paragraph (a)(3). An incidental release does not have the potential to become an emergency within a short time. If an incidental release occurs, employers do not need to implement HAZWOPER.

Lesson Summary

The OSH Act resulted in the creation of Occupational Safety and Health Administration, or OSHA. The OSH Act places responsibilities on employers and employees. Employers must ensure that workers have a safe and healthy environment. Citations can only be based on standards, regulations, and the General Duty Clause.

The National Institute for Occupational Safety and Health (NIOSH) was created as an occupational hazards research agency within the Department of Health and Human Services.

Under OSHA, employees have a right to know about the hazards they face, their rights and responsibilities, and OSHA standards. They are also entitled to training and the right to ask for an OSHA investigation of their workplace. Employees must comply with all applicable rules, regulations, and standards.

Employers must implement hazard elimination and control methods, report fatalities and injuries, and cooperate with OSHA compliance officers.

HAZWOPER Emergency Response would include the following:

- High concentrations of toxic substances
- Immediately Dangerous to Life and Health (IDLH) environments
- Oxygen-deficient atmospheres
- Fire or explosion hazards
- Situations that require an evacuation of the area
- Situations that require immediate attention because of the danger posed to employees in the area

Risks associated with identified hazardous substances must be recognized and communicated to the workers who may be exposed to them.

The steps in the hazard communication program include:

- Conduct a chemical inventory.
- Obtain and file a Safety Data Sheet (SDS) for each chemical.
- Make available a written document on the hazard communication program to all employees in a readily accessible location.
- Label all containers and storage containing chemicals with the required markings and warnings for the hazard.
- Educate employees about the hazards and proper use of the chemicals.

Lesson 2: Site Characterization

Lesson Focus

This lesson focuses on the following topics:

- Site Characterization Introduction
- Off-Site Characterization
- On-Site Survey
- On-Going Monitoring and Hazard Assessment
- Documentation

Site Characterization Introduction

When safety measures are built into the work process, workers are guaranteed constant protection. Such measures must be continually evaluated (inspected) and updated. Site characterization is the responsibility of the Project Team Leader, and it is a continuous process. At each phase of site characterization, information has to be obtained and evaluated to define hazards that can pose danger. All site personnel must be made aware of new information about site conditions

Site characterization, or continual gathering and evaluating conditions, has to be done because as work changes so do hazards. For example, when building a structure, the dangers are different when digging the foundation from those faced by workers doing roofing. There are three distinct phases of the site characterization process:

- The off-site survey
- The on-site survey
- On-going monitoring

Let's discuss each of these phases in further detail.

Off-Site Characterization

Off-site information can be obtained by using two methods:

- Interviews and research
- Perimeter reconnaissance

As much data as possible should be collected before any personnel are allowed to go on site. Try to obtain information through interviews and research.

If possible, the following information should be obtained:

- Precise location of the site
- Meteorological data (e.g., current weather and forecast, prevailing wind direction, precipitation levels, temperature profiles)
- A detailed account of the activity that occurred at the site
- Duration of the activity
- Geologic and hydrologic data
- Habitation and population centers; populations at risk
- Accessibility by air and roadways
- Pathways of dispersion

Interview and Research

Here are some examples of sources for interviews and research:

- Company records, receipts, logbooks, or ledgers
- Records from state and federal pollution control regulatory and enforcement agencies, state Attorney General's office, state occupational safety and health agencies, state Fire Marshal's office, waste storage inventories, and manifests or shipping paper
- Interviews with past and present personnel and their families
- Generator and transporter records

- Water department and sewage district records
- Interviews with nearby residents (note possible site-related medical problems)
- Local fire and police department records
- Court records
- Utility company records
- Media reports

Additionally, all information from the media and interviews should be validated.

Perimeter Reconnaissance

The second method for off-site characterization is *perimeter reconnaissance*. When site hazards are largely unknown or when there is no need to go on site immediately, perimeter reconnaissance should be done. From outside the boundary of the site, one can make visual observations, monitor for atmospheric concentrations of airborne pollutants, and collect soil and/or water samples. While these data are not clear-cut indicators of on-site conditions, they can assist in the preliminary evaluation. A preliminary site map should be developed with the locations of buildings, containers, impoundments, pits, ponds, and tanks.

Perimeter reconnaissance of a site should involve the following observations:

- The review of historical and current aerial photographs
- Disappearance of natural depressions, quarries, or pits
- Variation in reforestation of disturbed areas
- Mounding or uplift in disturbed areas or paved surfaces, or modifications in grade
- Changes in vegetation around buildings
- Changes in traffic patterns at the site

Site Safety Plan

The off-site survey is used to develop a site safety plan. The site safety plan outlines what must be accomplished and prescribes procedures to ensure the health and safety of the entry team.

During the research phase of offsite characterization, company receipts and records may identify various types of hazardous materials that may be present on-site. This information can be used in the development of the site safety plan for the site entry team to include hazard identification and level of Personal Protective Equipment to be worn by entry personnel.

The following are examples of hazardous materials properties that may be identified by company records:

- Oxidizer - A chemical which initiates combustion in other materials, causing fire either by itself or through the release of oxygen or other gases
- Flammable - Liquids which have a flashpoint below 100 degrees F
- Toxicity - The ability of a substance to cause a harmful effect
- Pyrophoric - A chemical that will ignite spontaneously in air at a temperature of 130 degrees F
- Combustible - Liquids which have a flashpoint at or above 100 degrees F
- Flashpoint - A minimum temperature at which a liquid produces enough vapor to form a flammable mixture
- Organic Peroxide - Any carbon containing compound with 2 Oxygen atoms joined together which can produce severe fire and explosion hazards

It is vital that personnel involved in collection and documentation of information are able to identify specific hazards associated with types of hazardous materials.

On-Site Survey

The purpose of an on-site survey is to verify and supplement information from the off-site characterization. Prior to going on site, the off-site characterization should be used to develop a site safety plan for site entry. The site safety plan outlines what needs to be accomplished and then prescribes procedures to protect the health and safety of the entry team. Careful evaluation of probable conditions enables the team leader to assess the hazards and to plan site activities. Because team members may be entering a largely unknown environment, the use of caution as well as conservative actions are necessary.

Protection of Entry Personnel

The project team leader uses information from interviews, records, research, and perimeter reconnaissance as the basis for selecting the protective equipment for the initial site survey. However, the leader should always err conservatively (i.e., use a higher level of protection unless exact conditions are known). In addition, the leader must consider the nature of the proposed work to be accomplished. For example, the purpose of the survey may be to inspect on-site conditions, to count containers, to measure the ambient air for "hot spots" (i.e., areas with high concentrations of toxic chemicals), and generally, to become familiar with the site. In these cases, the level of protection can be less stringent than if the entry team must open containers and take samples of unidentified materials.

The ensemble of clothing and equipment referred to as Level B protection is generally the minimum level recommended for an initial entry until the site hazards have been further identified.

Entry Team

The composition of the entry team depends on the site characteristics; however, the team should always consist of at least four persons: two workers who will enter the site and two outside support persons suited in personal protective equipment and prepared to enter the site in case of an emergency.

The entry team's responsibilities include:

- Monitoring air for IDLH conditions
- Monitoring for ionizing radiation
- Visually checking for dangerous conditions

On-going Monitoring and Hazard Assessment

Because site activities and weather conditions change, an ongoing air-monitoring program should be implemented after characterization has determined that the site is safe for the commencement of operations. After the project team leader determines from the above activities that the site is safe for the start-up of operations, an ongoing air-monitoring program should be started. An ongoing air-monitoring program is necessary because conditions can change.

When performing on-site surveys, it is important to:

- Note any safety hazards or indicators of potential exposure
- Note all types of containers (any labels or markings) and their condition
- Note physical properties of materials
- Note land features
- Identify natural wind barriers
- Determine potential pathways of dispersion
- Locate buried waste or contaminant plumes by using remote sensing if possible
- Collect and analyze air, water, and soil samples from the site.

On-going Monitoring

An on-going air monitoring program should be implemented by using a combination of methods, including:

- Stationary sampling equipment
- Personnel monitoring devices
- Periodic area monitoring

On-site and off-site survey data can be used to develop a plan detailing the procedures for monitoring conditions during clean-up operations. When site conditions change, the hazards should be reassessed. Some reassessment indicators include a new work phase, a change of season or weather, a change in job tasks during a work phase, or a change in levels of contamination.

Documentation

Good documentation will help to ensure accurate communication and ensure the quality of data collected. It can also provide reasons for safety decisions and support possible legal actions. Some examples of documentation include logbooks, graphs, photos, and analytical records.

Types of Documentation

Documentation can be accomplished by recording information pertinent to field activities, sample analysis, and site conditions in one of several ways. These include

- Logbooks
- Field data records
- Graphs
- Analytical records
- Photographs
- Sample labels
- Chain-of-custody forms

Document Control

These documents should be controlled to ensure that they are all accounted for when the project is completed. The task of document control should be assigned to one individual on the project team and should include the following responsibilities:

- Numbering each document
- Keeping a document inventory
- Collecting all documents at the end of each work period
- Making sure that all information is written in waterproof ink
- Placing all documents in a secured central file after completion

Documentation of On-Site Activities

Field personnel must record all on-site activities and observations in a field logbook (a bound book with consecutively numbered pages). Entries should be made during or just after completing a task to ensure thoroughness and accuracy.

Documentation for Photographs

Photographs can be an accurate, objective addition to a fieldworker's written observations. For each photograph taken, the following information should be recorded in the field logbook.

- Date, time, and name of site
- Name of the photographer
- Location of the subject within the site
- General compass direction of the orientation of the photograph
- General description of the subject
- Sequential number of the photograph and reference to it in a file.

Lesson Summary

When safety measures are built into the work process, workers are guaranteed constant protection. Such measures must be continually evaluated (inspected) and updated.

There are three distinct phases of the site characterization process:

- The off-site survey
- The on-site survey
- On-going monitoring

Off-site information can be obtained by using two methods:

- Interviews and research
- Perimeter reconnaissance

The purpose of an on-site survey is to verify and supplement information from the off-site characterization. Prior to going on site, the off-site characterization should be used to develop a site safety plan for site entry. Because site activities and weather conditions change, an ongoing air-monitoring program should be implemented after characterization has determined that the site is safe for the commencement of operations.

Good documentation will help to ensure accurate communication and ensure the quality of data collected. It can also provide reasons for safety decisions and support possible legal actions. Some examples of documentation include logbooks, graphs, photos, and analytical records. Documentation can be accomplished by recording information pertinent to field activities, sample analysis, and site conditions in one of several ways.

These include:

- Logbooks
- Field data records
- Graphs
- Analytical records
- Photographs
- Sample labels
- Chain-of-custody forms

Lesson 3: Toxicology

Lesson Focus

This lesson focuses on the following topics:

- Part 1: Exposure and Affects
 - Response to Chemical Exposure
 - Toxic Substance Forms
 - Target Organ Poisons
 - Routes of Exposure
 - Chemical Exposure
- Part 2: Exposure Limits and Protection
 - Chemical Interaction
 - Exposure Limits
 - Protection from Toxins

Toxicology

Toxicology is the study of chemical toxicity. Toxicity is the degree to which a chemical or substance has a harmful effect on us or other living things.

The Study of Toxicology

What we know today that the toxic effects of chemicals and other hazardous substances have been gathered from two sources of information:

1. Epidemiological studies

Epidemiological studies compare affected people in one group to affected people in another group. Sometimes the information comes from actual exposures to a chemical in the workplace.

2. Animal studies

Animal studies provide most of what we know about the dangerous effects of toxic substances. Humans may react differently than animals exposed to toxic materials.

MORE ABOUT

Thanks to hazardous chemical communication programs (HAZCOM or Right-to-Know), chemical toxicity information can be found in a number of ways. Today all hazardous chemicals in the workplace must be labeled to identify and warn against toxicity. More detailed information can be obtained from the Safety Data Sheet (SDS).

Response to Chemical Exposure

There are a number of factors that affect the severity of exposure to a chemical. These factors include the toxicity of the substance, route of exposure, chemical dose, interaction with other chemicals, and human response. We will cover each one of these points in detail.

Factors Affecting Response to Chemical Exposure

- **Toxicity:** Some chemicals produce immediate and dramatic biological effects. Others may produce no observable effects at all or the effects may be delayed.
- **Route of exposure:** Certain chemicals appear harmless in one form of contact, such as carbon monoxide on the skin. However, carbon monoxide inhaled into the lungs causes much more serious effects.
- **Chemical dose:** This is the amount and length of time that one is exposed to a chemical.
- **Human response:** Humans and other living animals vary in their response to chemical exposure. For some, a certain dose may produce symptoms of serious illness; for others only mild symptoms may appear; or there may be no noticeable effect at all. Often a prior exposure to a chemical affects the way that an individual

responds to being exposed at a later time. Thus, there is not only variation between different individuals; there may be different responses in the same individual at various exposures.

- **Interaction with other chemicals:** The study of the interaction of various chemicals with one another is known as chemistry. An example is the reaction between acids and bases. The physical and biochemistry of the living organism is much the same. Chemicals in combination can produce different biological responses than the responses seen when exposure is to one chemical alone.

Toxic Substance Forms

The physical forms in which chemicals are encountered at a waste site play a large role in routes of entry into the body.

Solids

The principal hazard is usually from dusts or fumes produced when solids change form. For example, polyurethane foam, when burned, gives off cyanide gas. Routes of entry in this case are inhalation, ingestion (saliva), and skin absorption.

Fumes

Tiny particles are produced from heating, volatilization, and condensation of metals (examples: zinc oxide fumes from welding of galvanized metal). Routes of entry are inhalation, ingestion (saliva), and skin absorption. It is important to note that toxic fumes may result from the burning of non-toxic substances.

Liquids, Vapors, and Mists

Examples of liquids include acids, organic solvents, and chlorinated organic solvents (examples: benzene, sulfuric acid, 1-trichloroethane). Routes of entry are inhalation and skin absorption.

Vapors are gases that result from the evaporation of liquids or the sublimation of solids (examples: gasoline, phosgene and iodine). Routes of entry are inhalation and skin absorption.

Mists are liquid droplets suspended in the air (examples: acid mists from electroplating processes or solvent mists from paint spraying operations). Routes of entry are inhalation and skin absorption.

Health Hazards

Although safety hazards related to the physical characteristics of a chemical can be objectively defined in terms of testing requirements, health hazard definitions are less precise and more subjective. Health hazards may cause measurable changes in the body such as decreased pulmonary function (breathing). These changes are generally indicated by the occurrence of signs and symptoms, such as shortness of breath, in exposed employees.

MORE ABOUT

However, shortness of breath is a non-measurable subjective feeling. The determination of occupational health hazards is complicated by the fact that many of the effects or signs and symptoms occur commonly in non-occupationally exposed populations, so that the effects of exposure are difficult to separate from normally occurring illnesses.

For our purposes, any chemical meeting any of the following definitions is a health hazard. However, this is not intended to be an exclusive categorization. If there is scientific data available that involves other animal species or test methods, they must also be evaluated to determine the applicability to employees. You need to understand that some chemicals may fall into more than one category and this information can be found on the SDS.

Asphyxiants are chemicals that deprive the victim's body tissues of oxygen. Asphyxiants interfere with the body's ability to transport or use the oxygen carried by the blood stream. Any gas heavier than air can easily displace oxygen and cause asphyxiation. Examples of chemical asphyxiants are carbon monoxide and hydrogen cyanide.

Carcinogens cause cancer in humans or in laboratory test animals. This is a chronic effect as there is a time period that elapses before a malignant tumor will appear. Examples of carcinogens are benzene, benzopyrene, asbestos fibers, acrylonitrile, and DBCP (dibromochloropropane).

Irritants aggravate various tissues causing redness, rashes, swelling, coughing, or even hemorrhaging. Chlorine and ammonia are two examples of irritants.

Sensitizers are allergens. These chemicals cause an allergic type of reaction due to sensitivity from prior exposure. An acute response may be swelling of the breathing tubes, which causes breathing difficulty. Sensitizers can cause chronic lung disease. Some common examples are epoxies, aromatic amines, formaldehyde, nickel metal, and malefic anhydride.

Mutagens cause alterations in the genes of a person exposed. The result may be malfunction of a specific organ or tissue, depending upon the type of cell affected. Gene damage can be passed on to children if the mutation occurred in either the sperm or the egg of the parents. Examples of mutagens are ethylene oxide, benzene, and hydrazine.

Teratogens damage or kill a developing fetus. This damage cannot be passed on to further generations, as it does not affect the genetic code. Examples of teratogens are thalidomide, dioxins, lead, and cadmium.

Target Organ Poisons

Many chemicals will target a particular tissue or organ and cause disease or damage at that location. They are called target organ poisons. Other types of chemical toxins target body systems and cause a variety of reactions. The following examples are listed according to the tissue or organ affected.

Lungs

Asbestos fibers and silica dust may cause a fibrosis effect in the lung tissue. Fibrosis is a condition in which the lung becomes scarred and inflexible, making the lung unable to expand and contract.

Skin

A condition called chloracne may be caused by repeated exposures to PCB's (polychlorinated biphenyls) and other chlorinated hydrocarbons.

Eyes

Lachrymators are chemicals that can cause instant tearing at low concentrations. Examples are tear gas and MACE. Other chemicals can cause cataracts, optic nerve damage, and retinal damage by circulating through the bloodstream and reaching the eye. Examples of these are naphthalene, methanol, and thallium.

Central Nervous System

Neurotoxins are chemicals that affect the brain and spinal cord.

Neurotoxins affect the neurons that carry electrical signals by inhibiting their function. This can cause behavior changes, emotional changes, alterations in walking, and loss of hand-eye coordination.

A condition called anoxia may occur, due to a lack of oxygen flow to the brain cells. Lack of oxygen results in cell death.

Examples of chemicals causing central nervous system effects are tetraethyl lead, chlorinated hydrocarbon pesticides (e.g., DDT), malathion, lead, and mercury.

Liver

Hepatotoxins are substances that are capable of damaging the liver. The liver is the main processing organ for toxins. It can convert toxins into nontoxic forms; however, the liver can generate a more toxic by-product, which can cause cellular and tissue damage.

Examples of hepatotoxins are carbon tetrachloride, chloroform, tannic acid, and trichloroethylene.

Examples of chemicals that cause cirrhosis (a fibrotic disease that results in liver dysfunction and jaundice) are carbon tetrachloride, alcohol, and aflatoxin.

Other effects can range from tumors to enlargement of the liver and fat accumulation.

Kidneys

The main function of the kidneys is to filter the blood and eliminate wastes. Because the waste gets concentrated in the process, toxins can be at much higher levels in the kidneys. Toxins that damage this organ are known as nephrotoxins.

Most heavy metals fall into this category, including mercury, arsenic, and lithium. Many halogenated (i.e., chlorinated) organic compounds are also nephrotoxins such as tetrachloroethylene, carbon tetrachloride, and chloroform. Other chemicals that damage the kidneys include carbon disulfide, methanol, toluene, and ethylene glycol

Blood

Substances capable of producing blood disorders are called hematotoxins. Examples of chemicals that affect the bone marrow, which is the source of most of the components of blood, are arsenic, bromine, methyl chloride, and benzene. Chemicals that affect platelets, which are cell fragments that help in the process of blood clotting, are aspirin, benzene, and tetrachloroethane.

Chemicals that affect white blood cells, which help the body defend against infection, are naphthalene and tetrachloroethane. Arsine, naphthalene, and warfarin can affect red blood cells, which carry oxygen throughout the body. Effects in the exposed individual may include reduced red blood count, or anemia; reduced white blood cell count, which leaves the victim susceptible to disease; and reduced platelet count, which increases the possibility of hemorrhaging.

Reproductive Organs

Reproductive toxins can cause sterility, infertility, or spontaneous abortions. They can also affect an individual's hormone levels and activity. Examples of male reproductive toxins are mercury, lead, DDT, PCBs, dioxin, benzene, toluene, and xylene. Examples of female reproductive toxins are DDT, PCBs, parathion, and diethylstilbestrol.

Routes of Exposure

Knowing how chemicals get into your body and how your body reacts is critical to personal protection. Toxic chemicals can enter the body in any one of, or in a combination of, four ways.

1. The most common toxic dose in the work environment comes through breathing, or inhalation.
2. The next most common route of entry is through eye and/or skin contact.
3. The most common route in the home is swallowing or ingestion.
4. A route that is more accidental is injection.

Inhalation

A toxic dose of chemicals can be inhaled in a number of ways: as a gas, mist, fumes, dust, or vapor.

The result is that a chemical enters the airways. The chemical may only make it as far as the mucous membranes of the nose, or it may reach the smallest cavities of the lungs. Anywhere along the way it can be absorbed and cause an adverse reaction. The reaction can be immediate, for example, as in the reaction to hydrogen sulfide gas.

On the other hand, the reaction can be delayed for years as in the reaction to asbestos fibers.

Skin and Eye Contact

Unlike inhalation, skin and eye contact with toxic chemicals normally results in damage at the point of contact. Sulfuric acid will do damage only at the point of contact, although there are a few toxic chemicals that can be absorbed through the skin and into the blood stream.

Once in the bloodstream, the chemical can move to any spot and do its damage. The most common chemicals that can be absorbed through the skin are in pesticides and herbicides. Just remember, if sweat can come out of your skin, chemicals can get into it.

Information on skin absorption is provided in the ACGIH publication: Threshold Limit Values for Chemical Substances and Physical Agents, OSHA Standard **29 CFR Part 1910.1000**, and other Standard references. These documents identify substances that can be readily absorbed through the skin, mucous membranes, and/or eyes by either airborne exposure or direct contact with a liquid.

This information, like most available information on skin absorption, is qualitative. It indicates whether, but not to what extent, a substance can pose a dermal hazard. Thus, decisions made concerning skin hazards are necessarily judgmental.

Quantitative data on eye irritation is not always available. Where a review of the literature indicates that a substance causes eye irritation, but no threshold is specified, one should have a competent health professional evaluate the data to determine the level of personal protection needed for workers.

Ingestion

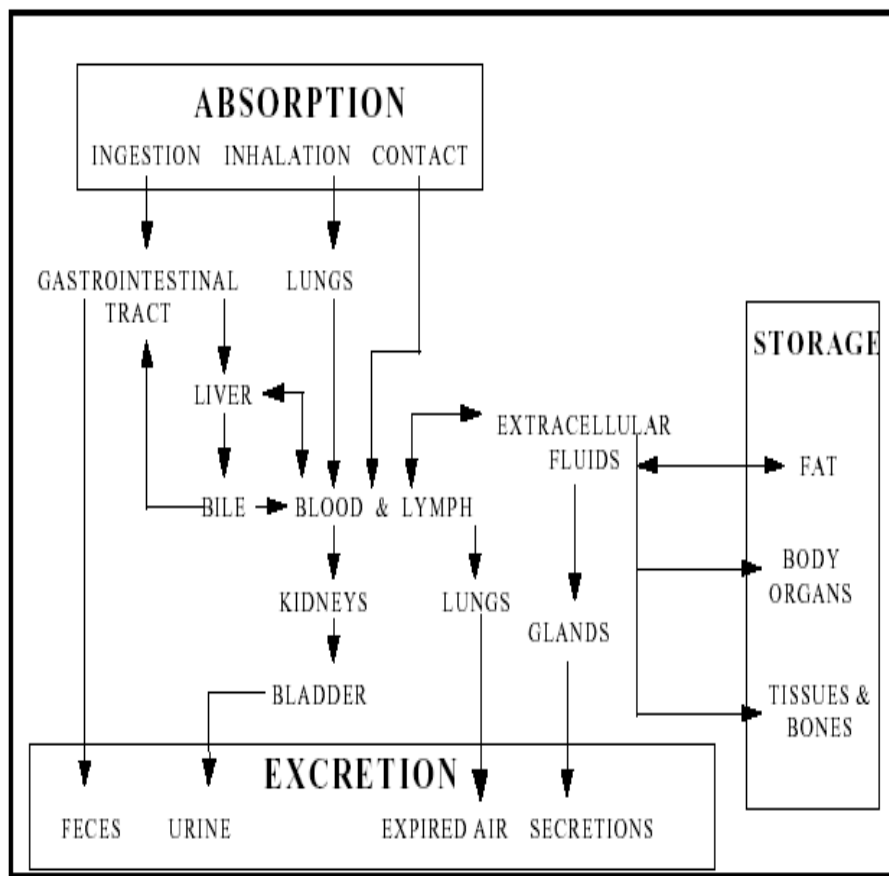
Ingestion is not a common route of transmission for industry workers unless personal hygiene is ignored or disregarded. Personnel that handle chemicals need to follow a hygiene program that will prevent the accidental ingestion of hazardous material, for example, properly washing hands before eating.

Injection

Chemicals can enter through the skin at any wound or injury site. This is commonly called injection. Injection can happen when handling a high-pressure hose with pinhole leaks because a chemical can seep through cracks in a worker's unprotected hands. Once the chemical enters the bloodstream, it has the potential to impact all organs and tissues.

The following flow chart shows chemical pathways upon entry of a chemical into the human body.

KEY ROUTES OF CHEMICAL ABSORPTION, DISTRIBUTION, AND EXCRETION



Many chemicals, particularly those that are soluble in oil or fat (lipids), are not removed from the body but are *stored* instead. Storage usually occurs in the adipose tissue or fat.

Chemical Exposure

Toxic chemicals can react in either an acute or chronic way.

Acute (immediate) responses are effects which occur shortly after exposure. Symptoms include headaches, dizziness, nausea and eye, skin, or respiratory damage. An acute response may cause unconsciousness or even death.

Chronic (delayed) responses are effects that occur a long time after exposure. Frequently, they are not reversible. Common organs affected are the liver, kidney, and lungs.

Chemical Dose

Whenever the term toxicity is used, dosage is generally indicated. Dosage is related to the quantity of a material and length of exposure. The higher the chemical dose, the greater the toxic reaction.

A toxic dose is often expressed as a lethal dose (LD), or lethal concentration (LC). A number given as an LD₅₀ or LC₁₀₀ refers to the specific amount of a particular material that results in a percentage of deaths of a sample group of laboratory animals. For example, an LC₅₀ of 25 mg/m³ means that a concentration of 25 mg/m³ of a certain substance is shown to have been fatal to 50 percent of a test group of animals.

Please note that the smaller the number (toxic dose) the more toxic the material.

Consider this example while referencing the table above. A 175-pound person given three drops (i.e., less than 1.0 mg/kg) of a dangerously toxic material can die. While three drops of a toxic material should not cause the death of a 175-pound person (dependent on personal response).

Toxic Concentration	
Quantity	Effect
<1.0mg/kg	Dangerously Toxic
1-50 mg/kg	Highly toxic
50-500 mg/kg	Toxic
500-5000 mg/kg	Moderately Toxic
5-50 g/kg	Low Toxicity

Highly Toxic

A chemical that is highly toxic has a LD50 (lethal dose 50%) of 50 milligrams of chemical per 1 kilogram of body weight administered orally; LD50 of 200 milligrams or less per kilogram of body weight when administered by continuous skin contact for 24 hours; LC50 (lethal concentration 50%) in air of 200 ppm by volume or less when administered by continuous inhalation for one hour or less; or 2 mg/kg per liter or less of mist, dust, or fume when continuously inhaled for one hour or less.

Toxic

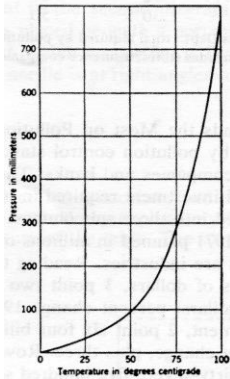
A chemical that is toxic has a LD50 (lethal dose 50%) of more than 50 milligrams/kilogram but less than 500 mg/kg of chemical per 1 kilogram of body weight administered orally (see chart on page 2-2); LD50 of more than 200 milligrams per kilogram of body weight but less than 1000 mg/kg when administered by continuous skin contact for 24 hours; LC50 (lethal concentration 50%) in air of more than 200 ppm by volume but less than 2000 when administered by continuous inhalation for one hour or less; or more than 2 mg/kg per liter but less than 20 mg/kg of mist, dust, or fume when continuously inhaled for one hour or less.

Chemical Interaction

When testing chemicals in the laboratory, toxicologists have learned that many chemicals act together in different ways on different biological systems. It is for this reason that $2 + 2$ does not always equal 4.

Additive Effect ($2 + 2 = 4$)

Some toxic chemicals add their effects together in producing a biological effect. In this case the effect is the same as being exposed to double the dose of either chemical alone. Example: acetaminophen and ibuprofen.



Synergistic Effect ($2 + 2 = 6$)

Synergism is the exposure to two different toxic chemicals that produce a more severe effect than simply doubling the dose of either one alone. An example is isopropyl alcohol and chloroform. The alcohol ties up the enzymes that would normally break down chloroform.

Potentiation ($0 + 2 = 10$)

In some cases a chemical without any known toxic effect can act together with a known toxic substance to make the toxic substance even more potent and thus more dangerous. Ethanol (ethyl alcohol) and chloroform together affect the liver in just such a manner.

Antagonism ($4 + 6 = 8$)

The interaction of two toxic chemicals can effectively produce less than the expected result, for example when combining Phenobarbital and benzopyrene. The Phenobarbital increases the enzyme activity that detoxifies the benzopyrene.

NOTE:

A prescription drug being taken by a worker may interact with a hazardous chemical encountered in the workplace. Therefore, a doctor needs to know that a worker can be exposed to specific chemicals on the job.

Individual Response

LD50 numbers represent the average response for a population because there is a great deal of variability caused by genetics. In other words, given the same dosage of a material, one worker's response can be at the opposite extreme of another worker's response.

Some individuals experience less of a response while others experience a greater response. To afford the greatest protection to the average worker, "exposure limits" have been developed. An exposure limit is the airborne concentration of a material to which nearly all individuals may be repeatedly exposed without adverse health effects.

Measurement

The two common units of measurement used in setting exposure limits are:

- Parts per million (ppm) or parts per billion (ppb)
- Milligrams per cubic meter of air (mg/m³)

The formula for converting ppm to mg/m³ is:

Molecular weight x ppm

mg/m³

The formula for converting mg/m³ to ppm is:

Mg/m³ x 24.45

Molecular weight

Ppm and ppb are volume to volume and mg/m³ is weight to volume.

Some examples to help imagine parts per million are:

- An ounce of chocolate in a million gallons of milk
- A drop of vermouth in a railcar full of vodka

An example of parts per billion is:

- One grain of sugar in a 10 pound bag of sugar.

For vapors or gases the constant of 24.5 liters vapor per mole of contaminant at 25° C and one atmosphere (760 mm Hg) of pressure is important. Some common examples of this constant are:

- Chlorine 1 ppm = 2.95 mg/m³
- Toluene 1 ppm = 3.83 mg/m³
- Trichloroethylene 1 ppm = 5.46 mg/m³

Exposure Limits

Published exposure level standards have been determined experimentally and in the workplace.

Exposure limits are expressed as permissible exposure limits, or PELs. PELs are enforceable standards promulgated by OSHA. In many cases they are derived from threshold limit values, or TLVs, published in 1968. The PEL for a substance is either an average exposure figured over an 8-hour work day (known as TWA or time weighted average) or a ceiling concentration (C), above which workers can not be exposed. Although personal protective equipment may not be required for exposures below the PEL, its use is advisable where there is a potential for overexposure.

Threshold Limit Value (Tlv)

The American Conference of Governmental Industrial Hygienists (ACGIH) publishes Threshold Limit Values for Chemical Substances and Physical Agents annually. The ACGIH has derived TLVs for many substances. TLVs are developed as guidelines to assist in the control of health hazards; for example, they may be used to determine the appropriate level of worker protection. TLVs are intended for use in the practice of industrial hygiene, not for use as legal standards. Rather, OSHA's PELs are the enforceable standards. TLVs are generally adopted as best practice by employers and are sometimes more restrictive than PELs.

The ACGIH defines three categories of TLVs:

- Time-weighted average (TWA)
- Short-term exposure limit (STEL)
- Ceiling (C)

All three categories can be useful in selecting levels of protection at a hazardous waste site. Refer to the Threshold Limit Values for Chemical Substances and Physical Agents publication for additional details.

Time-Weighted Average (TWA)

TWA is a dose measurement that is more chronic in nature. It is a time-weighted average concentration for a normal 8-hour day/40-hour week, to which nearly all workers can be repeatedly exposed, day after day, without adverse effect.

Short-term exposure limit (stel)

ACGIH defines this as a 15-minute TWA exposure that should not be exceeded at any time during the workday without the proper protective measures. Exposures above the TLV-TWA up to the STEL should not be longer than 15 minutes and should not occur more than four times per day with at least 60 minutes between exposures.

This is not a separate independent exposure limit. Instead, it supplements the TWA limit where there are recognized *acute* effects from a substance that normally has *chronic* effects.

Ceiling (C)

This is the level that can never be exceeded during any part of the working exposure without protective actions. It is not an average unless it cannot be measured any other way. Then, it is measured over a 15-minute time period.

Recommended Exposure Limit (REL)

A NIOSH recommended exposure limit, or REL, is the work-place exposure concentration recommended by NIOSH for promulgation by OSHA as a PEL. In some cases, NIOSH has described time-weighted average concentrations in terms of 10-hour, rather than 8-hour, averages. Exposures measured over any length of time can be compared to an 8-hour TWA through a mathematical formula.

Immediately Dangerous To Life and Health (IDLH)

IDLH exposure concentrations have been established by the NIOSH/OSHA Standards Completion Program (SCP) as a guideline for selecting respirators for some chemicals. The definition of IDLH varies depending on the source.

EXAMPLE:

The Mine Safety and Health Administration Standard defines IDLH conditions as those that pose an immediate threat to life or health or that pose an immediate threat of severe exposure to contaminants such as radioactive materials that are likely to have adverse cumulative or delayed effects on health.

The NIOSH Pocket Guide to Chemical Hazards defines IDLH concentration as the "...maximum level from which one could escape within 30 minutes without any escape-impairing symptoms or any irreversible health effects." The American National Standards Institute, Inc. (ANSI) defines IDLH as "...any atmosphere that poses an immediate hazard to life or produces immediate irreversible debilitating effects on health."

Regardless of their exact definition, all IDLH values indicate those concentrations of toxic substances from which escape is possible without irreversible harm should a worker's respiratory protective equipment fail.

IDLH concentrations should be assumed to represent concentrations above which only workers wearing respirators that provide the maximum protection (i.e., a positive pressure, full-face piece, self-contained breathing apparatus (SCBA) or a combination positive-pressure, full-face piece, supplied-air respirator with positive-pressure SCBA) are permitted. Specific IDLH values for many substances can be found in the [NIOSH Pocket Guide to Chemical Hazards](#).

Protection from Toxins

How do we protect ourselves from toxins? The human body can break down, detoxify, or eliminate many harmful chemicals if the dose is not too great. At the same time, exposure to toxins puts stress on the body.

Protect Yourself

You must know the potential for toxicity of the materials that you work with to adequately protect yourself from toxic exposure. Once the presence and the concentrations of specific chemicals or classes of chemicals have been established, the associated hazards should be determined.

Information on the chemical, physical, and toxicological properties of each chemical should be recorded on a Hazardous Substance Information Form. Health and safety personnel should then make the necessary information available in one place. New personnel should be briefed.

As many reference sources as possible should be used to fill out the sheets, because the information may vary from one source to another.

As data is gathered, a worst-case scenario should be developed so all necessary precautions may be taken. Safety data sheets (SDSs) provided by chemical manufacturers are one source for this information.

Protection Against Hazards

When the hazard is known, the hard part is over. One cannot protect from the unknown, but one can protect against the known hazard by doing the following:

1. Use the engineering controls provided, such as ventilation systems.

2. Use administrative controls.
3. Use common sense around chemicals. However, many chemical hazards can't be detected using only common sense, so never rely solely on it.
4. Select and use protective equipment based on SDS and safety officer recommendations.
5. Wash exposed areas thoroughly before eating, drinking, or smoking.

Lesson Summary

Toxicology is the study of chemical toxicity. Toxicity is the degree to which a chemical or substance has a harmful effect on us or other living things. The information for toxic effects of chemicals and other hazardous substances have been gathered from: Epidemiological studies and Animal studies. Some of the factors that affect the response to chemical exposure include: Toxicity, Route of exposure, Chemical dose, Human response and Interaction with other chemicals.

Knowing how chemicals get into your body and how your body reacts is critical to personal protection. Toxic chemicals can enter the body in any one of, or in a combination of, four ways.

1. The most common toxic dose in the work environment comes through breathing, or inhalation.
2. The next most common route of entry is through eye and/or skin contact.
3. The most common route in the home is swallowing or ingestion.
4. A route that is more accidental is injection.

Published exposure level standards have been determined experimentally and in the workplace. Exposure limits are expressed as permissible exposure limits, or PELs. PELs are enforceable standards promulgated by OSHA.

To protect yourself from known hazards of toxics, ensure that you:

1. Use the engineering controls, such as ventilation systems.
2. Use administrative controls.
3. Use common sense around chemicals. However, many chemical hazards can't be detected using only common sense, so never rely solely on it.
4. Select and use protective equipment based on SDS and safety officer recommendations.
5. Wash exposed areas thoroughly before eating, drinking, or smoking.